

SIEMENS

MAGNETOM

TD

Numaris - Service

Online - Help

TestTools

syngo MR 2004A

Fachliche Prüfung

Freigabe

Name/Abtlg.:

Name/Abtlg.:

Datum:

Datum:

Unterschrift:

Unterschrift:

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English

Doc. Gen. Date: 09/04

Version	Date	Author	Change and Reason of change
1.0	13.01.00	A. Orth	First edition
2.0	12.10.00	H. Appold	Release for Num/4 VA12A & VA12B
3.0	08.08.01	H. Appold	Release for Num/4 VA15A Disclaimer added and english correction.
4.0	08.02.02	H. Appold	Release for <i>syngo</i> MR 2002B (Num/4 VA21A) Added description for new tests and corrections, editorial changes.
5.0	14.02.03	H. Appold	Release for <i>syngo</i> MR 2003T (Num/4 VA23A) Updated description for tests and corrections, editorial changes.
6.0	17.09.04	H. Appold	Release for <i>syngo</i> MR 2004A (Num/4 VA25A) Added description for tests and corrections, editorial changes.

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General

Disclaimer

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Document revision level

The document corresponds to the version/revision level effective at the time of system delivery. Revisions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order current revision levels.

TestTools function

TestTools combines all hardware tests into a fully automated procedure which means that a full system function test can be done, including stability and linearity tests. However, you have the option of selecting individual components or FRUs for the test run. Testing is FRU-oriented, i.e. if possible the result of the test is the defective FRU only.

Test results are permanently saved to a log that can be viewed with the **Report** functions **Session History** and **Test Results**.

The TestTools platform lets you select the following:

1. Predefined test selections (problem section, left side)
2. Individual components or assemblies (component/FRU section, right side)

To save time, individual test types can be deselected, provided that the source for the error has been narrowed down to a practical level. After starting one of the test programs, a browser window with the test results (OK, FAILED, etc.) and detailed outputs including graphics is displayed. Detailed test information is readily available via HTML links. The window can be closed with the **Exit**-button in the top right or left open for further tests and can be updated with the **Update**-button in the bottom right part of the platform.

In case a test fails, a diagnostic table is displayed recommending the next steps.

NOTE	It is not possible to run any test and even not a normal sequence in case the licences for the Imager, GPA and the RFPA are invalid or missing. If it is not possible to start any of the tests, please check this also.
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NOTE	Please ensure that the system is switched on for at least 20 minutes before running any functional or interactive test to stabilize the temperature in the control cabinet.
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NOTE In case you have doubts if a test is correctly running or whether there is a general problem with that test, please have a look to the knowledge database (<http://www.tdkb.med.siemens.de/>) for further information. Usually the interesting entries there start with "TestTools" so it is possible to search for this word.

Additionally there is a general entry in the knowledge database with every new software release. The entry has the character of a speed-info and contains a hint list and an attached Excel-list (downloadable) showing all known bugs and errors.

And finally I can be contacted by mail: horst.appold@siemens.com.

NOTE If the test is aborted with "No results available" it is possible that the source of the problem is not the tested FRU. Please have a look to the event log for more details.

NOTE In case the system is equipped with a multi-nucleus option (spectroscopy, X-nucleus), also these tests are performed when a RF test is selected. If this is not intended or necessary please do not forget to de-select the test type "Multi Nuclei".

Everytime when such a test is to be performed it is necessary to select the nucleus/frequency prior to start the test. If the desired nucleus is once set it will stay until it is changed again. This is important to know during trouble-shooting.

NOTE After the end of the test run with some tests an **Expert_TestReport** is offered for further information. You will find additional error messages there.

When you return MMC FRUs, please attach the corresponding **Expert_TestReport** to the defective FRU. This log holds information that is of great value to the repair technicians in the factory.

In general all available information (error description, eventlog, etc.) are interesting for the factory and will help to improve the quality of the spare parts in stock.

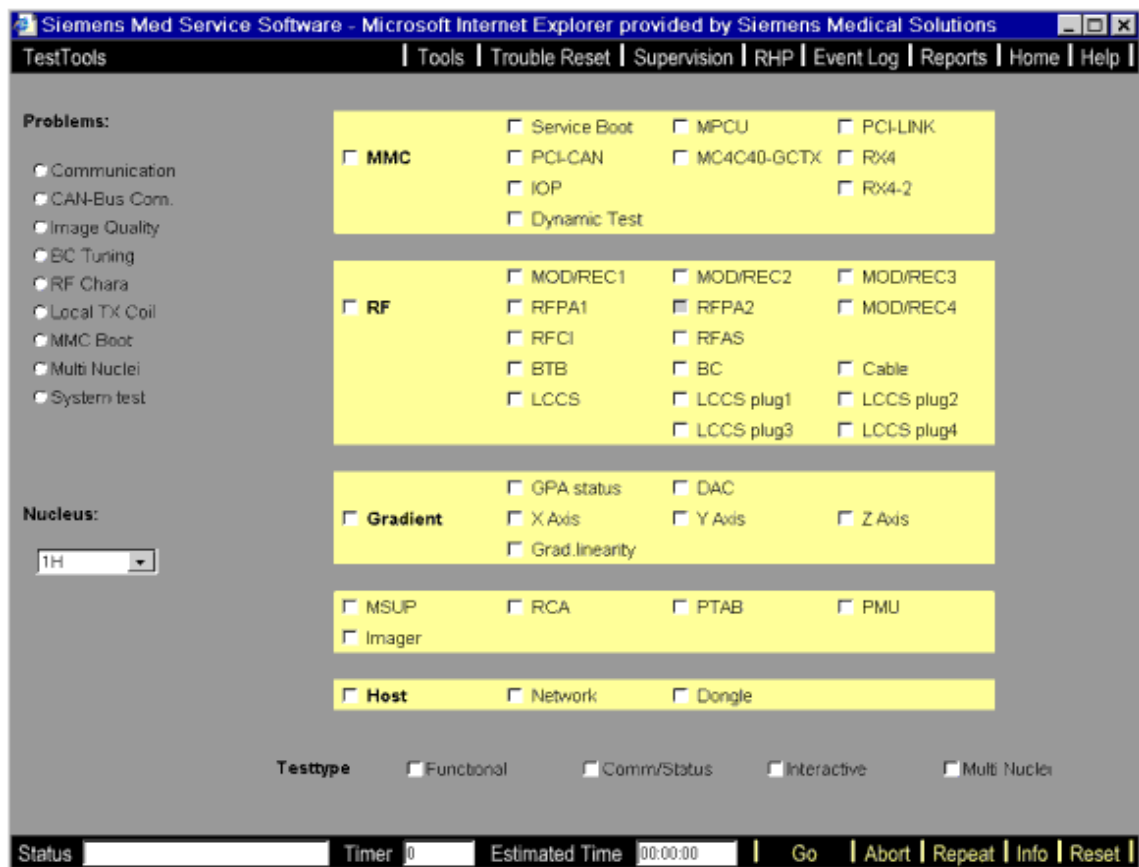
NOTE While running RF-tests or any test that is using the rf_pulse-sequence usually also the gradient system pulses with low amplitudes. This can be noticed by the normal "knocking" of the system and is normal.

In case you have problems both with RF and the gradient please keep this in mind as in that case the RF-tests might not work. The workaround is to change the supervision for the gradient system.

NOTE Many measurement results are internally stored as float values but displayed as integer. This may cause irritations in tables and graphics.

User Interface

Structure and Elements



The TestTools platform consists of a **Navigation Bar** (top), a **Function Frame** (center) and an **Action Bar** (bottom).

Navigation Bar

The **Navigation Bar** has eight buttons:

- **Help**

Help provides general information about the TestTools platform.

- **Home**

The Home button returns you to the Service Home menu. This is the only way to close the TestTools window safely. The close icon of the window should not be used, otherwise there is the danger that running test programs cannot be terminated anymore and the host has to be rebooted to be able to start the TestTools again.

• Report

The Report button opens a separate platform. You can select the corresponding output log file of the last system tests and step through all data files.

• Event Log

The Event Log button opens a separate platform where you can select information from the error logger. Each message ID has a hypertext link to an **Extended Error Message** explanation providing more information with regard to recommended actions.

• RHP

The RHP button opens a separate platform for generating and evaluating raw data.

• Supervision

Opens a new browser platform for modifying the supervision status of the CAN-components.

Unit	Supervision		Configuration		
	ON	OFF	Real	Virtual	Num
GPA	☒	☐	☒	☐	☐
MSIV/SIIM	☒	☐	☒	☐	☐
PTAB	☒	☐	☒	☐	☐
RCA	☒	☐	☒	☐	☐
RI CI	☒	☐	☒	☐	☐
BTB1	☒	☐	☒	☐	☐
BTB2	☒	☐	☒	☐	☐
IUI*	☒	☐	☒	☐	☐
RFAS	☒	☐	☒	☐	☐
RFPA	☒	☐	☒	☐	☐
TRA1	☒	☐	☒	☐	☐
PMU	☒	☐	☒	☐	☐
PCI INK	☒	☐	☒	☐	☐

Setting will be temporary saved, it will be returned to default configuration after Numaris restart

Save Settings Default

The supervision status should be changed whenever individual components should not be monitored during trouble shooting.

- **ON:** The component is supervised and the error messages from the peripheral units are monitored (normal setting).
- **OFF:** The error messages from the peripheral units are not monitored, but the data transfer and the acknowledge from the component is still supervised. Use this mode to suppress incoming error messages from the peripheral units.
- **Real:** The component is installed and supervised (normal setting).

- **Virtual:** The component is treated as existent, but the data transfer to and the acknowledge from the component is only simulated. This mode should be used for troubleshooting when a test measurement must be started without supervision of a certain component.
- **None:** The component is treated as non-existent by the software.

Clicking the **Default** button displays the default configuration. Click **Save Settings** to activate the settings temporarily. The default configuration will be restored after *syngo* MR is restarted.

• Trouble Reset

The Trouble Reset button opens a separate platform where you can restart the Host, the Imager and the Scanner. This is mainly interesting for Remote Diagnostics as the normal System Control is not reachable when not sitting at the user console.

• Tools

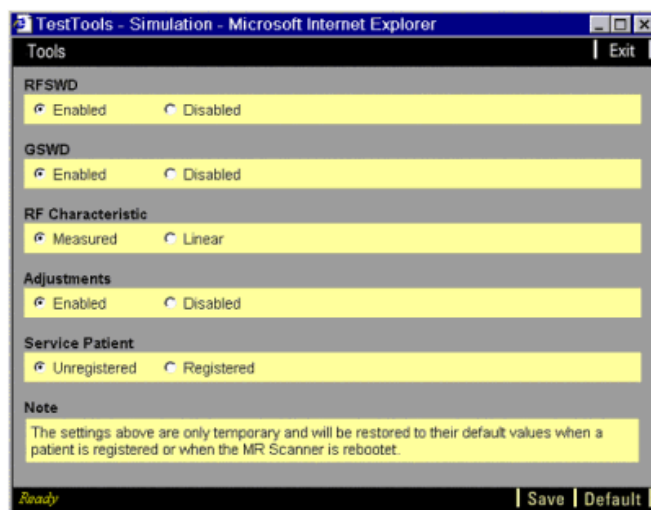
NOTE

It is possible that the message "Invalid value for Adjustments mode: Othermode" is displayed when entering the platform for the first time during a session. This "error message" is correct as besides the listed modes other modes exist for the Service Software but neither can be displayed nor selected.

Please set the mode according to your needs.

The setting of the RFSWD (enabled/disabled) has no influence in the TestTools measurements as the tests already check the necessary prerequisites.

The Tools button allows access to some special settings of the measurement system, previously possible only via separate passwords and using the operating system level. In this platform the watchdog programs for RF (RFSWD) and the gradient (GSWD) can be enabled or disabled. The same can be done for the adjustments. To change these settings might be very helpful during trouble-shooting. Additionally the RF characteristic can be toggled between the measured characteristic and the linear one, but this will affect only a few tests. And finally it is possible to register or unregister the Service Patient so that it is not necessary to run any test or QA to have this patient registered for further trouble shooting or service.



Action Bar

The **Action Bar** contains the following fields and buttons:

- **Status**



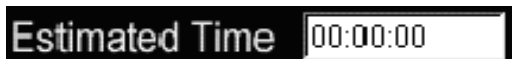
In this field the status of the currently running test is displayed. In this way the user always knows at which point the test is, even if nothing is displayed in the output window of the test.

- **Timer**



The timer is used while running measurements. In that case the remaining run time of the measurement is displayed. The time is only count down if the measurement time exceeds 10 seconds. Otherwise the timer is not started at all.

- **Estimated Time**



The Estimated Time field shows the time the system will need to perform the tests for the components and tests selected.

NOTE Especially with interactive tests this time can be not very precise as the time the user needs for the interactions cannot be calculated or determined. In that case the given time is a minimum time and can be longer when the test is really performed.

- **Go**

Clicking the Go button starts the selected tests and opens an output window.

- **Abort**

The Abort button is used to abort the tests or measurements in progress.

NOTE It may take a while until the tests are really stopped (approx. 50 seconds)- be patient.

Do not abort the MPCU tests, the MMC may remain in an undefined state. This can happen also with other tests, depending on which test step is aborted. In that case a reboot of the Scanner is recommended and necessary. Extremely dangerous is an abort of the imager tests. The imager may stay in a simulation mode where only a power off can help.

- Repeat



The Repeat button is used to set the number of test cycles. When selecting Repeat, a menu is displayed where you can set the repetition number, the default-value is always 10. Click the **OK**-button after entering the repetition number.

NOTE The repetition number is valid for one test run only and has to be set again before the next "Go".
Additionally interactive tests are performed only one time.

- Info

Click on the Info button to get a list of the individual test steps including the diagnoses for the selected tests.

If no test is selected a list of all available tests and their possible diagnose texts is generated.

The list itself always starts with a table of the tests and HTML-links to the detailed information later in the file.

- Reset

Reset deselects all settings in the platform.

NOTE With *syngo* MR 2002B (VA21A) the newly introduced selection for the nucleus is not reset to "1H" (hydrogen). Please pay attention to this when running RF tests.
This selection is only displayed when a multi-nucleus option (spectroscopy, X-nucleus) is installed.

Function Frame

The **Function Frame** is mainly divided into a **Problem section** (left) and a **Component or FRU section** (right):

- **Problem section**

The problem section includes predefined test sequences and the nucleus selection in case the system is equipped with a multi-nucleus option (spectroscopy, X-nucleus).

Predefined tests

With the predefined tests with a single click, e.g. the communication test and status request is performed for all components.

The following problems are available (depending on the system and the system configuration):

Communication	All communication tests will be performed.
CAN-Bus Com.	The CAN-nodes will be polled in their physical sequence in the bus-chain (refer to the test description for PCI-CAN -> Communication)
Image Quality	All functional and interactive tests for the RF- and gradient-FRUs will be performed.
BC Tuning	All functional and communication tests for the main Transmit-/Receive-path will be performed (Modulator, Receiver 1, RFPA, LCCS, RFAS, BTB and BC).
RF Chara	All functional and communication tests for the main Transmit-path will be performed (Modulator, Receiver 1, both RFPAs if available, RFCI, RFAS, BTB and BC).
Local TX Coil	All interactive and communication tests for the RFAS and RFCI will be performed.
MMC Boot	Only the functional Serviceboot test is selected, not the interactive part.
Multi Nuclei	All RF tests that are capable of other frequencies than that one for ¹ H will be performed (Modulator, Receiver, RFPAs, RFCI, RFAS and cable).
System test	All functional and communication tests of the system will be performed. Note that although all FRU's are selected, not for all of them tests exist.

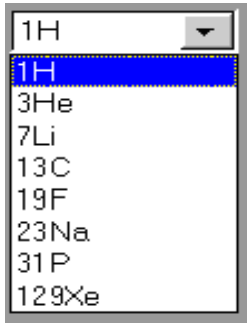
The last option is actually not a "problem": **System test**. With this button easily a full system test **without** the interactive or multi-nucleus tests is selectable, but only communication/status and functional tests are performed. If the interactive or the multi-nucleus tests are also necessary please select the test type **Interactive** or **Multi Nucleus** in the Test Types section accordingly.

Nucleus selection

Additionally in case the system is equipped with a multi-nucleus option (spectroscopy, X-nucleus) the nucleus which has to be used for the next RF tests can be selected in the problem section. Every nucleus corresponds to a certain frequency which will affect e.g. the automatic selection of the correct RFPA. The nuclei ^3He , ^{19}F and of course ^1H are using the DORA-amplifier, all other nuclei the second RFPA CORA (^7Li , ^{13}C , ^{23}Na , ^{31}P and ^{129}Xe) which is also used in the MAGNETOM Concerto.

With Trio and Allegra it is different as different amplifiers with different frequency ranges are used.

Generally (but not shown in this picture) the used RFPA (RFPA1, RFPA2) is displayed in the pull-down menu to ease the selection of the correct FRU.



NOTE The nucleus selection is only displayed when a multi-nucleus option (spectroscopy, X-nucleus) is installed.

- **Component/FRU section**

This section lets you select the assemblies to be tested. By selecting "parent" groups, e.g. MMC or RF, all components allocated to this group are automatically selected. Each individual component can also be selected or deselected.

The different tests are executed automatically by selecting one or several components. To save time, you can deselect test types as well.

NOTE You have to select a component first before you deselect one of the test types. This ensures that the component remains selected even if not all tests are selected anymore. However, if you retroactively change your selection of components, the types of tests selection will be reset.

Additionally a part of the Component/FRU section can be considered as

- **Test Types section**

After selecting one or more components, the tests available are automatically selected. If the error source has been narrowed down (e.g. communication problem) or if no interactive or multi-nucleus tests should be performed (e.g. Remote Access), you can deselect test types.

Four different test types are available for each component: Function, Communication/Status, Interactive and Multi Nuclei tests.

The Multi Nuclei tests are only available when a multi-nucleus option (spectroscopy, X-nucleus) is installed.

NOTE	First you have to select a component before you deselect one of the test types. This ensures that the component remains selected even if not all tests are selected anymore. However, if you retroactively change your selection of components, the types of tests selection will be reset.
-------------	---

Output Window

Structure

After starting one of the test programs, a browser window with the test results (OK, FAILED, etc.) and detailed outputs including graphics is displayed. Detailed test information are readily available via HTML links.

The **Output Window** is subdivided in two frames: Index frame and Result frame.

- **Index Frame**

The **Index Frame** contains a kind of index of all executed tests, including their results (OK, FAILED, etc.). With HTML links it is possible to jump to the detailed information in the result frame.

In case a test fails, a diagnostic table is displayed recommending the possible error source and next steps.

- **Result Frame**

Here the complete output information of the tests are displayed, including graphics, if available.

Sometimes also even more detailed test information are available by offering an additional HTML link. These information are very helpful e.g. for the factory.

Elements

The **Output Window** contains the following buttons:

- **Update**

The test results are displayed automatically at the end of the tests. Use the **Update**-button in the bottom right part of the platform to display the currently available results during the test run. An automatic update is only done at the end of the tests. Scrolling inside the window while the tests are running is allowed without any restriction.

- **Exit**

After the end of the tests in the top right corner a second button, the **Exit**-button.

The window can be closed using this button. It can be also left open for further tests and can be updated with the **Update**-button.

NOTE	Printing in the browser is possible by pressing <ctrl>-p.
-------------	---

Changes with VA25A

Structure

- The internal structure was changed by not using HTML anymore but XML. Thus it was possible to harmonize the output with Tune-Up/QA and to store all data in a XML-database which can be transferred to Erlangen for further evaluation (e.g. statistics). The disadvantage is that some of the well-known outputs are not available anymore (minor impact).
- Instead of using multiple pop-up's at the beginning and the end of the test they are now merged. Additionally they are automatically managed to reduce also the number of confirmation windows but mainly the number of changes at the system that are prerequisites for the tests (interactive tests). A pop-up that told you to reconnect a cable and one that tells you to disconnect the same cable again is history for two subsequent tests.

User interface and Tests

- The Serviceboot test was moved to the beginning of the MMC test section. This test is a prerequisite for almost all other test and it makes no sense to perform this test last.
- The FRU's MC4C40 and GCTX are merged as they are no longer as separate spare parts available.
- The RCA is now a separate FRU, instead of being part of the MSUP-test (communication only).
- Problem section: CAN-Bus communication - a new collection of old tests, checking the communication to the CAN-nodes in their physical sequence, first bus 1 and then bus 2.
- Nucleus selection: now also the used RFPA is displayed, the other RFPA is dimmed.
- Host Dongle: a new test that checks the dongle of the host.
- RFAS tests: it was notified that the directional coupler connections between LCCS and BCCS and not checked at all. In case of problems with the tuning of the system no helpful tests were available. This was changed by extending the already existing tests within RFAS to what was necessary.
- LCCS interactive: to reduce the trouble-shooting time the tests were split into the four coil sockets. Additionally the linearity and stability tests of socket 4 was moved to socket 1, which hopefully will reduce the number of re-seating the Service Plug.
- The Multi-Nucleus tests for Trio and Allegra were implemented.
- Minor changes in many other tests.

Components

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MMC

NOTE This section describes all MR systems.

This section describes the MMC hardware tests. The MMC tests allow you to diagnose the FRUs of the Modular Measurement Control: MPCU, PCI-Link, MC4C40-GCTX, MC1C40, RX4 (RX4-2), PCI-CAN, and IOP. The testing is FRU-based, i.e. the tests results in a list of FRUs with possible defects.

NOTE When you return MMC FRUs, please attach the corresponding **Expert TestReport** to the defective FRU. This log holds information that is of great value to the repair technicians in the factory.

The total test time for the MMC diagnostics is approximately **10 minutes**.

General

To diagnose the MMC components, select the MMC **checkbox**. Clicking **Go** will execute all available MMC tests. The tests are performed in a logical sequence, allowing for time-efficient testing and correct test results. If a test fails and the tested function is essential for subsequent tests, the test run is **stopped**.

NOTE Since the test sequence matters, test results are only valid when all MMC tests are selected for the test run.

However, you can also select individual FRUs for testing. If you choose to do so, make sure that the prerequisites are fulfilled.

The test results including a list of FRUs with possible defects will be displayed in a separate window. In addition, the MMC test results are permanently saved to a log file that can be viewed using the **Reports** function.

Service Boot

MMC Test Service Boot

The function enables you to boot the following FRUs step by step automatically:

- MPCU (PCI-Link)
- MC4C40-GCTX
- MC1C40
- RX4 (and RX4-2 if available)

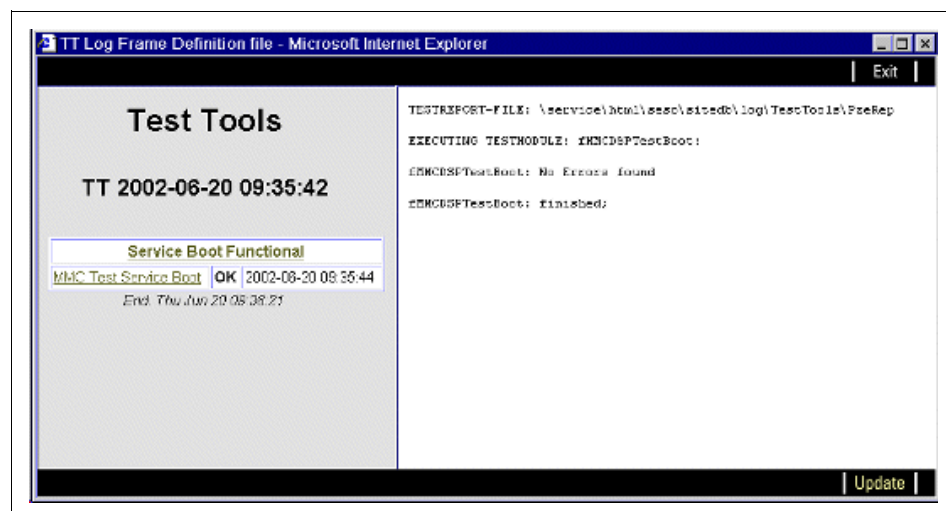
It can be started from MMC/Service Boot or from Problems section/MMC Boot.

NOTE **Start the interactive service boot test only when the previously run functional test failed. Otherwise the interactive test will always fail and show unreliable results.**

NOTE After the end of the test please have a look to the **Expert_TestReport** for further information. In case of errors you will find additional error messages there.

After the test, the result is presented in the test report.

Fig. 1 Result Service Boot - No errors



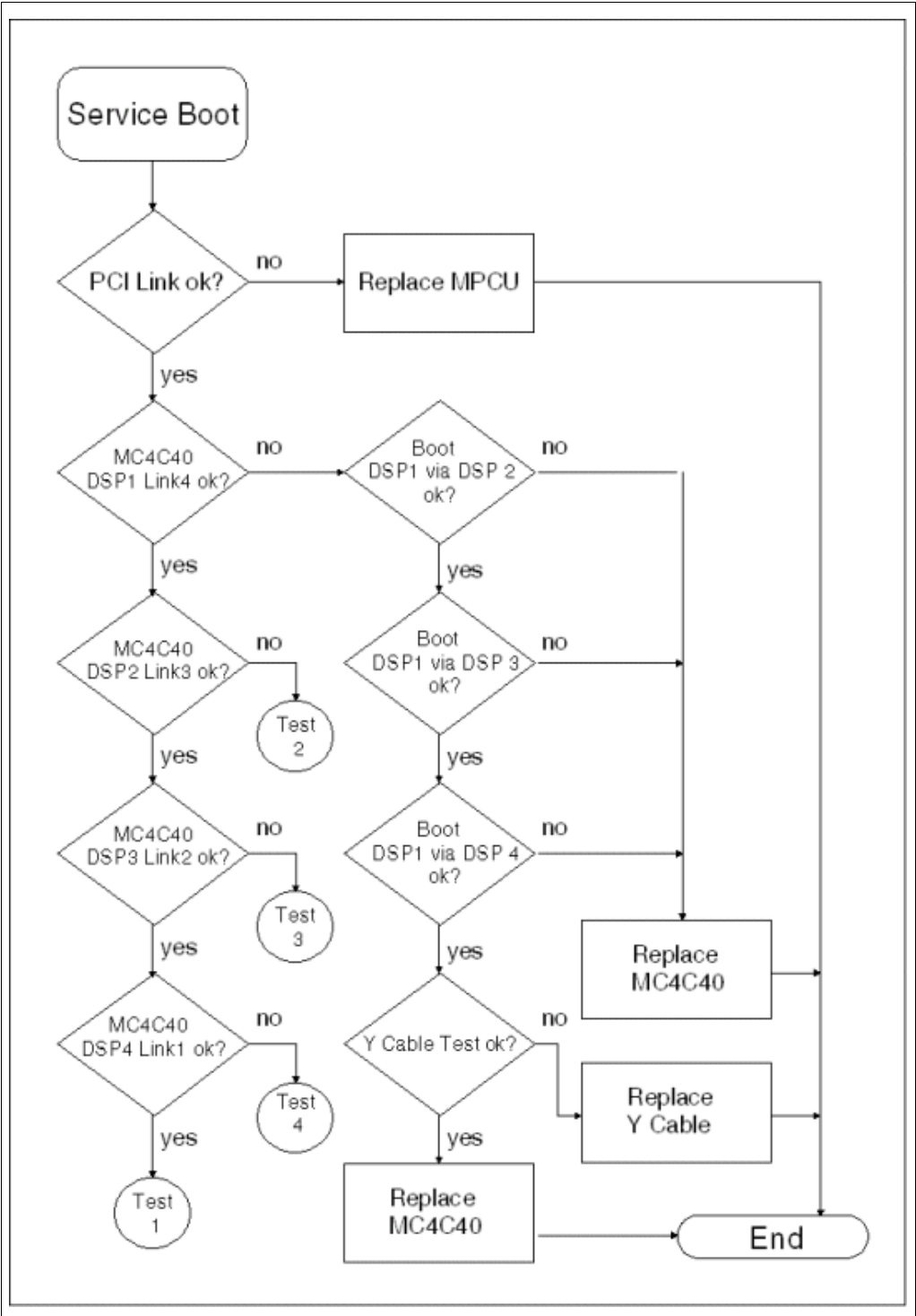
Application of other MMC tests make sense only after a successful boot of the FRUs listed above.

The following flowcharts show the test steps of the Function **Service Boot** for identifying the defective FRU.

If an error occurs during booting a DSP, an attempt will be made to boot the actual DSP via the other DSPs. If an error will be detected now, you have to replace the MC4C40.

If it is possible to boot the actual DSP via the other DSPs, you will be asked to start the interactive Y-cable test.

Fig. 2 Service boot - Test step 1



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Fig. 3 Service boot - Test step 2

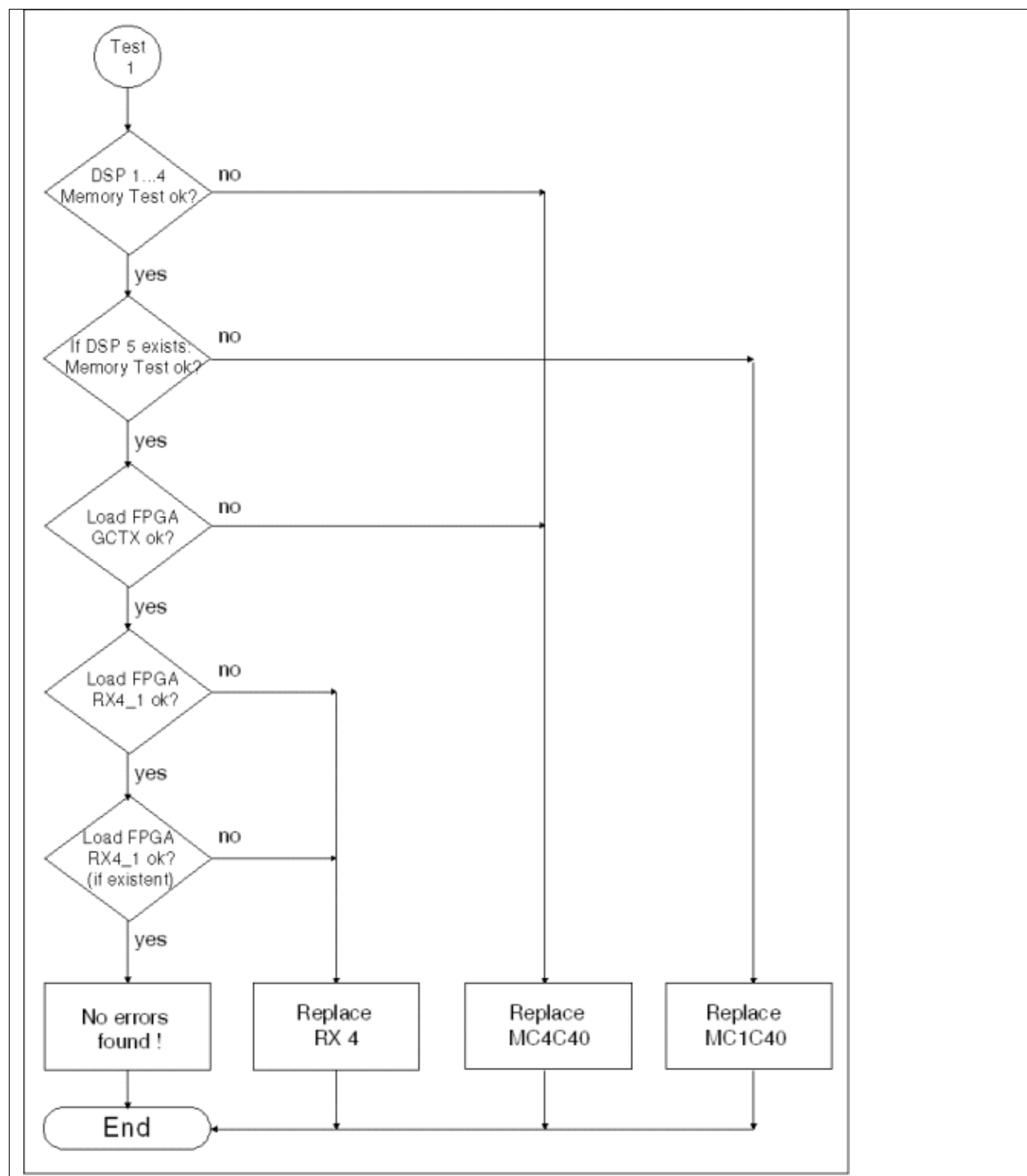
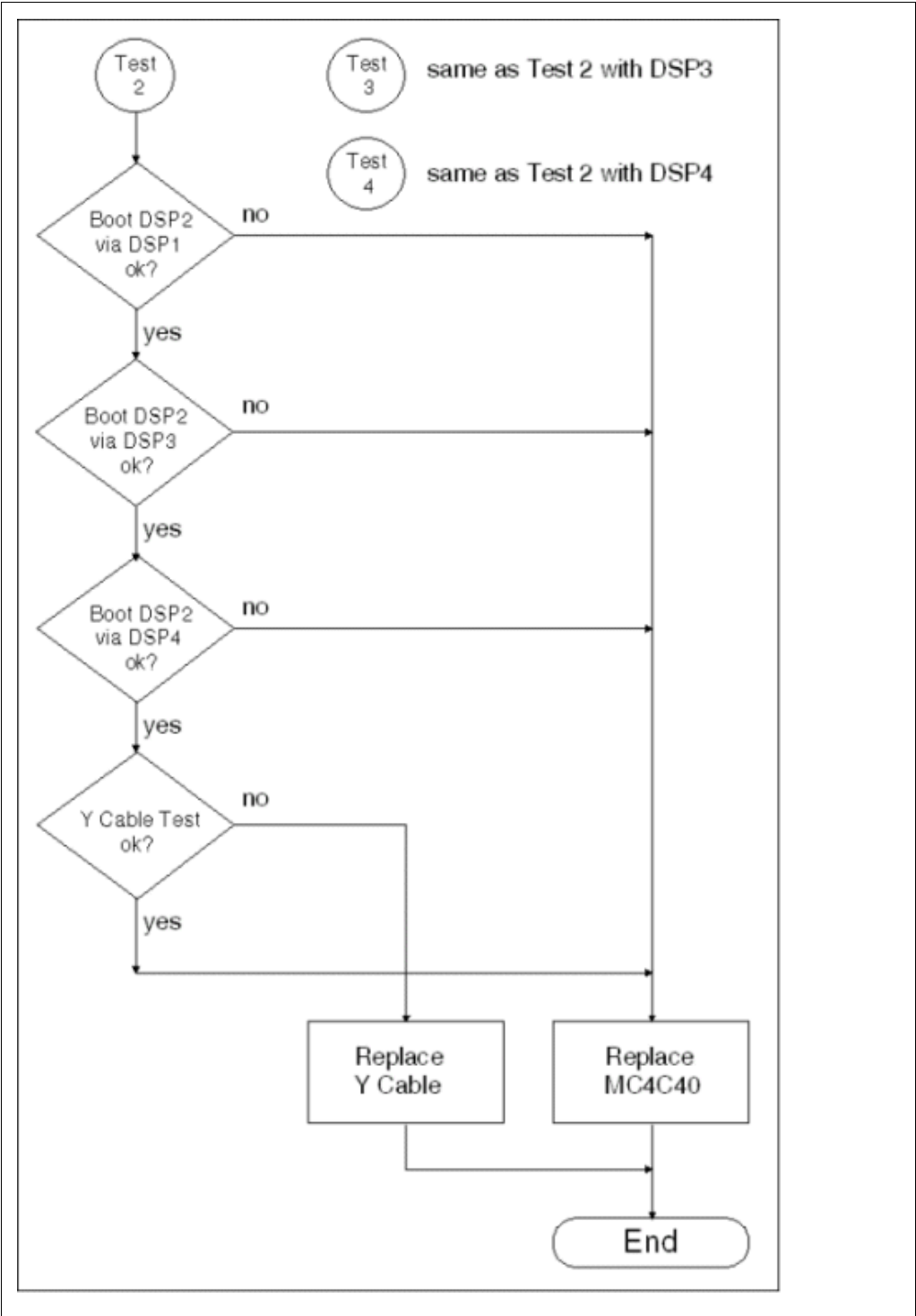


Fig. 4 Service boot - Test step 3



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Y-cable test

NOTE Start the interactive service boot test only when the previously run functional test failed. Otherwise the interactive test will always fail and show unreliable results.

A popup window (Fig. 5) gives you the necessary instructions.

If the Y-cable test fails you have to replace the cable.

Fig. 5 Popup window for Y-cable test (example)

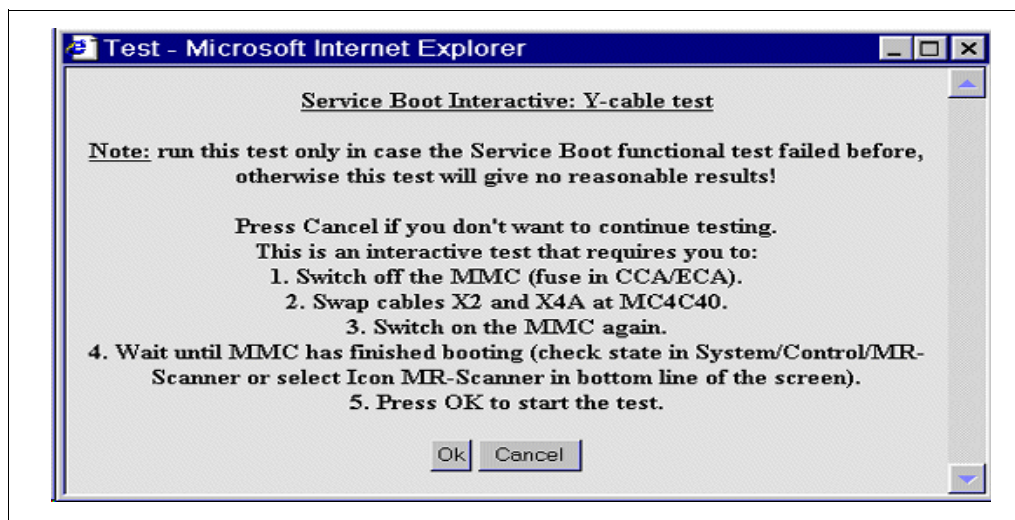
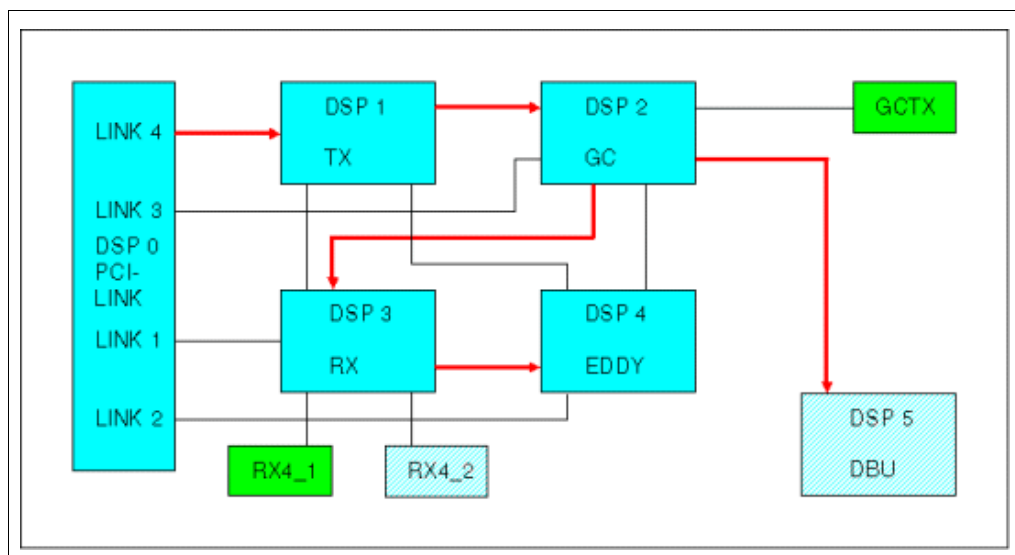


Fig. 6 shows the different connections of the DSPs and an example of a boot via DSP 1. On a similar way other links are used automatically for booting; on this way the defective DSP can be identified.

Fig. 6 Service Boot via DSP 1 TX



MPCU

The MPCU is tested using two communication/status tests, one functional and one interactive test.

MPCU Communication

MPCU Communication tests the communication between host and the MPCU. For this purpose, the boot status of the MPCU is requested. If this test fails, perform **MPCU Test** to find out about possible causes for the error.

Latest MPCU Selftest Status

Latest MPCU Selftest Status evaluates the boot status of the MPCU, i.e. the MPCU selftest results executed during power on of the MR system or warm restart of the MPCU:

```
MPCU_STATUS: RAM test                = ok
MPCU_STATUS: Ethernet controller    = ok
MPCU_STATUS: PCI-ISA bridge         = ok
MPCU_STATUS: Realtime clock        = ok
MPCU_STATUS: Parallel port         = ok
MPCU_STATUS: COM1, COM2            = ok
MPCU_STATUS: ISA bus               = ok
MPCU_STATUS: PCI bus              = ok
```

Communication from the host to the MPCU is a prerequisite for this test. If this test fails, perform **MPCU Test** to find out the possible causes for the error.

The selftest result is also displayed on the eight LED display attached to the parallel port of the MPCU (if attached).

MMC DSP Communication

MMC DSP Communication tests the communication between the MPCU and the DSPs which is essential for all other MMC tests. Without this communication the loading of the test software is not possible which results in that no MMC-test at all will run.

Serial Interface

This functional test performs an internal loopback test to check the serial interface to the PMU.

MPCU Test

MPCU Test tests the communication between host and the MPCU. For this purpose, the boot status of the MPCU is requested. If this test fails, you will enter a guided procedure during which possible causes for the error are determined step by step. If there is no fault in the beginning of the test (communication) the test will run just like a functional test without any interaction.

When switching off the MPCU, wait approximately 3 minutes to allow the power supply to discharge. Otherwise, the selftest will not be performed when switching the MPCU on again.

If this test fails, testing is stopped and the following tests will not be executed.

PCI-Link

The PCI-Link is tested using three functional tests.

These tests require a functional MPCU. Also, PCI-Link can only be tested completely when the MC4C40, GCTX and RX4 are available.

Boot DSPs

Boot DSPs tries to boot the DSPs on the MC4C40.

If this test fails, testing is stopped and the following tests will not be executed.

PCILINK Check Link Error

PCILINK Check Link Error checks the link error mechanism of the DSPs on the MC4C40.

If this test fails, you will enter a guided procedure during which possible causes for the error are determined step by step.

PCILINK Check Reset

PCILINK Check Reset tests the reset mechanism of the DSP network on the MC4C40.

If this test fails, testing is stopped and the following tests will not be executed.

PCI-CAN

The PCI-CAN is tested using one communication, two functional and one interactive test.

PCI-CAN Communication

All CAN-nodes are polled in the physical sequence of the components using, first CAN bus 1 (awake bus), then CAN bus 2 (sleep bus).

CAN bus 1 GPA, MSUP, SHIM, RCA

CAN bus 2 (sleep bus) IOP, TX-Cassette, RFCI, RFPA, PTAB, LCCS, BTB1, BTB2 (if installed)

CAN Controller 1

The CAN controller of the first CAN bus is tested.

If this test fails, testing is stopped and the following tests will not be executed.

CAN Controller 2

The CAN controller of the second CAN bus is tested.

If this test fails, testing is stopped and the following tests will not be executed.

External Loop

The two CAN controllers on the PCI/CAN board are checked via an external loopback test. Follow the instructions to perform this interactive test.

MC4C40-GCTX

The MC4C40 is tested using twelve, the GCTX is tested using ten functional tests. These tests require a functional MPCU and PCI-Link. In addition, MC4C40 can only be tested completely when GCTX and RX4 are available. In addition, GCTX can only be tested completely when DAC/GPA and RX4 are available.

The boards MC4C40 and GCTX are no longer separately available, like in the beginning of Harmony/Symphony. They are one FRU and have to be replaced together. Thus the tests were joined.

MC4C40 Link Test

MC4C40 Link Test checks the links from the MC4C40 to the PCI-Link and all link connections on MC4C40.

If this test fails, testing is stopped and the following tests will not be executed.

MC4C40 RAM Test

MC4C40 RAM Test tests the memory allocated to the four DSPs on the MC4C40.

If this test fails, testing is stopped and the following tests will not be executed.

MC4C40 Check BoardID & Location

MC4C40 Check BoardID & Location checks the coded board ID and the board location of board MC4C40.

MC4C40 Local RX_IO Test

MC4C40 Local RX_IO Test tests the RX_IO functionality.

If this test fails, you will enter a guided procedure during which possible causes for the error are determined step by step.

RX4 Interface Test

RX4 Interface Test checks the RX4 interface of the MC4C40.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX Bus Test

GCTX Bus Test checks the data and address bus between MC4C40 and GCTX.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX FPGA IDs

GCTX FPGA IDs checks the loadware ID to make sure that the two FPGAs on GCTX have been loaded properly.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX Clock Detect

GCTX Clock Detect checks the presence of the 10, 20 and 40 MHz clock signals generated in the RFSU. This test does not check the quality of the test signals.

If this test fails, testing is stopped and the following tests will not be executed.

20 MHz Clock Test

20 MHz Clock Test checks the 20 MHz clock signal generated in the RFSU. The 20 MHz signal is essential for all subsequent tests.

If this test fails, testing is stopped and the following tests will not be executed.

10 MHz Clock Test

10 MHz Clock Test checks the 10 MHz clock signal generated in the RFSU.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX Interrupt Test

GCTX Interrupt Test checks the control signal part of the bus between MC4C40 and GCTX.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX Link Error Test

GCTX Link Error Test checks the link error signal connection from GCTX via MC4C40 to PCI-Link.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX ED Control Word 2

The GCTX-specific control signals are checked.

If this test fails, testing is stopped and the following tests will not be executed.

Gradient Pipeline Test

This is an internal test for the GCTX interface to the gradient cabinet. However, the presence of the cable connection and of board DAC/GPA may cause this test to fail. In that case, you will enter a guided procedure during which the cause for the error is determined step by step.

Check Gradient Loop

The GCTX interface to the gradient cabinet is tested. For this purpose, data is sent to DAC/GPA and read back to compare with the initial data.

GCTX B0 Signals ED to TX

The GCTX-specific control signals are checked.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX TX Gain Outs

The GCTX-specific control signals are checked.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX Signals ED to TX

The GCTX-specific control signals are checked.

If this test fails, testing is stopped and the following tests will not be executed.

Load FPGA RX

It is checked whether the RX FPGA can be loaded.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX NCOs

It is checked whether the RX FPGA can be loaded.

If this test fails, testing is stopped and the following tests will not be executed.

GCTX NCO & Modulator

The GCTX-specific control signals are checked.

If this test fails, testing is stopped and the following tests will not be executed.

Modulator Loop Test

The GCTX interface to the RFSU is tested. For this purpose, data is sent to TX/RFSU and read back to compare with the initial data.

MC1C40: GPA-INT (not with MAGNETOM Concerto)

The MC1C40 is tested using five functional tests. These tests require a functional MPCU and PCI-Link.

DBU RAM

The memory allocated to the DSP on the MC1C40 is tested.

If this test fails, testing is stopped and the following tests will not be executed.

DBU GpaIntTestBus

The data and address bus between MC1C40 and GPA_INT is checked.

If this test fails, testing is stopped and the following tests will not be executed.

DBU GpaIntTestChin

The RS485 signals of the GPA_INT are checked.

If this test fails, testing is stopped and the following tests will not be executed.

DBU GpaIntTestReq

The request lines between MC1C40 and GPA_INT are checked.

If this test fails, testing is stopped and the following tests will not be executed.

DBU GpaIntTestTest

The signals SEQ_ON, SEQ_CLK and the DAC/ADC function block are checked.

If this test fails, testing is stopped and the following tests will not be executed.

MC1C40: EFI_DAS (MAGNETOM Concerto only)

The MC1C40/EFI_DAS is tested using five functional tests and one interactive test. These tests require a functional MPCU and PCI-Link.

EFI RAM

The memory allocated to the DSP on the MC1C40 is tested.

If this test fails, testing is stopped and the following tests will not be executed.

EFI EFITestBus

The data and address bus between MC1C40 and EFI_DAS is checked.

If this test fails, testing is stopped and the following tests will not be executed.

EFI EFITestReq

The interrupt signal at the DSP is checked. At the same time, the signals SEQ_CLK, CONVST and N_ADC_BUSY will be tested.

If this test fails, testing is stopped and the following tests will not be executed.

EFI EFITestSer

The serial interface of EFI_DAS is tested. All data written to SER_DAC_REG can be read in SER_DAC_TEST_REG after the end of transmission.

If this test fails, testing is stopped and the following tests will not be executed.

EFI EFITestRxOn

The RXON signal in the status register after transmission from ED_DSP via GCTX is checked.

If this test fails, testing is stopped and the following tests will not be executed.

EFI EFITestADC

The ADC offset by reading the ADC outputs is tested. The EFI probe cable must be removed from input D21/X4 during the test. Follow the instructions on screen to perform this interactive test.

RX4

The RX4 is tested using ten functional tests. These tests require a functional MPCU, PCI-Link and MC4C40. In addition, RX4 can only be tested completely when the GCTX is available.

Similar tests are available for the second RX4-board (RX4-2) if installed. A table showing the differences is located after the description of the RX4-tests.

RX4 RawLink Test Reg1

The data register 1 of the Raw Link (piggy board) on RX4 is tested.

RX4 RawLink TEST DAT_OVL

The function data overflow on the Raw Link (piggy board) on the RX4 is tested.

RX4 Memory Map Test

The data and address bus between the RX DSP and RX4 is tested. In addition, a RS484 loopback test is performed.

If this test fails, testing is stopped and the following tests will not be executed.

RX4 SRAM Test

The SRAM memory on RX4 is tested. For this purpose, the RX DSP accesses the memory on RX4 via RX bus.

If this test fails, testing is stopped and the following tests will not be executed.

RX4 Interrupt Test

The interrupt signal to the RX DSP are checked.

If this test fails, testing is stopped and the following tests will not be executed.

RX4 Link Error Test

The link error signal (PALI error) connection from RX4 via MC4C40 to PCI-Link is tested.

If this test fails, testing is stopped and the following tests will not be executed.

RX4 Mofi Reset

The internal reset signal from MC4C40 to MOFI and RFPA are checked.

RX4 Mofi Interface Test

The input channels ADC1 to ADC4 are tested. For this purpose, the GCTX writes data to RX4 via TX_IF_TEST bus. In addition, sequence control signals (TX_ON, etc.) are tested.

Receiver Loop Test

The RX interface to the RFSU is checked. For this purpose, data is written from RX/RFSU to RX4 and compared with the initial data.

RS485 Internal Loop Test

The RS485 interface is checked. The RS485 is used to transfer data to the second RX4 if the eight channel RF option is installed.

Tests for the RX4-2-board

RX4 RawLink Test Reg1	same as for RX4-1
RX4 RawLink Test DAT_OVL	same as for RX4-1
RX4_2 Memory Map Test	same as for RX4-1, but for RX4-2-board
RX4_2 SRAM Test	same as for RX4-1, but for RX4-2-board
RX4_2 Interrupt Test	same as for RX4-1, but for RX4-2-board
RX4_2 Link Error Test	same as for RX4-1, but for RX4-2-board
RX4_2 Mofi Reset	same as for RX4-1, but for RX4-2-board
RX4_2 Mofi Interface Test	same as for RX4-1, but for RX4-2-board
Receiver Loop Test	same as for RX4-1, but for RX4-2-board
RS485 Internal Loop Test	same as for RX4-1

IOP

The IOP is checked employing a communication/status test and an interactive test.

IOP Communication

The communication to the SLIO node on IOP is checked:

```
Starting Test GetIopStatus ... Communication with IOP is ok.
```

```
Starting Test GetIopVersion ... ok.
```

```
Version : V178
```

IOP Status

This test evaluates the status of the SLIO node on IOP, i.e. power supply voltages of MMC, reset button on IOP, and MPCU fan (at the MPCU processor):

```
Reset Iop Status ...
```

```
Starting Test GetIopStatus ... Status of IOP is ok.
```

```
IOP_STATUS: Reset on +3.3 V      = ok
```

```
IOP_STATUS: Reset on + 5 V       = ok
```

```
IOP_STATUS: Reset on - 5 V       = ok
```

```
IOP_STATUS: Reset on + 12 V      = ok
```

```
IOP_STATUS: Reset on + 12 V      = ok
```

```
IOP_STATUS: Reset by button      = ok
```

```
IOP_STATUS: MPCU fan supervision = ok
```

IOP Control Word Signals

The sequence control signals available on the fiber optic interface of the IOP are checked. Follow the instructions to perform this interactive test.

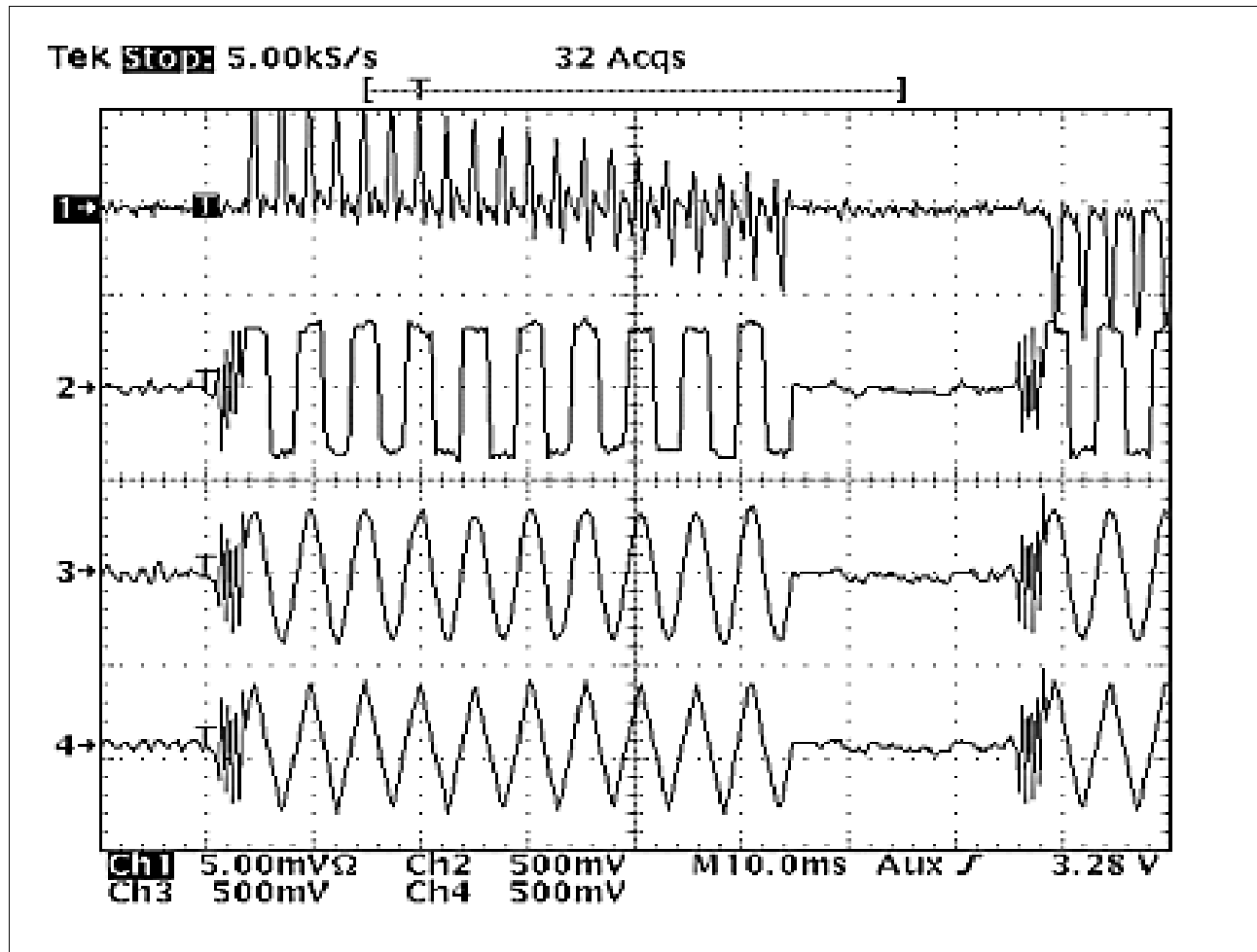
NOTE	This test can run indefinitely long to take into account that the electronics cabinets can be farer away from the user console. In that case the standard 30 seconds would be too short.
-------------	--

Dynamic Test

MMC Dynamic Test

This test runs a super-fast sequence that tests the dynamic behavior of the entire MMC, all boards are involved.

Fig. 7 Overview pulse diagram



Pulse sequence:

- Row 1: RF 20 RF pulses, pulse width 2.5 ms, pulse distance 20 μ s
- Row 2: x-gradient 20 rectangular pulses, pulse width 2.42 ms
- Row 3: y-gradient 20 sinus pulses, pulse width 2.57 ms
- Row 4: z-gradient 20 triangular pulses, pulse width 2.57 ms

Total scan duration 80 ms with TR=30 ms.

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RF - All systems

NOTE This section describes all MR systems except the MAGNETOM Allegra, Concerto and Trio Body.

The RF tests consists of a set of tests designed to verify the performance of the RF system and to assist you in locating defective components.

NOTE Beginning with *syngo* MR 2002B (NUMARIS/4 VA21A) all displayed graphs can be enlarged by clicking into the image and selecting (dragging) the region of interest. With "autoscale" the original view of the graph is restored.

General

The RF System hardware has been designed with several built-in loops to allow automated testing of RF components. In addition, level measurements are performed via the directional couplers (DICO) in the RFPA and the TALES.

The sequence used is common to all RF loop tests except the BTB test (body tune box). It is called `rf_loop` and contains six RF pulses of which the first two are relevant for TestTools and the others have zero amplitude. The first pulse is in sinc (function $\sin(x)/x$) form with constant amplitude and is used for level measurements and stability evaluations. The second pulse is also in sinc form and increases from 0 up to the maximum amplitude. It is used in linearity evaluations.

NOTICE The 50 Ohm termination of the RFPA is done automatically. The dummy load is located in the TAS at the filter plate and is water-cooled in series with the gradient/body coil cooling.
To prevent damage to the dummy load, do not perform RFPA tests without proper cooling.

NOTE With every measurement a unique ID is displayed in the output window (e.g. 4.0.67892). This so-called "Loid" can be used in conjunction with the Rawdata Handling Package RHP to display and evaluate the raw data of the measurement.

Additionally to this number also the used RFPA (RFPA1 or RFPA2) is printed out as an information. Depending on the test this allows it to identify any error or mistake in selecting the correct RFPA.

With every test that runs a measurement at the beginning the used nucleus is displayed. With all systems that are not equipped with the multi-nucleus option (spectroscopy, X-nucleus) this will be always "1H", otherwise it should be the selected nucleus, unless it is a test that always use 1H and is independent of the nucleus-selection.

NOTE In case the system is equipped with a multi-nucleus option (spectroscopy, X-nucleus), also these tests are performed when a RF test is selected. If this is not intended or necessary please de-select the test type "Multi Nuclei".

The nucleus which has to be used for the next RF tests can be selected in the problem section. Every nucleus corresponds to a certain frequency which will affect e.g the automatic selection of the correct RFPA. The nuclei ^3He , ^{19}F and of course ^1H are using the old DORA-amplifier, all other nuclei the second RFPA CORA (^7Li , ^{13}C , ^{23}Na , ^{31}P and ^{123}Xe).

The selection has to be done prior to the start of the test. If you realise that accidentally the wrong nucleus is selected please cancel the test immediately.

Everytime when such a test is to be performed it is necessary to select the nucleus/frequency prior to start the test.

Overview RF Tests

Test	Comm/Status	Functional	Interactive	Multi Nuclei ^{*)}
Mod/REC i	x	x	-	x
RFPA 1/2	x	x (RFPA 1/ DORA)	x (RFPA 2/ CORA)	x(RFPA 2/ CORA)
RFAS	x	x	x	x
LCCS	x	x	x	x
RFCI	x	x	x	-
BTB	x	x	x	-
BC	x	x	x	-
Cable	-	-	x	x (LC-TX only)

^{*)} Only available with spectroscopy option.

Stability

The first pulse of the sequence with a constant RF amplitude is used. The resulting magnitude data is the evaluation of the stability of the center column which is located in the center of the RF pulse. The numerical result is the maximum value minus the minimum value divided by the mean value. Instabilities normally cause phase ghosting or poor S/N values.

An overlaid graphic represents an average of the samples close to the centre column. This allows to differentiate between noise and instabilities.

Linearity

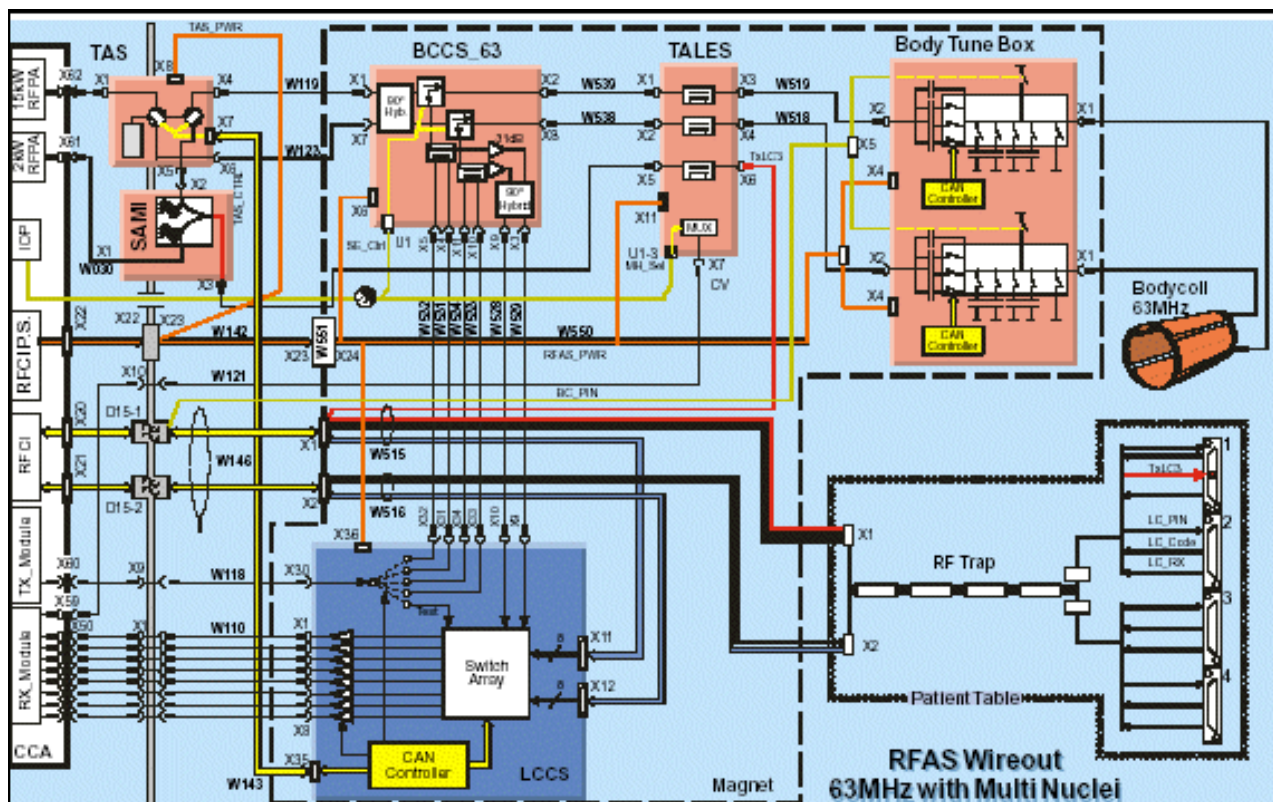
For linearity evaluations, the second RF amplitude is stepped from 0 to a positive pulse amplitude. The center column of each line (maximum of the single pulses) is fitted to a linear function. The numerical result is the maximum deviation from the linear fit divided by the maximum value making up the data set. Non-linearities result in poor slice profiles and reduced S/N.

NOTE

With many tests the first measurements are not used for the final evaluation. So don't be worried if the graphical display and the numerical differ for this.

Overview RF system Symphony with Spectroscopy Option (NUMARIS/4 VA25A)

Fig. 8 Overview RF system Symphony with Spectroscopy Option (NUMARIS/4 VA25A)



MOD/REC

- TX-Cassette Communication (former Modulator Communication)
- Modulator-Receiver Loop (Functional/Multi Nuclei)
- Receiver A/B symmetry (Functional/Multi Nuclei)
- Reference Voltage (TALES)

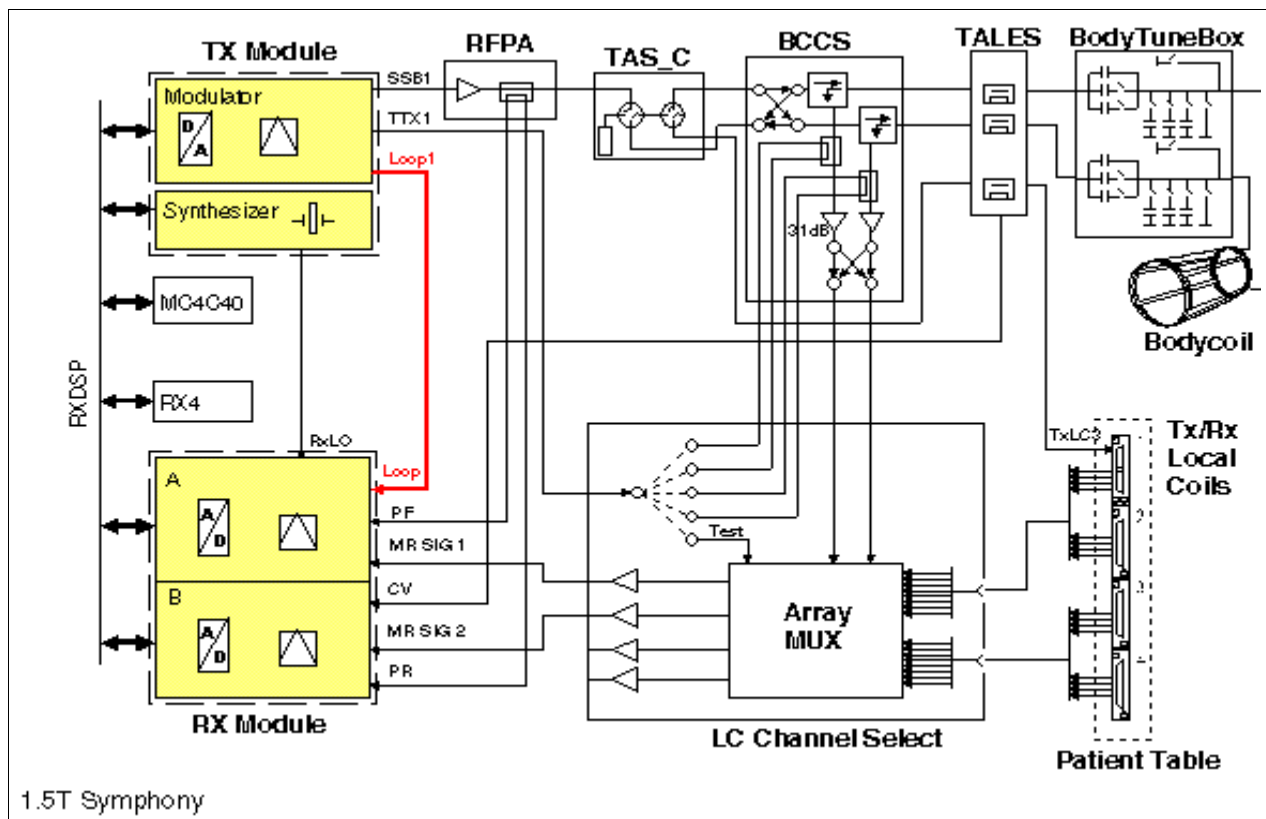
TX-Cassette Communication

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The test was formerly named Modulator Communication.

Modulator-Receiver Loop

Fig. 9 Modulator-Receiver Loop 1.5T



[illegible]

The rf_loop sequence is started and the output signal of the modulator is fed to the receiver module 1 A/B.

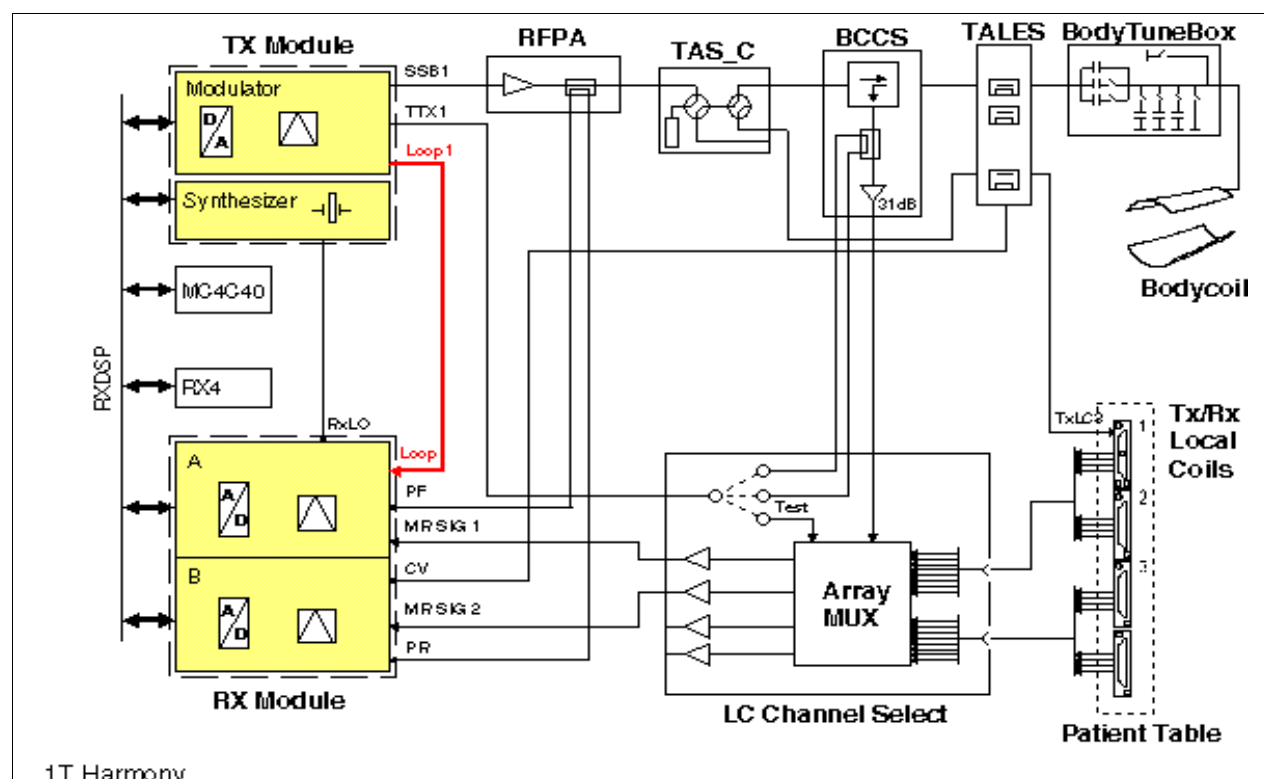
The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Successful completion of this test is the prerequisite for all additional RF tests. If the test results are satisfactory, it can be assumed that the controller provides the transmitter with data as specified, that the data is modulated correctly in the modulator, received correctly by the receiver and then forwarded to the MMC and Imager.

NOTE Since transmitting/receiving are simultaneous activities, both the transmit and the receive signal are affected by signal fluctuations in the synthesizer. As a result, fluctuations remain undetected, a fact that has to be taken into account during troubleshooting.

Fig. 11 Receiver A/B symmetry 1.5T

Fig. 11 Receiver A/B symmetry 1.5T



Signal Path

The output signal of the modulator is fed to the receiver module A/B.

Evaluation

The raw data of receivers A and B are compared for symmetry.

Conclusions

With respect to the signal path and data acquisition, the test is identical to the Modulator-Receiver loop. The only difference is that only one raw data line is being measured and only the first pulse will be evaluated to check the symmetric signal processing of both receivers.

Reference Voltage (TALES)

Fig. 13 Reference voltage 1.5T

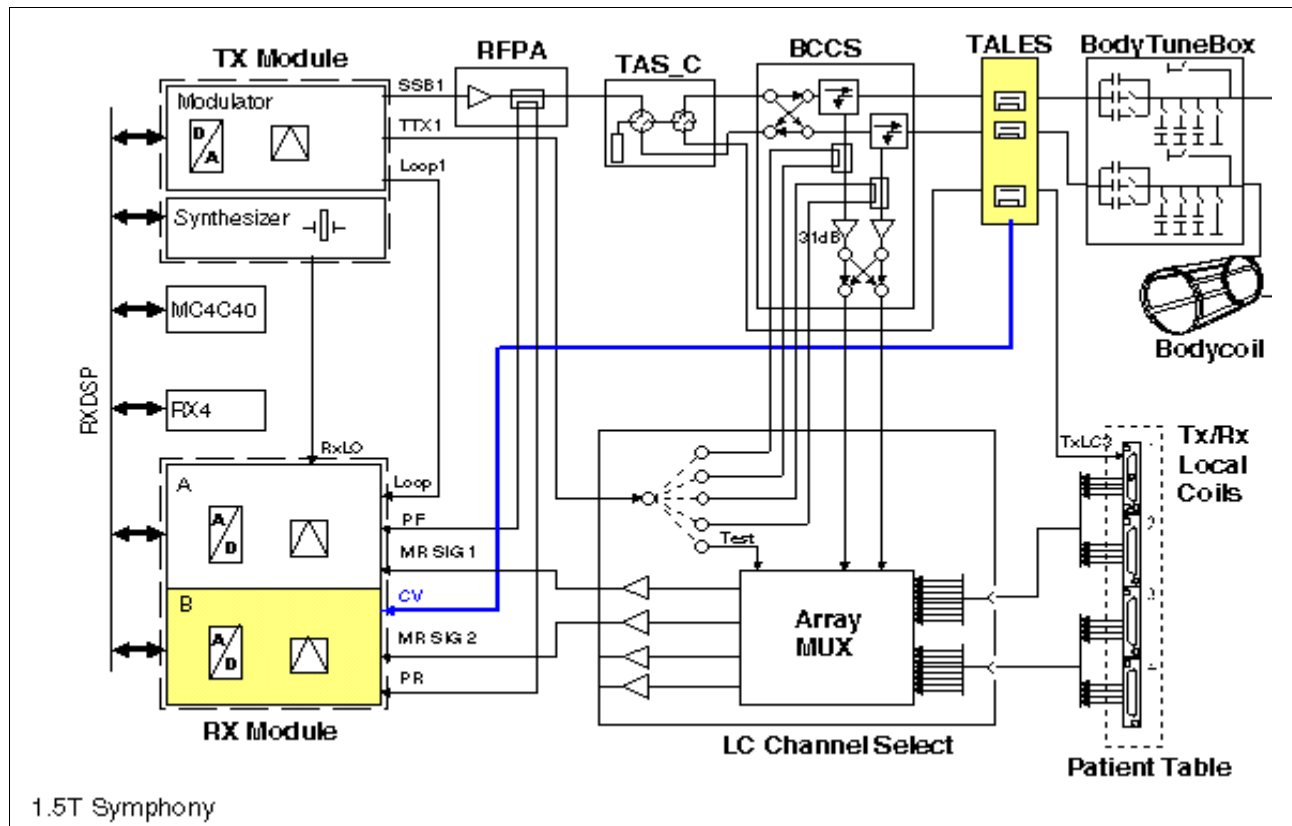
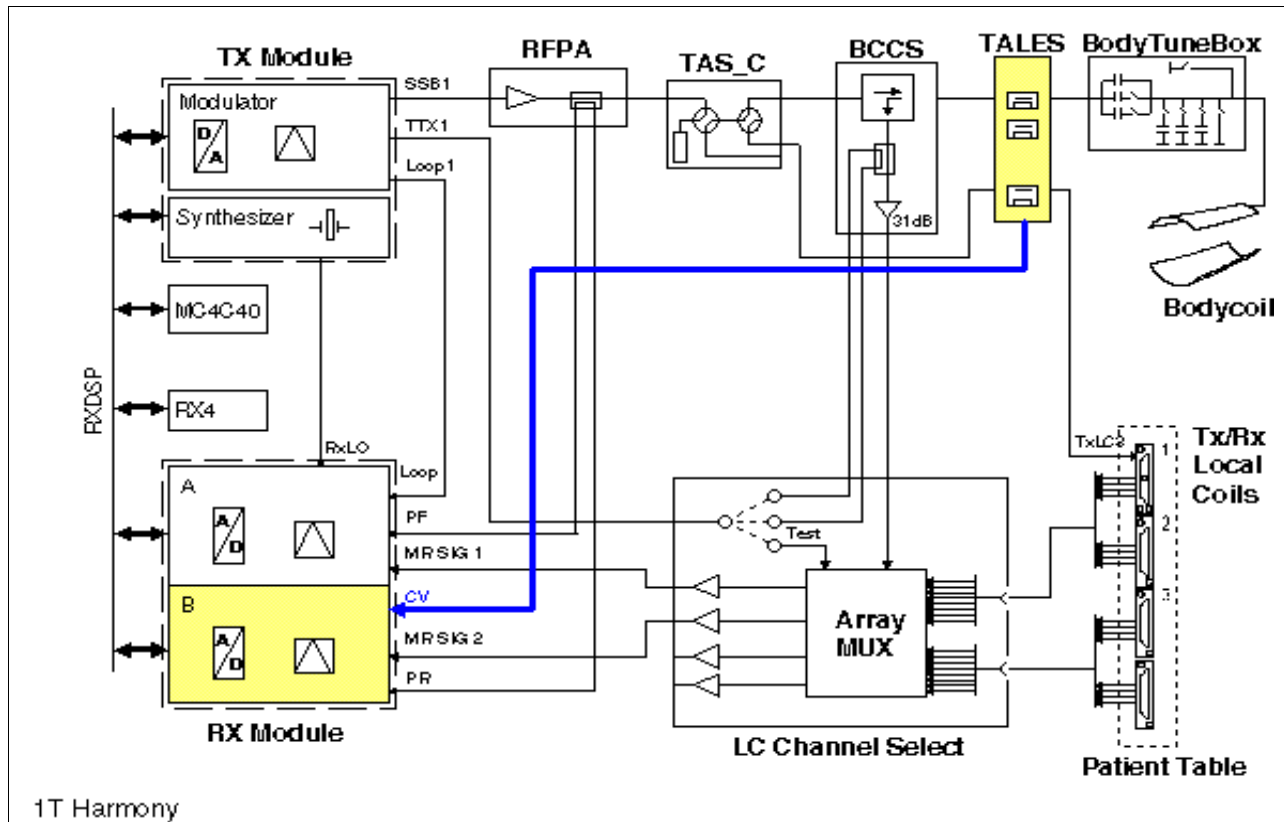


Fig. 14 Reference voltage 1T



Signal Path

The TALES output signals are fed to the receiver 1B (CV-Input).

Evaluation

Verification that the TALES reference voltage is 500 mV.

Conclusions

The TALES continuously supplies all internal directional coupler signals triggered by the controller. In addition, it supplies the signal of an integrated reference voltage source. The DICO signals and the reference voltage are multiplexed and fed to the receiver CV input. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the receiver and signal processing during data acquisition. In case of error, the 500 mV reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

RFPA 1/2

- RFPA Communication/Status
- RFPA Power Test
- RFPA Loop
- RFPA-TALES
- RFPA Long Time Stability

RFPA Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

In case the system is equipped with a multi-nuclear option (spectroscopy, X-nucleus), the communication of both RFPAs is tested and both status are returned.

RFPA Power Test

NOTE The test requires a valid RF characteristic.

Fig. 15 RFPA Power Test 1.5T

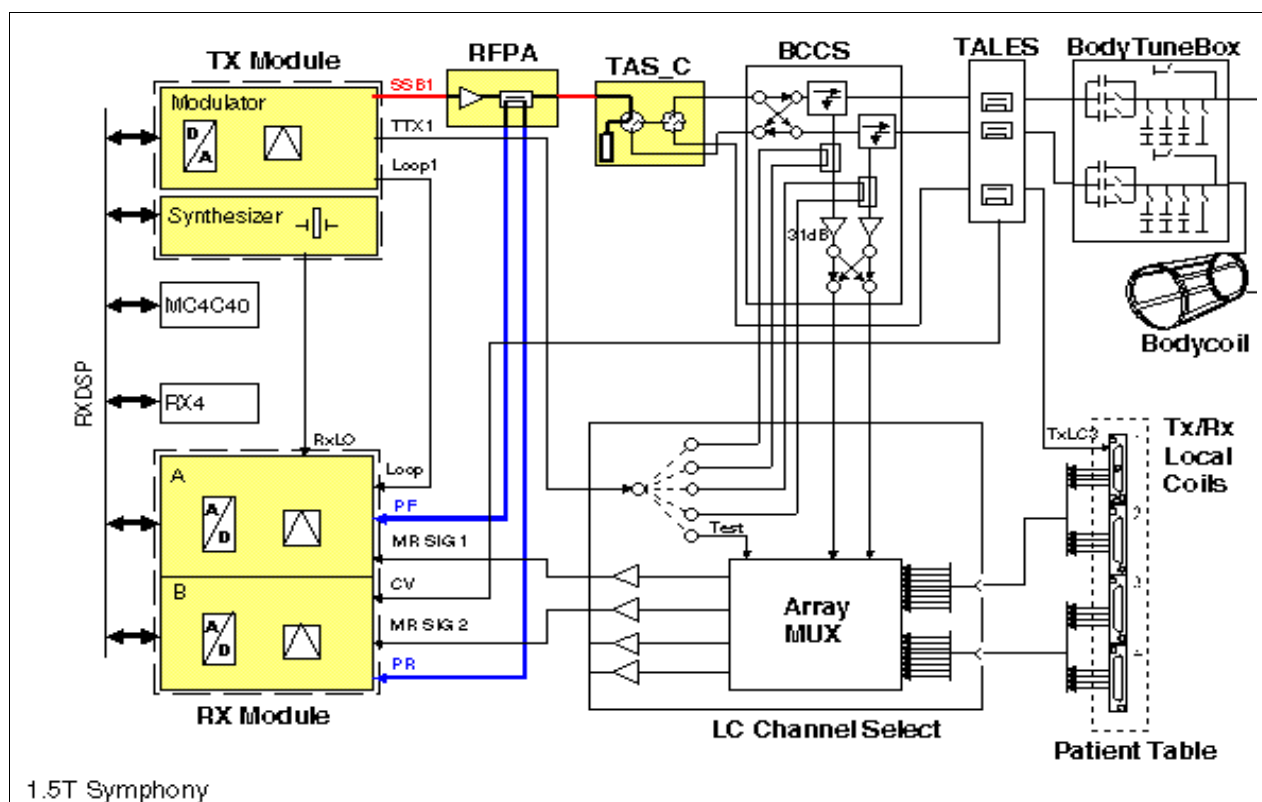
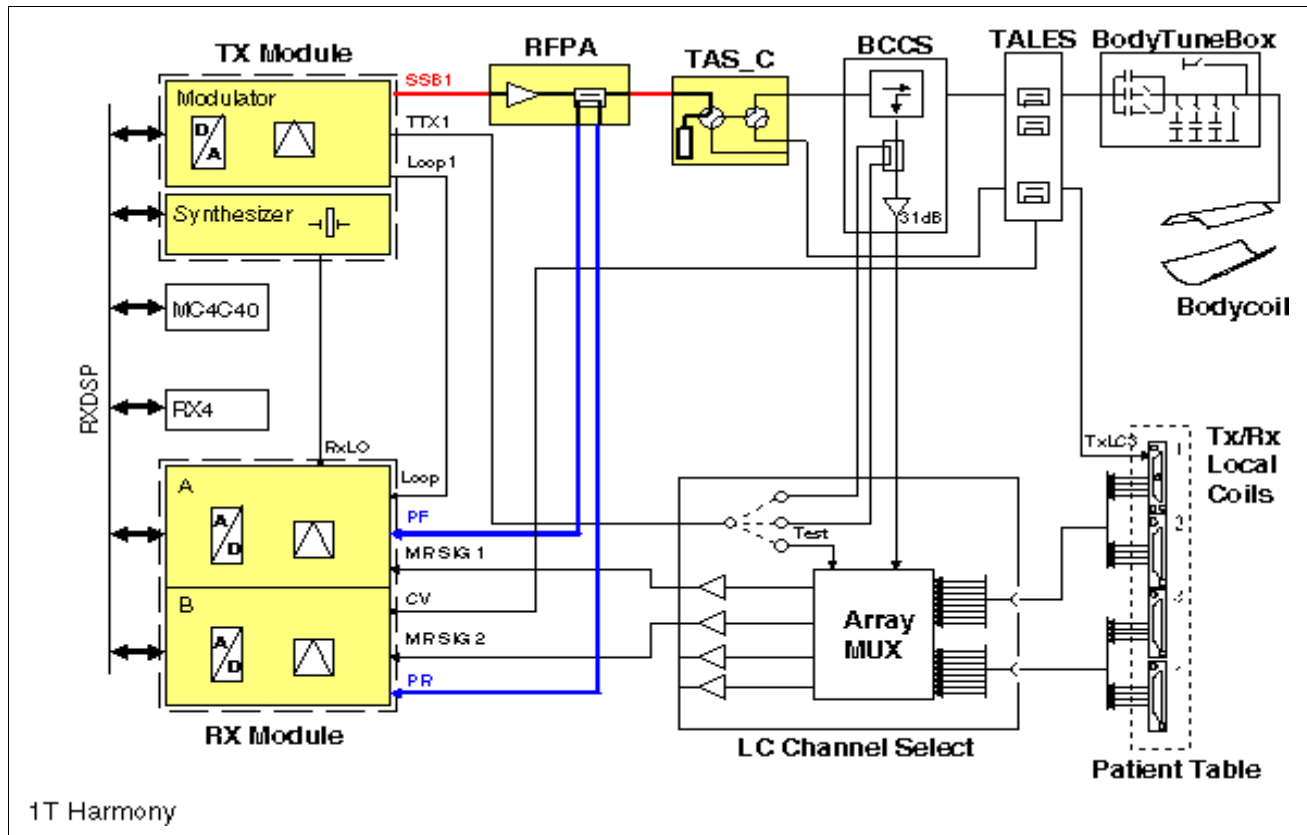


Fig. 16 RFPA Power Test 1T



Signal Path

The output signal SSB1 of the modulator is fed via the RFPA 1/DORA to the dummy load of the TAS_C. The Forward_Power Signal is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF characteristic data.

During the multi nuclei test, the output signal SSB2 of the modulator is fed via the RFPA 2/CORA to SAMI and TALES X5.

The cable to BCCS/X7 must be connected to TALES/X6 and the service plug or a spectroscopy/multi nucleus coil must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

Evaluation

- The actual forward voltage (measured via directional coupler) is compared to the nominal value read from the RF characteristic measurement.
- The ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.

Conclusions

The objective of the test is to ensure that the RFPA operates at peak performance and that the reflected voltage remains below 20% of the forward voltage at a dummy load of 50 Ohm.

Peak performance can no longer be achieved when the total gain (signal processing in the RFSU and gain in the RFPA) and, as a result, the characteristic curve has changed due to aging components. If the deviation lies outside the acceptable tolerance, the RF characteristic must be reexamined. If peak performance still cannot be achieved, the output level of the modulator and the directional coupler in the RFPA should be checked to determine the cause of the error. For further information, refer to the Troubleshooting Guide.

If there is an increase in the reflected voltage, check all cable connections from the RFPA to the Receiver and the transmission cable to the TAS assembly. You should also check with the LEDs if the TAS relay is set correctly. A defective dummy load would cause high voltage reflection, too.

RFPA Loop

The RFPA Loop test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

Fig. 17 RFPA Loop 1.5T

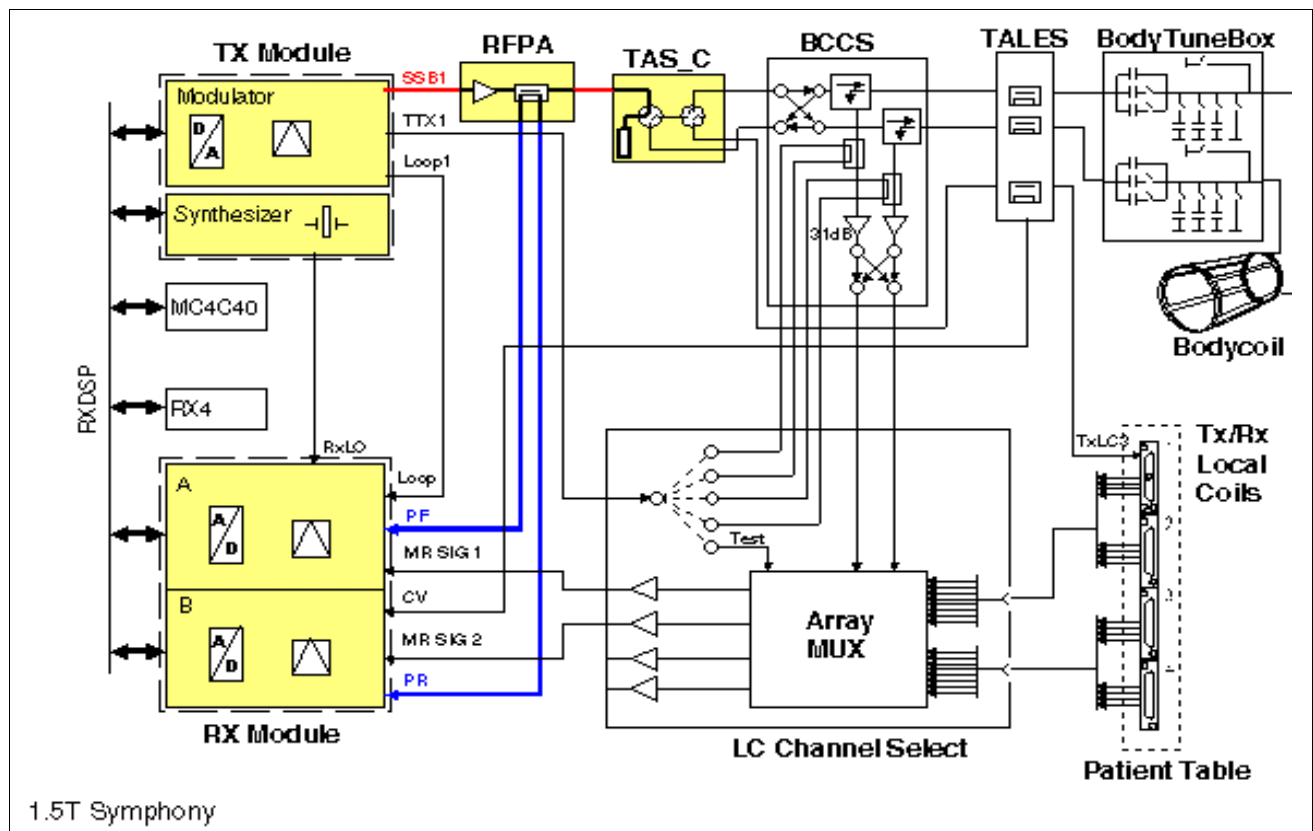
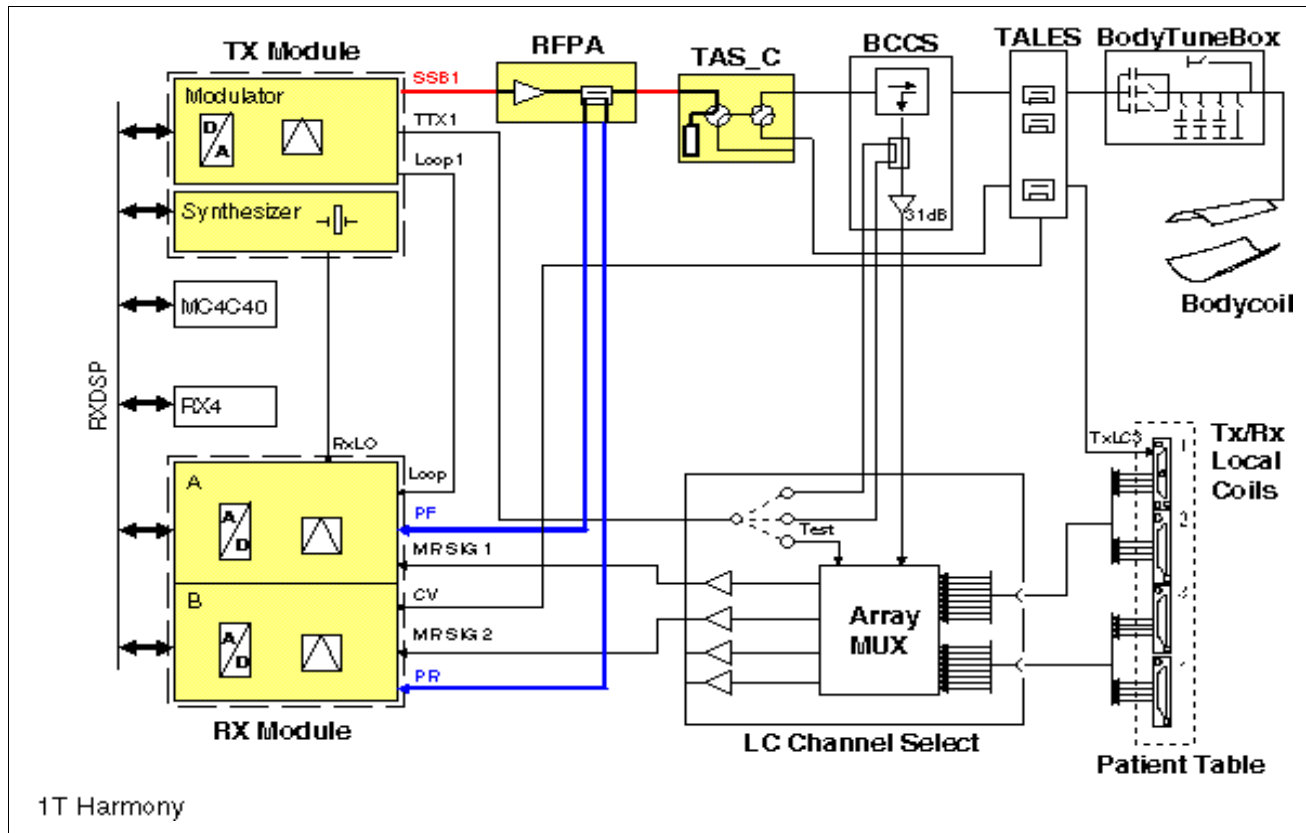


Fig. 18 RFPA Loop 1T



Signal Path

The output signal of the modulator is fed via the RFPA 1/DORA to the dummy load of the TAS_C. The forward signal is measured via the directional coupler in the RFPA.

During the multi nuclei test (RFPA 2), the output signal SSB2 of the modulator is fed via the RFPA 2/ CORA to SAMI and TALES X5.

The cable to BCCS/X7 must be connected to TALES/X6 and the service plug or a spectroscopy/multi nucleus coil must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability. The raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

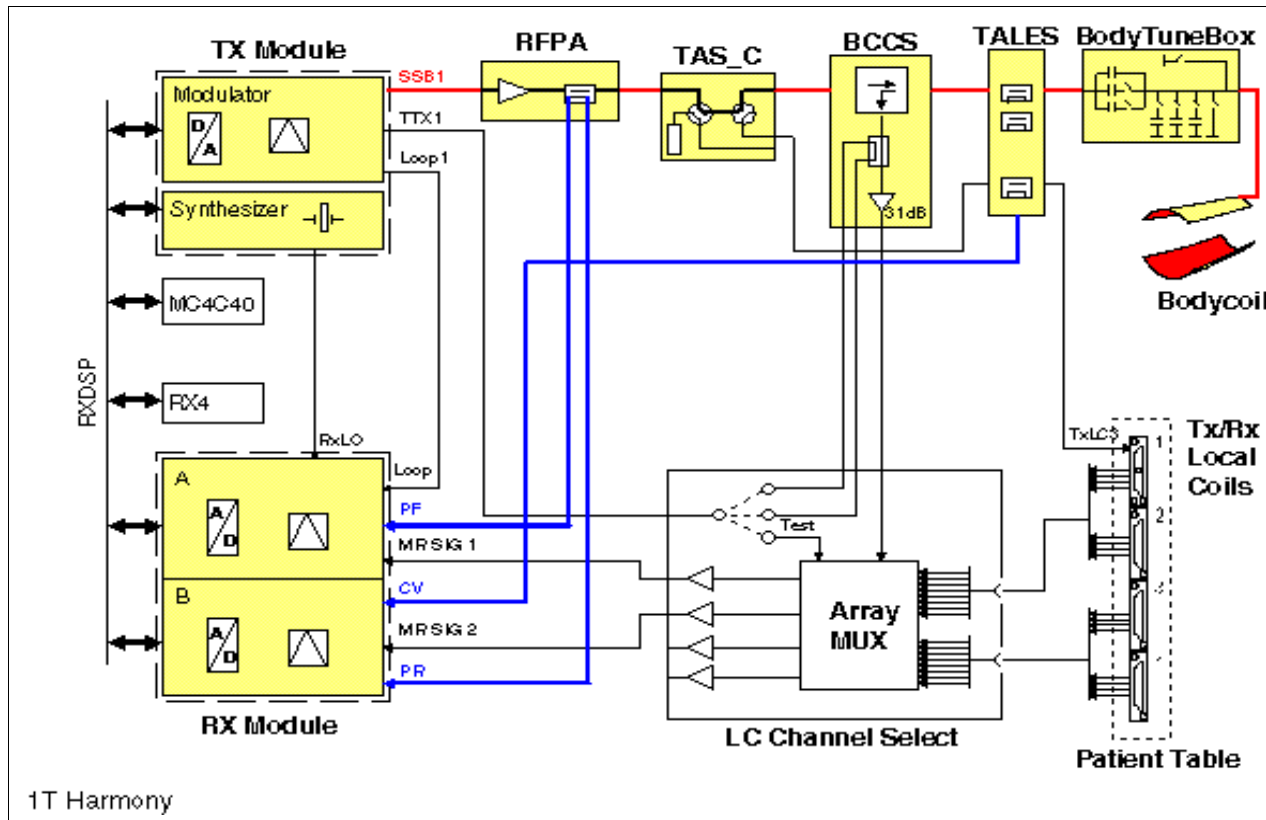
This test is used primarily to check the stability and linearity of the RFPA. Modulator control, data acquisition via receiver and RF characteristic acquisition must be intact and correct to evaluate the results.

The RFPA TALES test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

The diagram illustrates the transmit chain architecture for the 1.5T Symphony system. Key components and their interconnections include:

- TX Module:** Contains a Modulator, Synthesizer, and MC4C40. It outputs TTX1 and Loop1 signals.
- RX Module:** Contains two channels (A and B) with mixers and amplifiers. It receives RxLO and outputs Loop, PF, MR SIG 1, CV, MR SIG 2, and PR signals.
- RFPA (Radio Frequency Power Amplifier):** Receives TTX1 and Loop1 signals.
- TAS_C (Transmit/Receive Switch):** Receives signals from RFPA and BCCS.
- BCCS (Bandpass Filter and Channel Selection):** Receives signals from TAS_C and outputs to TALES.
- TALES (Transmit/Receive Switch):** Receives signals from BCCS and outputs to BodyTuneBox.
- BodyTuneBox:** Contains a Body Coil Tuning circuit and a Bodycoil (Body Coil) on the Patient Table.
- LC Channel Select:** Receives signals from the RX Module and outputs to the Tx/Rx Local Coils.
- Patient Table:** Contains the Tx/Rx Local Coils, which are connected to the Bodycoil.

Fig. 20 RFPA-TALES 1T



NOTICE Ensure that the loader with spherical phantom is positioned in the body coil.

Signal Path

1. The output signal of the modulator SSB1 is fed via the RFPA 1-TAS_C-BCCS-TALES-BTB to the body coil.
2. In the interactive test the output signal SSB2 of the modulator is fed via the RFPA 2/CORA to SAMI and TALES X5. The cable to BCCS/X7 must be connected to TALES/X6 and the service plug or a spectroscopy/multi nucleus coil must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

The body coil must be tuned for RFPA 1 - TALES (functional) to evaluate the signals. For this reason, coil tuning is automatically initiated at the beginning of the test.

Evaluations

- Forward Voltage from the TALES within tolerance
- Reflected Voltage from the TALES below limit
- Deviation of the forward voltage from the TALES and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks almost the entire transmission component of the system.

The results of the forward voltage at the TALES provide information on whether the actual value acquired by the TALES corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALES is a gauge for the 50 Ohm termination through the body coil.

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltages measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. On the one hand, it is determined by the coil tuning, and on the other hand by the characteristics of the components that follow the RFPA in the transmit path such as the TX cable, BCCS, TALES, BTB, etc.

RFPA (Long Time Stability)

The test is identical to the RFPA Loop test, but runs approximately 10 minutes. The test is only available on system with fMRI functionality.

RFAS

- RFAS Communication/Status
- Reference Voltage (TALES)
- Transmitter-Receiver-Switch
- BCCS Hybrid (Symphony only)
- RFPA-TALES
- BCCS (preamp1) - LCCS Loop
- BCCS (preamp1 2) - LCCS Loop (Symphony only)
- BC-TX-path (reflection)
- BC-TX-path (voltage)
- Local Coil TX Path (interactive/multi nuclei)

RFAS Communication/Status

The LCCS-SLIO controls the TAS component in order to verify communication to the LCCS as well.

The communication test consists of sending a command (status and version query) to an individual component which, if successfully completed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

Reference Voltage (TALES)

Fig. 21 Reference voltage 1.5T

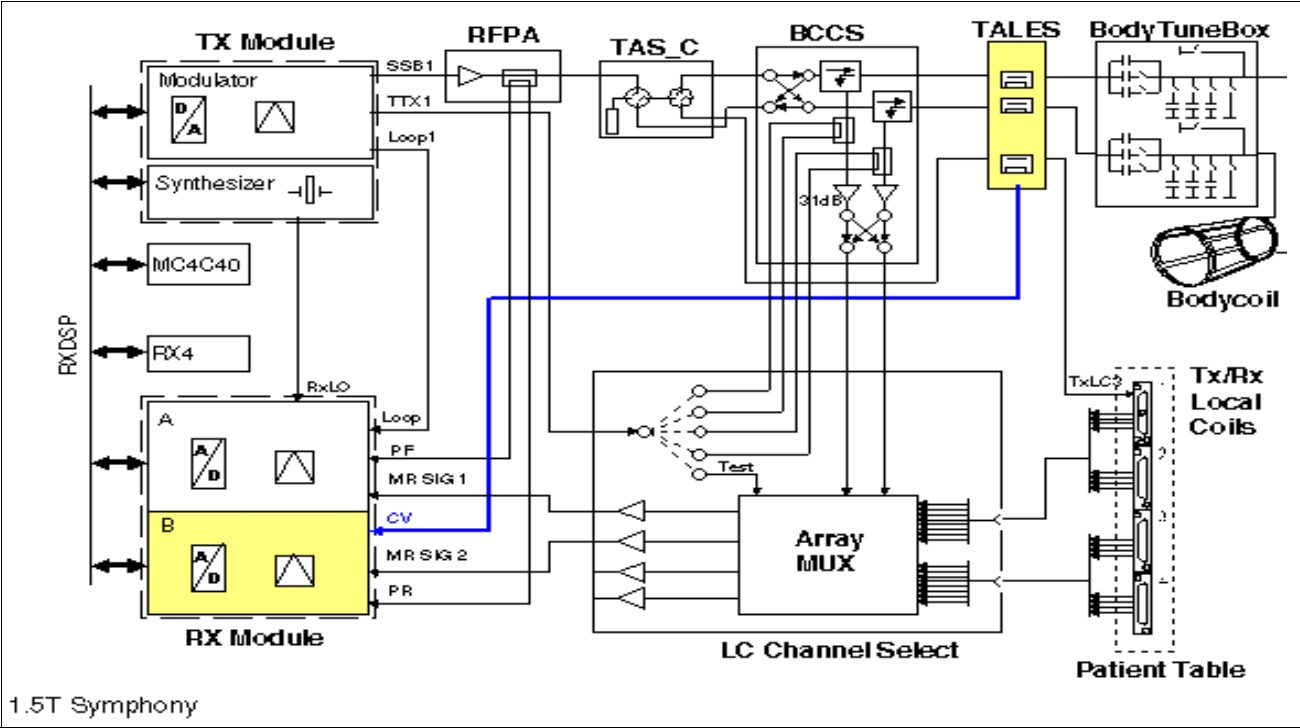
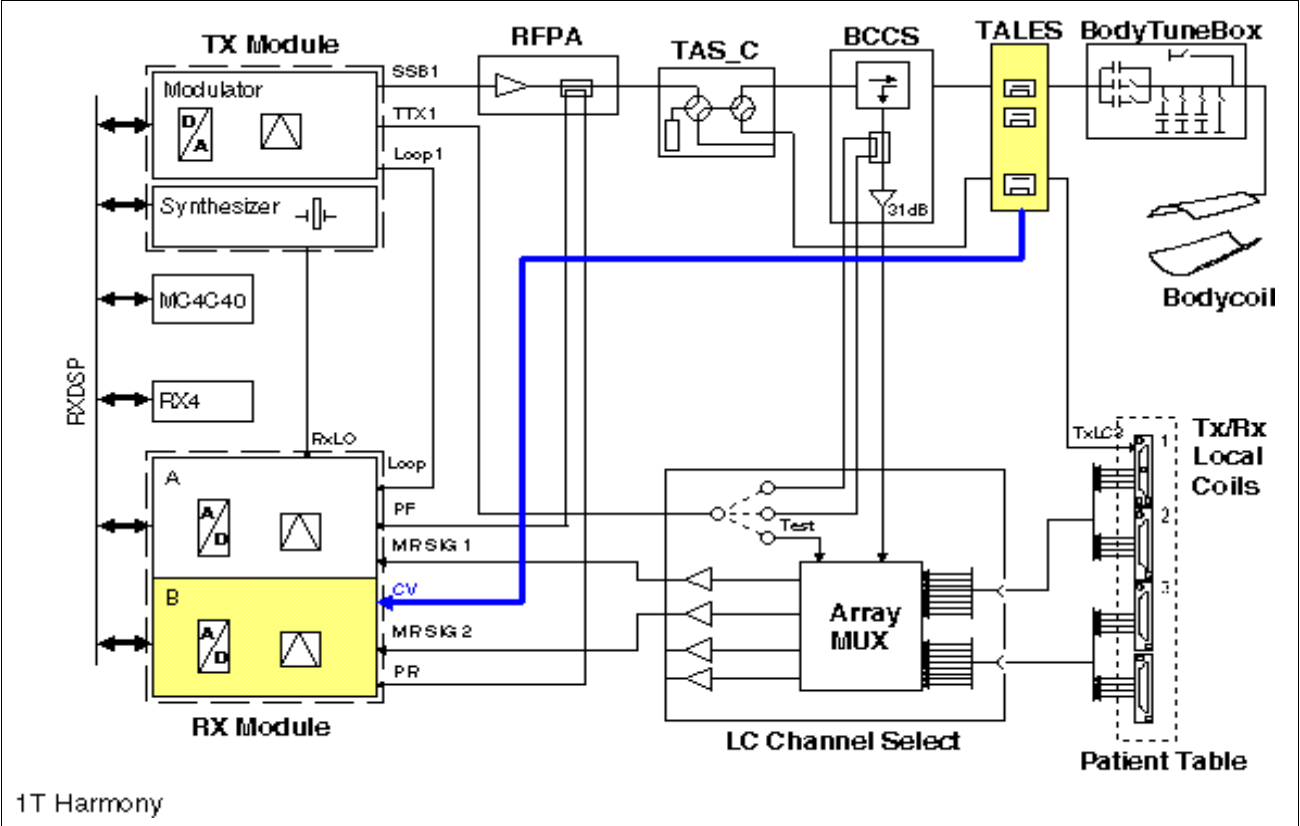


Fig. 22 Reference voltage 1T



Signal Path

The TALES output signals are fed to the receiver 1B (CV-Input).

Evaluation

Verification that the TALES reference voltage is 500 mV.

Conclusions

The TALES continuously supplies all internal directional coupler signals triggered by the controller via fiber optic cable. In addition, the TALES supplies the signal of an integrated reference voltage source. The directional coupler signals and the reference voltage are multiplexed and fed to the receiver CV input via a coaxial cable. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the receiver and signal processing during data acquisition. In case of error, the 500 mV reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

Transmitter-Receiver-Switch

Signal Path

Two measurements with different TX-paths will be performed using the service sequence graddelay with RF-amplitude set to zero:

- First measurement RFPA to the TAS-C/dummy load
- Second measurement RFPA to the Body Coil.

The receive path is always via the TX/RX-switch on BCCS to LCCS/SGA1 and finally to the receiver.

Evaluation

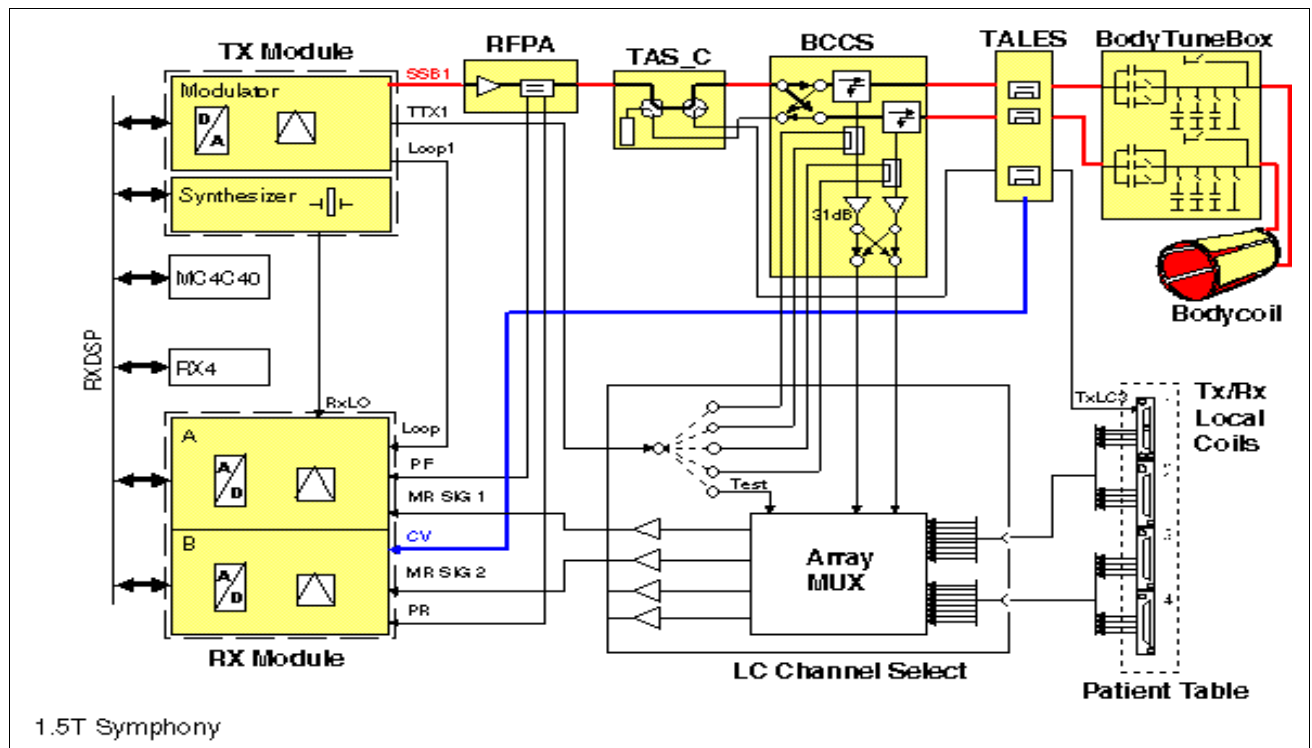
Two noise measurements will be compared. During the first measurement, the output of the RFPA is disconnected from the TX/RX-switch. During the second measurement, the output of the RFPA is connected to the TX/RX-switch and its noise (even unblank is activated) will be received if the switch does not work properly.

Conclusions

The TX/RX-switch or the switch control (from IOP U9 to BCCS U1) is probably defective if the additional noise from the RFPA during the second measurement exceeds the specification.

BCCS Hybrid (Symphony only)

Fig. 23 BCCS Hybrid



NOTICE Ensure that the loader with the spherical phantom is positioned in the Body Coil.

Signal Path

The output signal of the modulator is fed via the RFPA-TAS_C-BCCS-TALES-BTB to the Body Coil.

Evaluation

- The forward voltages of both transmitter channels (90° and 0°) measured at the TALES are compared.
- Ratio of the reflected and forward voltage from the TALES.

Conclusions

On the one hand, the deviation of the forward voltage for 90° and 0° serves as a gauge for symmetrical tuning of the 90° and 0° systems of the body coil. On the other hand, the test results should provide information about the BCCS functionality (Hybrid). If the results are not within specifications, first check the tuning of the body coil on both systems (tuning ratio of both systems).

RFPA 1 -TALES (functional) /RFPA 2 - TALES (interactive)

Fig. 24 RFPA-TALES 1.5T

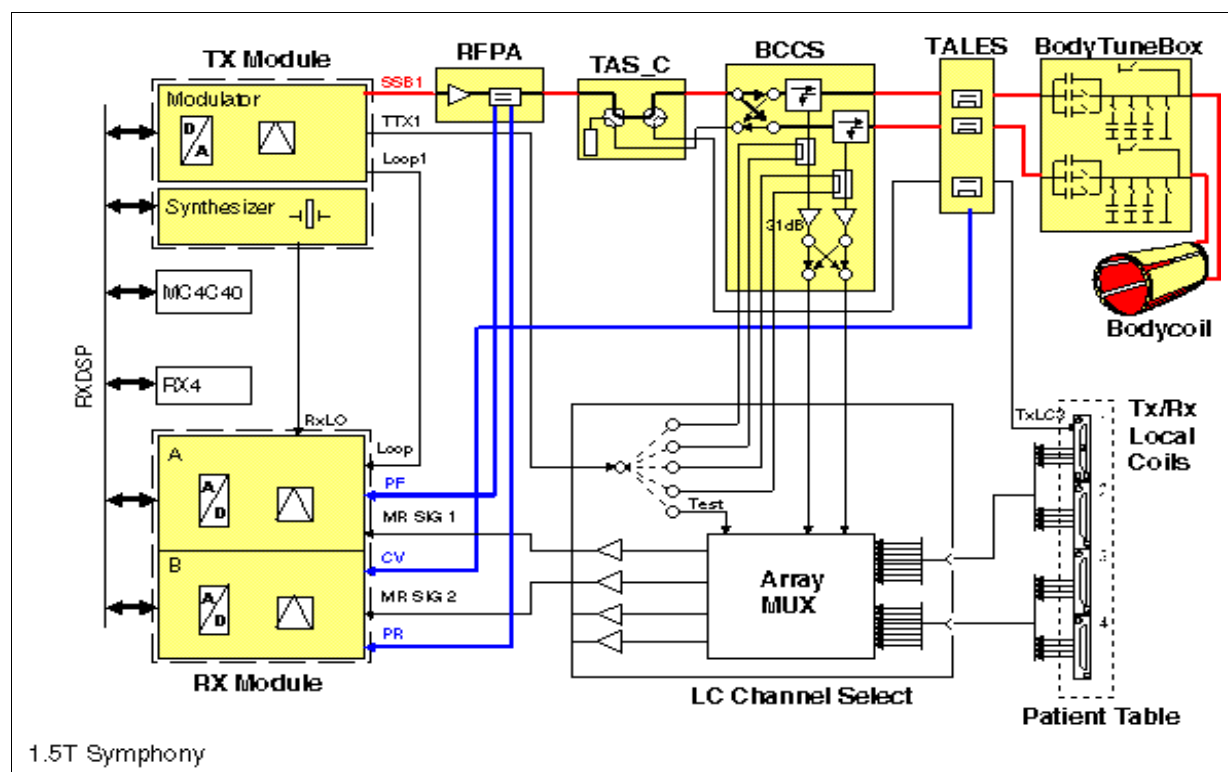
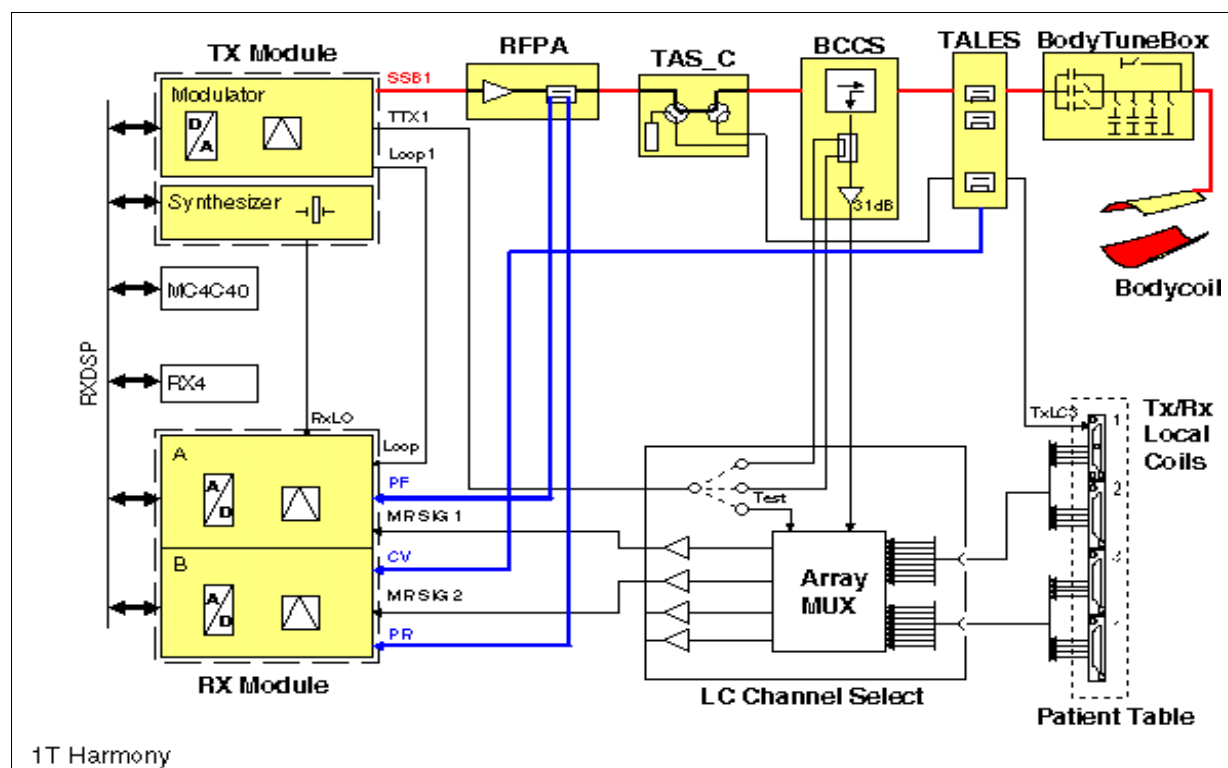


Fig. 25 RFPA-TALES 1T



NOTICE Ensure that the loader with spherical phantom is positioned in the body coil.

Signal Path

1. The output signal of the modulator SSB1 is fed via the RFPA 1-TAS_C-BCCS-TALES-BTB to the body coil.
2. In the interactive test the output signal SSB2 of the modulator is fed via the RFPA 2/CORA to SAMI and TALES X5. The cable to BCCS/X7 must be connected to TALES/X6 and the service plug or a spectroscopy/multi nucleus coil must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

The test consists of two measurements. One RF pulse is transmitted for each measurement. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

The body coil must be tuned for RFPA 1 - TALES (functional) to evaluate the signals. For this reason, coil tuning is automatically initiated at the beginning of the test.

Evaluations

- Forward Voltage from the TALES within tolerance
- Reflected Voltage from the TALES below limit
- Deviation of the forward voltage from the TALES and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks practically the entire transmit path of the system.

The results of the forward voltage at the TALES provide information on whether the actual value acquired by the TALES corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALES is a gauge for the 50 Ohm adjustment through the body coil.

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltage measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm adjustment of the RFPA. On the one hand, it is determined by the coil tuning, and on the other hand by the characteristics of the components that follow the RFPA in the transmit path such as the TX cable, BCCS, TALES, BTB, etc.

BCCS (preamp) - LCCS Loop

Fig. 26 BCCS (preamp) - LCCS 1.5T

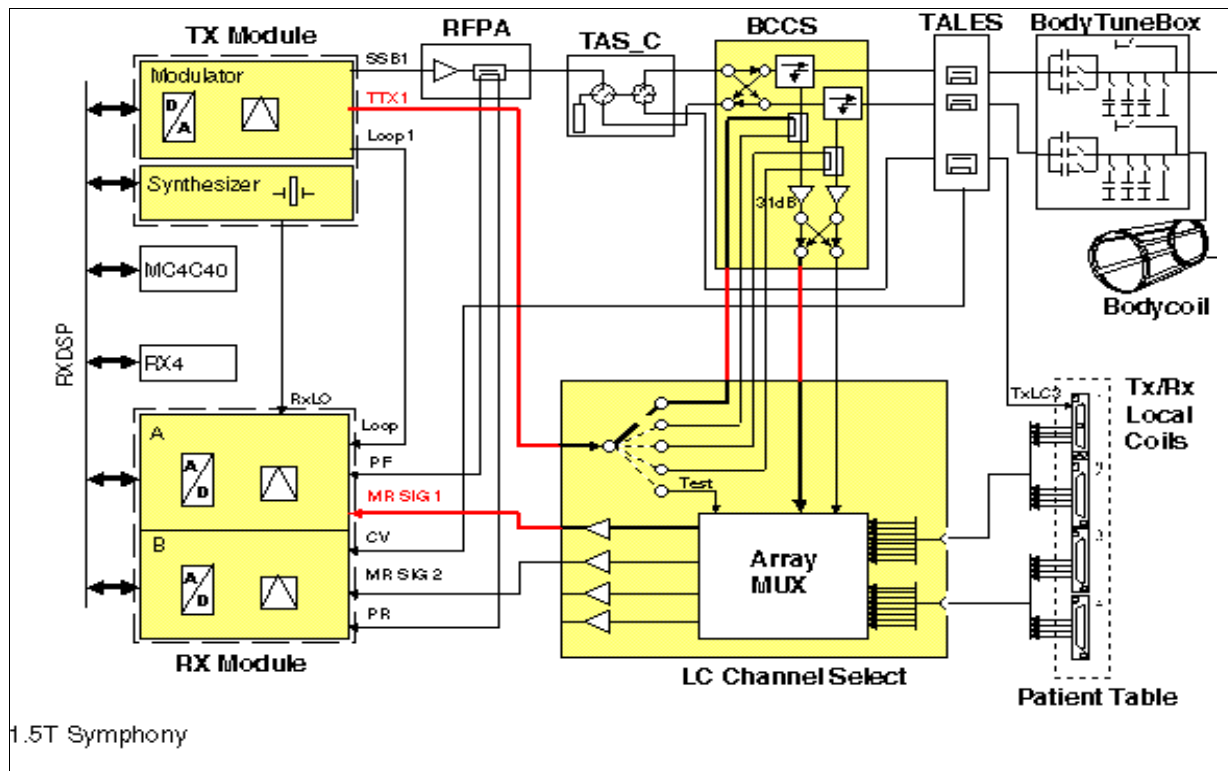
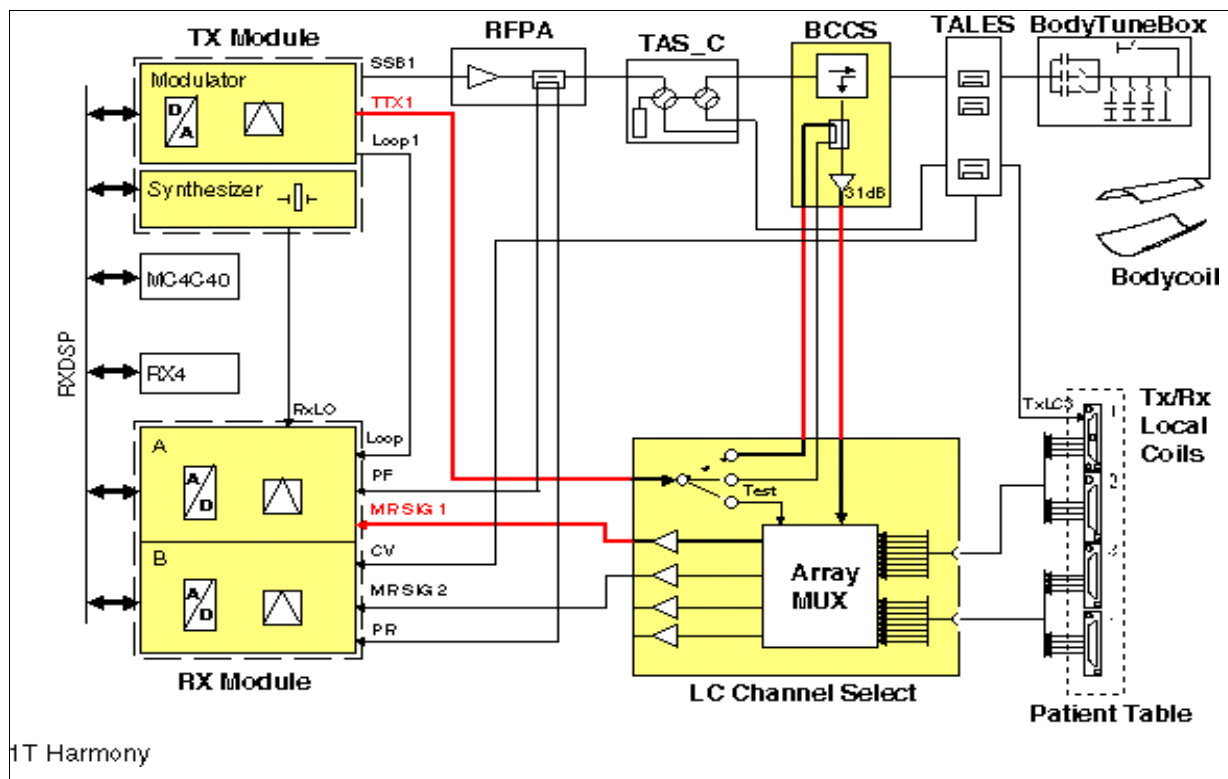


Fig. 27 BCCS (preamp) - LCCS 1T



Signal Path

The TTX output signal of the modulator is fed via the LCCS forward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver 1A/2A via the LCCS. The tests are performed with high gain and low gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

In a second measurement the TTX output signal of the modulator is fed via the LCCS backward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver 1A/2A via the LCCS. The test is performed with high gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

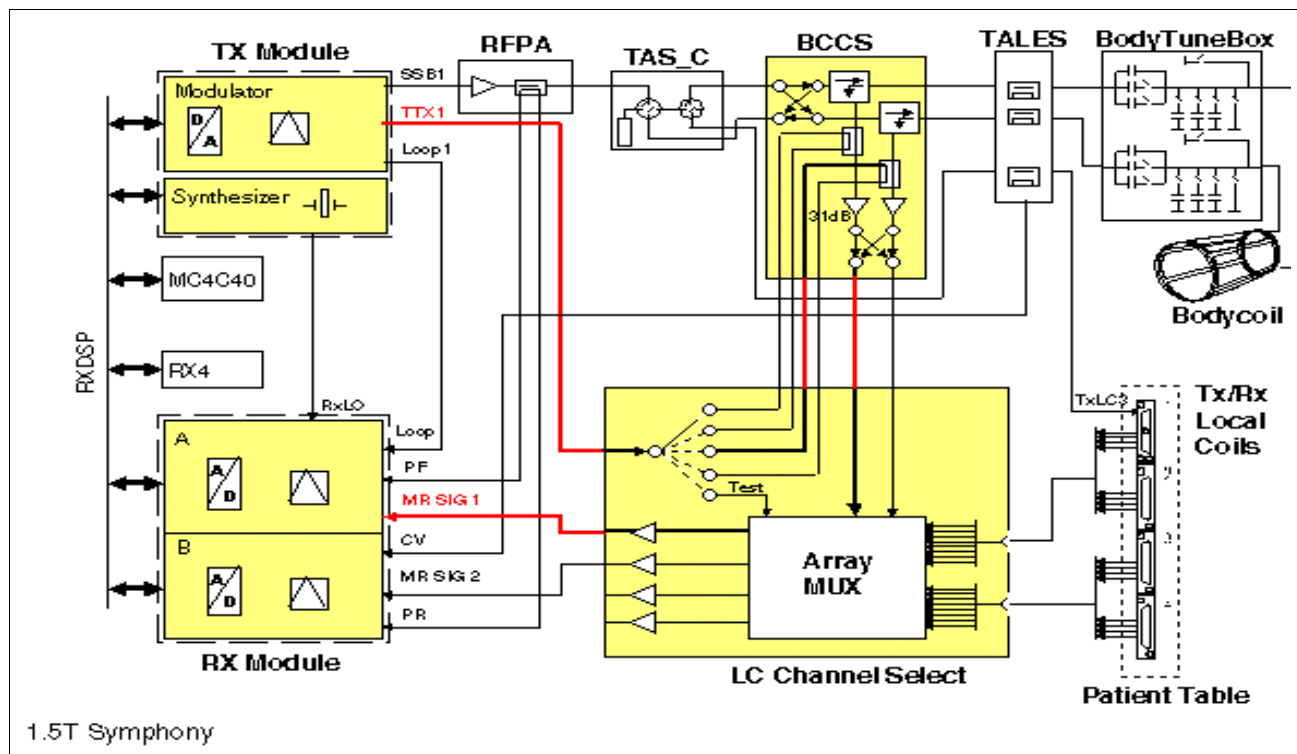
Conclusions

This test is used primarily to check the stability and linearity of the preamplifier in the BCCS and the LCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

Additional the cables and connections between BCCS and LCCS (<W531>, <W532> and <W529>) are checked.

BCCS (preamp1 2) - LCCS Loop (Symphony only)

Fig. 28 BCCS (preamp1 2) - LCCS



This test is similar to the previous test, but the signal is fed into the second preamplifier of the BCCS. Only the high gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS is used.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

This test is used primarily to check the stability and linearity of the preamplifiers in the BCCS and the LCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

Additional the cables and connections between BCCS and LCCS (<W533>, <W534> and <W528>) are checked.

Fig. 29 BC-TX-path (reflection) 1.5T

[illegible]

The diagram illustrates the 1T Harmony system architecture, showing the flow of signals between various components. The TX Module (Transmit) includes a Modulator, Synthesizer, and MC4G40. The RX Module (Receive) includes two channels (A and B) with mixers and amplifiers. The system also includes an RPA (Radio Frequency Power Amplifier), TAS_C (Transmit/Receive Switch), BCCS (Body Coil Control System), TALES (Transmit/Receive Switch), and BodyTuneBox. The Body Coil is shown as a red and yellow structure. The system is connected to a Patient Table via Tx/Rx Local Coils. The diagram shows the flow of signals between these components, including TX1, Loop1, Loop, PF, MRSKs 1, CV, MRSKs 2, and PR.

NOTICE Ensure that the loader with spherical phantom is positioned in the Body Coil.

Signal path

The output signal of the RFPA is fed via the BTB to the TX coil. The forward and reflected power is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from the minimum to the maximum.

Evaluation

- The actual forward and reflected voltage (measured via directional coupler) at maximum reflection is displayed.
- The maximum ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.
- The first diagram shows the measured values (voltage) of power forward and power reflected over the transmitter voltage.
- The second diagram shows the DICO reflection (power reflected/power forward (%)) over the transmitter voltage.

Conclusions

The objective of the test to ensure that the TX-path works properly without excessive reflections caused by defective connectors and cables.

If this test fails, perform the interactive **Cable** test BC-TX-path (reflection) (for experts only).

BC-TX-path (voltage)

Fig. 31 RFPA-TALES 1.5T

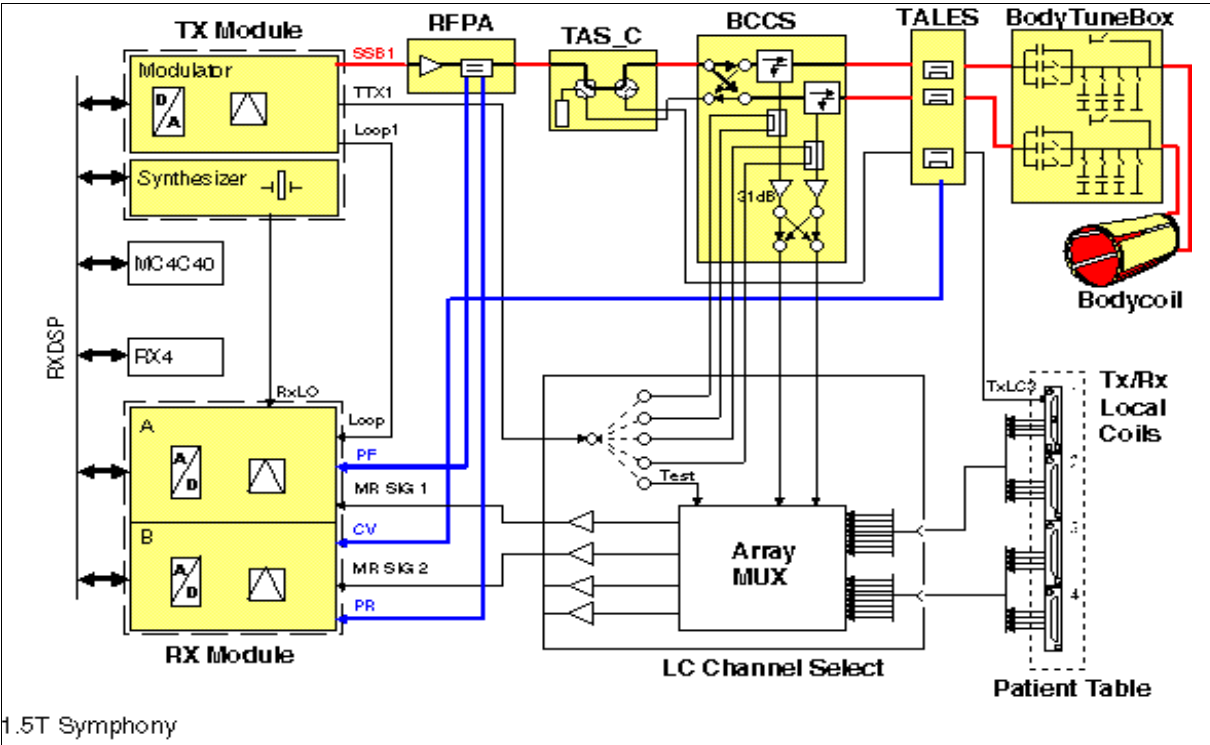
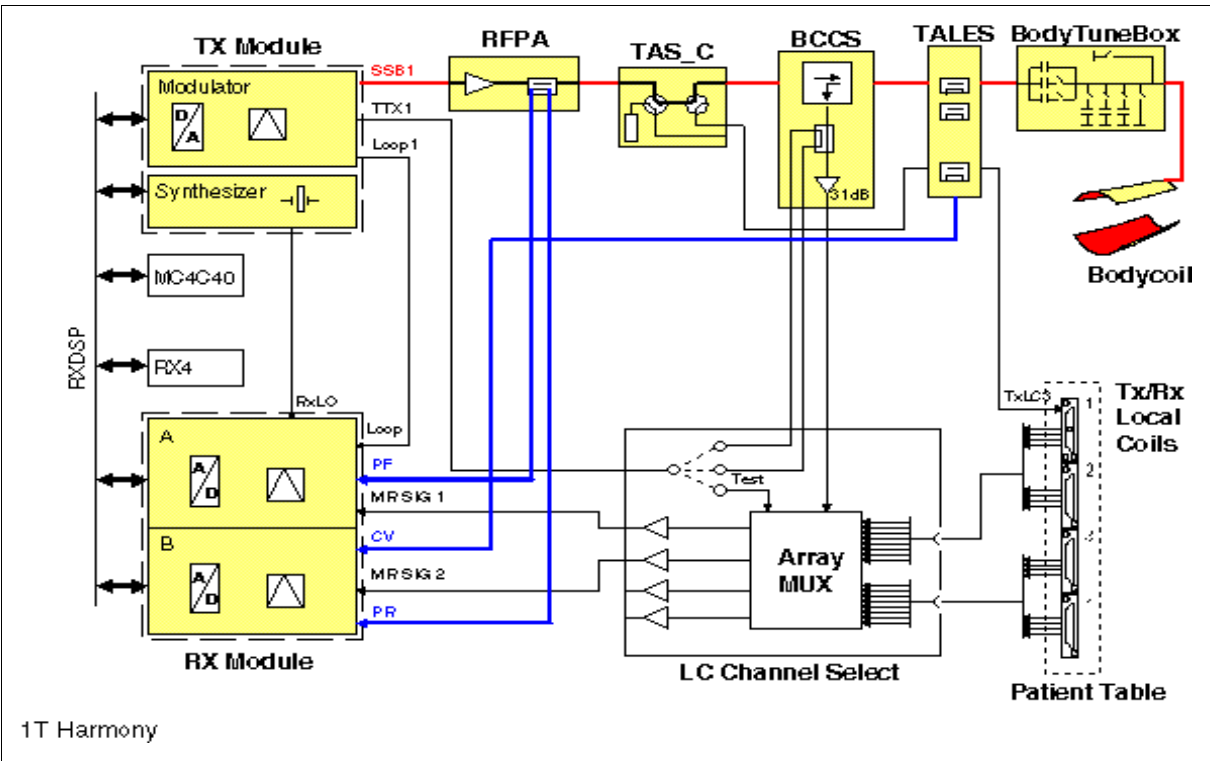


Fig. 32 RFPA-TALES 1T



NOTICE Ensure that the loader with spherical phantom is positioned in the Body Coil.

Signal path

The output signal of the RFPA is fed via the BTB to the TX coil. The forward and reflected power is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from the minimum to the maximum.

Evaluation

- The actual forward and reflected voltage (measured via DICO and TALES) at maximum deviation is displayed.
- The maximum deviation of the forward voltage (measured via DICO and TALES) is calculated.
- The first diagram shows the measured power forward (voltage, measured via DICO and TALES) over the transmitter voltage.
- The second diagram shows the deviation (%) of DICO and TALES power forward over the transmitter voltage.

Conclusions

If the deviation of DICO and TALES voltages is out of tolerance repeat first the RF characteristic measurement.

Significant higher voltage values measured by TALES could be caused by an RFPA oscillations on other than the MR-frequency: the DICO measurement is selective at the MR frequency, but the TALES is sensitive for a wider frequency band.

Perform the interactive **Cable** test BC-TX-path (voltage) (for experts only).

Local Coil TX Path (interactive/Multi Nuclei)

Fig. 33 Local Coil TX Path 1.5T

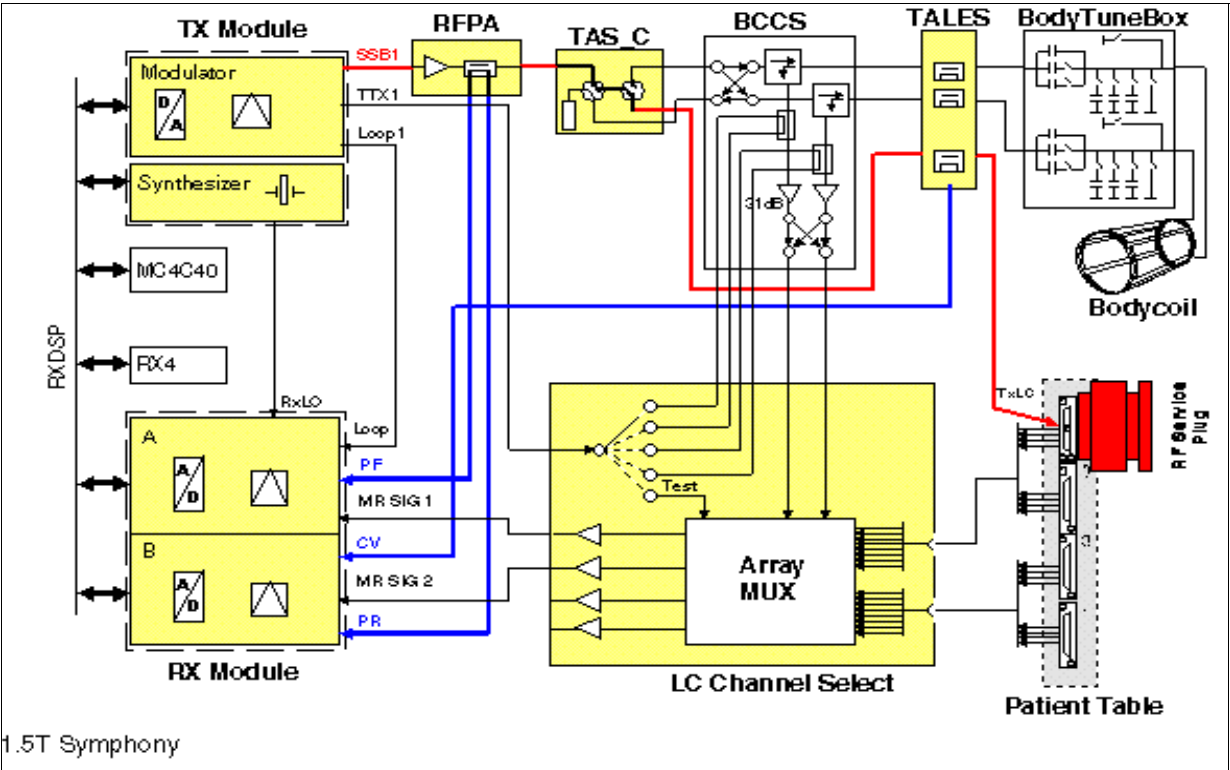
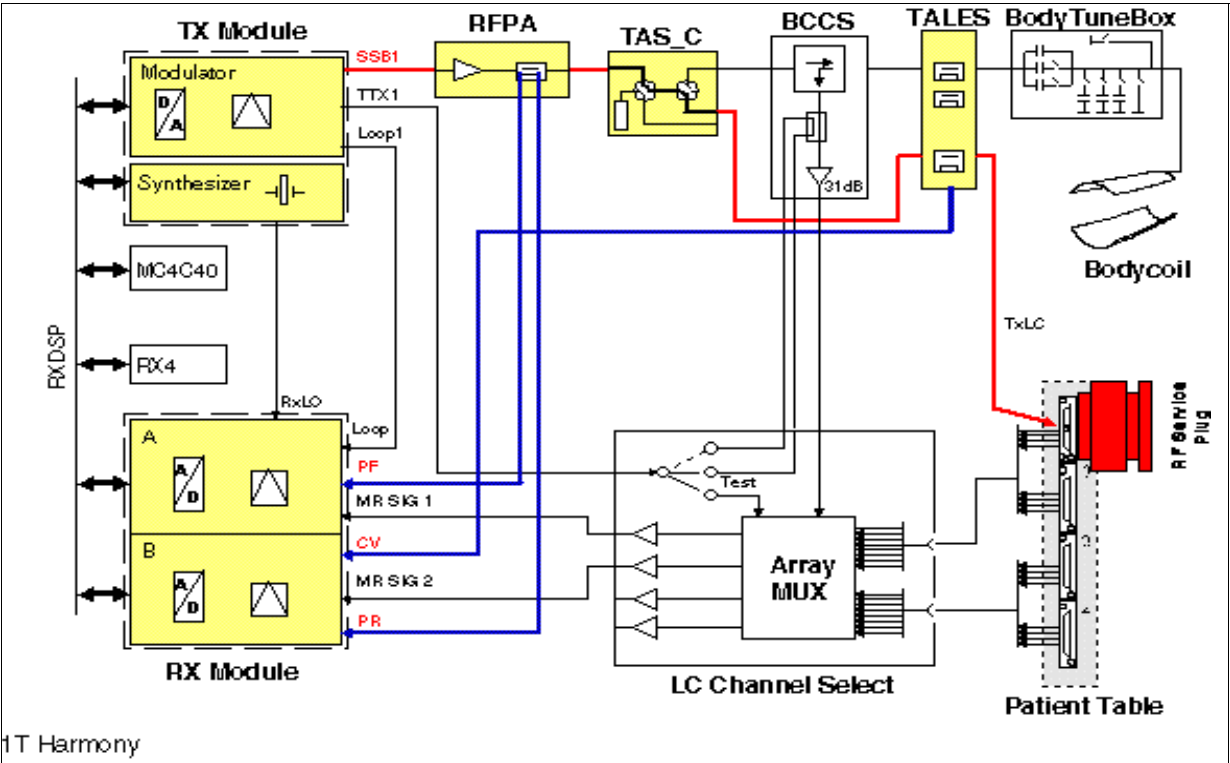


Fig. 34 Local Coil TX Path 1T



Preparation

The RF Service Plug must be connected to coil connector #1 at the patient table (connector for the extremity coil). Follow the instructions provided during the test run.

Signal Path

The output signal of the modulator is fed to the RFPA-TAS_C-TALES-Extremity Coil (local coil connector #1). The TX path of the TAS_C is switched to extremity coil mode.

This test requires that the transmission cable to the extremity coil be terminated at 50 Ohm. The connection must be terminated using the 50 Ohm resistor in the RF Service Plug.

The test consists of two measurements. One RF pulse is transmitted for each measurement. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Evaluation

- Deviation of the forward voltage from the DICO and the TALES
- Ratio of the reflected and forward voltage from the TALES

Conclusions

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for converting the ADC values from the receiver into voltage values), a valid characteristic curve must be present.

With regard to the linear RF-characteristic, i.e. after the software has been installed and before the tune-up has been performed, the deviation may be significantly above the specification.

The ratio of the reflected voltage and the forward voltage measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. In this test, it is determined by the 50 Ohm resistor in the RF Service Plug. If the ratio lies outside of the specifications, the cause can probably be found in the cable connection from the TAS via Tales to coil connector #1 (movable cable in the table) or in the TAS itself (for this test, the TAS is switched to extremity coil path).

If the forward voltage does not exceed a certain level (approx. 100 V), the RF safety watchdog evaluates the measured values within the most sensitive range of the TALES. In this range, the reflected voltage of the TALES can no longer be produced and the value 0 is displayed. In this case, the evaluation of return / forward voltage is no longer valid.

BTB

- BTB Communication/Status
- BTB Test (functional + interactive)

BTB Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

BTB Tests (functional)

NOTICE	Ensure that the loader with the large spherical phantom is precisely positioned in the Body Coil.
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The BTB tests are based on the analysis of reflective voltage factors for the different switching modes of the short circuit switch, the PIN diodes and the tuning capacitors in the BTB.

Prerequisite for this test is the successful tuning calibration of the TALES (tuning calibration is performed during Tune-up, the successful BTB calibration is not necessary for this test). Otherwise, the results of this test are invalid.

The test consists of several individual tests each including a reference measurement and the actual level measurements followed by an evaluation: Static Detune, Dynamic Detune, test of Serial Capacitors and Relays and Parallel Capacitors and Relays.

The first measurement is performed as a "reference" with an abbreviated path, i.e. without using the BTB and the BC. The other measurements are done with a specified setting of the switch in the BTB.

After these tests the measurements are repeated for the BTB 2 if installed.

Signal Path for the reference measurement

The TTX output signal of the modulator is fed to the LCCS-BCCS (preamplifier)-LCCS and to the receiver.

The level measured is used as the reference.

Fig. 35 BTB Test, Measurement for Reference 1.5T

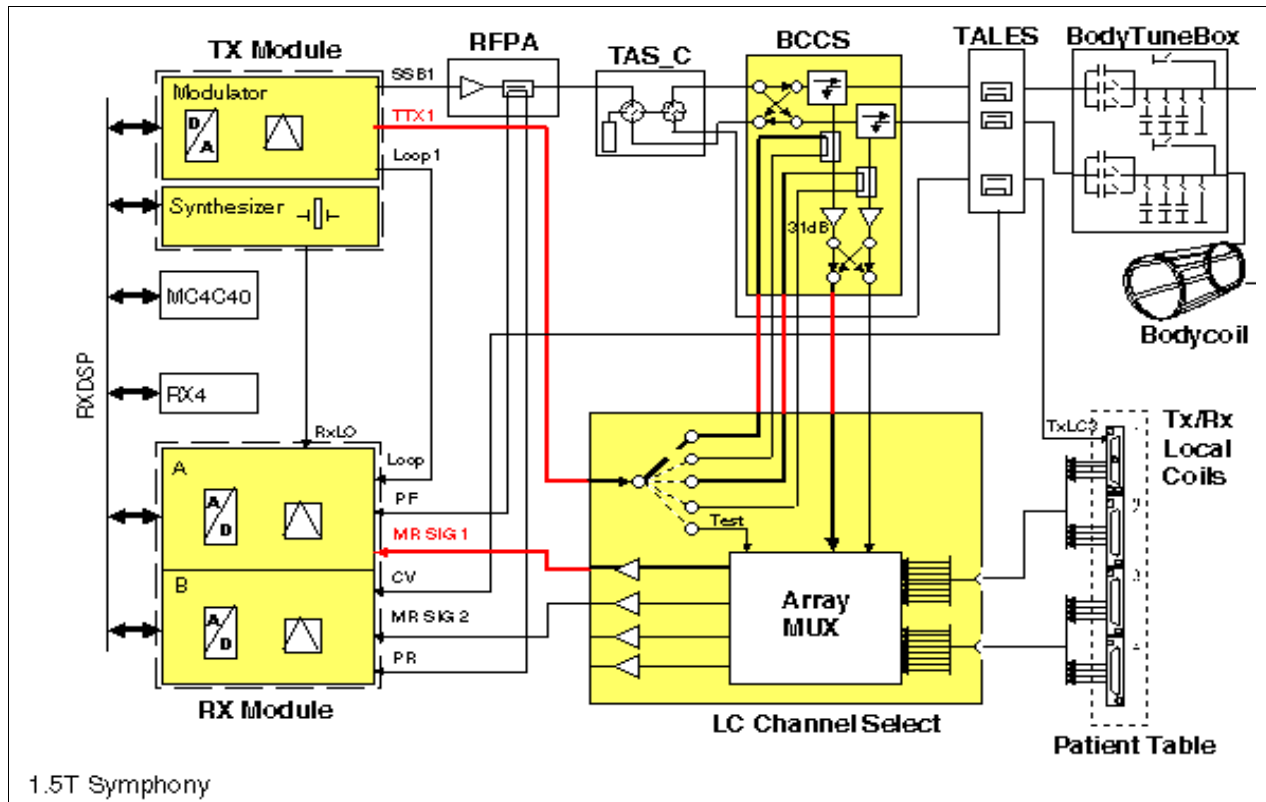
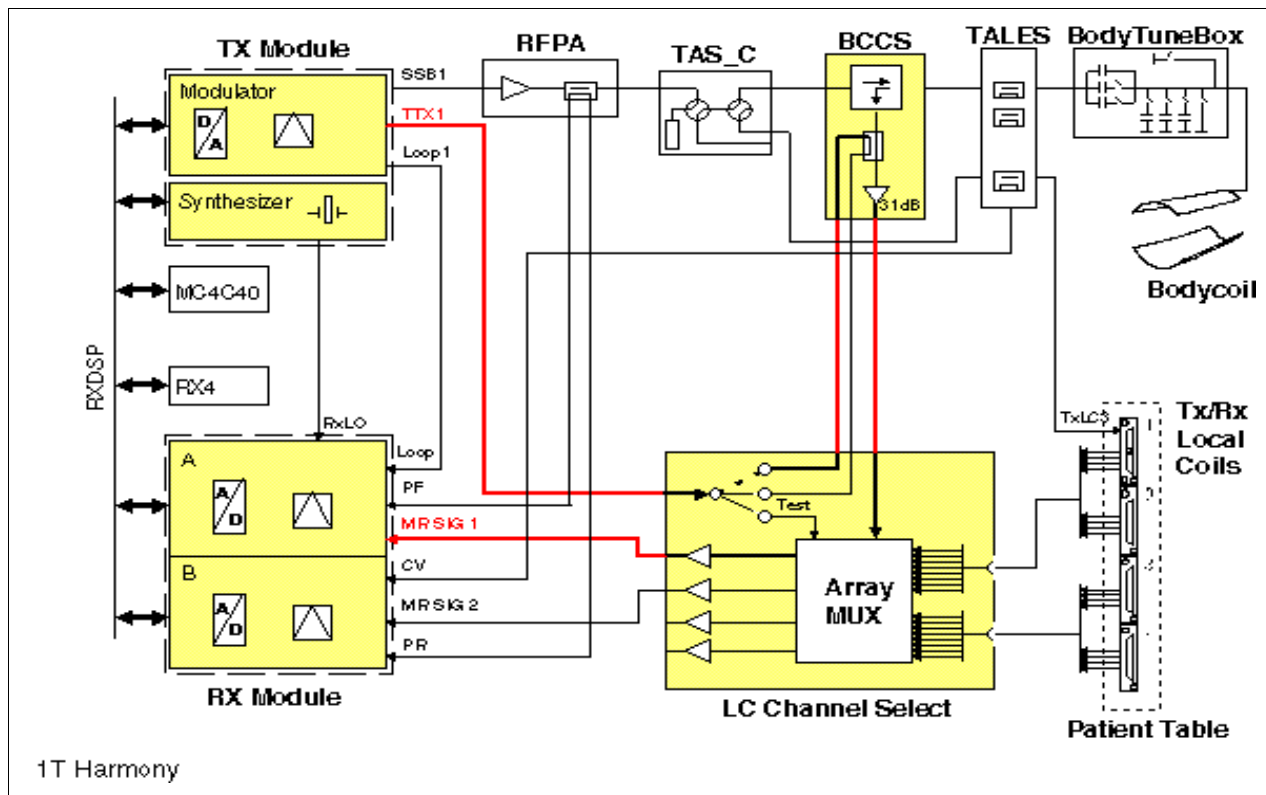


Fig. 36 BTB Test, Measurement for Reference 1T



Signal Path for the measurement

The TTX output signal of the modulator is fed to the LCCS-BCCS-TALES-BTB-BC and reflected back to the BTB-TALES-BCCS-LCCS and receiver. The short circuit switch, the PIN diodes and the capacitors in the BTB are switched in sequence.

Test 1: Static detuning active, the serial capacitors are switched in sequence.

Test 2: Dynamic detuning active, the serial capacitors are switched in sequence.

Test 3: The serial capacitors are switched in sequence.

Test 4: The parallel capacitors are switched in sequence.

The tests are repeated for the BTB2 (Symphony only).

NOTE Some tests are sensitive to a malfunction of the Body Coil as well.

Fig. 37 BTB Test, Measurement 1.5T

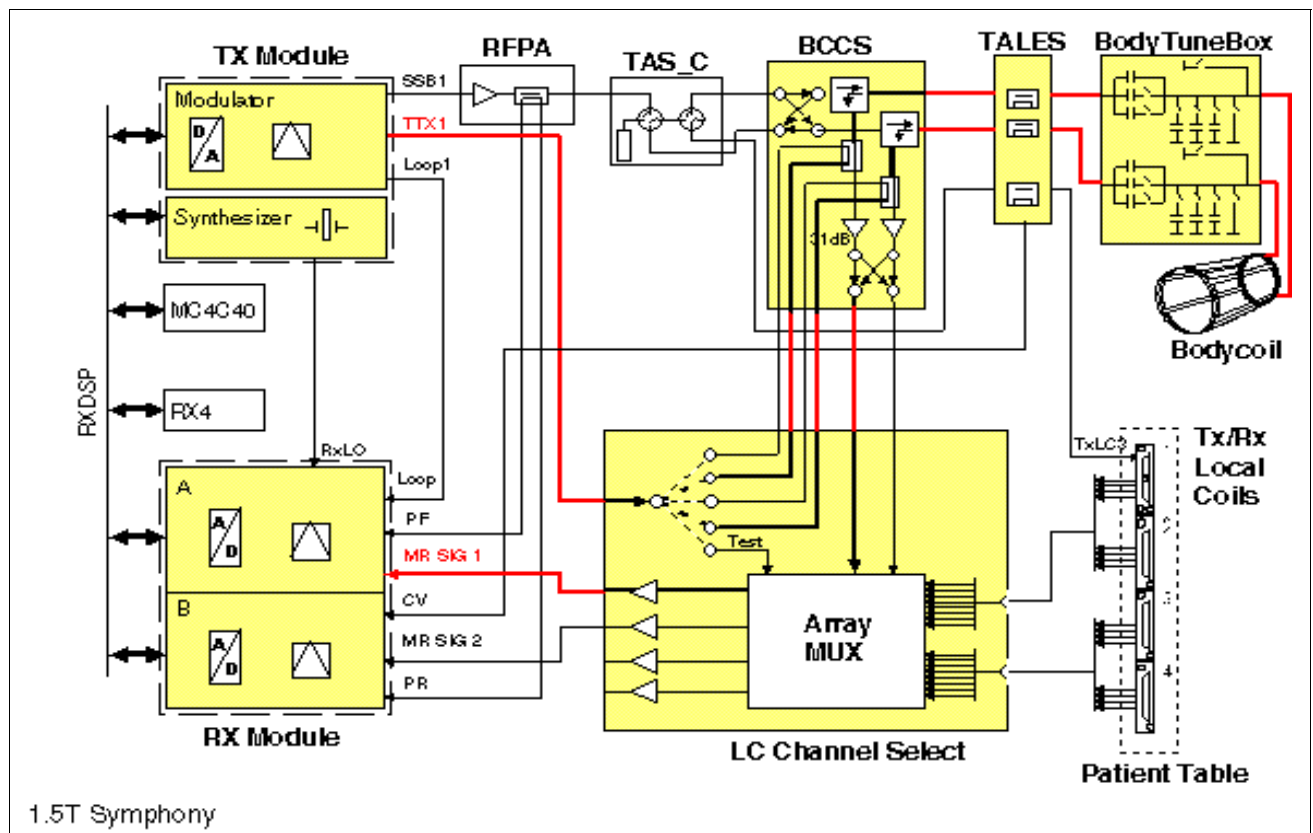
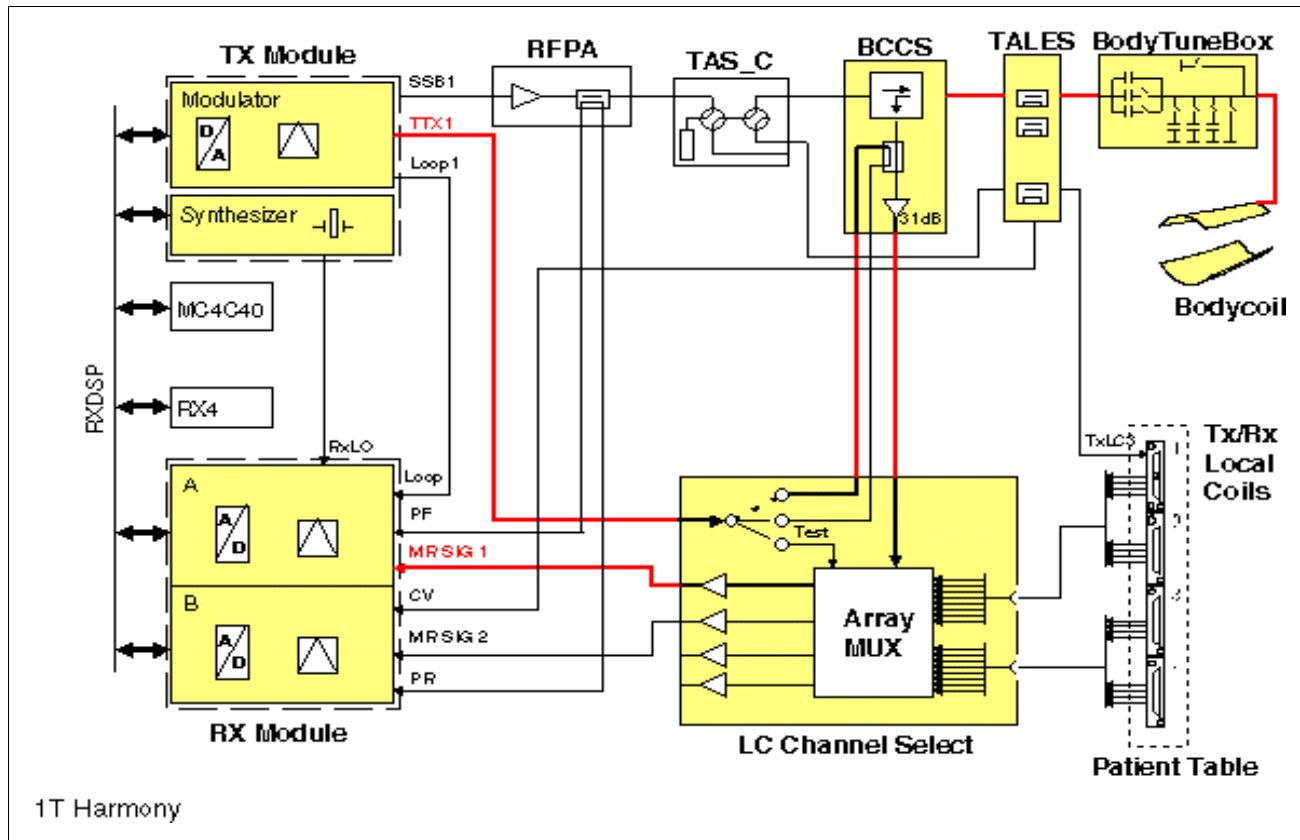


Fig. 38 BTB Test, Measurement 1T



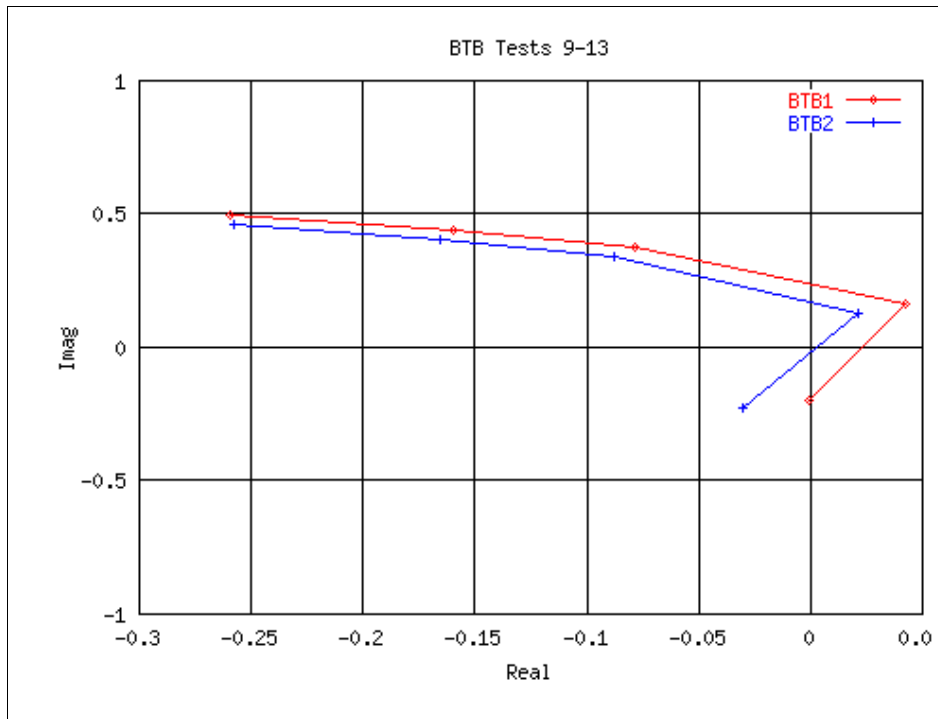
Evaluation

The reflection factors (real and imaginary) are generated for the individual switch settings and checked against the specifications.

Afterwards, several relative values from the individual results are determined. They allow a more detailed evaluation of the BTB functionality.

The results of some measurements are displayed graphically in a x-y diagram (real part on x-axis, imaginary part on y-axis). From deviations of the measurement compared with the normal situation shown in Fig. 27 the error can be analyzed. For further information refer to the Troubleshooting Guide.

Fig. 39 Example of measurement result (for Symphony)



Conclusions

In the HTML output, the BTB Diagnosis, possible errors of the BTB/BC are displayed.

In case of doubt whether to replace either the Body Coil or the body tune box (whichever is defective), please contact the MR Hotline (HSC) for further instructions.

In addition, perform the RFCI tests for checking the PIN diodes of the BTB.

BTB Tests (interactive)

For better error localization (TALES/BTB) two interactive tests are used. Pop-up windows inform you about to establish the correct measurement setup (no transmit signal to the body coil).

- Check of Tuning Calibration (TALES outputs X3, (X4) open)
- BTB Reflection factors (BTB outputs X1/1, (X1/2) shorted)

The Tune-up procedure "Tuning calibration" will be checked by measurement of a reflection factor and comparison with a reference value.

Mesurement of reflection factors of the BTB systems checks the parallel capacitors C_s and relays of the BTB.

BC

- Body Coil (functional)
- Body Coil (interactive)

Body Coil (functional)

NOTICE Ensure that the loader with the large spherical phantom is **precisely** positioned in the Body Coil.

The body coil test is based on the analysis of reflective voltage factors that depend on the frequency. The test simulates a network analyzer that provides the frequency for the smallest reflection factor and the resonance curve of the body coil as results. The available frequency range of approximately 1MHz is stepped through in 10kHz steps. The respective reflection factors are produced and entered in a frequency reflection factor diagram. The frequency with the smallest reflection factor is checked against the specifications. The Q-factor of the coil will be evaluated and checked against specification.

Additionally a power test with increasing transmitter amplitude is performed. The TALES reflection values over the transmitter amplitudes are graphically displayed.

The test is available in two variations:

- As an automatic test with the loader and phantom positioned (functional)
- Load-free, i.e. the table with loader and phantom must be outside the magnet (interactive test)

Fig. 40 Body Coil 1.5T

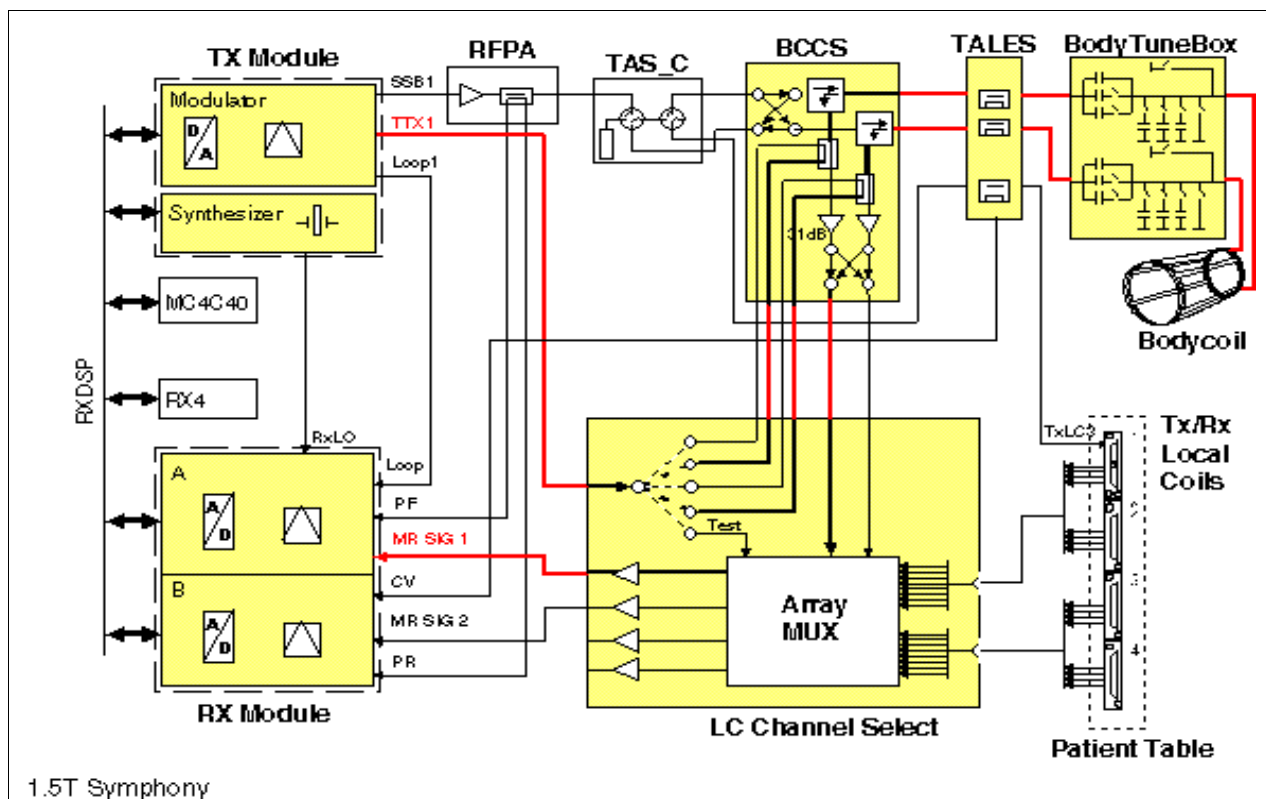
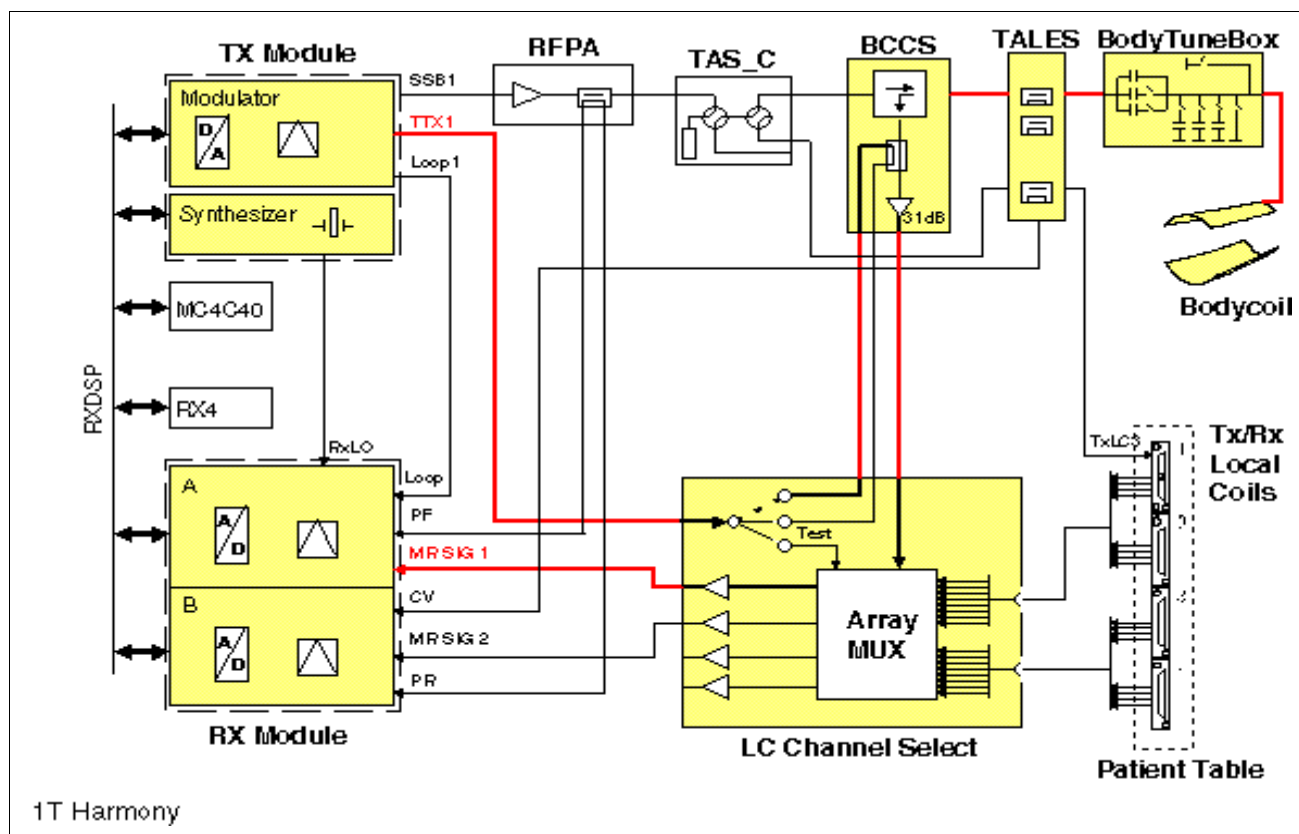


Fig. 41 Body Coil 1T



Body Coil (interactive/without load)

The first part of the measurement is identically to the functional measurement. Additionally the decoupling of the body coil elements (transmission measurement only for Symphony) is evaluated.

Fig. 42 BC - Reflection Evaluation

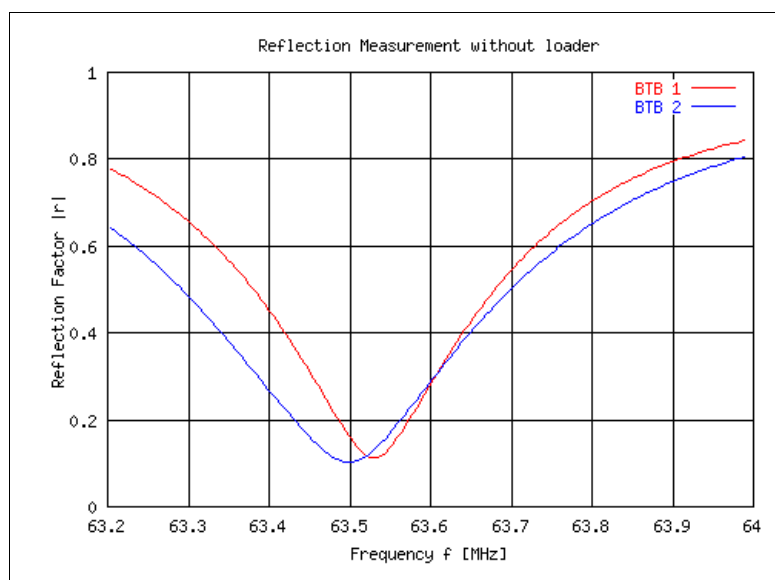
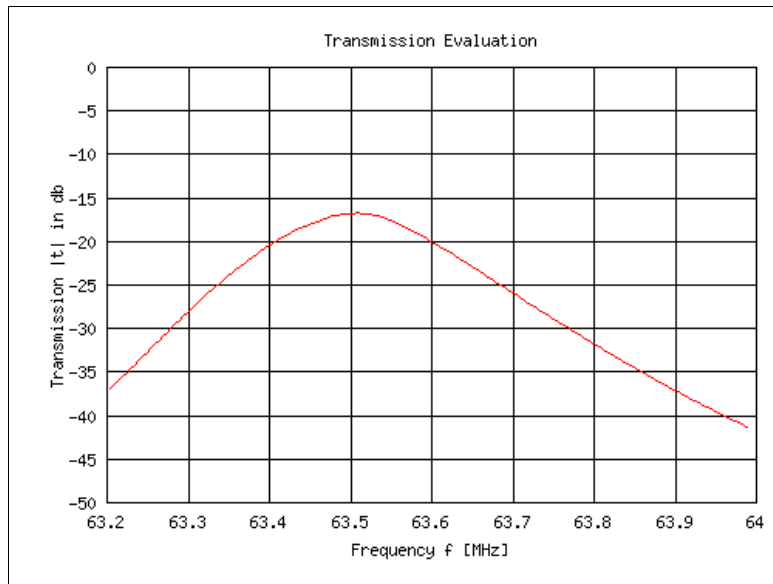


Fig. 43 BC - Transmission Evaluation



Also a power test with increasing transmitter amplitude is performed. The TALES reflection values over the transmitter amplitudes are graphically displayed.

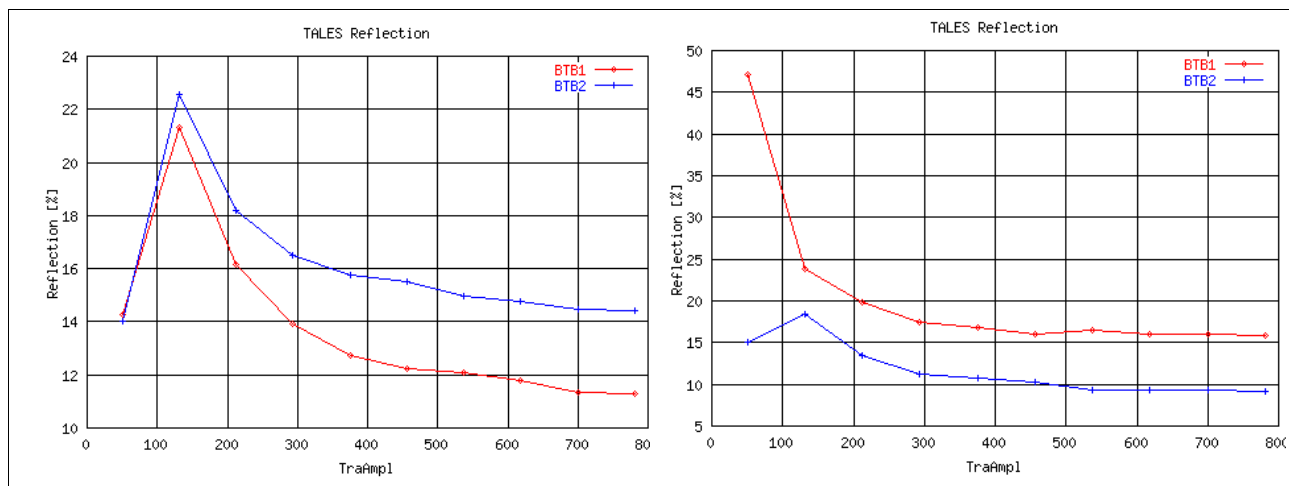
Conclusions

If the tolerances of the reflection measurement evaluation are exceeded and the BTB tests are in specification, the problem is probably located in the BC.

The power test checks the disruptive strength of the BC. The test should run without shut down to the maximum voltage. The reflection of TALES should be approximately constant at voltages higher than 200 V. Refer to the figure below.

The power test without load is the worst case, i.e. the reflected voltages are significantly higher.

Fig. 44 BC power test - Left: TALES reflection with load, Right: TALES reflection without load (example of Symphony)



If the results exceed the specification, in case of doubts, please contact the MR Hotline (HSC) for further instructions.

Cable

- BC-TX-path (refl.)
- BC-TX-path (volt.)
- LC-TX-path

NOTE **These tests are for experts only!**
Additionally special equipment is necessary for the tests.

These advanced tests enable you to check the complete TX-path to BC or LC and to make a systematical troubleshooting in the TX-path by connecting a power 50 Ohm terminator at various locations in the TX-path and/or configuring of particular loops.

The test should be used by experienced service engineers only. Special tools (power dummies, RF-cables, connectors, adapters and attenuators) are needed.

Popup windows inform you about the actual measurement configuration.

NOTE The tests require a valid RF characteristic or the linear RF characteristic.
Ensure that the spherical phantom is positioned in the TX-coil.

NOTE When using the linear characteristic it might be possible that the used transmitter amplitude exceeds the allowed range resulting in an abort of the measurement. This is normal and not a problem of the system.

NOTE For the first measurement of each test also the reflection factors are evaluated and analyzed. If the reflection exceeds 90% the test is aborted with the error message Error: DICO reflection out of spec for first measurement.
In that case check the cabling and the Service Plug/50 Ohm power dummy (if used).

Safety

WARNING **High RF voltage**

Contact with RF conductors during transmitting of the RFPA will result in death or serious physical injury.

Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input cable at X1 and the unblank line U1 from the RFPA.

BC-TX-path (refl.) (interactive)

NOTICE When transmitting to the Body Coil, ensure that the loader with spherical phantom is positioned in the Body Coil.
The test requires a valid RF characteristic or the linear RF characteristic.
For spectroscopy coils select the checkbox Multi Nuclei and the desired nucleus.

The objective of the test to check the components of the TX-path to the body coil with maximum transmitter amplitude for disruptive strength step by step. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (read from the RF characteristic data) are performed.

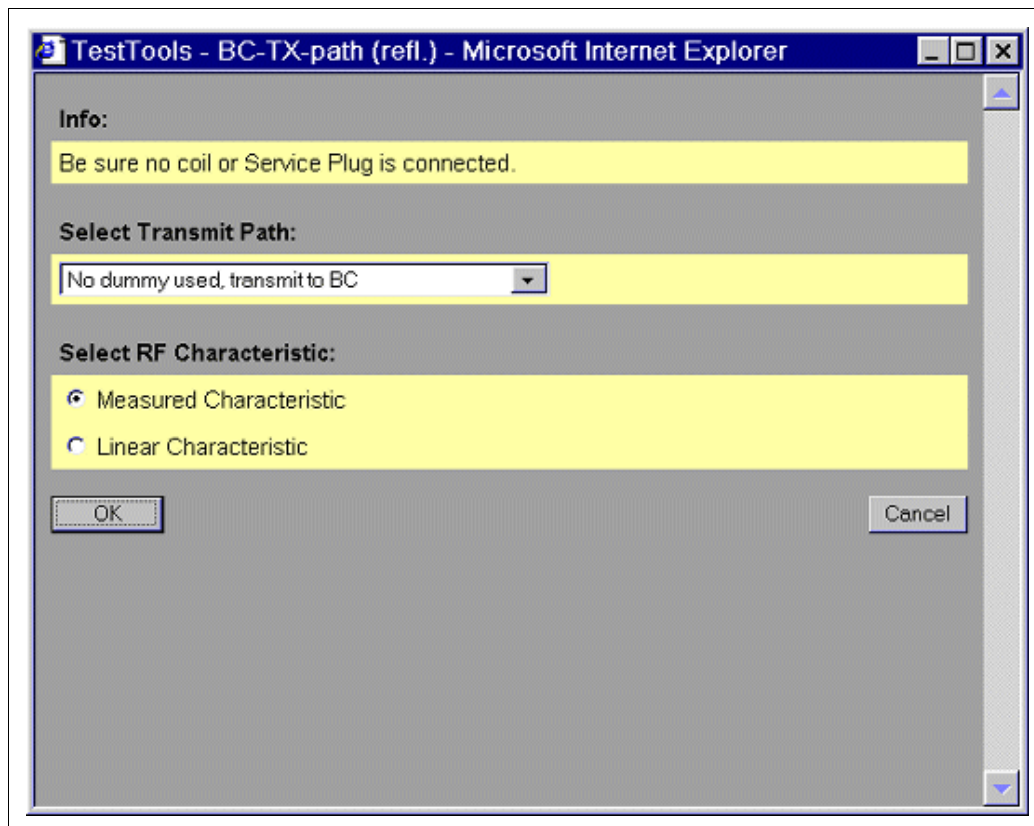
Signal path

The normal TX-path to the body coil is used.

The TX-path can be opened at several locations and terminated by one or two 50 Ohm dummies. In the appearing popup window you can select the desired TX-path (refer to [Fig. 45](#) and [Fig. 46](#)).

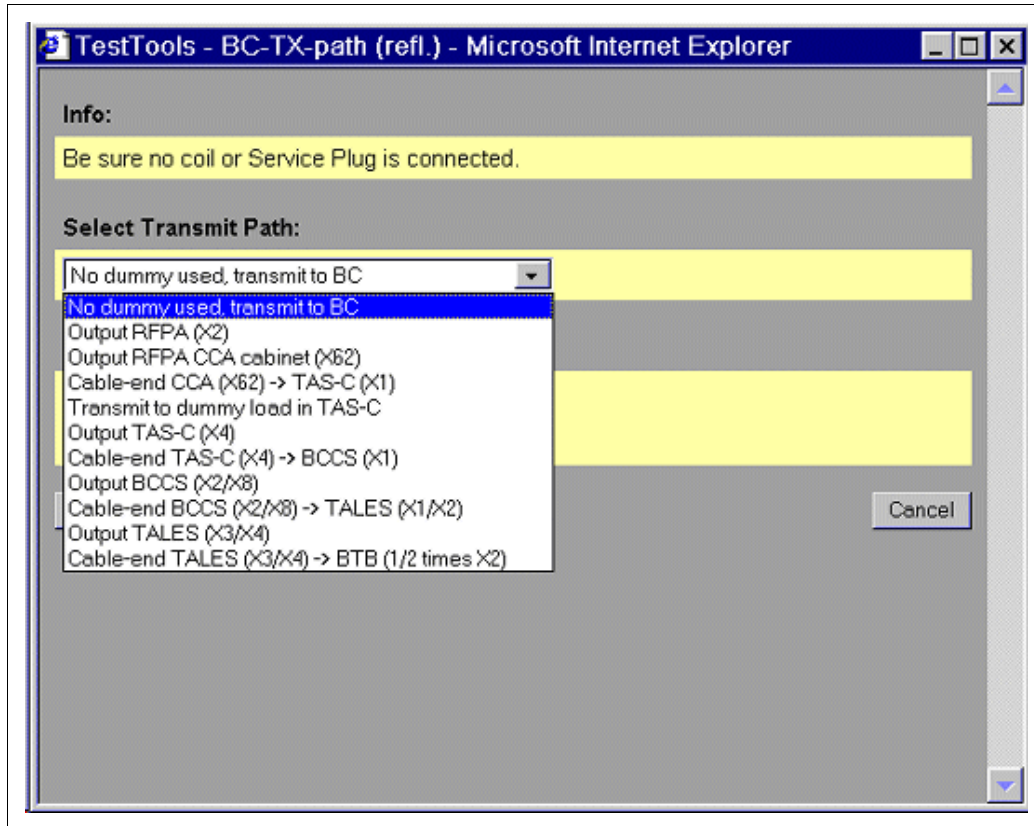
The selection of the location of the termination of the TX path with a dummy load makes no influence on the measurement but selects the heading of the HTML output table. This is e.g. helpful when sending the results to the HSC.

Fig. 45 Pop up window BC-TX-path (refl.) - Selection of used RF Characteristic



The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the popup window.

Fig. 46 Pop up window BC-TX-path (refl.)
Selection of TX-path termination location by 50 Ohm dummies



Evaluation

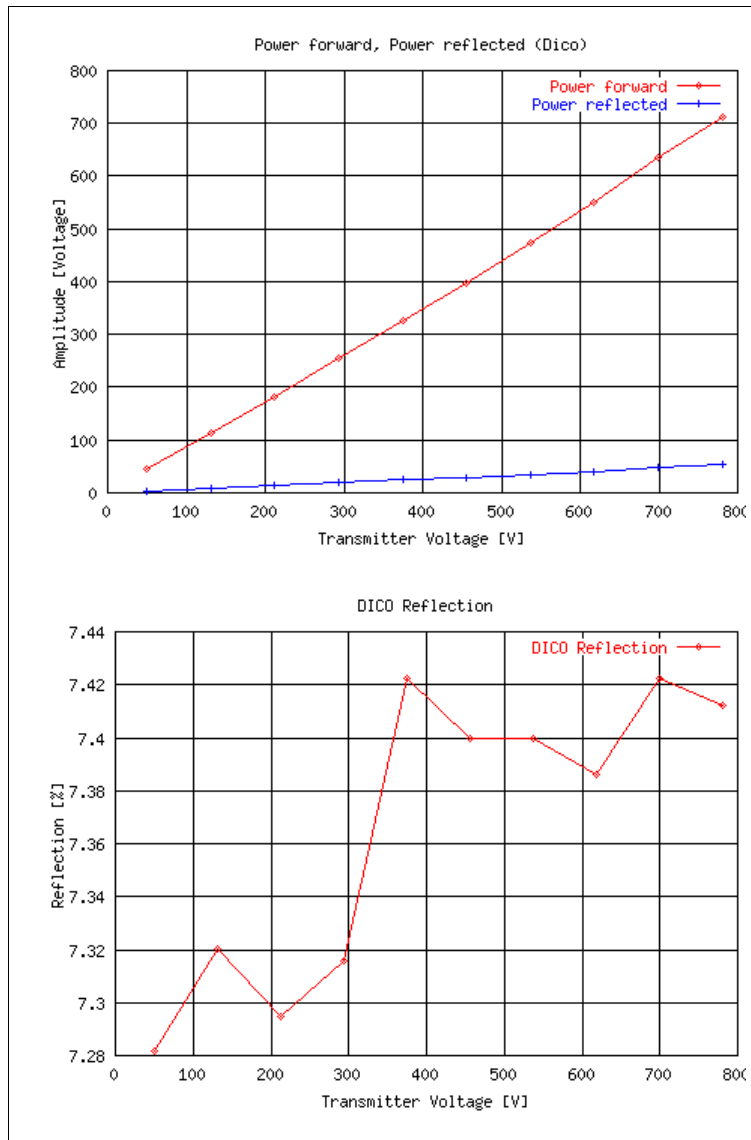
HTML output table:

- PF (DICO) at maximum reflection (Volt)
- PR (DICO) at maximum reflection (Volt)
- PR/PF (DICO) Reflection at maximum ratio (%)

Graphical output:

- PF, PR (DICO/Volt) over TRA voltage
- PR/PF DICO Reflection (%) over TRA voltage

Fig. 47 Example of graphical output



Conclusions

If there are too high reflections in the BC-TX-path you can locate the defective component in the TX-path by repeating the test after connecting the dummy load to a component before it's current position. In particular check for loose connectors.

Examples for errors are low power of RFP, arcing of capacitors in the BTB or BC, defective cables and/or connectors.

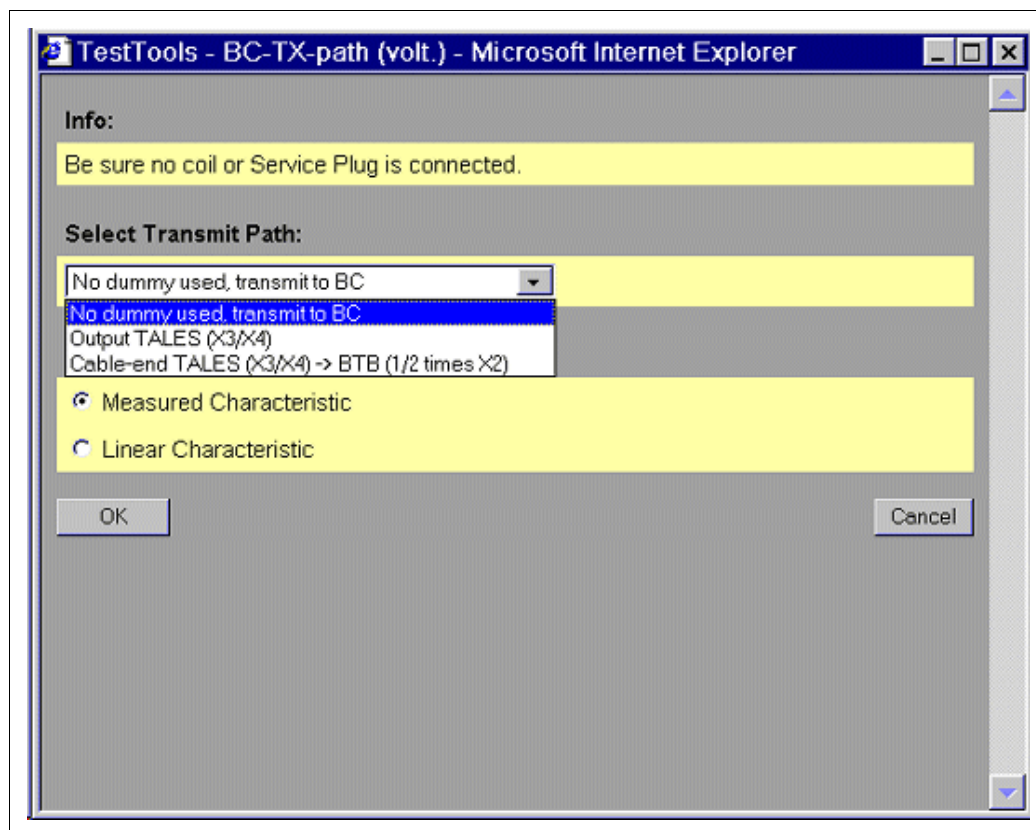
The DICO-Reflection should be lower than 50%. Sporadic reflections up to 70% may be caused by arcing in the BTB (switching of capacitors), reproducible high reflections may be caused by arcing in the Body Coil.

BC-TX-path (volt.) (interactive)

NOTICE When transmitting to the Body Coil, ensure that the loader with spherical phantom is positioned in the Body Coil.
The test requires a valid RF characteristic or the linear RF characteristic.

The objective of the test is to detect oscillations of the RFPA. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (read from the RF characteristic data) are performed.

Fig. 48 Pop up window BC-TX-path (volt.)
Selection of TX-path termination location by a 50 Ohm dummy

**Signal path**

The normal TX-path to the Body Coil is used.

The TX-path can be opened at locations between TALES and BTB and terminated by (one or two) dummy loads. In the appearing popup window you can select the desired TX-path (refer to Fig. 48). The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the popup window.

The selection of the location of the interruption in the TX-path makes no influence of the measurement but selects the heading of the output table. If the interruption is between TALES and RFPA, the TALES values can not be measured and displayed.

Evaluation

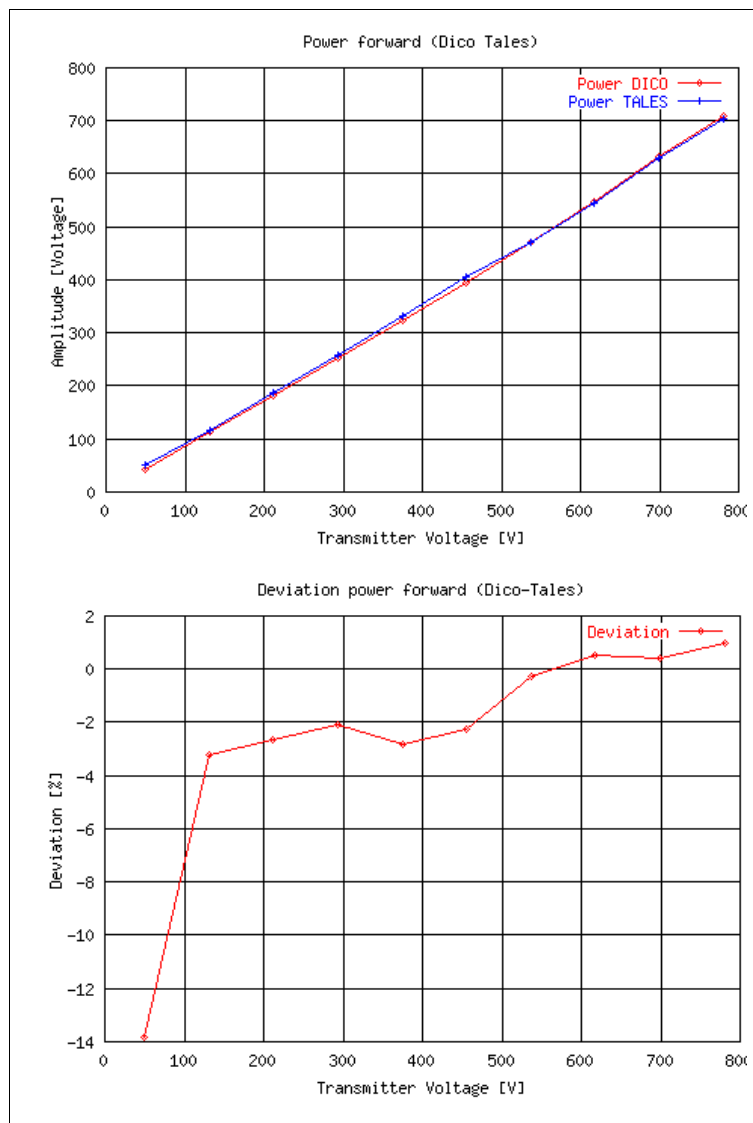
HTML output table:

- PF (DICO) at maximum deviation (Volt)
- PF (TALES) at maximum deviation (Volt)
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ maximum deviation (%)

Graphical output:

- PF(DICO/Volt) and PF(TALES/Volt) over TRA voltage
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ deviation (%) over TRA voltage

Fig. 49 Example of graphical output



Conclusions

If the deviations between DICO and TALES are out of specification, repeat the RF characteristic measurement.

Locate the defective component in the TX-path by repeating the test after connecting the dummy load to a component before it's current position.

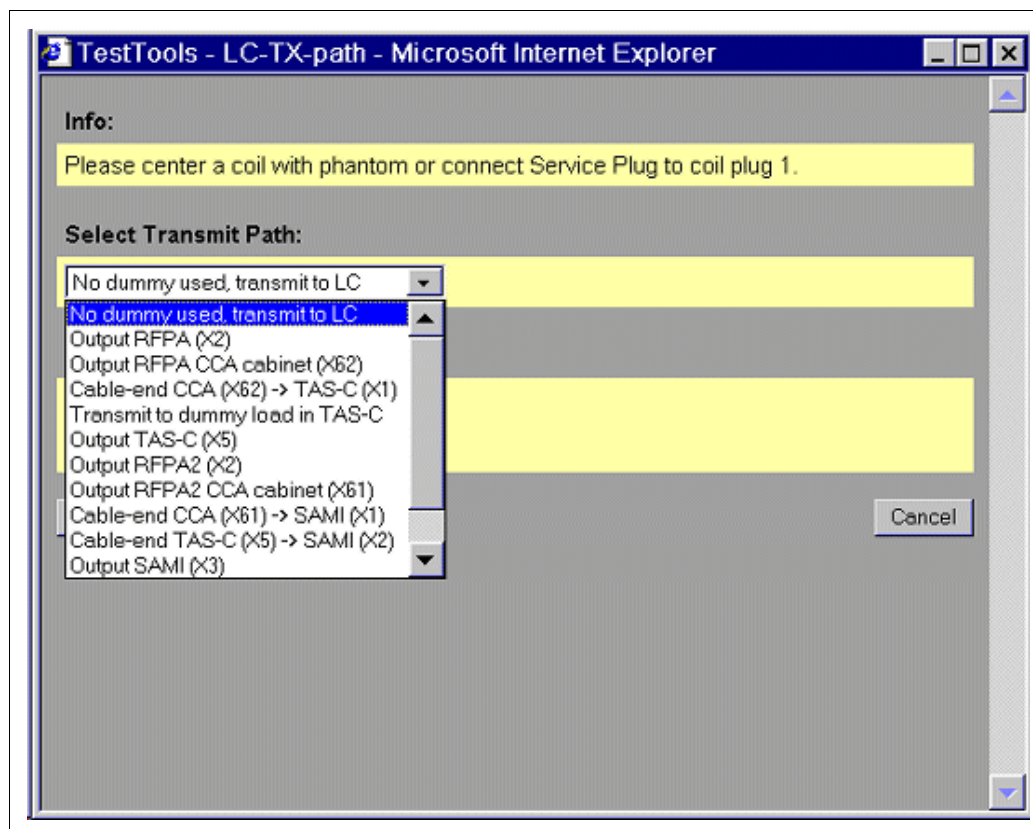
The reason of significantly higher reflection measured by the TALES in comparison to DICO can be oscillations of the RFPA, transmitted on an other frequency besides the MR-frequency.

LC-TX-path (interactive/Multi Nuclei)

NOTICE When transmitting to the Local Coil transmit path, ensure that either the service plug or a local TxRx-coil is connected to coil plug 1.
The test requires a valid RF characteristic or the linear RF characteristic.
For spectroscopy/multi nucleus coils select the checkbox Multi Nuclei and the desired nucleus.

The objective of the test is to check the components of the LC-TX-path. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (95 V for service plug and maximum local coil voltage read from the coil file) are performed.

Fig. 50 Pop up window LC-TX-path
Selection of TX-path termination location by a 50 Ohm dummy



Signal path

The TX-path for local coils is used. The used RFPA (CORA/DORA) depends on the selection of the nucleus.

The TX-path can be opened at locations and terminated by a 50 Ohm dummy. In the appearing popup window you can select the desired TX-path (refer to [Fig. 50](#)). The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the popup window.

The selection of the location of the interruption in the TX-path makes no influence of the measurement but selects the heading of the output table. If the TX-path is terminated by a 50 Ohm dummy between TALES and RFPA, the TALES values can not be measured and displayed.

NOTE	Whenever a dummy load is used to interrupt the TX-path BEFORE the TALES, the comparison of DICO- and TALES-values will not be successful as there are no TALES-values available and thus the test will fail. This is normal and not a problem of the system.
-------------	---

Evaluation

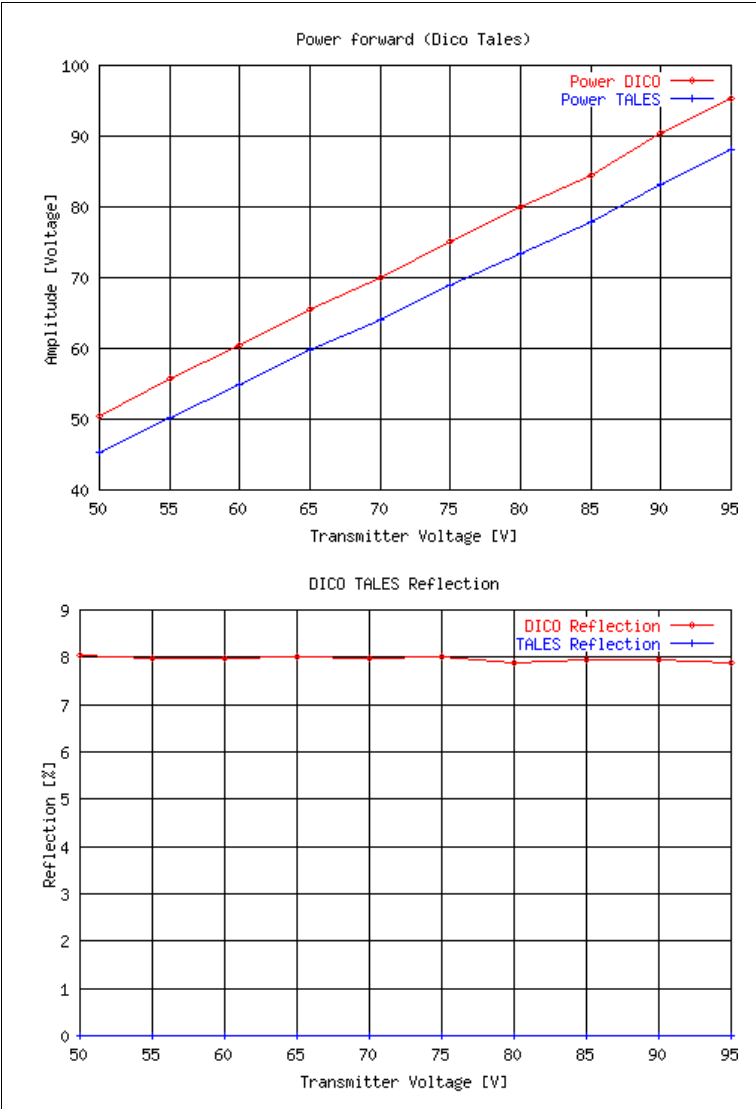
HTML output table:

- PF (DICO) at maximum deviation (Volt)
- PF (TALES) at maximum deviation (Volt)
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ maximum deviation (%)

Graphical output:

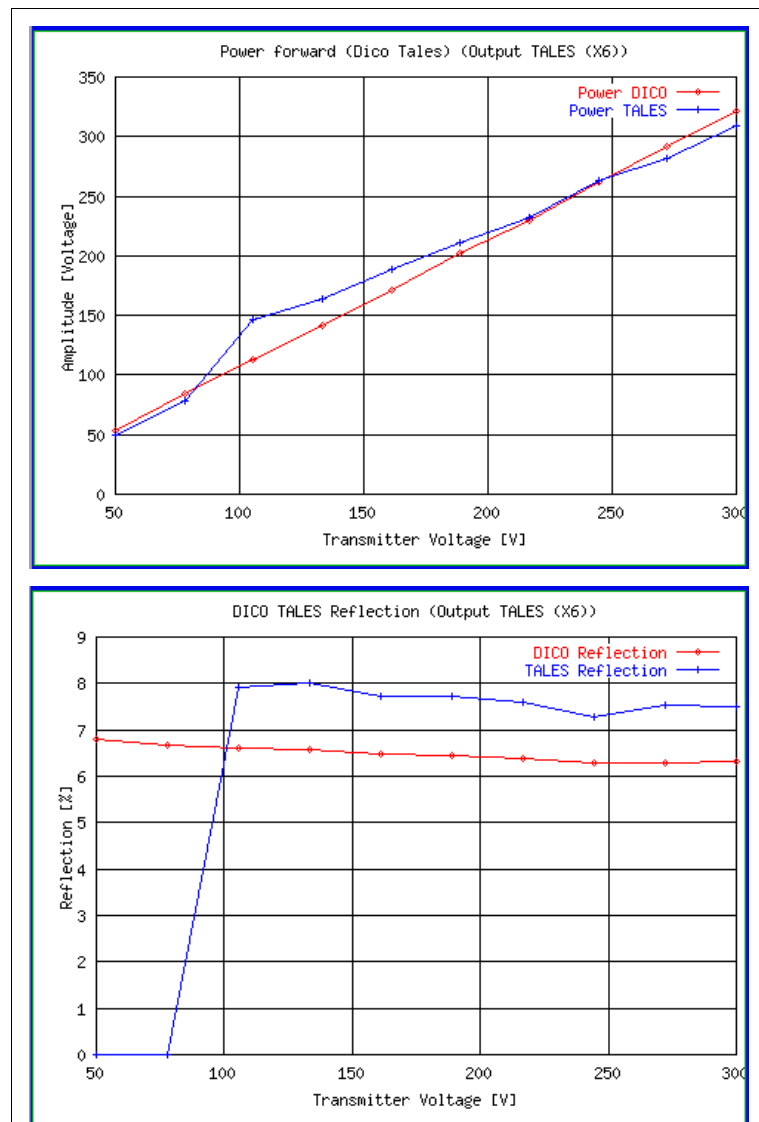
- PF(TALES/Volt) and PF(DICO/Volt) over TRA voltage
- PR/PF DICO reflection (%) and PR/PF TALES reflection (%) over TRA voltage

Fig. 51 Example: Transmit to Service plug



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Fig. 52 Example:
Transmit to Extremity Coil (Dummy connected to TALEs X6)



Conclusions

NOTE For reflected voltages below 100 V the measured/displayed TALEs value is 0 V.

If the deviations between DICO and TALEs (for voltages higher than 100 V) are out of specification, repeat the RF characteristic measurement for the actual nucleus.

Locate the defective component in the TX-path by repeating the test after connecting the 50 Ohm dummy to a component before it's current position.

LCCS (Functional, Comm/Status)

- LCCS Communication/Status
- BCCS (preamp1) - LCCS Loop
- BCCS (preamp1 2) - LCCS Loop (Symphony only)

LCCS plug 1,2,3,4 (Interactive, Multi Nuclei)

- LCCS (SGA x) Service Plug, pos n
- LCCS SGA x Loop (High Gain/Low Gain, plug 4 only)

LCCS Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

BCCS (preamp1) - LCCS Loop

Fig. 53 BCCS (preampl) - LCCS 1.5T

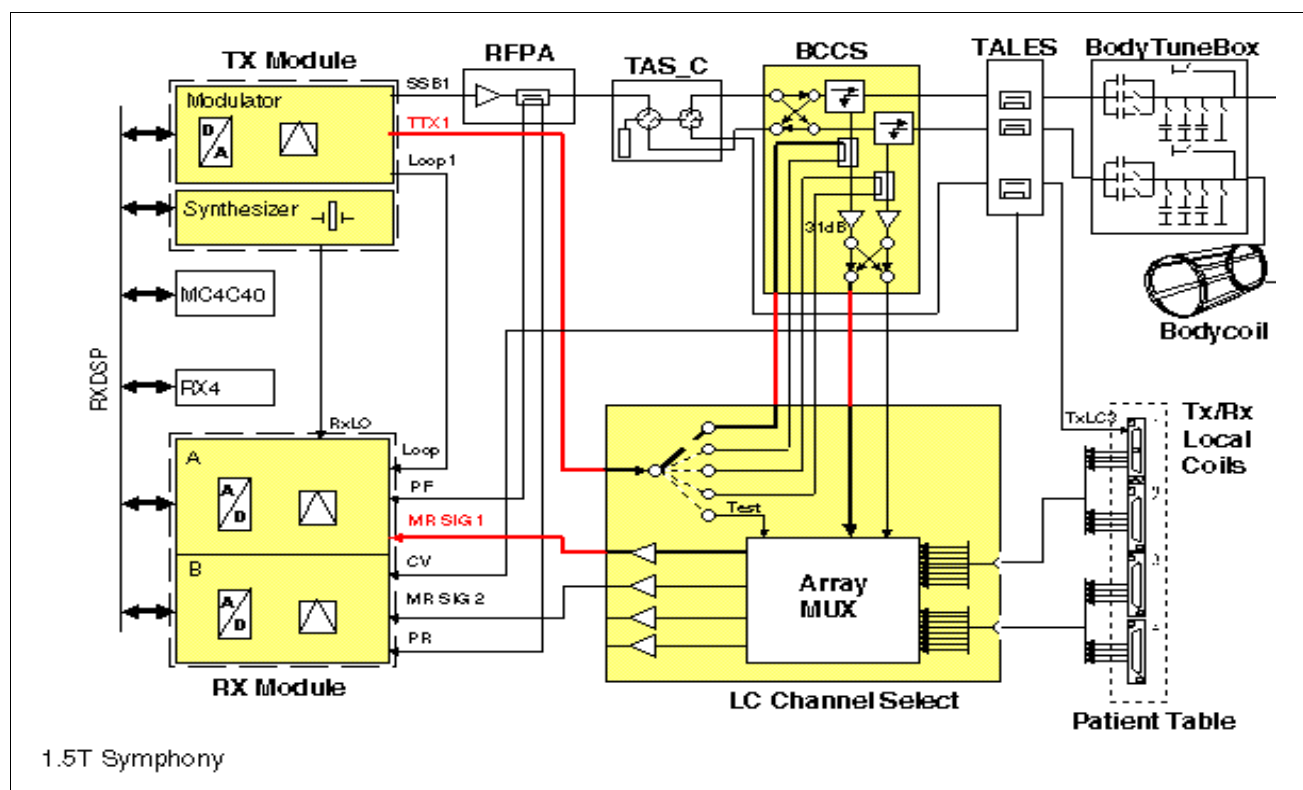
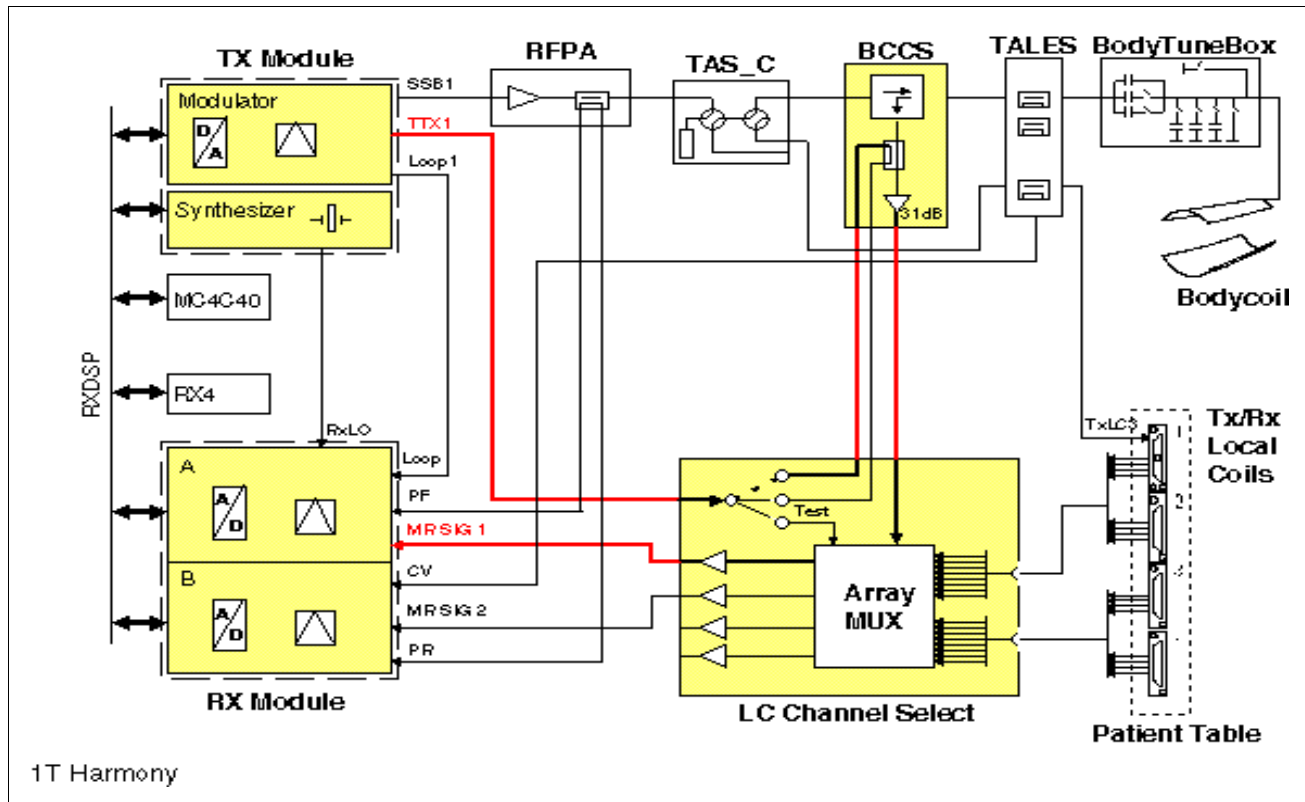


Fig. 54 BCCS (preamp) - LCCS 1T



Signal Path

The TTX output signal of the modulator is fed via the LCCS forward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver A/B (MRSIG1/MRSIG2) via the LCCS. The tests are performed with high gain and low gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

In a second measurement the TTX output signal of the modulator is fed via the LCCS backward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver 1A/2A via the LCCS. The test is performed with high gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

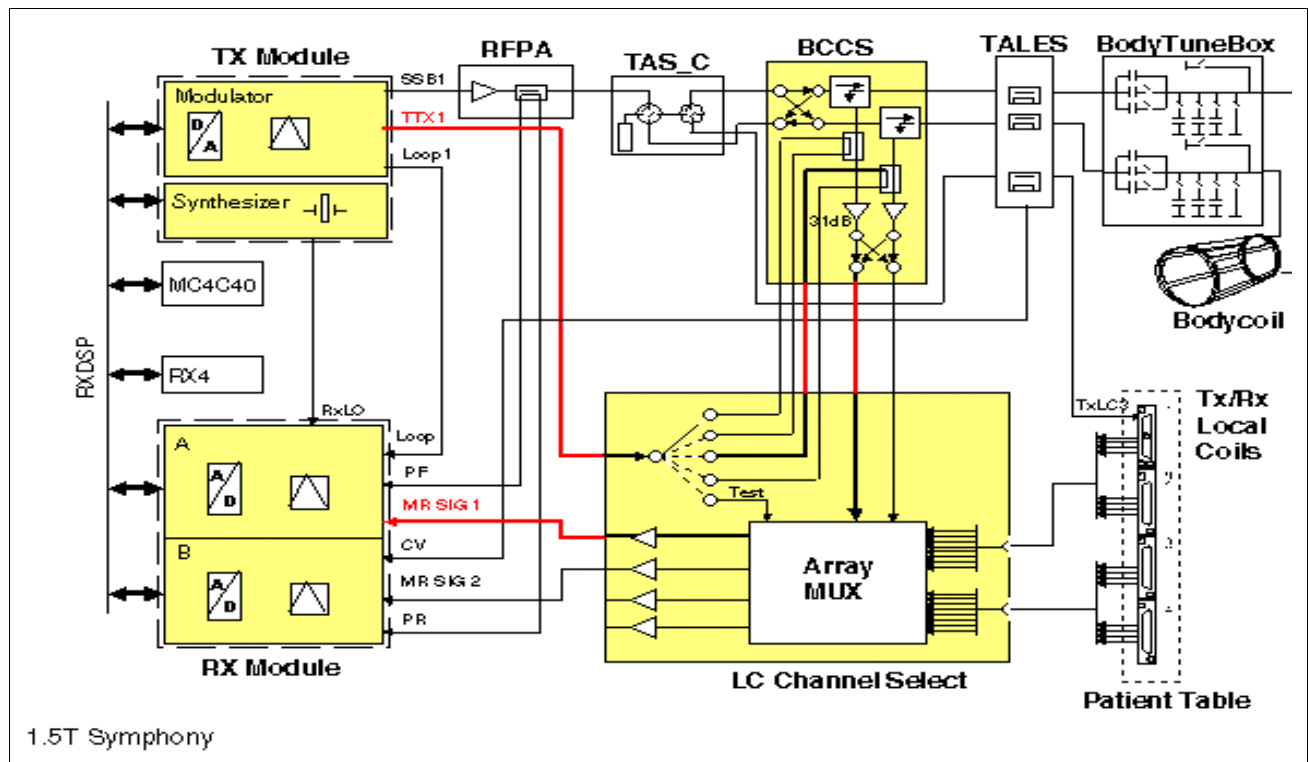
This test is used primarily to check the stability and linearity of the preamplifier in the BCCS and the LCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

The SGA amplifiers (high gain und low gain) are located in the LCCS.

Additional the cables and connections between BCCS and LCCS (<W531>, <W532> and <W529>) are checked.

BCCS (preamp1 2) - LCCS Loop (Symphony only)

Fig. 55 BCCS (preamp1 2) - LCCS



This test is similar to the previous test, but the signal is fed into the second preamplifier of the BCCS. Only the high gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS is used.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

This test is used primarily to check the stability and linearity of the preamplifiers in the BCCS and the LCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

The SGA amplifiers (high gain und low gain) are located in the LCCS.

Additional the cables and connections between BCCS and LCCS (<W533>, <W534> and <W528>) are checked.

LCCS plug 1,2,3,4 - LCCS (SGA x) Service Plug, pos n (interactive)

Fig. 56 LCCS interactive Test 1.5T

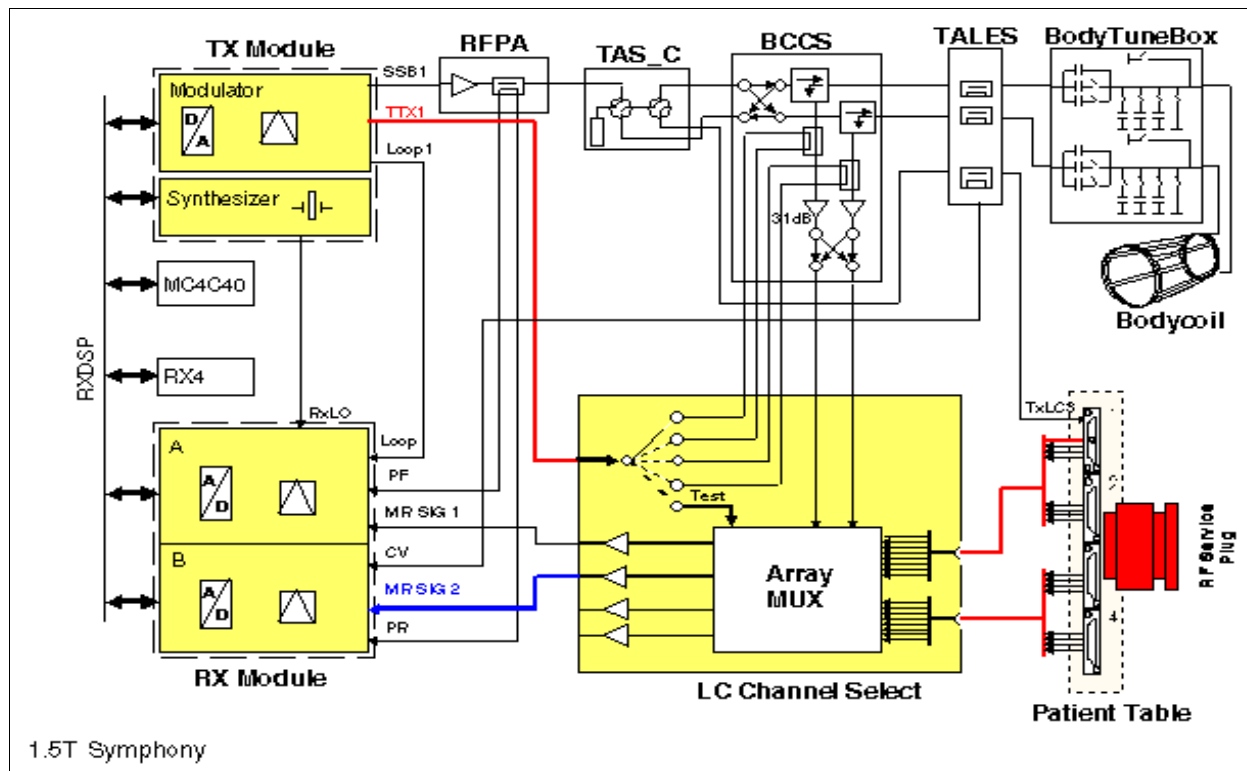
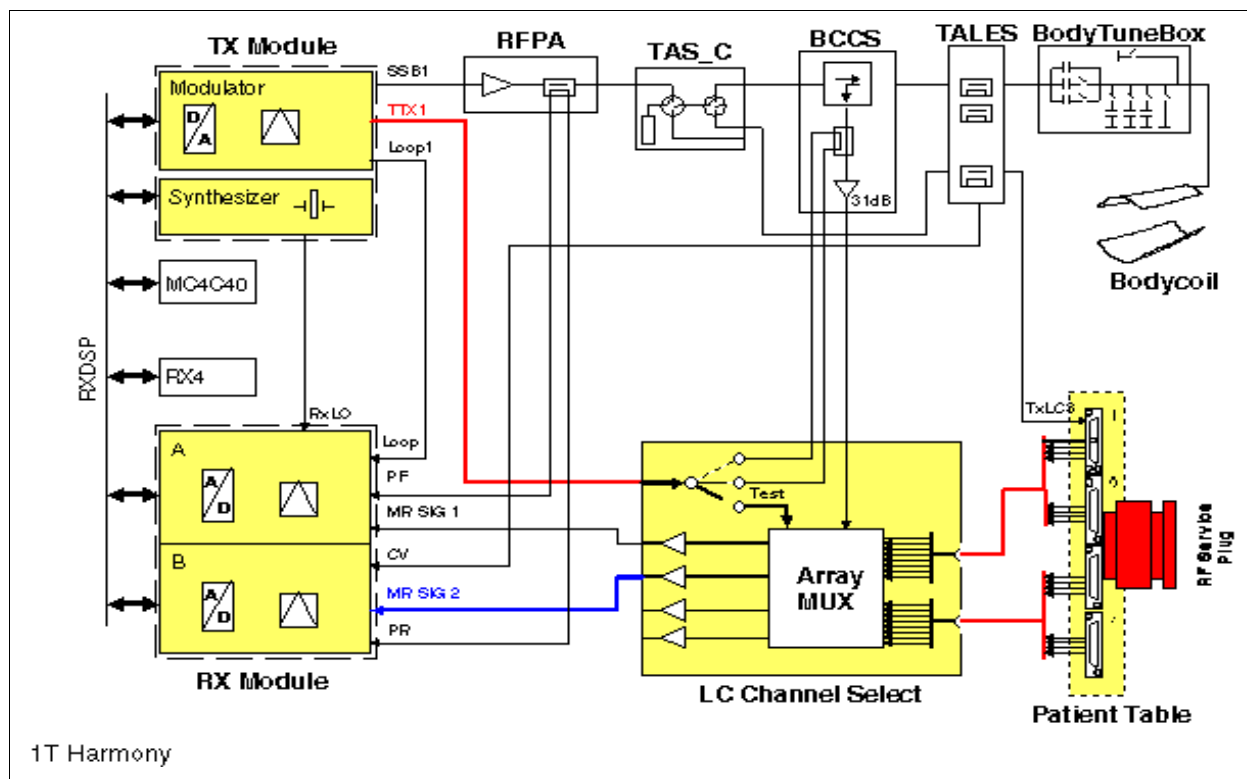


Fig. 57 LCCS interactive Test 1T



Preparation

The RF Service Plug must be connected to the coil connector 1,2,3 or 4 at the patient table. Follow the instructions provided during the test run.

Signal Path

The TTX output signal of the modulator is fed via LCCS to the RF Service Plug (attenuator network) and back to the LCCS (via switching matrix) and to the receiver module.

The TTX RF signal from the modulator is applied to one port of the receiver via the internal multiplexer of the LCCS, attenuated via the attenuator network in the Service Plug, and forwarded back to the LCCS and to the receiver module. In the process, the signal path in the LCCS is selected so that all switching points of the matrix are checked.

The service plug position has to be changed accordingly for the local coil receptacles by following the instructions in the interactive dialog.

NOTE	The functional combination of two receiver signals in the LCCS cannot be automatically tested at this time.
-------------	---

Evaluation

The TTX signal is attenuated in the attenuator network of the RF Service Plug in such a way that the signal level of the first channel is 100%, the second approx. 50%, the third approx. 25% and the fourth approx. 12.5%. The signal levels are checked against the specifications. The graduated levels allow complete testing of the switch matrix as well as the cabling from the local coil plug to the LCCS and continuing to the receiver.

Conclusions

In case of an error, the cause could be the LCCS. However, it could also result from incorrect or defective cabling from the local coil receptacle to the LCCS (movable cable in the patient table) and the receiver module. Before replacing the LCCS, you should test the RF receiver cable first (local coil receptacle - LCCS).

LCCS plug 1,2,3,4 - SGA x Loop (Low Gain only at plug 4 /High Gain) (interactive)

Fig. 58 LCCS interactive Test 1.5T

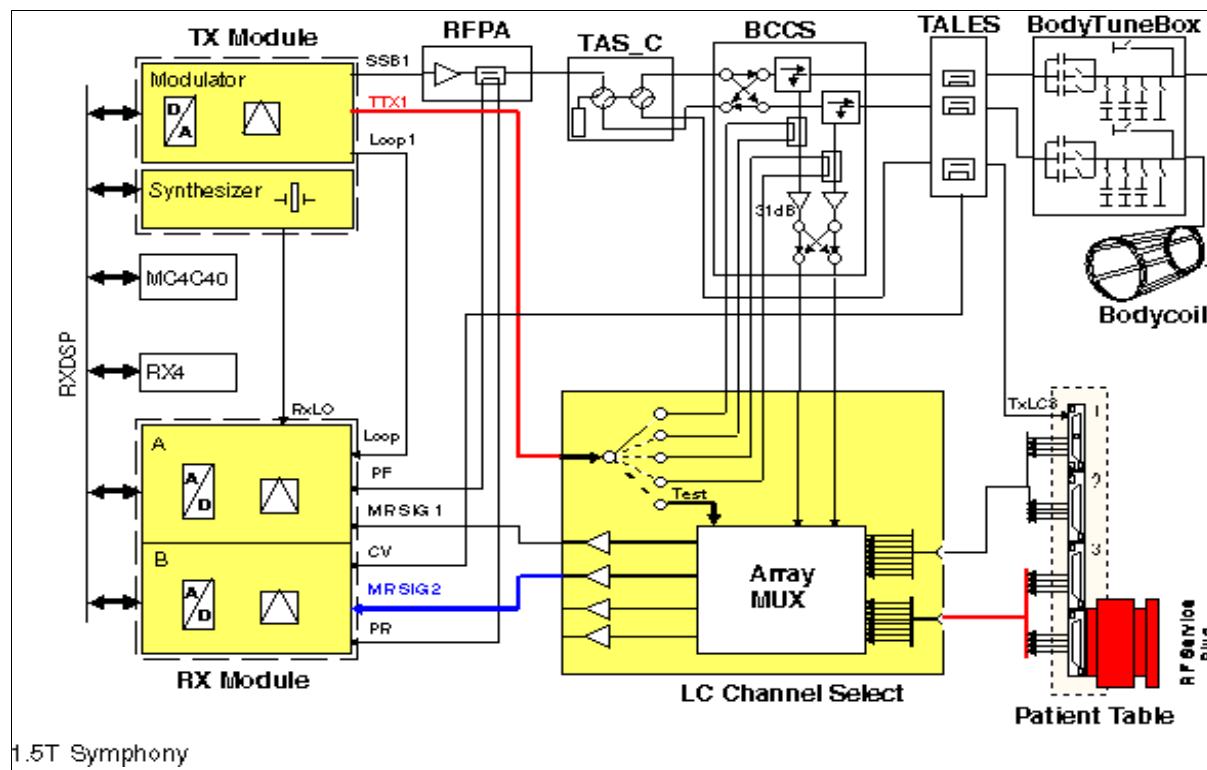
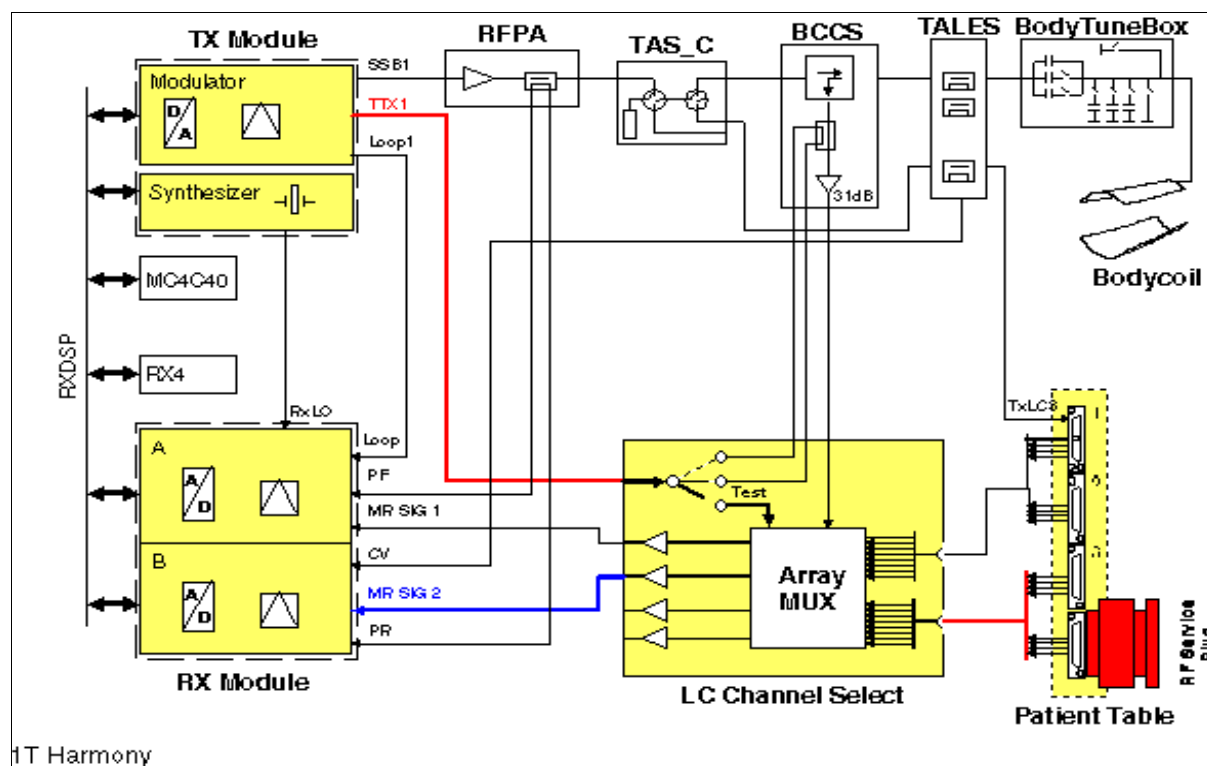


Fig. 59 LCCS interactive Test 1T



Preparation

The RF Service Plug must be connected to the coil connector 1,2,3 or 4 at the patient table. Follow the instructions provided during the test run.

Signal Path

The TTX output signal of the modulator is fed via the LCCS to the RF Service Plug and back to the LCCS (SGA) and to the receiver module.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

At coil plug 4 the measurement is done with high and low gain setting of the SGA, at coil plug 1,2,3 only with high gain setting.

Conclusions

The test is used primarily to check the SGA stability in the LCCS. Because of the internal circuitry, the SGA 1 cannot be checked using this test. It can only be tested indirectly via the body coil preamplifier in the BCCS (BCCS preampl - LCCS Loop).

LCCS plug 1,2,3,4 (interactive)

This test is identical to the previous LCCS tests, but **only** the selected plug will be tested

RFCI

- RFCI Communication/Status
- RFCI PIN Diodes (Local/Body Coil)
- RFCI PIN Diodes Service Plug, position (interactive)
- RFCI PIN Diodes DVM (interactive)

RFCI Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message. As long as no measurement was done with the connected coil the status will be "setting not loaded" and the respective PIN channels and mask "not set". So this is no error condition.

RFCI PIN Diodes (Local/Body Coil)

NOTICE The local coils and the loaded phantom must not be positioned in the magnet for this test.
If additional RF tests are performed afterwards, the local coils must be removed and the loaded phantom positioned in the magnet center to prevent damage to the resonator or to the local coils.

The test of the forward current (Body Coil: 600 mA, local coils: 100 mA) and reverse voltage (-30 V) supplied and measured by the RFCI for the PIN diodes of the Body Coil (located in the body tune box) and all PIN diodes in the connected local coils is done automatically.

NOTE If the PIN diodes in the head coil should be checked, the upper part of the Head Coil must be connected. Without the upper part, the test will show an error on the 3rd channel because of the missing PIN diode in the upper part.

Conclusions

This test verifies the function of the RFCI, the cables and the PIN diodes in the Body Coil and the local coils. In case of an error, the interactive tests should be performed with the service plug to differentiate between the coil and the controller.

RFCI PIN Diodes Service Plug, pos n (interactive)**Preparation**

The RF Service Plug must be connected to the coil connector at the patient table. Follow the instructions provided during the test run.

NOTE	The test does not use the normal communication between the MPCU and the RFCI, but the CAN-bus. In this way it is possible that a defect of the backplane is not detected.
-------------	--

Evaluation

This interactive test checks the RFCI forward current (100 mA) and the reverse voltage (-30 V) driven through the 4 diodes in the RF Service Plug. The plug position has to be changed by following the instructions in the interactive dialog.

Conclusions

If defective PIN diode channels are detected during the test, the measurements with the voltmeter in the following test will provide more detailed information regarding the possible cause of the error.

RFCI PIN Diodes Volt Meter (interactive)

The RFCI switches between current and voltage for all PIN diode channels including the body coil channel in a fixed time pattern of approx. 5 seconds. The RF Service Plug can be connected to the 4 plug positions in sequence without interrupting the test. The voltage drop across the PIN diodes in the service plug and at the test jacks of the RFCI service board is measured using the DVM.

NOTE	The test does not use the normal communication between the MPCU and the RFCI, but the CAN-bus. In this way it is possible that a defect of the backplane is not detected.
-------------	--

Service Plug

- Reverse voltage (UR) at PIN diode: - 27 ... - 33 V DC, typically -30.5 V
- Forward current voltage drop (IF): + 1.9 ... + 2.3 V DC, typically 2.1 V (corresponds to 100 mA)

In addition to the measurement at the PIN diodes, the voltage supply of the internal RX preamplifier (+10 V +/- 1 V) of the local coils can be checked at the service plug. The voltage is continuously present independent of the running test, but influenced by the PIN diode switching. For that reason, abort the PIN diode test for an accurate measurement of the preamplifier supply voltage.

RFCI Service Board

Local coil PIN diode control:

- Reverse voltage (UR): - 27 ... - 33 V DC, typically -30.5 V

Forward current voltage drop (IF): + 2 ... + 6 V DC, typically 2.6 V (depending on the cable length between CCA and patient table)

- ⇒ Error: Open circuit voltage (IF current regulator at limit):
approx. + 14 V DC

Body coil PIN diode control:

- Forward current voltage drop (IF): + 3 ... + 8 V DC
(according to a forward current of 600 mA, dependent on the cable length between CCA and body tune box).
- ⇒ Error: Open circuit voltage (IF current regulator at limit):
approx. + 15 V DC

Test points at the RFCI

- J10 left side TP 1-4 --> coil connector 1, PIN diode 1-4
- J10 right side TP 5-8 --> coil connector 2, PIN diode 1-4
- J9 left side TP 9-12 --> coil connector 3, PIN diode 1-4
- J9 right side TP 13-16 --> coil connector 4, PIN diode 1-4
- J7 left top TP --> body coil, PIN diode

LEDs at the RFCI

During the test, the LEDs at the RFCI service board light up alternately. There are 4 x 4 groups of LEDs for each of the four PIN diode connections of every coil connector.

- U_ON_I_OF_1 = on: PIN diode control UR = on
- UR_ON_x = on: UR present
- IF_ON_x = on: IF present

Conclusions

Measurement at RF Service Plug

UR	IF	Conclusion
OK	OK	OK.
NOT OK	OK	Error RFCI.
OK	NOT OK	Error RFCI; PIN diode of Service Plug may be defective.
10 V DC (LC preamplifier) not OK		1. Broken or shorted cable between LCCS (X11/12) and coil connector. 2. 15 V to 10 V conversion on LCCS defective. Check error messages!

Measurement at RFCI Service Board

UR	IF	Conclusion
NOT OK	OK	Error RFCI.
OK	NOT OK	Broken cable between RFCI and LC connector.
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and LC connector.

Measurement at RFCI Service Board (body coil)

UR	IF	Conclusion
OK	OK	OK
NOT OK	OK	Error RFCI/defective PIN diode of body coil (short circuit).
OK	NOT OK	Defective PIN diode of Body Coil or broken cable between RFCI and body tune box (BTB). Perform TestTools/RF/BTB/BC/Functional.
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and LC connector or defective PIN diode of body coil. Perform TestTools/RF/BTB/BC/Functional.

In addition, you can measure several supply voltages at the RFCI service board.

RF - MAGNETOM Allegra

NOTE This section describes the MR system MAGNETOM Allegra only.

The RF tests consists of a set of tests designed to verify the performance of the RF system and to assist you in locating defective components.

NOTE Beginning with *syngo* MR 2002B (NUMARIS/4 VA21A) all displayed graphs can be enlarged by clicking into the image and selecting (dragging) the region of interest. With "autoscale" the original view of the graph is restored.

General

The RF System hardware has been designed with several built-in loops to allow automated testing of RF components. In addition, level measurements are performed via the directional couplers (DICO) in the RFPA and the TALES.

The sequence used is common to all RF loop tests. It is called *rf_loop* and contains six RF pulses of which the first two are relevant for TestTools and the others have zero amplitude. The first pulse is in sinc (function $\sin(x)/x$) form with constant amplitude and is used for level measurements and stability evaluations. The second pulse is also in sinc form and increases from 0 up to the maximum amplitude. It is used in linearity evaluations.

NOTICE The 50 Ohm termination of the RFPA is **not** done automatically. You have to connect the dummy load manually by connecting the cable to TALES X3 via an adapter to TALES X6 (refer to [Fig. 61](#)). The TxRx (head) coil or the service plug must be connected to coil plug 1. Follow the instruction of the confirmation popup windows. The dummy load is located in the TAS at the filter plate and is water-cooled in series with the gradient/body coil cooling. To prevent damage to the dummy load, do not perform RFPA tests without proper cooling.

NOTE With every measurement a unique ID is displayed in the output window (e.g. 4.0.67892). This so-called "Loid" can be used in conjunction with the Rawdata Handling Package RHP to display and evaluate the raw data of the measurement.

Additionally to this number also the used RFPA (RFPA1 or RFPA2) is printed out as an information. Depending on the test this allows it to identify any error or mistake in selecting the correct RFPA.

With every test that runs a measurement at the beginning the used nucleus is displayed. With all systems that are not equipped with the multi-nucleus option (spectroscopy, X-nucleus) this will be always "1H", otherwise it should be the selected nucleus, unless it is a test that always use 1H and is independent of the nucleus-selection.

NOTE In case the system is equipped with a multi-nucleus option (spectroscopy, X-nucleus), also these tests are performed when a RF test is selected. If this is not intended or necessary please de-select the test type "Multi Nuclei".

The nucleus which has to be used for the next RF tests can be selected in the problem section.

The selection has to be done prior to the start of the test. If you realize that accidentally the wrong nucleus is selected please cancel the test immediately.

Everytime when such a test is to be performed it is necessary to select the nucleus/frequency prior to start the test.

Overview RF Tests

Test	Comm/Status	Functional	Interactive	Multi Nuclei
Mod/REC i	x	x	-	x
RFPA1/2	x	x	x	x
RFAS	x	x	x	x
LCCS	x	x	x	x
RFCI	x	x	x	-
Cable	-	-	x	x (LC-TX only)

Stability

The first pulse of the sequence with a constant RF amplitude is used. The resulting magnitude data is the evaluation of the stability of the center column which is located in the center of the RF pulse. The numerical result is the maximum value minus the minimum value divided by the mean value. Instabilities normally cause phase ghosting or poor S/N values.

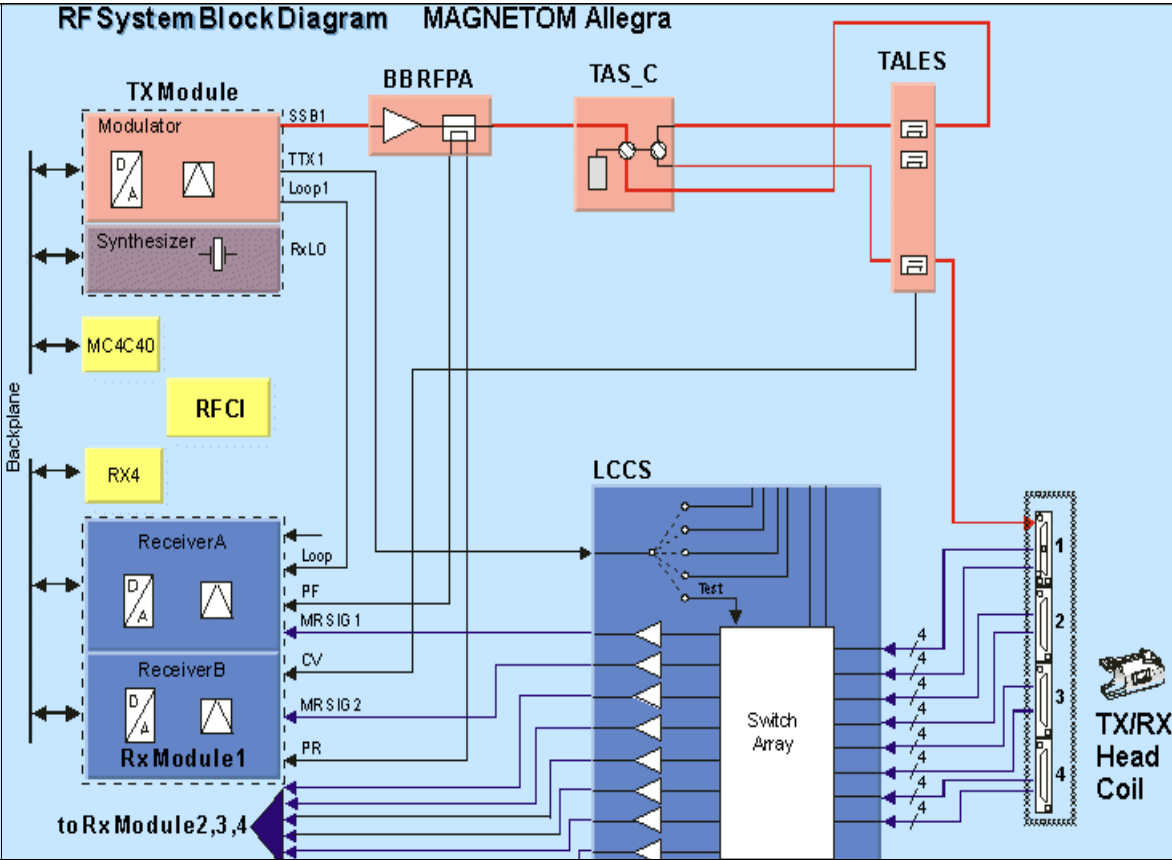
An overlayed graphic represents an average of the samples close to the centre column. This allows to differentiate between noise and instabilities.

Linearity

For linearity evaluations, the second RF amplitude is stepped from 0 to a positive pulse amplitude. The center column of each line (maximum of the single pulses) is fitted to a linear function. The numerical result is the maximum deviation from the linear fit divided by the maximum value making up the data set. Non-linearities result in poor slice profiles and reduced S/N.

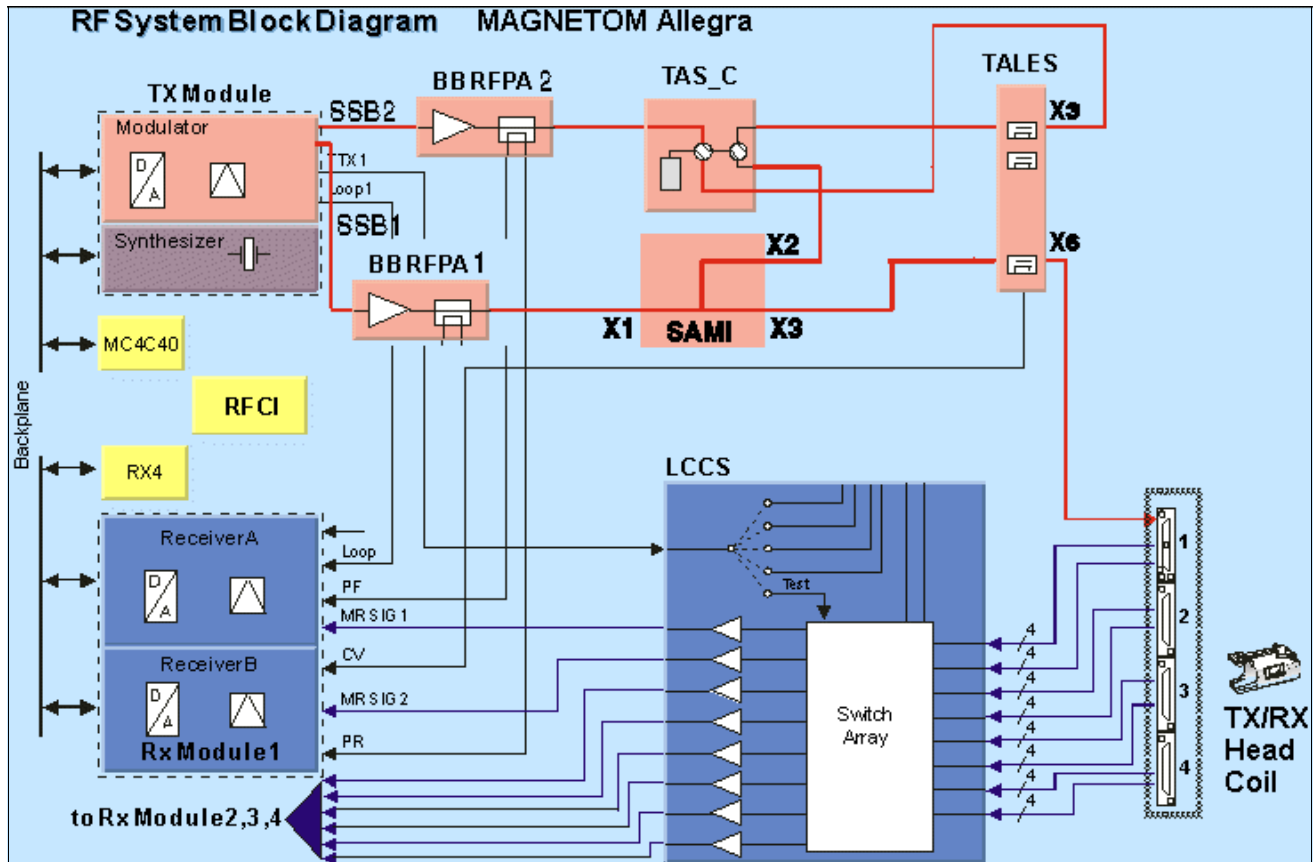
Overview RF-system/RFAS Allegra

Fig. 60 Overview RF System Allegra without Multinuclei Option



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Fig. 61 Overview RF System Allegra with Multinuclei Option



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MOD/REC

- TX-Cassette Communication (former Modulator Communication)
- Modulator-Receiver Loop (Functional/Multi Nuclei)
- Receiver A/B symmetry (Functional/Multi Nuclei)
- Reference Voltage (TALES)

TX-Cassette Communication

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The test was formerly named Modulator Communication.

Modulator-Receiver Loop

Signal Path

The rf_loop sequence is started and the output signal of the modulator is fed to the receiver module 1 A/B.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

Successful completion of this test is the prerequisite for all additional RF tests. If the test results are satisfactory, it can be assumed that the controller provides the transmitter with data as specified, that the data is modulated correctly in the modulator, received correctly by the receiver and then forwarded to the MMC and Imager.

The test results are, however, not indicative of the signal processing functionality at the receiver inputs CV and DC. Similarly, the signal switch-over in the transmitter to SSB1 or TTX is not part of this test. Instead, the signal paths are tested using the RFPA (SSB1) and LCCS tests (TTX).

NOTE	Since transmitting/receiving are simultaneous activities, both the transmit and the receive signal are affected by signal fluctuations in the synthesizer. As a result, fluctuations remain undetected, a fact that has to be taken into account during troubleshooting Receiver A/B symmetry.
-------------	--

Receiver A/B Symmetry

Signal Path

The output signal of the modulator is fed to the receiver module A/B.

Evaluation

The raw data of receivers A and B are compared for symmetry.

Conclusions

With respect to the signal path and data acquisition, the test is identical to the Modulator-Receiver loop. The only difference is that only one raw data line is being measured and only the first pulse will be evaluated to check the symmetric signal processing of both receivers.

Reference Voltage (TALES)

Signal Path

The TALES output signals are fed to the receiver 1B (CV-Input).

Evaluation

Verification that the TALES reference voltage is 500 mV.

Conclusions

The TALES continuously supplies all internal directional coupler signals triggered by the controller. In addition, it supplies the signal of an integrated reference voltage source. The DICO signals and the reference voltage are multiplexed and fed to the receiver CV input. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the receiver and signal processing during data acquisition. In case of error, the 500 mV reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

RFPA 1/2

- RFPA Communication/Status
- RFPA Power Test
- RFPA Loop
- RFPA-TALES
- RFPA Long Time Stability

RFPA Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

In case the system is equipped with a multi-nuclear option (spectroscopy, X-nucleus), the communication of both RFPAs is tested and both stati are returned.

RFPA Power Test

NOTE The test requires a valid RF characteristic.

Signal Path (one RFPA only)

If a local coil or the service plug is connected to the coil plug, the output signal SSB1 of the modulator is fed via the RFPA and TAS to TALEX X5/6. The dummy load of the TAS_C must be manually connected, i.e. cable to TALEX X3 via adapter to TALEX X6.

If no coil or service plug is connected to the coil plug, the RFPA transmits to the dummy load in the TAS.

The Forward_Power Signal is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF characteristic data.

Signal Path /Multi Nuclei

The output signal SSB2 of the modulator is fed via the RFPA2 and TAS/SAMI X2 to TALEX X5/6. The output signal SSB1 of the modulator is fed via the RFPA1 and SAMI X1 to TALEX X5/6. The dummy load of the TAS_C must be manually connected, i.e. cable to TALEX X3 via adapter to TALEX X6. The Forward_Power Signal is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF characteristic data.

Evaluation

- The actual forward voltage (measured via directional coupler) is compared to the nominal value read from the RF characteristic measurement.
- The ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.

Conclusions

The objective of the test is to ensure that the RFPA1/2 operates at peak performance and that the reflected voltage remains below a value (dependent on the nucleus) of the forward voltage at a dummy load of 50 Ohm.

Peak performance can no longer be achieved when the total gain (signal processing in the RFSU and gain in the RFPA) and, as a result, the characteristic curve has changed due to aging components. If the deviation lies outside the acceptable tolerance, the RF characteristic must be reexamined. If peak performance still cannot be achieved, the output level of the modulator and the directional coupler in the RFPA should be checked to determine the cause of the error. For further information, refer to the Troubleshooting Guide.

If there is an increase in the reflected voltage, check all cable connections from the RFPA to the Receiver and the transmission cable to the TAS assembly. You should also check with the LEDs if the TAS relay is set correctly. A defective dummy load would cause high voltage reflection, too.

RFPA Loop

The RFPA Loop test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

Signal Path

The output signal of the modulator is fed via the RFPA1/2 and SAMI to TALES X5/6. The dummy load of the TAS_C must be manually connected, i.e. cable to TALES X3 via adapter to TALES X6. The forward signal is measured via the directional coupler in the RFPA.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability. The raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

This test is used primarily to check the stability and linearity of the RFPA's. Modulator control, data acquisition via receiver and RF characteristic acquisition must be intact and correct to evaluate the results.

RFPA-TALES

The RFPA TALES test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

Signal Path (one RFPA only)

If a local coil or the service plug is connected to the coil plug, the output signal SSB1 of the modulator is fed via the RFPA and TAS to TALES X5/6. The dummy load of the TAS_C must be manually connected, i.e. cable to TALES X3 via adapter to TALES X6.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Note:

The DICO reflection may be much higher than the TALES reflection because the content of harmonics of the RFPA output will be reflected in the SAMI.

Signal Path /Multi Nuclei

The output signal SSB2 of the modulator is fed via the RFPA2 and TAS/SAMI X2 to TALES X5/6. The output signal SSB1 of the modulator is fed via the RFPA1 and SAMI X1 to TALES X5/6. The dummy load of the TAS_C must be manually connected, i.e. cable to TALES X3 via adapter to TALES X6.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Evaluations

- Forward Voltage from the TALES/DICO within tolerance
- Reflected Voltage from the TALES/DICO below limit
- Deviation of the forward voltage from the TALES and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks almost the entire transmission component of the system.

The results of the forward voltage at the TALES provide information on whether the actual value acquired by the TALES corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALES is a gauge for the 50 Ohm termination through the 50 Ohm dummy of TAS_C.

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltages measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. It is determined by the characteristics of the components that follow the RFPA in the transmit path such as the TX cable, TALES, etc.

RFPA (Long Time Stability)

The test is identical to the RFPA Loop test, but runs approximately 10 minutes. The test is only available on system with fMRI functionality.

RFAS

- RFAS Communication/Status
- Reference Voltage (TALES)
- RFPA-TALES
- Local Coil TX-path (interactive/multi nuclei)

RFAS Communication/Status

The LCCS-SLIO controls the TAS component in order to verify communication to the LCCS as well.

The communication test consists of sending a command (status and version query) to an individual component which, if successfully completed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

Reference Voltage (TALES)

Signal Path

The TALES output signals are fed to the receiver 1B (CV-Input).

Evaluation

Verification that the TALES reference voltage is 500 mV.

Conclusions

The TALES continuously supplies all internal directional coupler signals triggered by the controller via fiber optic cable. In addition, the TALES supplies the signal of an integrated reference voltage source. The directional coupler signals and the reference voltage are multiplexed and fed to the receiver CV input via a coaxial cable. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the receiver and signal processing during data acquisition. In case of error, the 500 mV reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

RFPA-TALES

The RFPA TALEs test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

Signal Path

The output signal of the modulator is fed via the RFPA 1/2 and SAMI to TALEs X5/6. The dummy load of the TAS_C must be manually connected, i.e. cable to TALEs X3 via adapter to TALEs X6.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALEs.

Evaluations

- Forward Voltage from the TALEs/DICO within tolerance
- Reflected Voltage from the TALEs/DICO below limit
- Deviation of the forward voltage from the TALEs and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks almost the entire transmission component of the system.

The results of the forward voltage at the TALEs provide information on whether the actual value acquired by the TALEs corresponds to the nominal value determined by the RF characteristic.

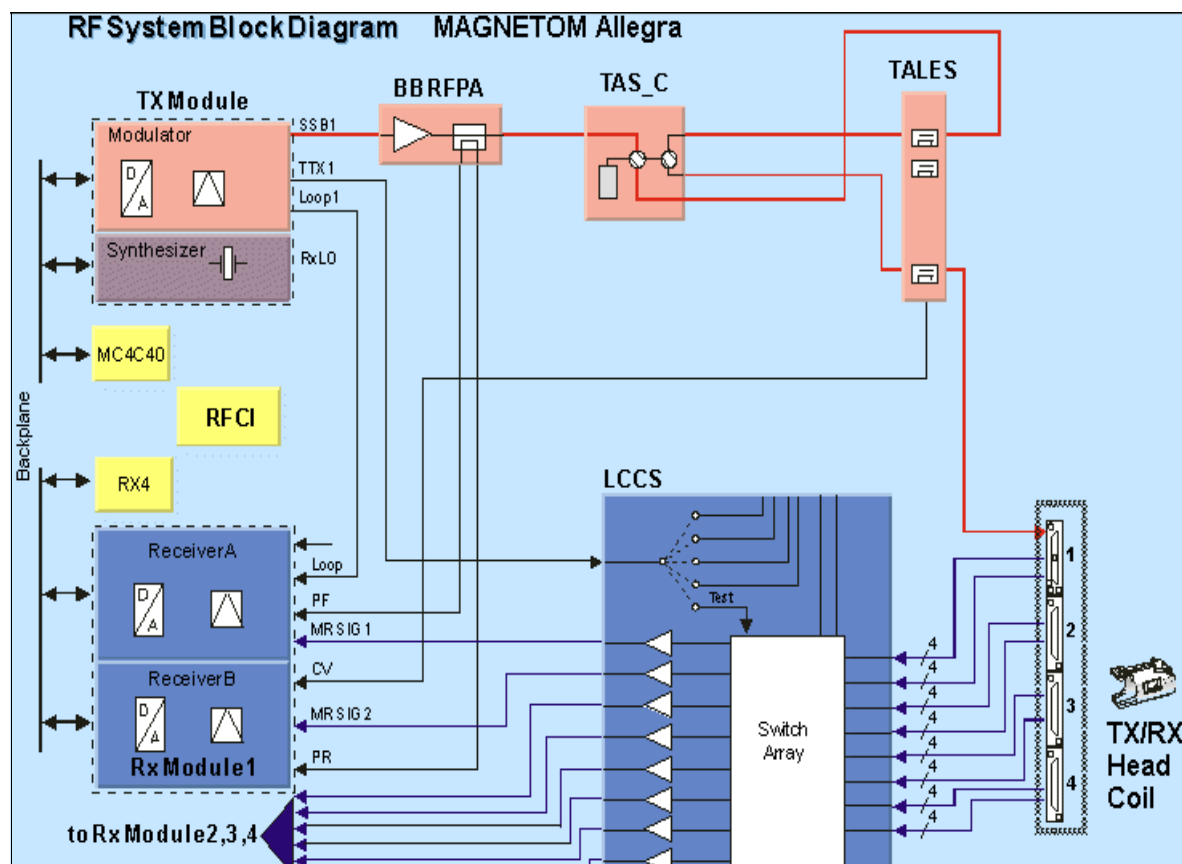
The value of the reflected voltage at the TALEs is a gauge for the 50 Ohm termination through the 50 Ohm dummy of TAS_C.

The forward voltage measured via DICO must approximately match that of the TALEs. Since the correlation between the DICO value and the TALEs voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltages measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. It is determined by the characteristics of the components that follow the RFPA in the transmit path such as the TX cable, TALEs, etc.

Local Coil TX-path (interactive/Multi Nuclei)

Fig. 62 RF System Allegra without Multinuclei Option



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RF System Block Diagram MAGNETOM Allegra

The diagram illustrates the RF system architecture for the MAGNETOM Allegra. It shows the flow of signals between various components, including the Backplane, TX Module, BBRFPA 1 and 2, TAS_C, SAMI, TALES, LCCS, and the TX/Rx Head Coil.

Backplane: The central hub for the system, connected to the TX Module, RX Module, and LCCS.

TX Module: Contains a Modulator and Synthesizer. It outputs TX1 and Loop1 signals to BBRFPA 2. It also receives control signals from the Backplane (MC4C40, RF CI, RX4).

BBRFPA 1 and 2: Base Band Radio Frequency Power Amplifiers. BBRFPA 1 outputs X1 and X2 signals to SAMI. BBRFPA 2 outputs X3 and X6 signals to SAMI and TALES.

TAS_C: Transmitter and Receiver Control. It receives X2 and X3 signals from SAMI and outputs X1 and X2 signals to SAMI.

SAMI: Signal Amplifier Module. It receives X1 and X2 signals from BBRFPA 1 and BBRFPA 2, and outputs X3 and X6 signals to TALES.

TALES: Transmitter and Receiver Amplifier. It receives X3 and X6 signals from SAMI and outputs X1 and X2 signals to the LCCS.

LCCS: Low Noise Amplifier. It receives X1 and X2 signals from SAMI and X3 and X6 signals from TALES. It outputs signals to the Switch Array.

Switch Array: A multi-channel switch that routes signals from the LCCS to the TX/Rx Head Coil.

TX/Rx Head Coil: The final output of the system, connected to the Switch Array.

Rx Module: Contains ReceiverA and ReceiverB. It receives signals from the LCCS and outputs signals to the Backplane (Loop, PF, MRSIG1, CV, MRSIG2, PR).

- Deviation of the forward voltage from the DICO and the TALES
- Ratio of the reflected and forward voltage from the TALES

Conclusions

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for converting the ADC values from the receiver into voltage values), a valid characteristic curve must be present.

With regard to the linear RF-characteristic, i.e. after the software has been installed and before the tune-up has been performed, the deviation may be significantly above the specification.

The ratio of the reflected voltage and the forward voltage measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. In this test, it is determined by the 50 Ohm resistor in the RF Service Plug. If the ratio lies outside of the specifications, the cause can probably be found in the cable connection from the TAS via Tales to coil connector #1 (movable cable in the table) or in the TAS itself (for this test, the TAS is switched to local coil path).

If the forward voltage does not exceed a certain level (approx. 100 V), the RF safety watchdog evaluates the measured values within the most sensitive range of the TALES. In this range, the reflected voltage of the TALES can no longer be produced and the value 0 is displayed. In this case, the evaluation of return / forward voltage is no longer valid.

Cable

- LC-TX-path (interactive/multi nuclei)

NOTE	These tests are for experts only! Additionally special equipment is necessary for the tests.
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These advanced tests enable you to check the complete TX-path to LC and to make a systematic troubleshooting in the TX-path by connecting a power 50 Ohm terminator at various locations in the TX-path and/or to configure particular loops.

The test should be used by experienced service engineers only. Special tools (power dummies, rf-cables, connectors, adapters and attenuators) are needed.

Popup windows inform you about the actual measurement configuration.

Safety

WARNING	High RF voltage
----------------	------------------------

Contact with RF conductors during transmitting of the RFPA will result in death or serious physical injury.

Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input cable and the unblank line U1 from the RFPA.

Allegra with Multi-Nucleus option

Contrary to the MAGNETOM Symphony with Multi-Nucleus option (spectroscopy, X-nucleus) with the MAGNETOM Allegra the nucleus selection in the TestTools-platform is not enough when testing the Local Coil Transmit Path for the 1H-frequency.

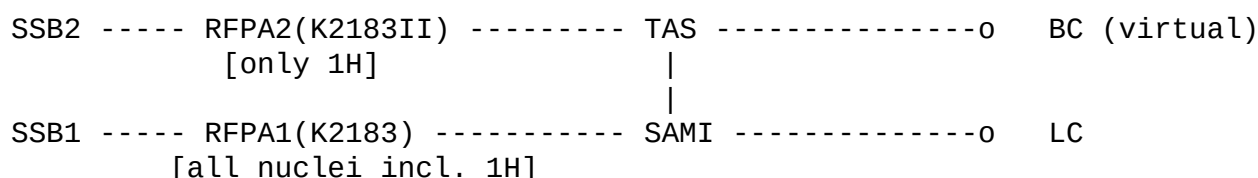
The reason is that the used RFPA's of the Allegra do not have completely different frequency ranges: both RFPA's support the 1H-frequency. Further frequencies are not used with RFPA2, although the RFPA's are identical.

With the Allegra also the plugged in coil is relevant:

- With a Multi-Nucleus coil capable also of the 1H-frequency (e.g. Service Plug) always the K2183II RFPA is used (RFPA2).
- With a mono-resonant coil capable only for 1H (e.g. TxRx Headcoil) always the K2183RFPA is used (RFPA1).

For all other frequencies always RFPA1 and the Service Plug or Multi-Nucleus Coil is used.

A simplified scheme can explain this a little bit:



This knowledge is important to use always the correct RFPA when testing the different TX-paths with 1H, especially the path RFPA2-TAS-SAMI-LC-path.

The behaviour and the scheme is basically the same like with the MAGNETOM Trio. It is just that with the Allegra the order of the RFPA's was defined in the opposite way: RFPA1 is RFPA2 and vice versa. Also the RFPA's of the MAGNETOM Symphony use the Trio order.

As you can see in the scheme it is also possible to test the BC-TX-path with the MAGNETOM Allegra. For this it is necessary to have no coil or Service Plug connected. The Body Coil is only simulated with it's coil code, the transmit path will finally end up in the dummy of the TAS.

The "virtual" BC-TX-path is used during system start for the PALI-test when no TxRx-Coil is connected. In this way it is important to have this test capability also to find errors in this path.

NOTICE The tests require a valid RF characteristic or the linear RF characteristic.
Ensure that the spherical phantom is positioned in the TX-coil.

NOTE When using the linear characteristic it might be possible that the used transmitter amplitude exceeds the allowed range resulting in an abort of the measurement. This is normal and not a problem of the system.

NOTE For the first measurement of each test also the reflection factors are evaluated and analyzed. If the reflection exceeds 90% the test is aborted with the error message Error: DICO reflection out of spec for first measurement.
In that case check the cabling and the Service Plug/50 Ohm power dummy (if used).

LC-TX-path (interactive/Multi Nuclei)

NOTICE If the test is started without connected Local Coil, the RFPA1 transmits to the TAS-dummy load via TALEX X1/X3.
For transmitting to the Local Coil, ensure that either the service plug or a local TX-coil is connected to coil plug 1.
The test requires a valid RF characteristic or the linear RF characteristic.
For spectroscopy coils select the checkbox Multi Nuclei and the desired nucleus.

The objective of the test is to check the components of the LC-TX-path. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (95 V for service plug and maximum local coil voltage read from the coil file) are performed.

Fig. 64 Pop up window LC-TX-path
Selection of used RF Characteristic

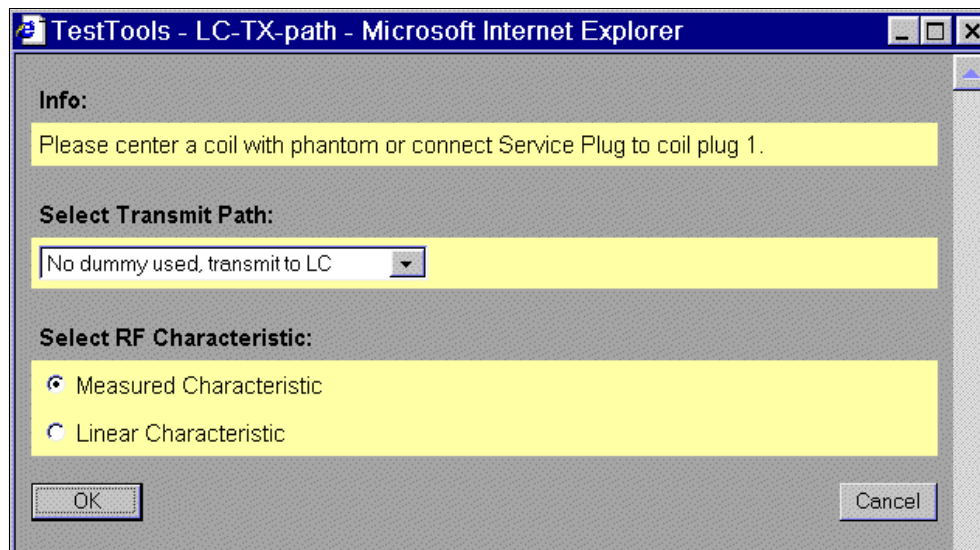
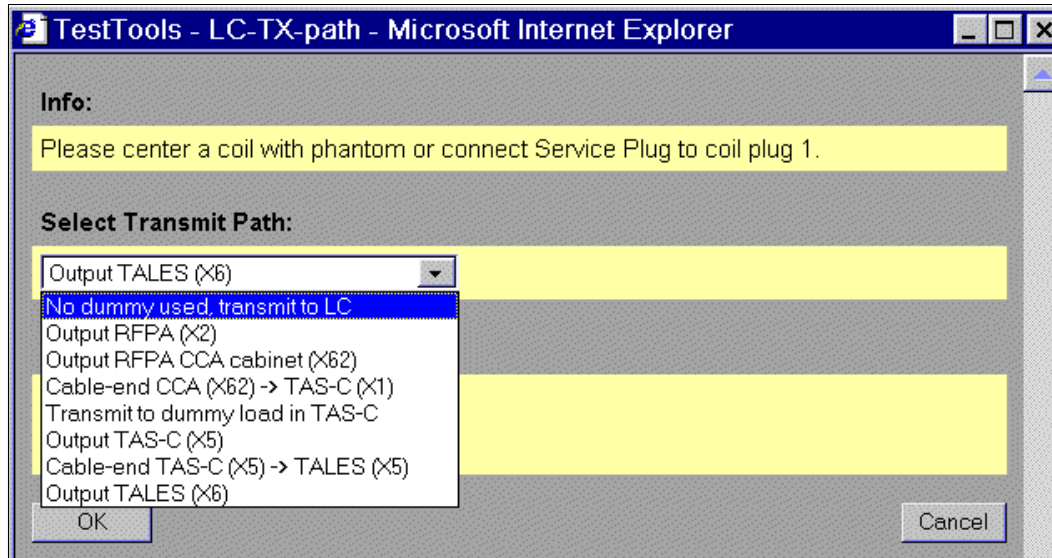


Fig. 65 Pop up window LC-TX-path
Selection of TX-path termination location by a 50 Ohm dummy



Signal path

The TX-path for local coils is used. The used RFPA depends on the selection of the nucleus.

The TX-path can be opened at locations and terminated by a 50 Ohm dummy. In the appearing popup window you can select the desired TX-path (refer to Fig. 65, example only. The location of interruption points depends on the system options).

The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the popup window.

The selection of the location of the interruption in the TX-path makes no influence of the measurement but selects the heading of the output table. If the TX-path is terminated by a 50 Ohm dummy between TALES and RFPA, the TALES values can not be measured and displayed.

NOTE	Whenever a dummy load is used to interrupt the TX-path BEFORE the TALES, the comparison of DICO- and TALES-values will not be successful as there are no TALES-values available and thus the test will fail. This is normal and not a problem of the system.
-------------	---

Evaluation

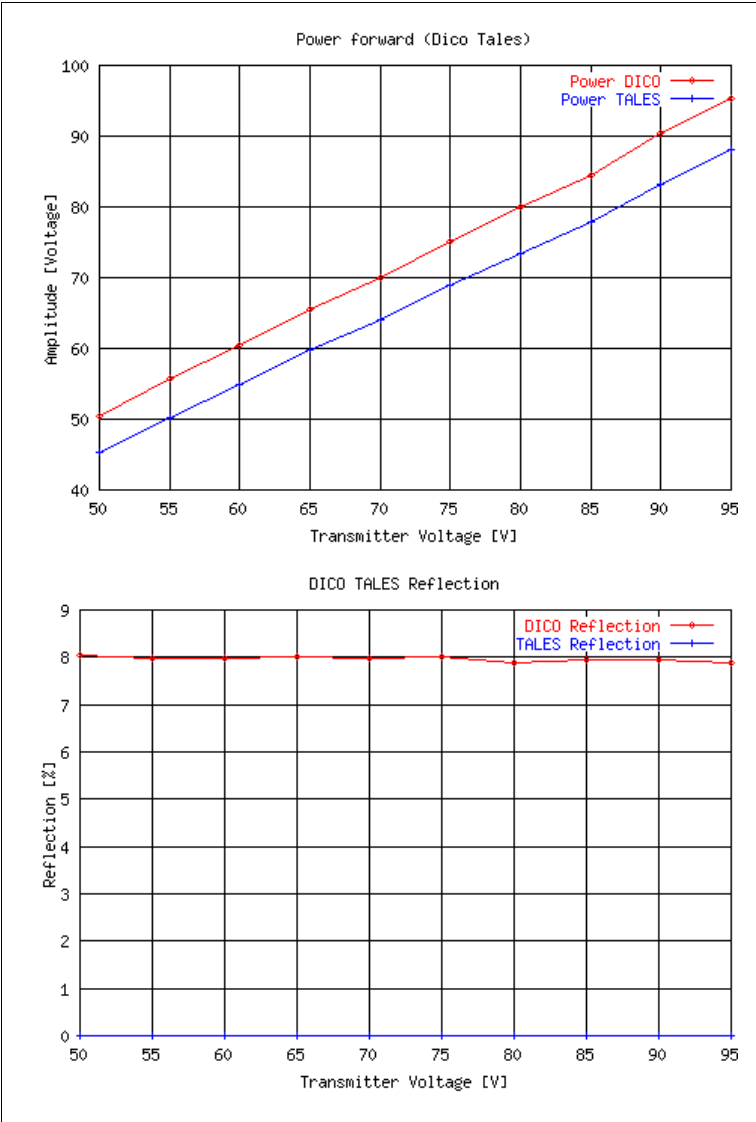
HTML output table:

- PF (DICO) at maximum deviation (Volt)
- PF (TALES) at maximum deviation (Volt)
- (PF(TALES) - PF (DICO))/ PF(TALES) maximum deviation (%)

Graphical output:

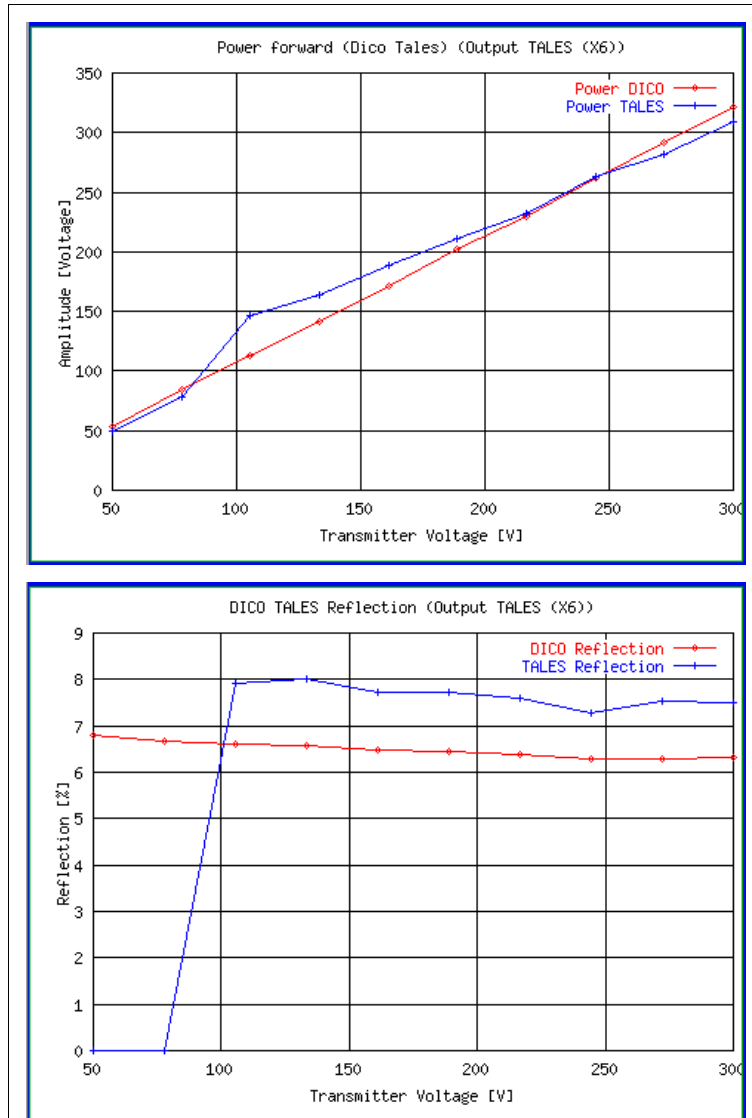
- PF(TALES/Volt) and PF(DICO/Volt) over TRA voltage
- PR/PF DICO reflection (%) and PR/PF TALES reflection (%) over TRA voltage

Fig. 66 Example: Transmit to Service plug



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Fig. 67 Example:
Transmit to LC TX coil (Dummy connected to TALEs X6)



Conclusions

NOTE For reflected voltages below 100 V the measured/displayed TALEs value is 0 V.

If the deviations between DICO and TALEs (for voltages higher than 100 V) are out of specification, repeat the RF characteristic measurement for the actual nucleus.

Locate the defective component in the TX-path by repeating the test after connecting the 50 Ohm dummy to a component before it's current position.

LCCS

- LCCS Communication/Status
- LCCS (SGA x) Service Plug, pos 1(interactive/multi nuclei))
- LCCS SGA x Loop (Low Gain/High Gain) (interactive/multi nuclei)

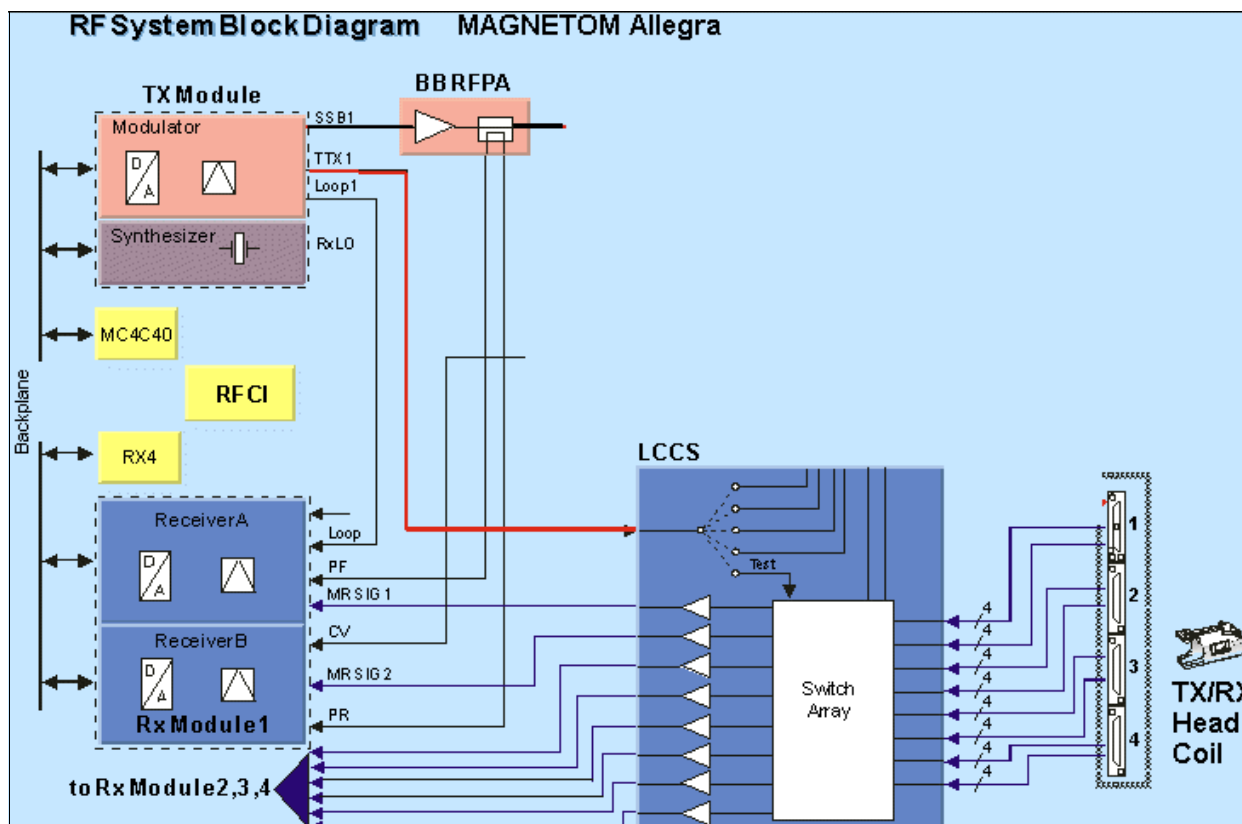
LCCS Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

LCCS (SGA x) Service Plug, pos 1 (interactive/multinuclei)

Fig. 68 LCCS Interactive Test



Preparation

The RF Service Plug must be connected to the coil connector at the patient table. Follow the instructions provided during the test run.

Signal Path

The TTX output signal of the modulator is fed via LLCS to the RF Service Plug (attenuator network) and back to the LCCS (via switching matrix) and to the receiver module.

The TTX RF signal from the modulator is applied to one port of the receiver via the internal multiplexer of the LCCS, attenuated via the attenuator network in the Service Plug, and forwarded back to the LCCS and to the receiver module. In the process, the signal path in the LCCS is selected so that all switching points of the matrix are checked.

NOTE	The functional combination of two receiver signals in the LCCS cannot be automatically tested at this time.
-------------	---

Evaluation

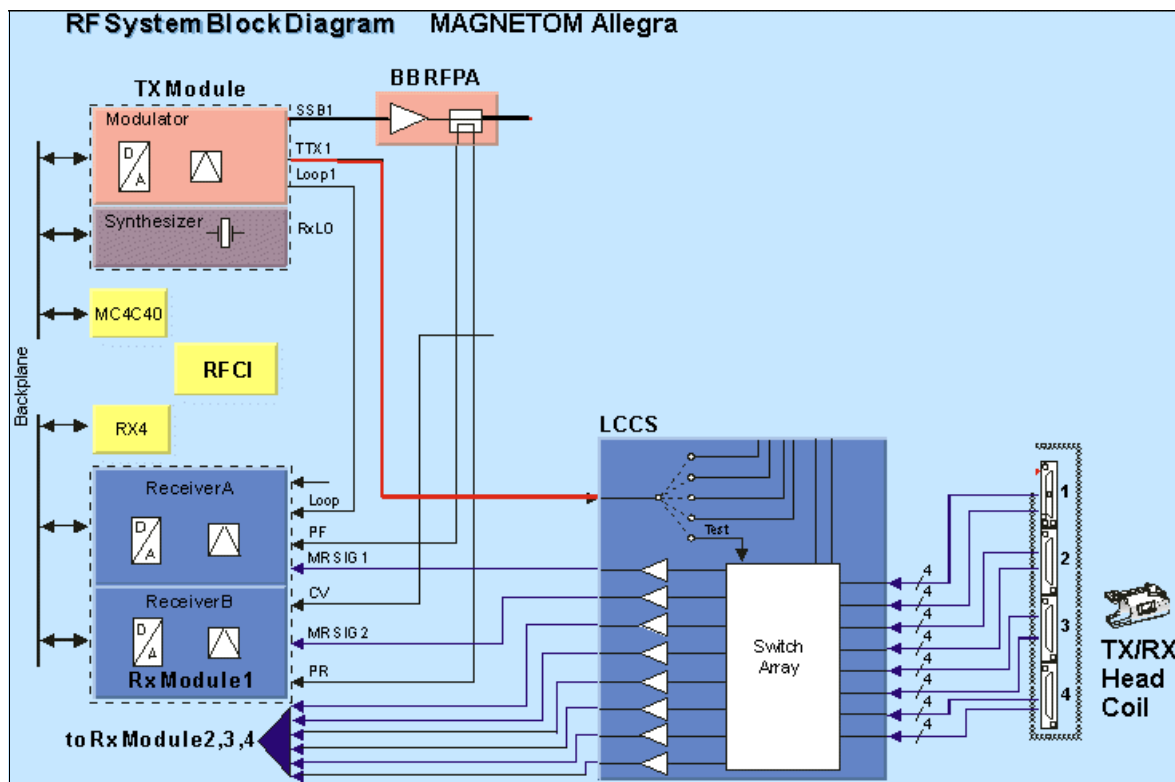
The TTX signal is attenuated in the attenuator network of the RF Service Plug in such a way that the signal level of the first channel is 100%, the second approx. 50%, the third approx. 25% and the fourth approx. 12.5%. The signal levels are checked against the specifications. The graduated levels allow complete testing of the switch matrix as well as the cabling from the local coil plug to the LCCS and continuing to the receiver.

Conclusions

In case of an error, the cause could be the LCCS. However, it could also result from incorrect or defective cabling from the local coil receptacle to the LCCS (movable cable in the patient table) and the receiver module. Before replacing the LCCS, you should test the RF receiver cable first (local coil receptacle - LCCS).

LCCS SGA x Loop (Low Gain/High Gain) (interactive/multinuclei)

Fig. 69 LCCS Interactive Test

**Preparation**

The RF Service Plug must be connected to the coil connector at the patient table. Follow the instructions provided during the test run.

Signal Path

The TTX output signal of the modulator is fed via the LCCS to the RF Service Plug and back to the LCCS (SGA) and to the receiver module.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

The test is used primarily to check the SGA stability in the LCCS. Because of the internal circuitry, the SGA 1 cannot be checked using this test.

RFCI

- RFCI Communication/Status
- RFCI PIN Diodes (Local Coil)
- RFCI PIN Diodes Service Plug, pos 1(interactive)
- RFCI PIN Diodes DVM (interactive)

RFCI Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message. As long as no measurement was done with the connected coil the status will be "setting not loaded" and the respective PIN channels and mask "not set". So this is no error condition.

RFCI PIN Diodes (Local Coil)

NOTICE The local coils must not be positioned in the magnet for this test.

The test of the forward current (Local coils: 100 mA) and reverse voltage (-30 V) supplied and measured by the RFCI for the PIN diodes of all PIN diodes in the connected local coil is done automatically.

NOTE If the PIN diodes in the head coil should be checked, the upper part of the Head Coil must be connected. Without the upper part, the test will show an error on the 3rd channel because of the missing PIN diode in the upper part.

Conclusions

This test verifies the function of the RFCI, the cables and the PIN diodes in the local coils. In case of an error, the interactive tests should be performed with the service plug to differentiate between the coil and the controller.

RFCI PIN Diodes Service Plug, pos 1(interactive)**Preparation**

The RF Service Plug must be connected to the coil connector at the patient table. Follow the instructions provided during the test run.

NOTE	The test does not use the normal communication between the MPCU and the RFCI, but the CAN-bus. In this way it is possible that a defect of the backplane is not detected.
-------------	--

Evaluation

This interactive test checks the RFCI forward current (100 mA) and the reverse voltage (-30 V) driven through the 4 diodes in the RF Service Plug. The plug position has to be changed by following the instructions in the interactive dialog.

Conclusions

If defective PIN diode channels are detected during the test, the measurements with the voltmeter in the following test will provide more detailed information regarding the possible cause of the error.

RFCI PIN Diodes DVM (interactive)

The RFCI switches between current and voltage for all PIN diode channels in a fixed time pattern of approximately 5 seconds. The RF Service Plug must be connected to the plug position #1. The voltage drop across the PIN diodes in the service plug and at the test jacks of the RFCI service board is measured using the DVM.

Service Plug

- Reverse voltage (UR) at PIN diode: - 27 ... - 33 V DC, typically -30.5 V
- Forward current voltage drop (IF): + 1.9 ... + 2.3 V DC, typically 2.1 V (corresponds to 100 mA)

In addition to the measurement at the PIN diodes, the voltage supply of the internal RX preamplifier (+10 V +/-1 V) of the local coils can be checked at the service plug. The voltage is continuously present independent of the running test, but influenced by the PIN diode switching. For that reason, abort the PIN diode test for an accurate measurement of the preamplifier supply voltage.

RFCI Service Board

Local coil PIN diode control:

- Reverse voltage (UR): - 27 ... - 33 V DC, typically -30.5 V

Forward current voltage drop (IF): + 2 ... + 6 V DC, typically 2.6 V (depending on the cable length between CCA and patient table)

⇒ Error: Open circuit voltage (IF current regulator at limit):
approx. + 14 V DC

Test points at the RFCI

- J10 left side TP 1-4 --> coil connector 1, PIN diode 1-4

LEDs at the RFCI

During the test, the LEDs at the RFCI service board light up alternately. There is 1x4 group of LEDs for each of the four PIN diode connections of the first coil connector.

- U_ON_I_OF_1 = on: PIN diode control UR = on
- UR_ON_x = on: UR present
- IF_ON_x = on: IF present

Conclusions

Measurement at RF Service Plug

UR	IF	Conclusion
OK	OK	OK.
NOT OK	OK	Error RFCI.
OK	NOT OK	Error RFCI; PIN diode of Service Plug may be defective.
10 V DC (LC preamplifier) not OK		1. Broken or shorted cable between LCCS (X11/12) and coil connector. 2. 15 V to 10 V conversion on LCCS defective. Check error messages!

Measurement at RFCI Service Board

UR	IF	Conclusion
NOT OK	OK	Error RFCI.
OK	NOT OK	Broken cable between RFCI and LC connector.
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and LC connector.

In addition, you can measure several supply voltages at the RFCI service board.

RF - MAGNETOM Concerto

NOTE This section describes the MAGNETOM Concerto only.

The RF tests consists of a set of tests designed to verify the performance of the RF system and to assist you in locating defective components.

NOTE Beginning with *syngo* MR 2002B (NUMARIS/4 VA21A) all displayed graphs can be enlarged by clicking into the image and selecting (dragging) the region of interest. With "autoscale" the original view of the graph is restored.

General

The RF System hardware has been designed with several built-in loops to allow automated testing of RF components. In addition, level measurements are performed via the directional couplers (COD-ICO) in the RFPA and the Local Coil Amplifiers (LOCA). The RFPA used with the MAGNETOM Concerto is the CORA (Cost Optimized Rf Amplifier) from the MAGNETOM Open/Open Viva.

The sequence used is common to all RF loop tests except the BTB test (body tune box). It is called `rf_loop` and contains six RF pulses of which the first two are relevant for TestTools and the others have zero amplitude. The first pulse is in sinc (function $\sin(x)/x$) form with constant amplitude and is used for level measurements and stability evaluations. The second pulse is also in sinc form and increases from 0 up to the maximum amplitude. It is used in linearity evaluations.

NOTICE There is no automatic 50 Ohm termination of the RFPA like with the high field systems. The dummy load which is located close to the magnet is just used for the hybrid and has to be manually connected to the RFPA if necessary. For many tests, this is not done, but the RF-power is instead guided to the TX-coil. Most of the tests require a valid coil code. For this reason, a local coil has to be connected to the system and usually positioned outside the magnet.

NOTE With every measurement a unique ID is displayed in the output window (e.g. 4.0.67892). This so-called "Loid" can be used in conjunction with the Rawdata Handling Package RHP to display and evaluate the raw data of the measurement.

Additionally to this number also the used RFPA (RFPA1 or RFPA2) is printed out as an information. Depending on the test this allows it to identify any error or mistake in selecting the correct RFPA.

With every test that runs a measurement at the beginning the used nucleus is displayed. With the Concerto this will be always "1H".

Stability

The first pulse of the sequence with a constant RF amplitude is used. The resulting magnitude data is the evaluation of the stability of the center column which is located in the center of the RF pulse. The numerical result is the maximum value minus the minimum value divided by the mean value. Instabilities normally cause phase ghosting or poor S/N values.

Linearity

For linearity evaluations, the second RF amplitude is stepped from 0 to a positive pulse amplitude. The center column of each line (maximum of the single pulses) is fitted to a linear function. The numerical result is the maximum deviation from the linear fit divided by the maximum value making up the data set. Non-linearities result in poor slice profiles and reduced S/N.

NOTE	With many tests the first measurements are not used for the final evaluation. So don't be worried if the graphical display and the numerical differ for this.
-------------	---

MOD/REC

- TX-Cassette Communication (former Modulator Communication)
- Modulator-Receiver Loop
- Receiver A/B symmetry

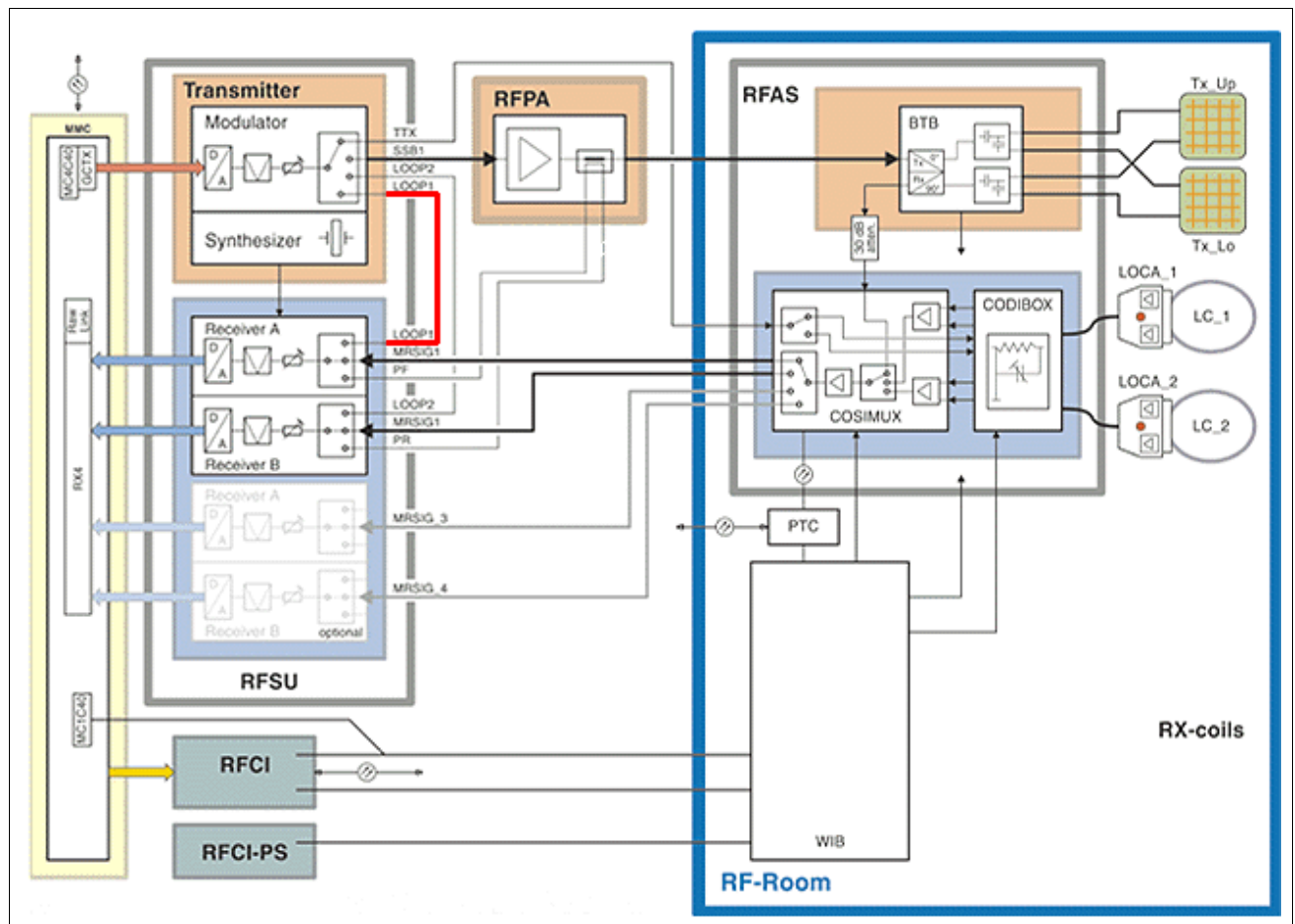
TX-Cassette Communication

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The test was formerly named Modulator Communication.

Modulator-Receiver Loop

Fig. 70 Modulator-Receiver Loop



Signal Path

The rf_loop sequence is started and the output signal of the modulator is fed to the receiver module A/B.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse for linearity.

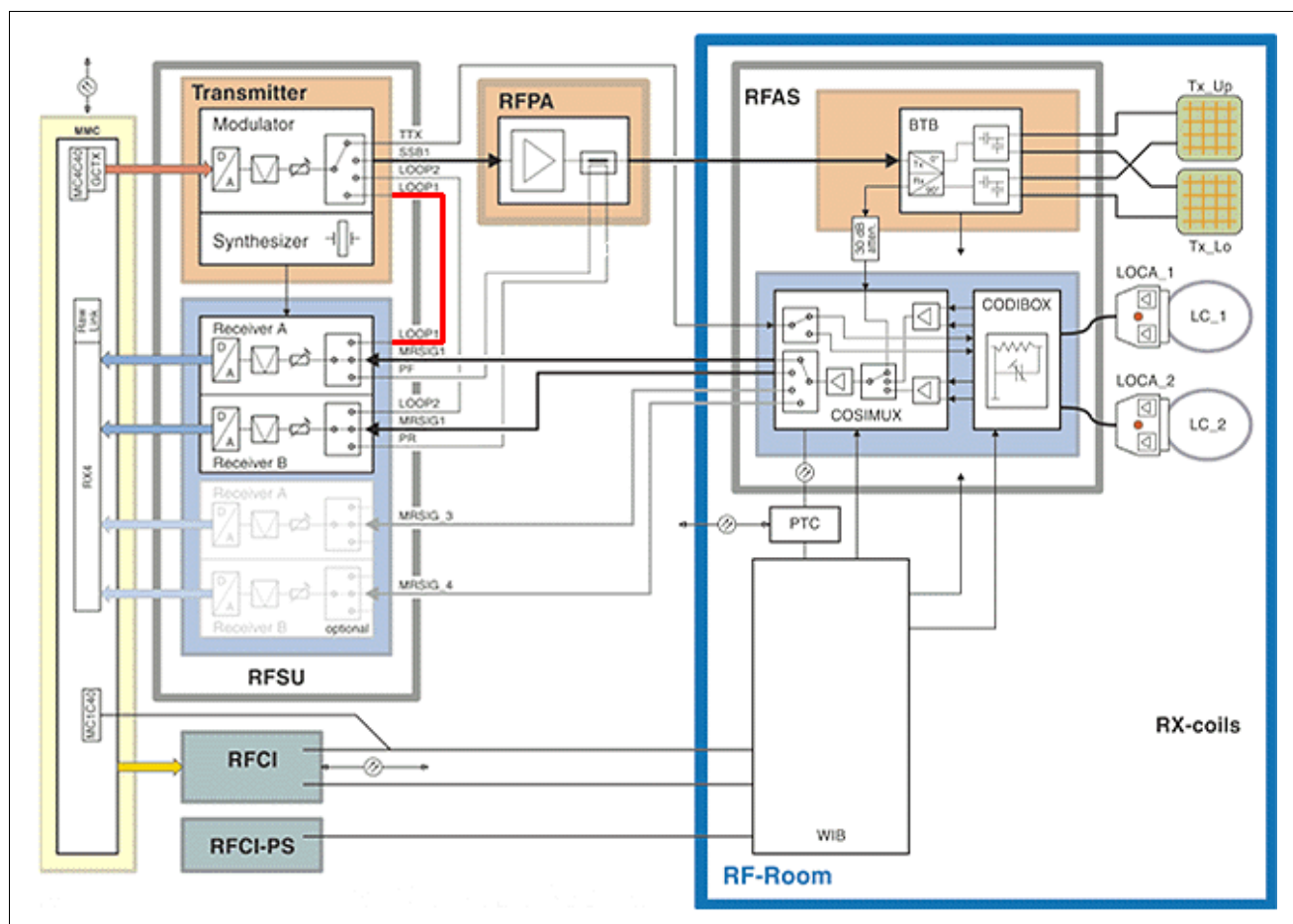
Conclusions

Successful completion of this test is the prerequisite for all additional RF tests. If the test results are satisfactory, it can be assumed that the controller provides the transmitter with data as specified, that the data is modulated correctly in the modulator, received correctly by the receiver and then forwarded to the MMC and Imager.

NOTE Since transmitting/receiving are simultaneous activities, both the transmit and the receive signal are affected by signal fluctuations in the synthesizer. As a result, fluctuations remain undetected, a fact that has to be taken into account during troubleshooting.

Receiver A/B symmetry

Fig. 71 Receiver A/B symmetry



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Signal Path

The rf_loop sequence is started and the output signal of the modulator is fed to the receiver module A/B.

Evaluation

The raw data of receivers A and B are compared for symmetry.

Conclusions

With respect to the signal path and data acquisition, the test is identical to the Modulator-Receiver loop. The only difference is that only one raw data line is being measured and only the first pulse will be evaluated.

The objective of the test is to check the symmetric signal processing of both receivers.

For further information, refer to the Troubleshooting Guide.

RFPA

- RFPA Communication/Status
- RFPA Power Test
- RFPA Loop
- CAPTICON-RFPA Status Test (interactive)

RFPA Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

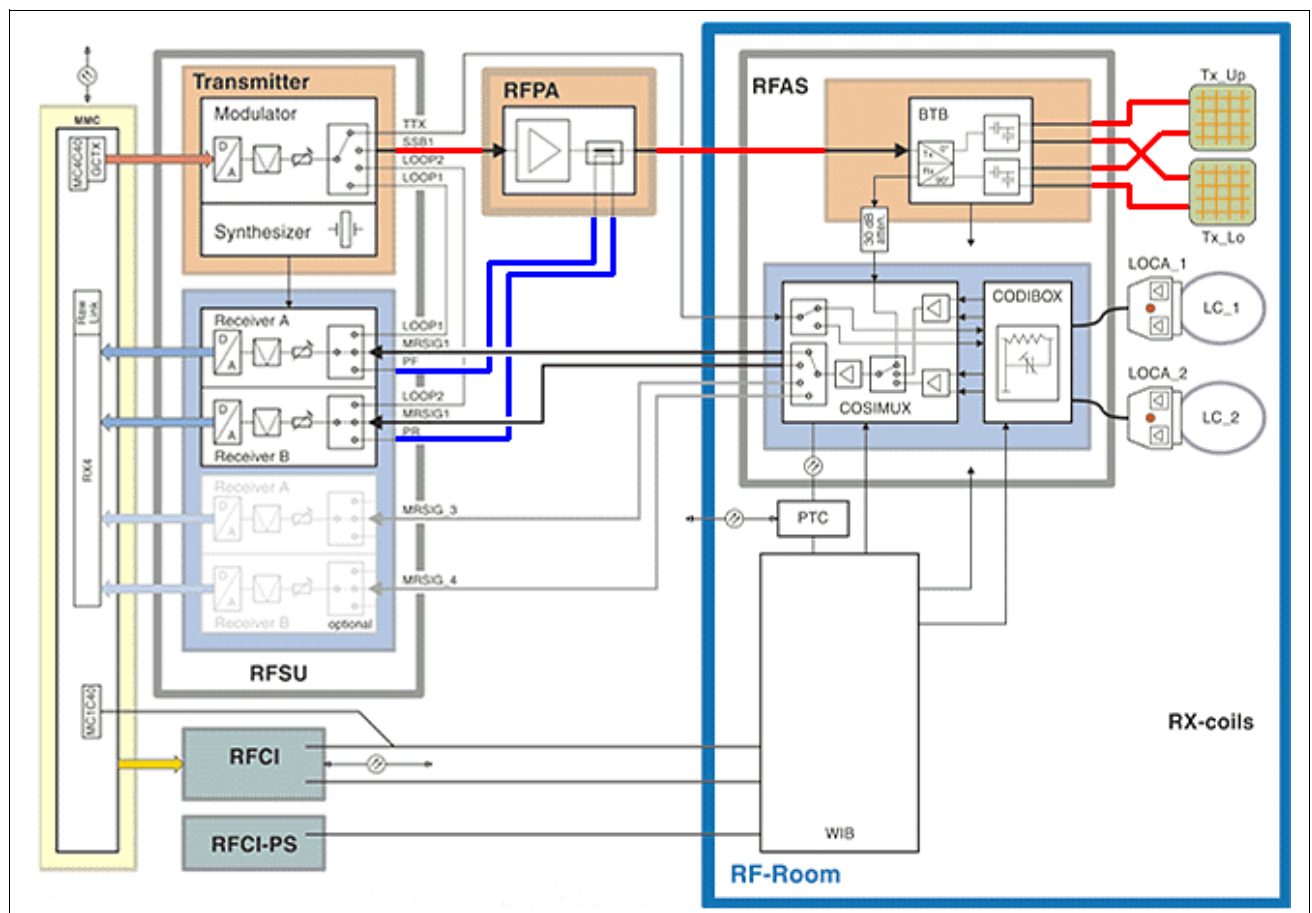
The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

RFPA Power Test

With the MAGNETOM Concerto, the power test is in fact a functional test. Contrary to Harmony/Symphony where only a single "shot" with the maximum amplitude is done, several measurements with an increasing amplitude starting from the minimum to the maximum voltage are performed.

NOTICE The test requires a valid RF characteristic.
Ensure that the spherical phantom is positioned in the TX-coil.

Fig. 72 RFPA Power Test



Signal Path

The output signal of the RFPA is fed via the BTB to the TX-coil coil. The forward and reflected power is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from the minimum to the maximum.

Evaluation

- The actual forward voltage (measured via directional coupler) is compared to the nominal value read from the RF characteristic measurement.
- The ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.
- In the graphical output the measured values (voltage) of power forward and power reflected are displayed over the transmitter voltage.

Conclusions

The objective of the test is to ensure that the RFPA operates at peak performance and that the reflected voltage remains below 20% of the forward voltage at the tuned TX-coil (ideally 50 Ohm).

Peak performance can no longer be achieved when the total gain (signal processing in the RFSU and gain in the RFPA) and, as a result, the characteristic curve has changed due to aging components. If the deviation lies outside the acceptable tolerance, the RF characteristic must be reexamined. If peak performance still cannot be achieved, the output level of the modulator and the directional coupler in the RFPA should be checked to determine the cause of the error. For further information, refer to the Troubleshooting Guide.

In addition, the varying amplitude allows to determine at which amplitude the RFPA trips because at that point the RFPA switches off and the test is aborted. If only one amplitude was used, no information about the occurrence of an arcing would be available.

If there is an increase in the reflected voltage, repeat the TX calibration. If that does not help, check all cable connections from the RFPA (CODICO) to the receiver and the transmission cables to the RFPA, BTB and TX-coil.

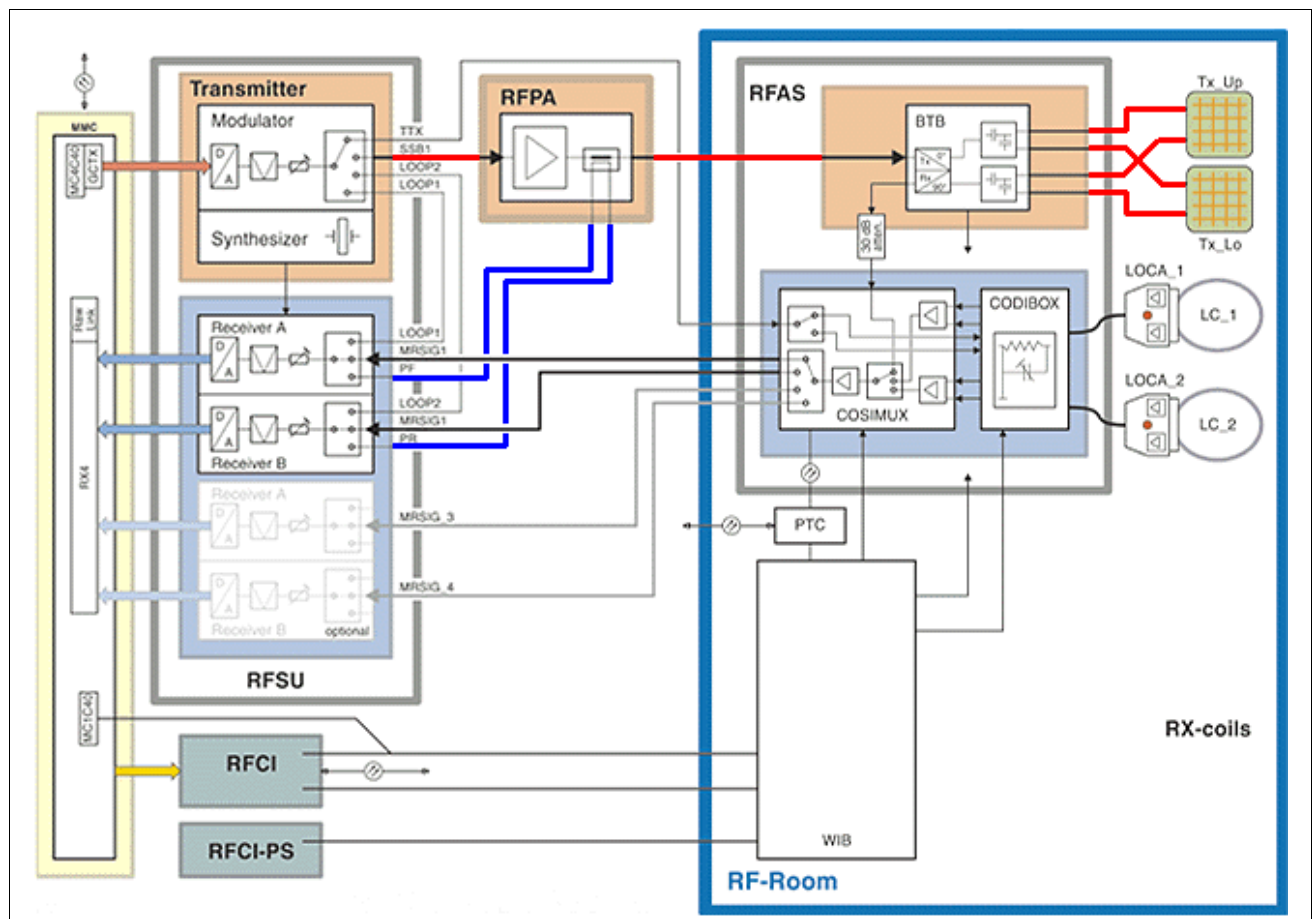
For further information, refer to the Troubleshooting Guide.

RFPA Loop

The RFPA Loop test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

NOTICE Ensure that the spherical phantom is positioned in the TX-coil.

Fig. 73 RFPA Loop



Signal Path

The output signal of the modulator is fed via the BTB to the TX-coil. The forward and reflected power is measured via the directional coupler in the RFPA (CODICO).

Evaluation

- The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse for linearity.
- The average of the forward power is checked against a minimum, the average of the reflected power against a maximum specification.

Conclusions

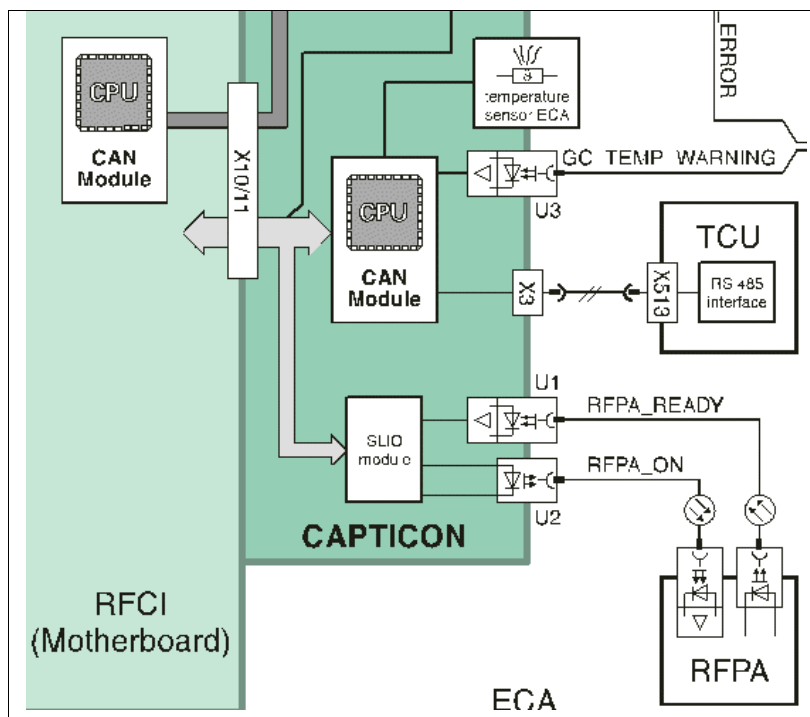
This test is used primarily to check the stability and linearity of the RFPA. Modulator control, data acquisition via receiver and RF characteristic acquisition must be intact and correct to evaluate the results.

For further information, refer to the Troubleshooting Guide.

CAPTICON-RFPA Status Test (interactive)

This interactive test checks the RFPA switch on and switch off sequence and the status of the RFPA. During the test you have to watch the LEDs V701 (PA_ON) and V702 (READY) at the RFPA. They will be switched on and off according to the description in the pop-up window.

Fig. 74 CAPTICON-RFPA Status Test



Signal Path

The signals RFPA_ON and RFPA_READY are routed via FOC from the SLIO module at Capticon (RFCI) to the RFPA.

Conclusions

If the test fails, check/replace:

1. FOC between Capticon and RFPA
2. Capticon
3. RFPA
4. RFCI motherboard.

RFAS

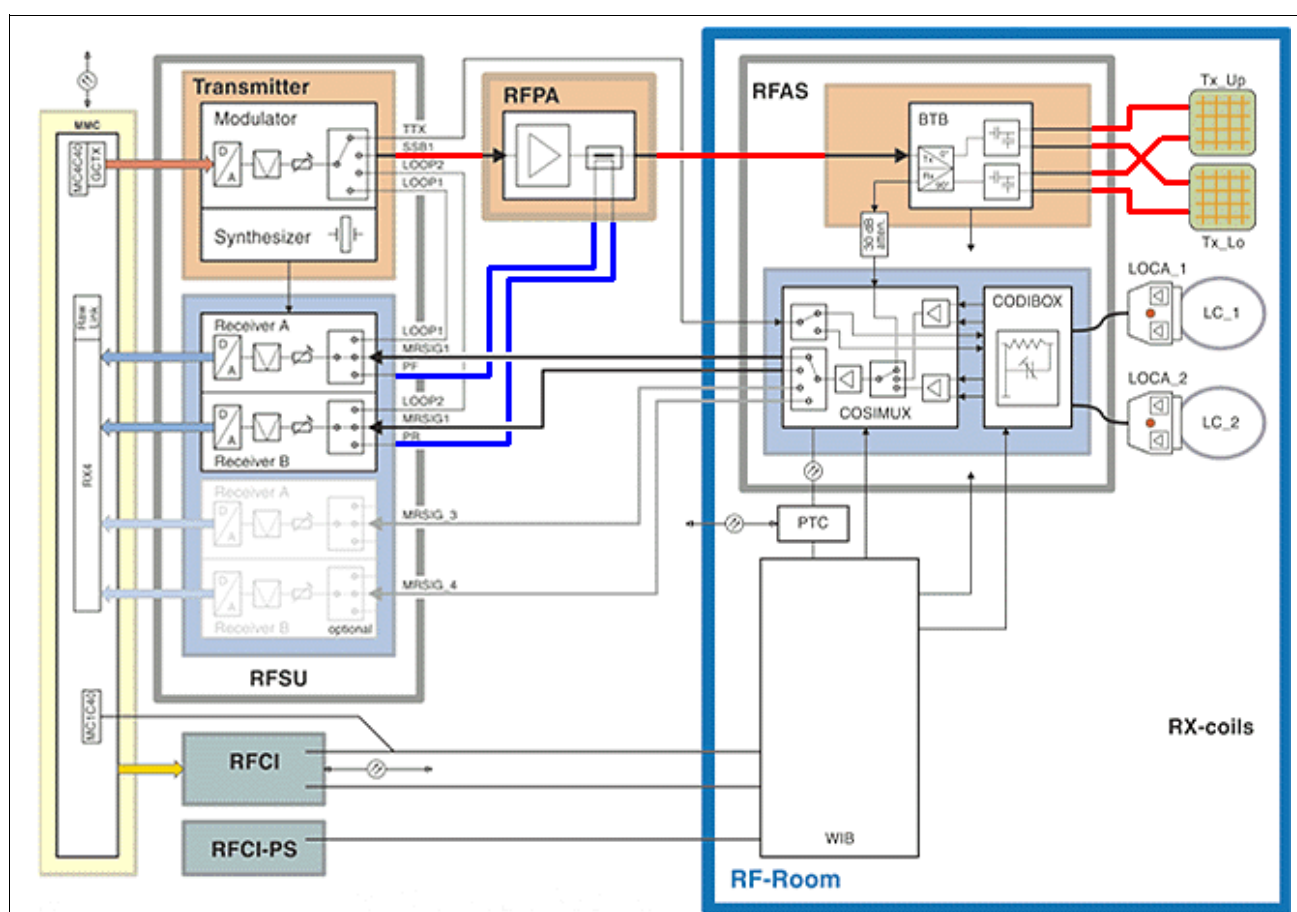
- TX-path (Reflection)

TX-path (Reflection)

This test is nearly identical to the RFPA Power test. Several measurements with an increasing amplitude starting from the minimum to the maximum voltage are performed.

NOTICE The test requires a valid RF characteristic or the linear RF characteristic.

Fig. 75 RFAS TX-path (Reflection)



Signal Path

The output signal of the RFPA is fed via the BTB to the TX-coil. The forward and reflected power is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from the minimum to the maximum.

Evaluation

- The actual forward and reflected voltage (measured via directional coupler) at maximum reflection is displayed.
- The maximum ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.
- The first diagram shows the measured values (voltage) of power forward and power reflected over the transmitter voltage.
- The second diagram shows the DICO reflection (power reflected/power forward(%)) over the transmitter voltage.

Conclusion

The objective of the test to ensure that the TX-path works properly without excessive reflections caused by defective connectors and cables.

If this test fails, perform the interactive **Cable** test TX-path (reflection).

Cable

- TX-path (Reflection)

NOTE	These tests are for experts only! Additionally special equipment is necessary for the tests.
-------------	---

This advanced test enables you to check the complete TX-path to the TX-coil and to make a systematic troubleshooting in the TX-path by connecting a power 50 Ohm terminator at various locations in the TX-path and/or configuring of particular loops.

The test should be used by experienced service engineers only. Special tools (power dummies, RF-cables, connectors, adapters and attenuators) are needed.

Popup windows inform you about the actual measurement configuration.

NOTICE	The tests require a valid RF characteristic or the linear RF characteristic.
---------------	--

NOTE	When using the linear characteristic it might be possible that the used transmitter amplitude exceeds the allowed range resulting in an abort of the measurement. This is normal and not a problem of the system.
-------------	---

NOTE	For the first measurement of each test also the reflection factors are evaluated and analyzed. If the reflection exceeds 90% the test is aborted with the error message Error: DICO reflection out of spec for first measurement. In that case check the cabling and the 50 Ohm power dummy (if used).
-------------	---

Safety

WARNING	High RF voltage
----------------	------------------------

Contact with RF conductors during transmitting of the RFPA will result in death or serious physical injury.

Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input cable at X1 and the unblank line U1 from the RFPA.

TX-path (Reflection)

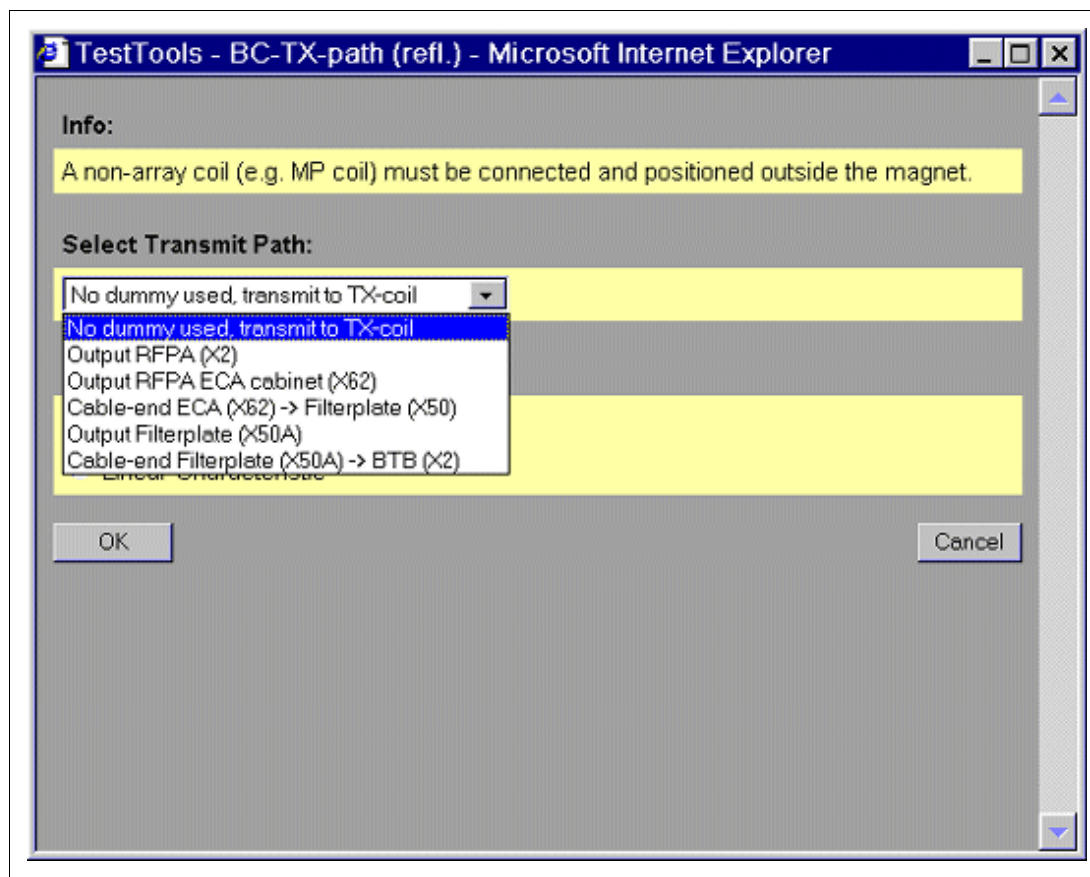
This test is nearly identical to the RFAS test but is interactive. The test enables you to check the TX-path at different locations between the RFPA and the BTB. Several measurements with an increasing amplitude starting from the minimum to the maximum voltage are performed.

Signal Path

The output signal of the RFPA is fed via the BTB into the TX-coil. The TX-path can be opened at several locations (refer to [Fig. 76](#)) and terminated with a 50 Ohm power dummy.

The forward and reflected power is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from the minimum to the maximum.

Fig. 76 Selection menu - transmit path



Evaluation

- The actual forward and reflected voltage (measured via directional coupler) at maximum reflection is displayed.
- The maximum ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.
- The first diagram shows the measured values (voltage) of power forward and power reflected over the transmitter voltage.
- The second diagram shows the DICO reflection (% power reflected/power forward) over the transmitter voltage.

Conclusion

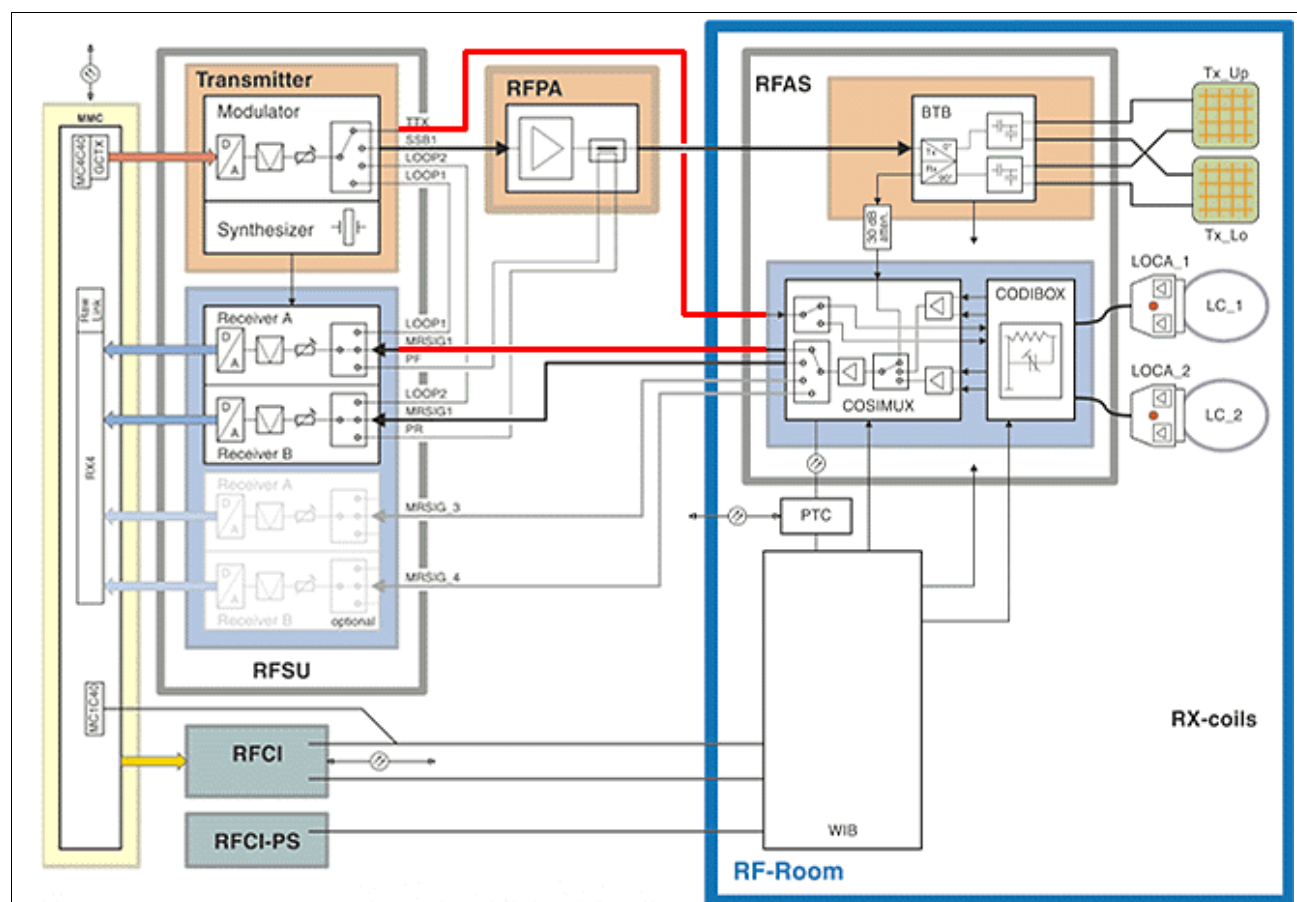
The objective of the test is to ensure that distinct parts of the TX-path work properly without excessive reflections. Step by step connect the dummy load to the different possible locations in the TX-path.

Step	Test step: Dummy connected to	Possible error sources (if test fails and previous test ok)
1	Output RFPA (X2)	RFPA
2	Output RFPA ECA cabinet (X62)	Connector output RFPA X2, RF-cable in the ECA, connector X62
3	Cable-end ECA (X62) -> Filterplate (X50)	RF-cable ECA/Filterplate, connector X50
4	Output Filterplate (X50A)	Connector X50
5	Cable-end Filterplate (X50A) -> BTB (X2)	RF-cable Filterplate BTB (X2)
6	No dummy used, transmit to TX-coil	BTB, TX-coil

- COSIMUX - SGA (receiver 1A/1B/2A/2B)
- COSIMUX linear
- COSIMUX circular
- COSIMUX LED test

This test is used to check the stability and linearity of the amplifiers of the COSIMUX, both in low and high gain. The amplifiers 1, 2 and 3 are switched to output channel 3, amplifier 4 uses channel 4. A linear characteristic is used for this test.

Fig. 77 COSIMUX - SGA



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Signal Path

The RFAS tune path is fed to the SGAs, the signals after the amplification to the different receiver channels.

Evaluations

- The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse for linearity.
- The signal amplitude is checked against a minimum specification.

Conclusions

This test is used primarily to check the stability and linearity of the SGAs of the COSIMUX. Modulator control and data acquisition via receiver must be intact and correct to evaluate the results.

For further information, refer to the Troubleshooting Guide.

Conclusions

This test is used to check the symmetry and amplification of the SGAs of the COSIMUX. Modulator control and data acquisition via receiver must be intact and correct to evaluate the results.

For further information, refer to the Troubleshooting Guide.

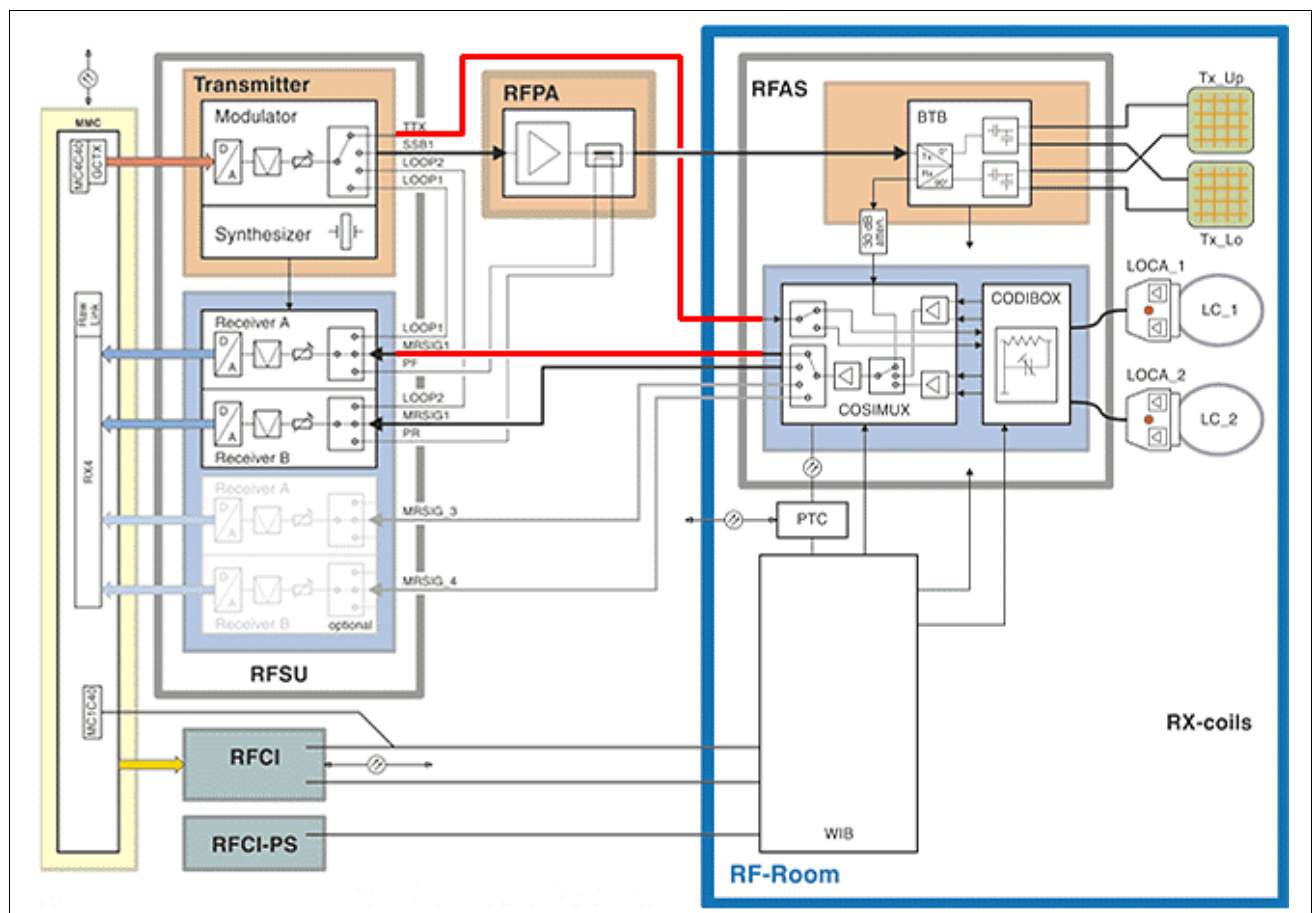
COSIMUX circular

The switchable gain amplifiers (SGA) in the COSIMUX are tested using the different output channels of the board. Two inputs (either 1/2 or 3/4) are always connected to one output. The optional second receiver is also tested if installed (but only channel 3, not channel 4).

A linear characteristic is used for this test.

NOTE Remove the LOCA cables X1, X2, X5 and X6 at the COSIMUX.

Fig. 79 COSIMUX circular



Signal Path

The RFAS tune path is fed to the SGAs, the signals after the amplification to receiver channel 1A.

Evaluations

- The combination SGA 1 - output channel 1 is used as a reference for the other measurements.
- The signal amplitudes of a pair of measurements are checked for symmetry, the phase for 90 degrees. Both are compared to a specification.
- All signal amplitudes are checked against a minimum specification.

Conclusions

This test is used primarily to check the CP symmetry of the SGAs of the COSIMUX. Modulator control and data acquisition via receiver must be intact and correct to evaluate the results.

COSIMUX LED test

This is an interactive test that requires you to monitor the LEDs at the COSIMUX indicated at the Display Tune Panel (DTP). The correct settings of the COSIMUX are visually inspected by stepping through the LEDs using the button LAMP of the DTP.

As the number of digits of the DTP is limited only the last two digits of the tested LED are displayed:.

Display at DTP

"V31 30 41 40 on"
 "V31 30 41 40 off"
 "0 10 34 47 44 46"
 "1 34 16 47 44 46"
 "2 36 34 37 44 22"
 "3 36 34 37 24 44"
 "4 11 47 44 46"
 "5 341216 474446"
 "6 36343744131722"
 "7 36 34 37 21 14"
 "8 36 34 37 20 23"
 "9 36343715212444"

Meaning

LEDs V731, V730, V741 and V740 are on
 LEDs V731, V730, V741 and V740 are off
 Test step 0: LEDs V710, V734, V747, V744 and V746 are on
 Test step 1: LEDs V734, V716, V747, V744 and V746 are on
 Test step 2: LEDs V736, V734, V737, V744 and V722 are on
 Test step 3: LEDs V736, V734, V737, V724 and V744 are on
 Test step 4: LEDs V711, V747, V744 and V746 are on
 Test step 5: LEDs V734, V712, V716, V747, V744 and V746 are on
 Test step 6: LEDs V736, V734, V737, V744, V713, V717 and V722 are on
 Test step 7: LEDs V736, V734, V737, V721 and V714 are on
 Test step 8: LEDs V736, V734, V737, V720 and V723 are on
 Test step 9: LEDs V736, V734, V737, V715, V721, V724 and V744 are on

For the test itself it is wise to have a printout of this page.

For the meaning of the LED please refer to the Trouble Shooting Guide, part 6 RF-System, chapter COSIMUX.

If the test is not successful, possible error causes are the RFCI, the COSIMUX or the cabling between them.

- LOCA - amplifier

This test is used to check the stability and linearity of the preamplifier chain of the COSIMUX and the LOCAs. The preamplifiers of the LOCAs are always tested together with the corresponding SGA of the COSIMUX. Both is done in low and high gain.

A linear characteristic is used for this test.

NOTE Attach the Service Adapter to the LOCA which is under evaluation.

The diagram illustrates the RF-System architecture, divided into three main functional areas: Transmitter, Receiver, and RF-Room.

- Transmitter (Top Left):** Contains a Modulator and a Synthesizer. The Modulator is connected to the Synthesizer and the RFPA. The Synthesizer is connected to the Modulator and the RFPA. The RFPA (Radio Frequency Power Amplifier) is connected to the Modulator and the RFAS.
- Receiver (Bottom Left):** Contains Receiver A and Receiver B. Receiver A is connected to the RFPA and the RFAS. Receiver B is connected to the RFPA and the RFAS. The RFAS (Radio Frequency Amplifier System) is connected to the Receiver and the RF-Room.
- RF-Room (Right):** Contains the RFAS, which includes a BTB (Bandpass Filter) and a CODIBOX. The BTB is connected to the RFPA and the CODIBOX. The CODIBOX is connected to the RFPA and the RFAS. The RFAS is connected to the RF-Room. The RF-Room also includes a PTC (Power Transfer Coefficient) and a WIB (Wireless Interface Board). The PTC is connected to the RFAS and the WIB. The WIB is connected to the RFAS and the RF-Room. The RF-Room also includes RX-coils, which are connected to the RFAS and the RF-Room.

The diagram shows the flow of signals and power between these components, with various interconnections and feedback loops.

The RFAS tune path is fed to the preamplifiers in the LOCAs via COSIMUX and CODIBOX, the signals after the amplification to the different receiver channels.

Evaluations

- The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse for linearity.
- The signal amplitude is checked against a minimum specification.

Conclusions

This test is used primarily to check the stability and linearity of the preamplifiers of the LOCAs, but also the matching of the preamplifier chain of COSIMUX and LOCA. Modulator control and data acquisition via receiver must be intact and correct to evaluate the results, and the tests for COSIMUX itself must have run successfully.

RFCI

- RFCI Communication/Status
- RFCI PIN Diodes (Local/TX Coil)
- RFCISetVaricapTest
- RFCIGetCoilsTest
- RFCIStatTest
- RFCIDynTest

RFCI Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message. As long as no measurement was done with the connected coil the status will be "setting not loaded" and the respective PIN channels and mask "not set". So this is no error condition.

RFCI PIN Diodes (Local/TX Coil)

The test of the forward current (TX-coil: 600 mA, local coils: 200 mA) and reverse voltage (-30 V) supplied and measured by the RFCI for the PIN diodes of the TX-coil (located in the body tune box) and all PIN diodes in the connected local coils is done automatically.

NOTE	If coil parts contain PIN diodes and these coil parts are not connected to the part which is attached to the LOCA, the PIN diodes are not correctly identified.
-------------	---

NOTICE	The local coils must not be positioned in the magnet for this test. For safety reasons, it is recommended to position the spherical phantom in the magnet center to prevent damage to the resonator or to the local coils.
---------------	---

Conclusions

This test verifies the function of the RFCI, the cables and the PIN diodes in the TX-coil (BTB) and the connected local coils.

For further information, refer to the Troubleshooting Guide.

RFCISetVaricapTest

This is an interactive test that requires you to connect the Service Plug to LOCA1 and a voltmeter (DVM) to the test points of the service plug (VAR1/VAR2 - GND) and monitor the reading. The test will start with approx. 5 mV and double the voltage from then on until 20 V are reached. Afterwards the voltage will run down again. As soon as 0 V are reached connect the voltmeter to the next test point. Between VAR1 and VAR2 are 10 seconds waiting time, between LOCA1 and LOCA2 20 seconds.

NOTE	This test requires that the Service Plug for the MAGNETOM Concerto is connected to LOCA1. First the voltage from VAR1 of LOCA1 is varied, then VAR2 and afterwards VAR and VAR2 of LOCA2 are tested. After the end of the test it is necessary to connect the Service Plug to LOCA again, otherwise the next test (RFCIGetCoilsTest) will fail.
-------------	---

Conclusions

Possible error causes are the RFCI itself, the cabling to the LOCAs or the service plug.

For further information, refer to the Troubleshooting Guide.

RFCIGetCoilsTest

This test automatically checks the correct recognition of the coil code of the service plug.

NOTE	This test requires that the Service Plug for the MAGNETOM Concerto is connected to LOCA 1. With LOCA 2 it is not working.
-------------	---

Conclusions

Possible error causes are the RFCI, the cabling to the LOCA or the service plug itself.

For further information, refer to the Troubleshooting Guide.

RFCIStatTest

This test checks the CAN-controlled (static) detuning for the local coils and the TX-coil. The status of the RFCI is read back and compared to the nominal values.

NOTE	This test requires that the Service Plug for the MAGNETOM Concerto is connected to LOCA 1.
-------------	--

Conclusions

Possible error causes are the RFCI itself, the CAN-bus, the cabling to the LOCAs or the service plug.

For further information, refer to the Troubleshooting Guide.

RF - MAGNETOM Trio Body

NOTE This section describes the MR system MAGNETOM Trio Body only.

The RF tests consists of a set of tests designed to verify the performance of the RF system and to assist you in locating defective components.

NOTE Beginning with *syngo* MR 2002B (NUMARIS/4 VA21A) all displayed graphs can be enlarged by clicking into the image and selecting (dragging) the region of interest. With "autoscale" the original view of the graph is restored.

General

The RF System hardware has been designed with several built-in loops to allow automated testing of RF components. In addition, level measurements are performed via the directional couplers (DICO) in the RFPA and the TALES.

The sequence used is common to all RF loop tests except the BTB and BC test (body tune box and body coil). It is called *rf_loop* and contains six RF pulses of which the first two are relevant for TestTools and the others have zero amplitude. The first pulse is in sinc (function $\sin(x)/x$) form with constant amplitude and is used for level measurements and stability evaluations. The second pulse is also in sinc form and increases from 0 up to the maximum amplitude. It is used in linearity evaluations.

The system uses two different RFPA's:

- 35 kW Astex RFPA transmits only to the body coil.
- 8 kW Dressler RFPA (broad band) transmits only to the local TX coils.

NOTE In the user interface of the TestTools only one FRU is available. For the tests please keep the following in mind:
If the functional tests are selected the 35 kW Astex RFPA 1 is tested, with interactive tests the 8 kW Dressler RFPA 2 is used.

NOTICE The 50 Ohm termination of the 35 kW RFPA is done automatically. The dummy load is located in the TAS_3T at the filter plate and is water-cooled in series with the gradient coil cooling.
To prevent damage to the dummy load, do not perform RFPA tests without proper cooling.

When using the 8 kW RFPA (interactive tests), the cable to BCCS/X7 must be connected to TALES/X6 and a coil or the service plug must be connected to coil plug 1.

NOTE With every measurement a unique ID is displayed in the output window (e.g. 4.0.67892). This so-called "Loid" can be used in conjunction with the Rawdata Handling Package RHP to display and evaluate the raw data of the measurement.

Additionally to this number also the used RFPA (RFPA1 or RFPA2) is printed out as an information. Depending on the test this allows it to identify any error or mistake in selecting the correct RFPA.

With every test that runs a measurement at the beginning the used nucleus is displayed. With all systems that are not equipped with the multi-nucleus option (spectroscopy, X-nucleus) this will be always "1H", otherwise it should be the selected nucleus, unless it is a test that always use 1H and is independent of the nucleus-selection.

NOTE In case the system is equipped with a multi-nucleus option (spectroscopy, X-nucleus), also these tests are performed when a RF test is selected. If this is not intended or necessary please de-select the test type "Multi Nuclei".

The nucleus which has to be used for the next RF tests can be selected in the problem section.

The selection has to be done prior to the start of the test. If you realise that accidentally the wrong nucleus is selected please cancel the test immediately.

Everytime when such a test is to be performed it is necessary to select the nucleus/frequency prior to start the test.

Overview RF Tests

Test	Comm/Status	Functional	Interactive	Multi Nuclei
Mod/REC i	x	x	-	x
RFPA	x	RFPA 1 (35 kW)	RFPA 2 (8 kW)	RFPA 1 (8 kW)
RFAS	x	x	x	x
LCCS	x	x	x	x
RFCI	x	x	x	-
BTB	x	x	x	-
BC	x	x	x	-
Cable	-	-	x	x (LC-TX only)

Stability

The first pulse of the sequence with a constant RF amplitude is used. The resulting magnitude data is the evaluation of the stability of the center column which is located in the center of the RF pulse. The numerical result is the maximum value minus the minimum value divided by the mean value. Instabilities normally cause phase ghosting or poor S/N values.

An overlaid graphic represents an average of the samples close to the centre column. This allows to differentiate between noise and instabilities.

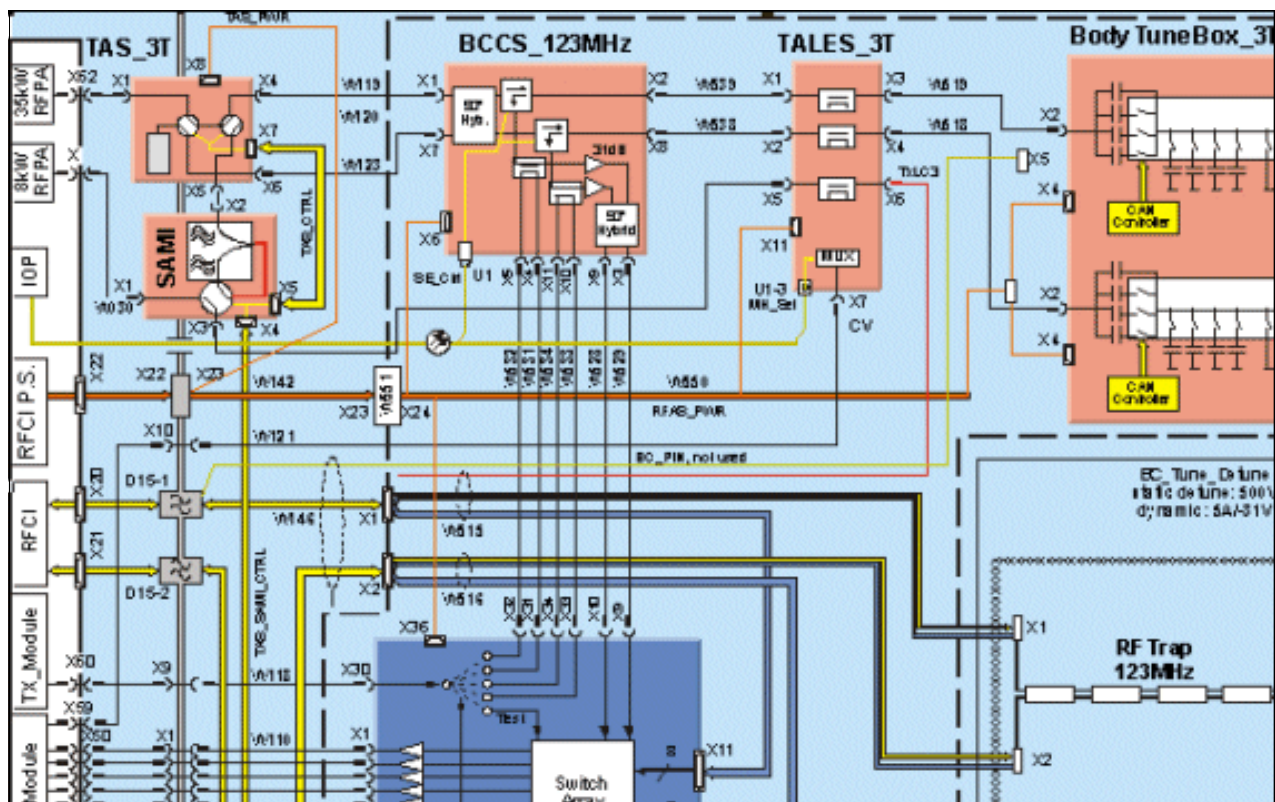
Linearity

For linearity evaluations, the second RF amplitude is stepped from 0 to a positive pulse amplitude. The center column of each line (maximum of the single pulses) is fitted to a linear function. The numerical result is the maximum deviation from the linear fit divided by the maximum value making up the data set. Non-linearities result in poor slice profiles and reduced S/N.

NOTE With many tests the first measurements are not used for the final evaluation. So don't be worried if the graphical display and the numerical differ for this.

Overview RF system Trio Body (NUMARIS/4 VA25A)

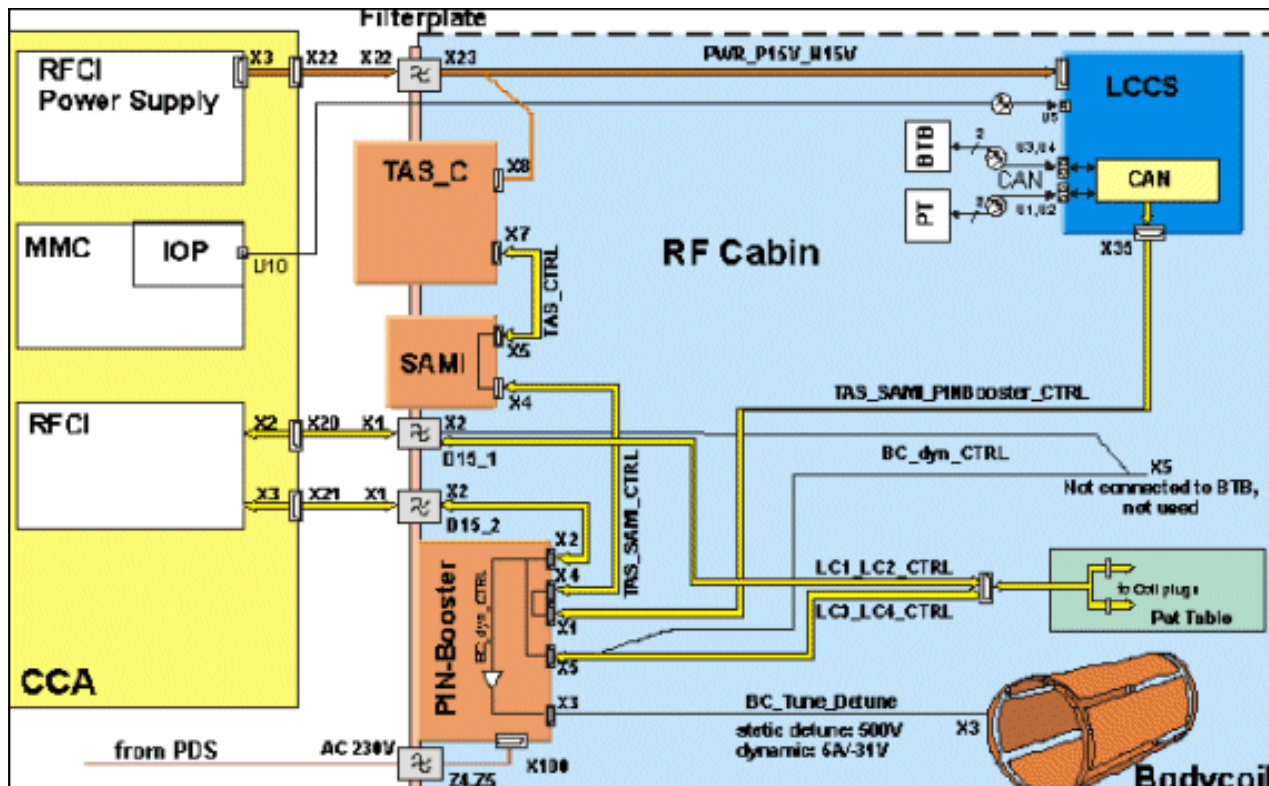
Fig. 81 Overview RF system Trio Body (NUMARIS/4 VA25A)



NOTE Systems with *syngo* MR 2003T (NUMARIS/4 **VA23A**) do not possess a SAMI. The TX-path from the 8 kW RFPA 2 is directly routed through a dummy plate (connectors X1/X3) through the filter plate to TALES X5 (LC-TX-path). The 35 kW RFPA 1 only transmits to the Body Coil.

Overview RFAS Trio Body (NUMARIS/4 VA25A)

Fig. 82 Overview RFAS Trio Body (NUMARIS/4 VA25A)

**NOTE**

Systems with *syngo* MR 2003T (NUMARIS/4 VA23A) do not possess a SAMI. The TX-path from the 8 kW RFA 2 is directly routed through a dummy plate (connectors X1/X3) through the filter plate to TALEX X5 (LC-TX-path). The 35 kW RFA 1 only transmits to the Body Coil.

- TX-Cassette Communication (former Modulator Communication)
- Modulator-Receiver Loop (Functional)
- Receiver A/B symmetry (Functional)
- Reference Voltage (TALES)

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

Modulator-Receiver Loop

The diagram illustrates the MR system architecture, showing the flow of signals between various components:

- TX Module:** Contains a Modulator and a Synthesizer. It is connected to the RXDSP and the MIC4C40. The Modulator outputs to the Synthesizer, which outputs to the MIC4C40. The Synthesizer also outputs to the RX Module.
- RX Module:** Contains two channels, A and B. Channel A outputs to the MIC4C40 and the RXDSP. Channel B outputs to the RXDSP. The RX Module also outputs to the RXDSP.
- RFP A (35kW):** A power amplifier that receives signals from the TX Module and the RX Module. It outputs to the TAS_3T.
- TAS_3T:** A transmit/receive switch that routes signals between the RFP A and the BCCS.
- BCCS:** A baseband control and conditioning system that receives signals from the TAS_3T and the RX Module. It outputs to the TALES.
- TALES:** A transmit/receive switch that routes signals between the BCCS and the BodyTuneBox.
- BodyTuneBox:** A system that receives signals from the TALES and outputs to the Bodycoil.
- Bodycoil:** A coil used for imaging, connected to the BodyTuneBox.
- LC Channel Select:** A system that receives signals from the BCCS and the Bodycoil. It outputs to the Array MUX.
- Array MUX:** A multiplexer that receives signals from the LC Channel Select and outputs to the RXDSP.
- RXDSP:** The receiver digital signal processor, which receives signals from the Array MUX and the RX Module.
- Patient Table:** The table on which the patient is positioned, connected to the Bodycoil.

The rf_loop sequence is started and the output signal of the modulator is fed to the receiver module x A/B.

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

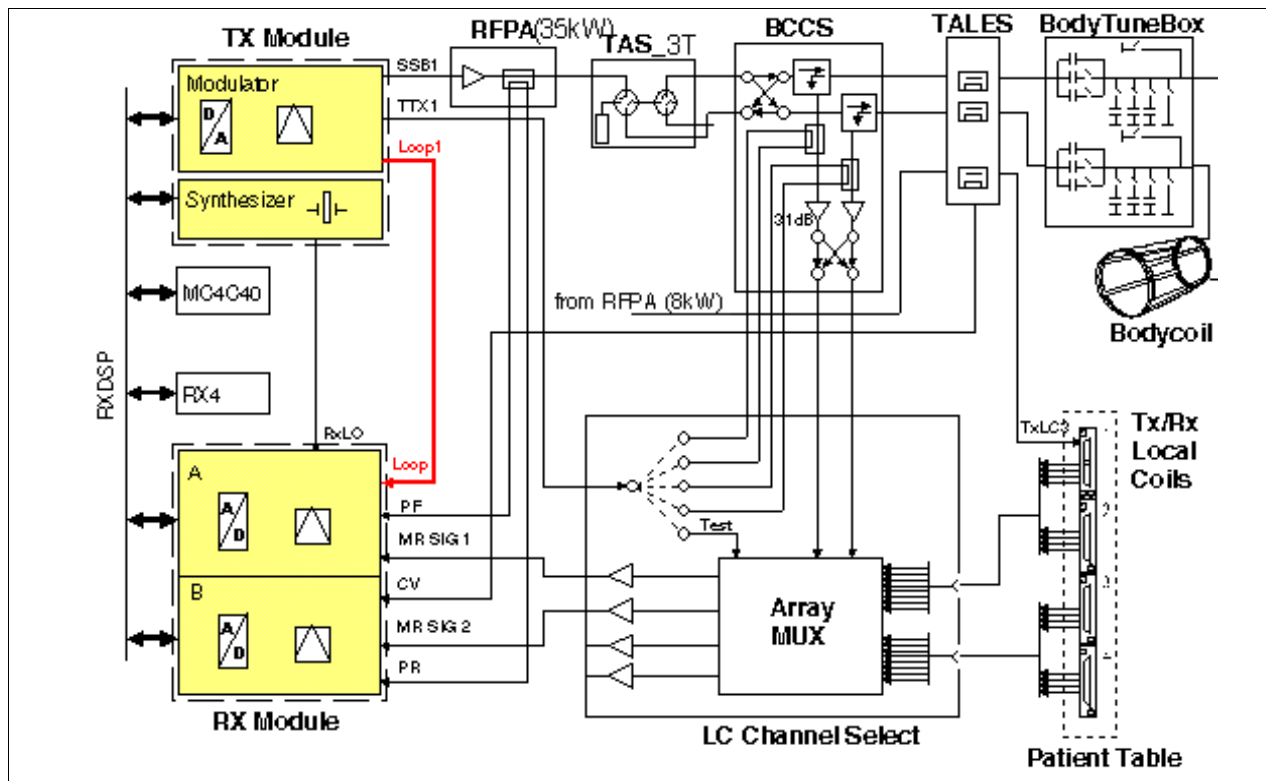
Successful completion of this test is the prerequisite for all additional RF tests. If the test results are satisfactory, it can be assumed that the controller provides the transmitter with data as specified, that the data is modulated correctly in the modulator, received correctly by the receiver and then forwarded to the MMC and Imager.

The test results are, however, not indicative of the signal processing functionality at the receiver inputs CV and DC. Similarly, the signal switch-over in the transmitter to SSB1/2 or TTX is not part of this test. Instead, the signal paths are tested using the RFPA (SSB1/2) and LCCS tests (TTX).

NOTE Since transmitting/receiving are simultaneous activities, both the transmit and the receive signal are affected by signal fluctuations in the synthesizer. As a result, fluctuations remain undetected, a fact that has to be taken into account during troubleshooting.

Receiver A/B symmetry

Fig. 84 Receiver A/B symmetry



Signal Path

The output signal of the modulator is fed to the receiver module A/B.

Evaluation

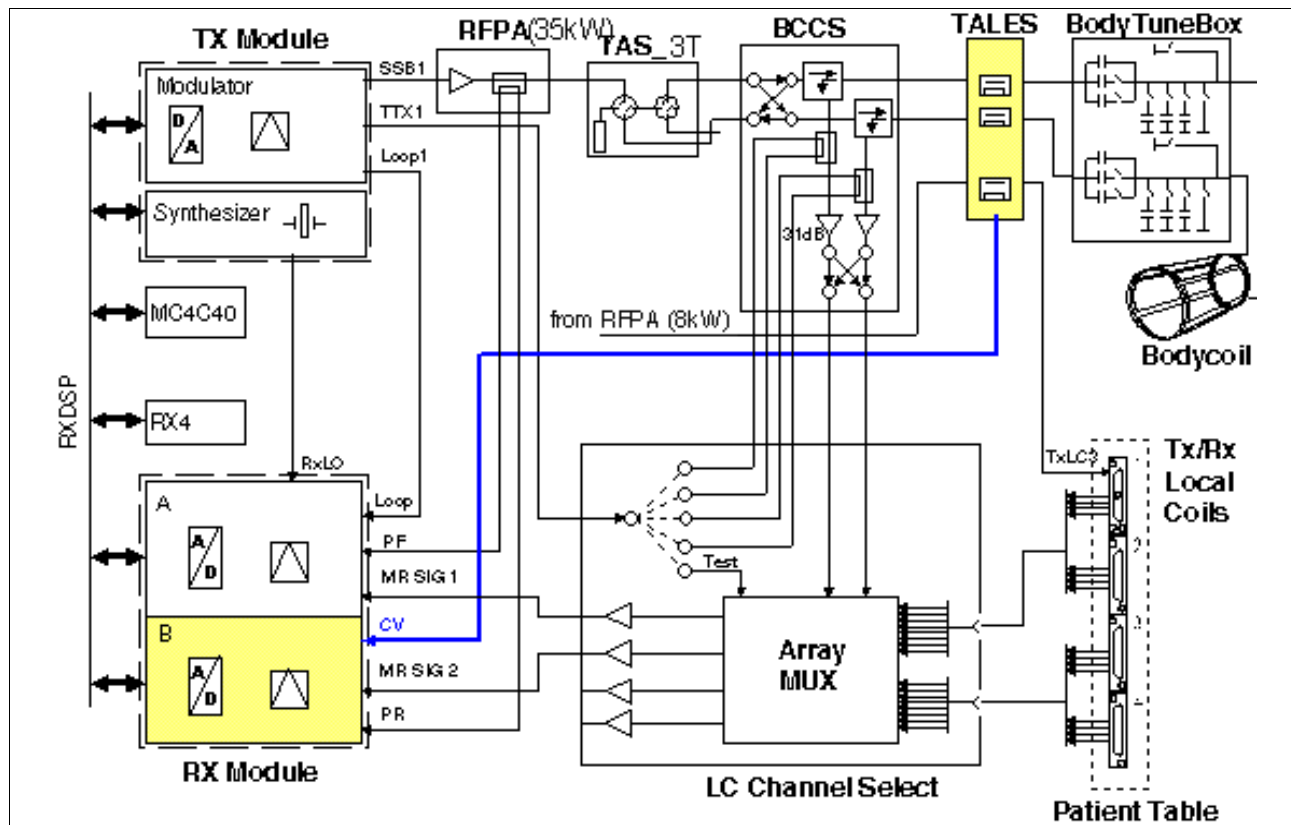
The raw data of receivers A and B are compared for symmetry.

Conclusions

With respect to the signal path and data acquisition, the test is identical to the Modulator-Receiver loop. The only difference is that only one raw data line is being measured and only the first pulse will be evaluated to check the symmetric signal processing of both receivers.

Reference Voltage (TALES)

Fig. 85 Reference voltage



Signal Path

The TALES output signals are fed to the receiver 1B (CV-Input).

Evaluation

Verification that the TALES reference voltage is 500 mV.

Conclusions

The TALES continuously supplies all internal directional coupler signals triggered by the controller. In addition, it supplies the signal of an integrated reference voltage source. The DICO signals and the reference voltage are multiplexed and fed to the receiver CV input. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the receiver and signal processing during data acquisition. In case of error, the 500 mV reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

RFPA 1/2

- RFPA Communication/Status
- RFPA Power Test
- RFPA Loop
- RFPA-TALES
- RFPA Long Time Stability

RFPA Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

The communication of both RFPAs is tested and both stati are returned.

NOTE	In the user interface of the TestTools both FRU's are available. For the tests please keep the following in mind: If the functional tests are selected the 35 kW Astex RFPA 1 is tested, with interactive tests the 8 kW Dressler RFPA 2 is used.
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RFPA Power Test (functional test with RFPA 1 - 35 kW Astex; interactive multi nuclei with RFPA 2 - 8 kW Dressler)

NOTE The test requires a valid RF characteristic.

Fig. 86 RFPA 1 (35 kW) Power Test

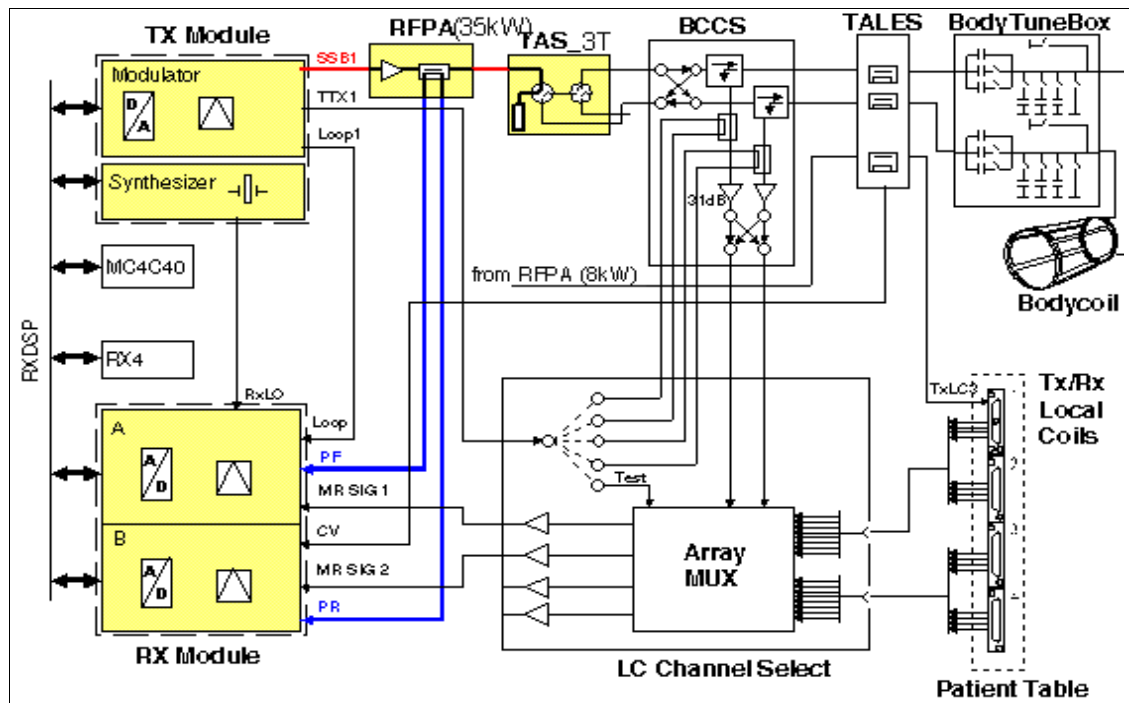
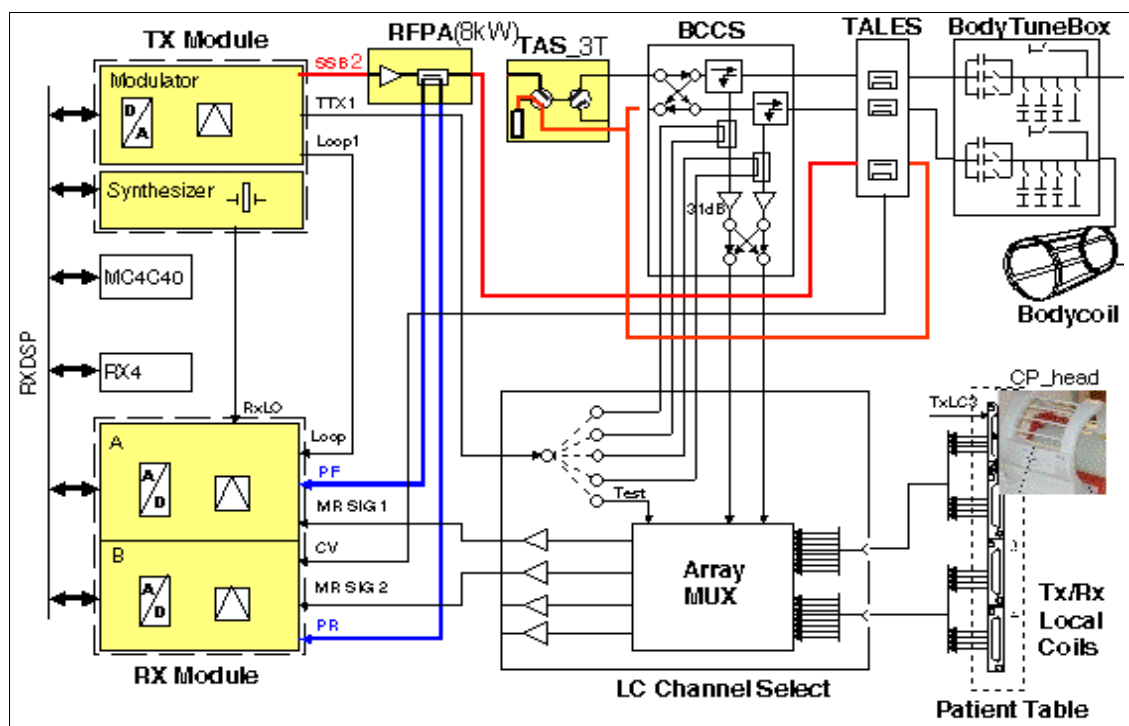


Fig. 87 RFPA 2 (8 kW) Power Test



Signal Path

1. The output signal SSB1 of the modulator is fed via the RFPA 1(35 kW) to the dummy load of the TAS_3T. The Forward_Power Signal is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF characteristic data.
2. The output signal SSB2 of the modulator is fed via the RFPA 2 (8 kW) and SAMI X1/X3 to TALES X5. The cable to BCCS/X7 must be connected to TALES/X6 and the service plug must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

NOTE If the service plug is connected to coil plug 1, the RX voltage is limited to 250 V. Only the graphical output would be useful for test evaluation.

Evaluation

- The actual forward voltage (measured via directional coupler) is compared to the nominal value read from the RF characteristic measurement.
- The ratio of the reflected voltage to the forward voltage (measured via directional coupler) is calculated.

Conclusions

The objective of the test is to ensure that the RFPA operates at peak performance and that the reflected voltage remains below 30 % of the forward voltage at a dummy load of 50 Ohm.

Peak performance can no longer be achieved when the total gain (signal processing in the RFSU and gain in the RFPA) and, as a result, the characteristic curve has changed due to aging components. If the deviation lies outside the acceptable tolerance, the RF characteristic must be reexamined. If peak performance still cannot be achieved, the output level of the modulator and the directional coupler in the RFPA should be checked to determine the cause of the error. For further information, refer to the Troubleshooting Guide.

If there is an increase in the reflected voltage, check all cable connections from the RFPA to the Receiver and the transmission cable to the TAS assembly. You should also check with the LEDs if the TAS relay is set correctly. A defective dummy load would cause high voltage reflection, too.

RFPA Loop (functional test with RFPA 1 - 35 kW Astex; interactive with RFPA 2 - 8 kW Dressler)

The RFPA Loop test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

Fig. 88 RFPA 1(35 kW) Loop

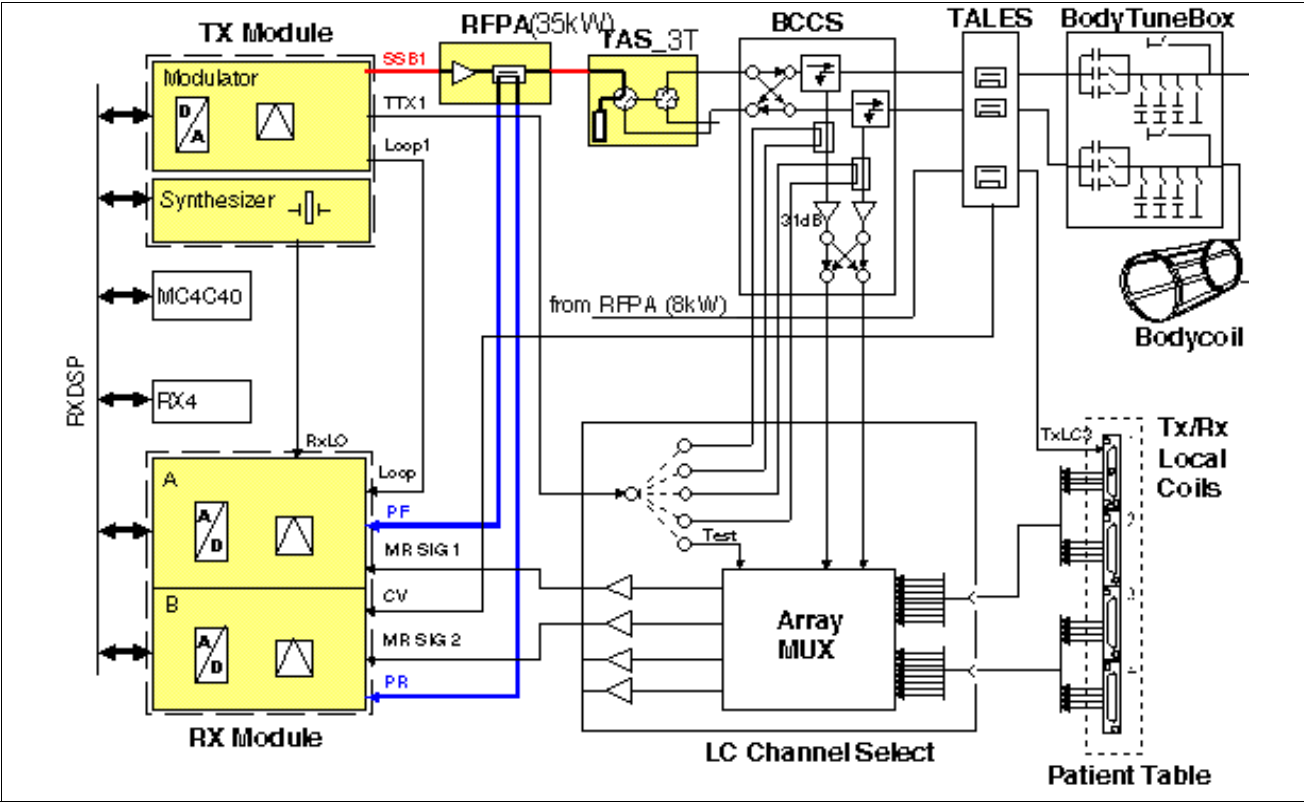
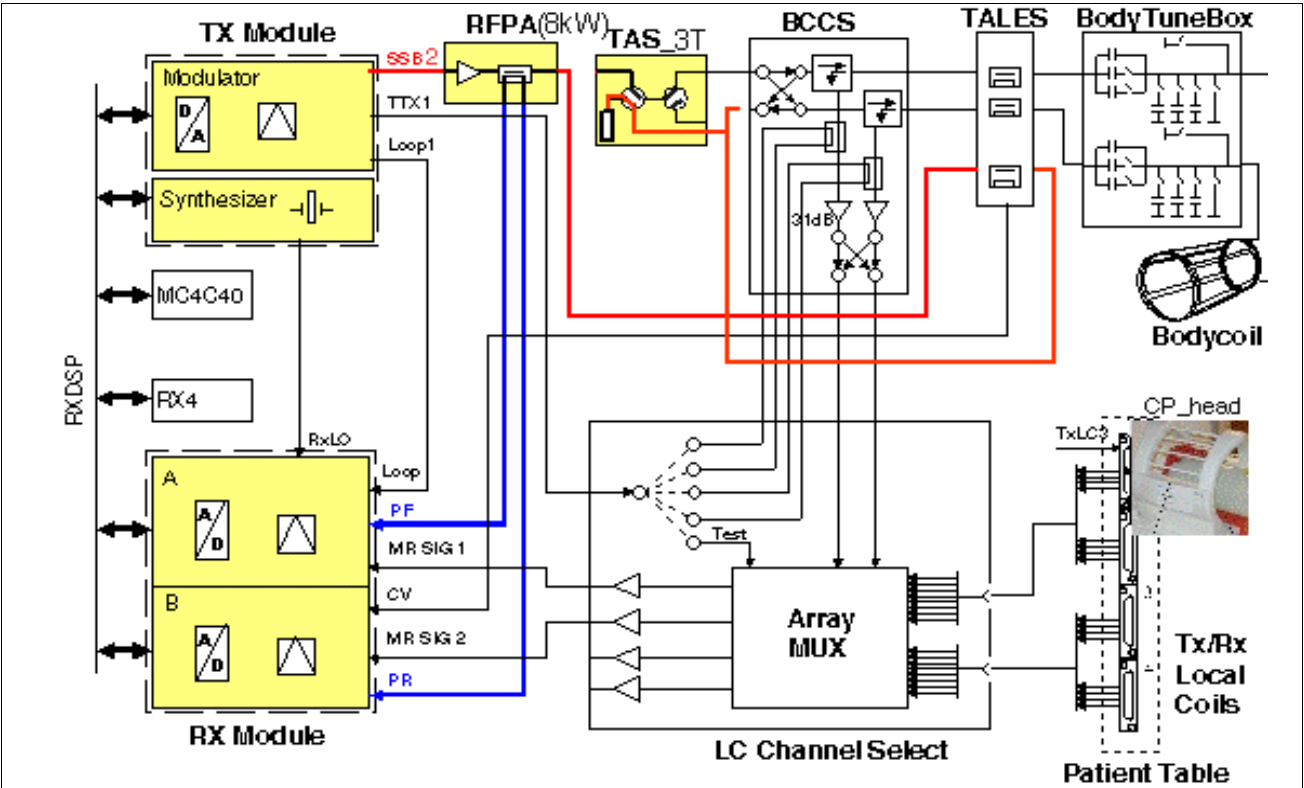


Fig. 89 RFPA 2 (8 kW) Loop



Signal Path

1. The output signal of the modulator is fed via the RFPA 1 (35 kW) to the dummy load of the TAS_3T. The forward signal is measured via the directional coupler in the RFPA.
2. The output signal SSB2 of the modulator is fed via the RFPA 2 (8 kW) and SAMI X1/X3 to TALES X5. The cable to BCCS/X7 must be connected to TALES/X6 and the service plug or the CP head coil must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability. The raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

This test is used primarily to check the stability and linearity of the RFPA. Modulator control, data acquisition via receiver and RF characteristic acquisition must be intact and correct to evaluate the results.

RFPA-TALES (functional test with RFPA 1 - 35 kW Astex; interactive with RFPA 2 - 8 kW Dressler)

The RFPA TALEs test is done in two steps. The first measurement is performed with the valid RF characteristic, the second measurement with a predefined linear characteristic, but with different transmitter amplitudes and specifications. If there is no valid characteristic available, i.e. after the software has been installed and before the tune-up has been performed, the linear characteristic is used instead of the valid characteristic for the first measurement as well. In this case, the deviation may be significantly above the specification or the test may even abort.

NOTICE	Ensure that the loader with spherical phantom is positioned in the body coil for the RFPA1 35 kW test.
---------------	--

Fig. 90 RFPA 1 -TALES (35 kW RFPA)

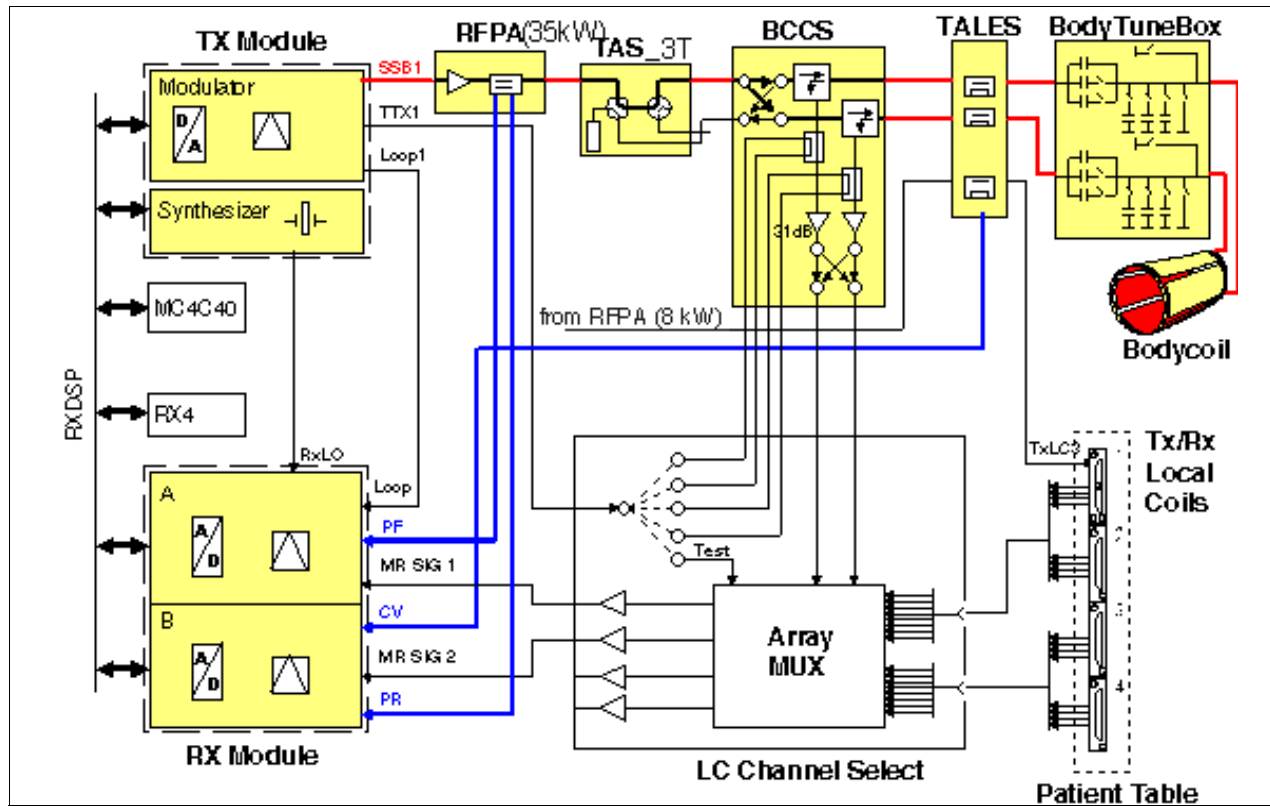
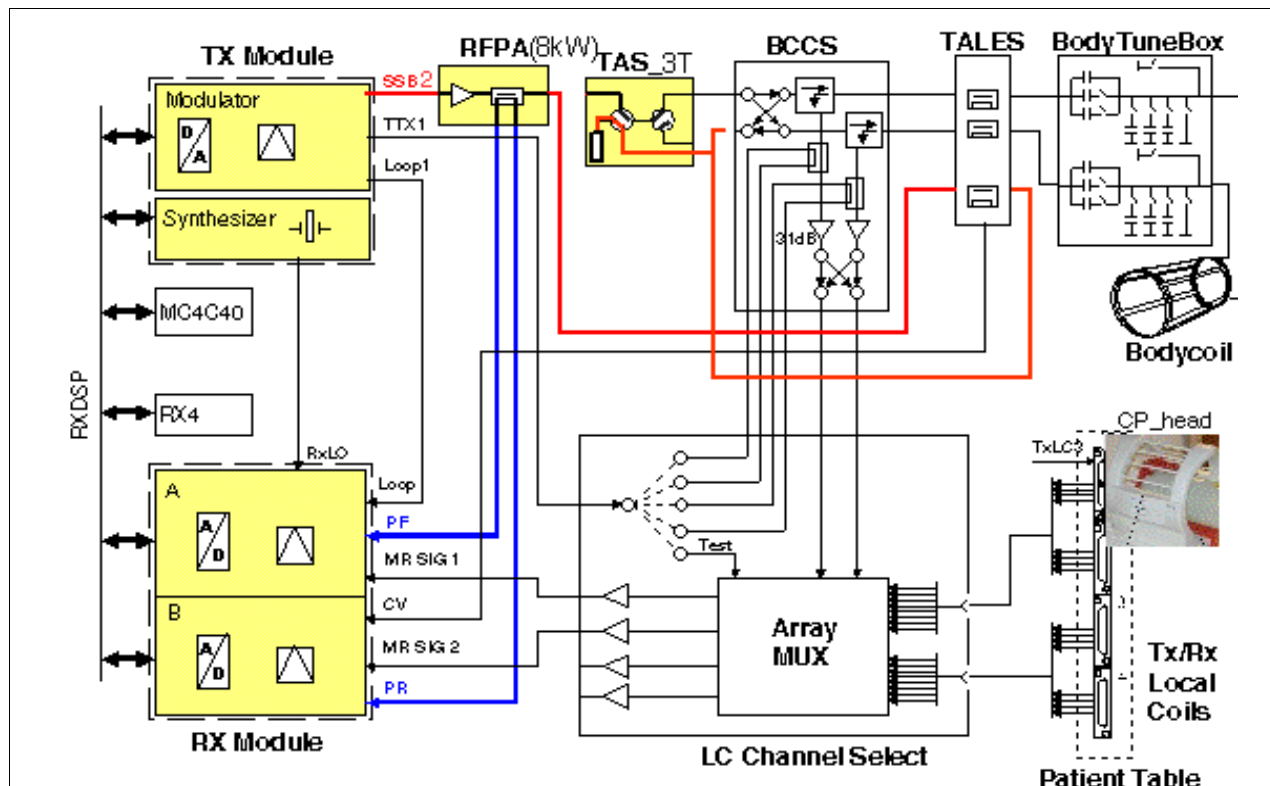


Fig. 91 RFPA 2 -TALES (8 kW RFPA)



Signal Path

1. The output signal of the modulator is fed via the RFPA 1(35 kW)-TAS_3T-BCCS-TALES-BTB to the body coil.
2. The output signal SSB2 of the modulator is fed via the RFPA 2(8 kW) and SAMI X1/X3 to TALEs X5. The cable to BCCS/X7 must be connected to TALEs/X6 and the service plug or the CP head coil must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

The test consists of two measurements. One RF pulse is transmitted for each. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALEs.

The body coil must be tuned to evaluate the signals. For this reason, coil tuning is automatically initiated at the beginning of the test.

Evaluations

- Forward Voltage from the TALEs within tolerance
- Reflected Voltage from the TALEs below limit
- Deviation of the forward voltage from the TALEs and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks almost the entire transmission component of the system.

The results of the forward voltage at the TALEs provide information on whether the actual value acquired by the TALEs corresponds to the nominal value determined by the RF characteristic.

The value of the reflected voltage at the TALEs is a gauge for the 50 Ohm termination through the body coil.

The forward voltage measured via DICO must approximately match that of the TALEs. Since the correlation between the DICO value and the TALEs voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltages measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. On the one hand, it is determined by the coil tuning, and on the other hand by the characteristics of the components that follow the RFPA in the transmit path such as the TX cable, BCCS, TALEs, BTB, etc.

RFPA (Long Time Stability)

The test is identical to the RFPA Loop test, but runs approximately 10 minutes. The test is only available on system with fMRI functionality.

RFAS

- Communication/Status
- Reference Voltage (TALES)
- Transmitter-Receiver-Switch
- BCCS Hybrid
- RFPA-TALES
- BCCS (preamp1) - LCCS Loop
- BCCS (preamp2) - LCCS Loop
- BC-TX-path (reflection)
- BC-TX-path (voltage)
- Local Coil TX Path (interactive)

RFAS Communication/Status

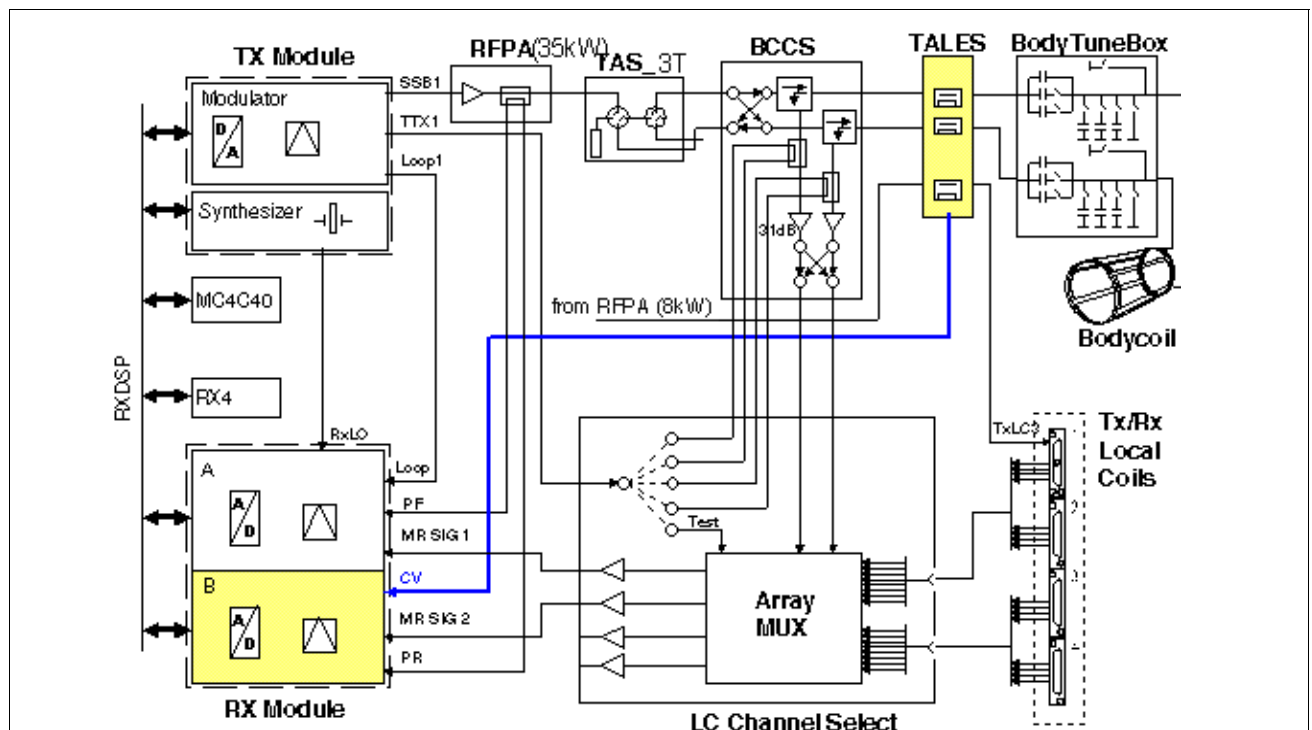
The LCCS-SLIO controls the TAS via the PIN-Booster component in order to verify communication to the LCCS as well.

The communication test consists of sending a command (status and version query) to an individual component which, if successfully completed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

Reference Voltage (TALES)

Fig. 92 Reference voltage



Signal Path

The TALES output signals are fed to the receiver 1B (CV-Input).

Evaluation

Verification that the TALES reference voltage is 500 mV.

Conclusions

The TALES continuously supplies all internal directional coupler signals triggered by the controller via fiber optic cable. In addition, the TALES supplies the signal of an integrated reference voltage source. The directional coupler signals and the reference voltage are multiplexed and fed to the receiver CV input via a coaxial cable. The reference voltage test provides information regarding correct control of the TALES, intact cabling to the receiver and signal processing during data acquisition. In case of error, the 500 mV reference voltage signal can be measured with a digital voltmeter. For further information, refer to the Troubleshooting Guide.

Transmitter-Receiver-Switch

Signal Path

Two measurements with different TX-paths will be performed using the service sequence graddelay with RF-amplitude set to zero:

- First measurement RFPA to the TAS-C/dummy load
- Second measurement RFPA to the Body Coil.

The receive path is always via the TX/RX-switch on BCCS to LCCS/SGA1 and finally to the receiver.

Evaluation

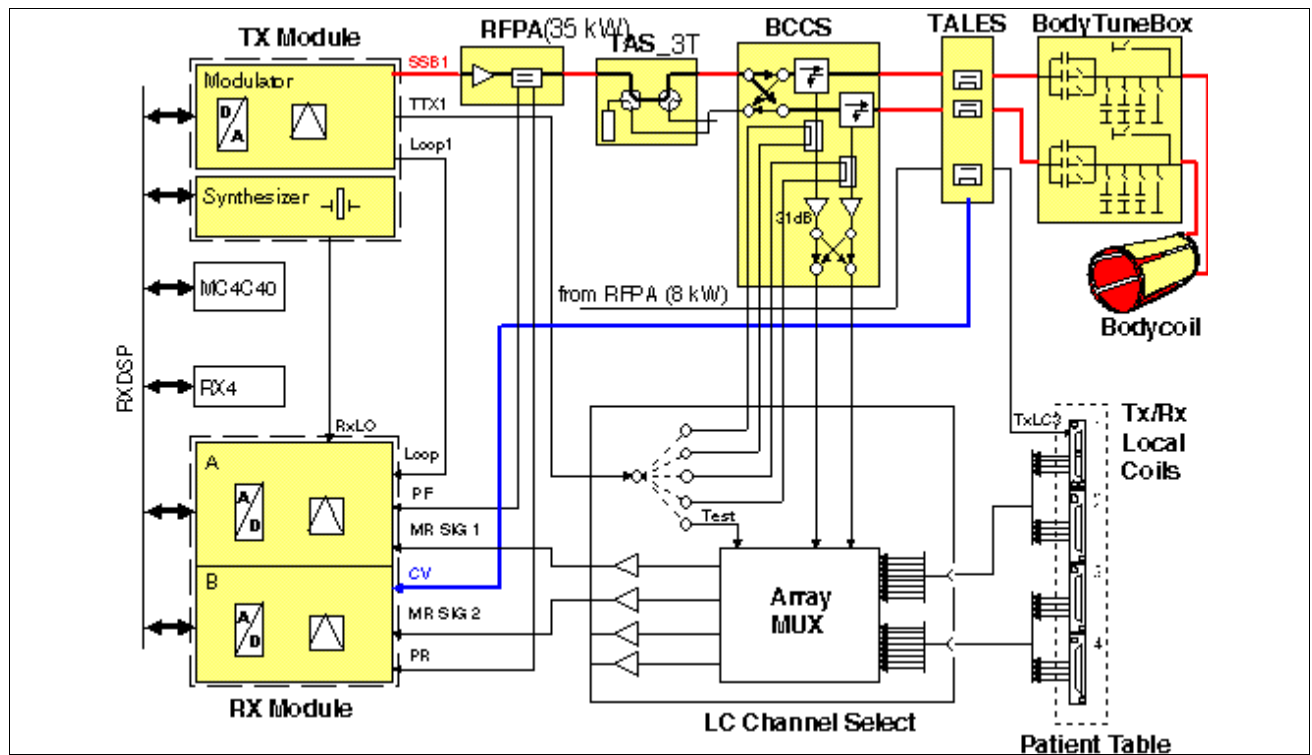
Two noise measurements will be compared. During the first measurement, the output of the RFPA is disconnected from the TX/RX-switch. During the second measurement, the output of the RFPA is connected to the TX/RX-switch and its noise (even unblank is activated) will be received if the switch does not work properly.

Conclusions

The TX/RX-switch or the switch control (from IOP U9 to BCCS U1) is probably defective if the additional noise from the RFPA during the second measurement exceeds the specification.

BCCS Hybrid

Fig. 93 BCCS Hybrid



NOTICE Ensure that the loader with the spherical phantom is positioned in the Body Coil.

Signal Path

The output signal of the modulator is fed via the RFPA-TAS_3T-BCCS-TALES-BTB to the Body Coil.

Evaluation

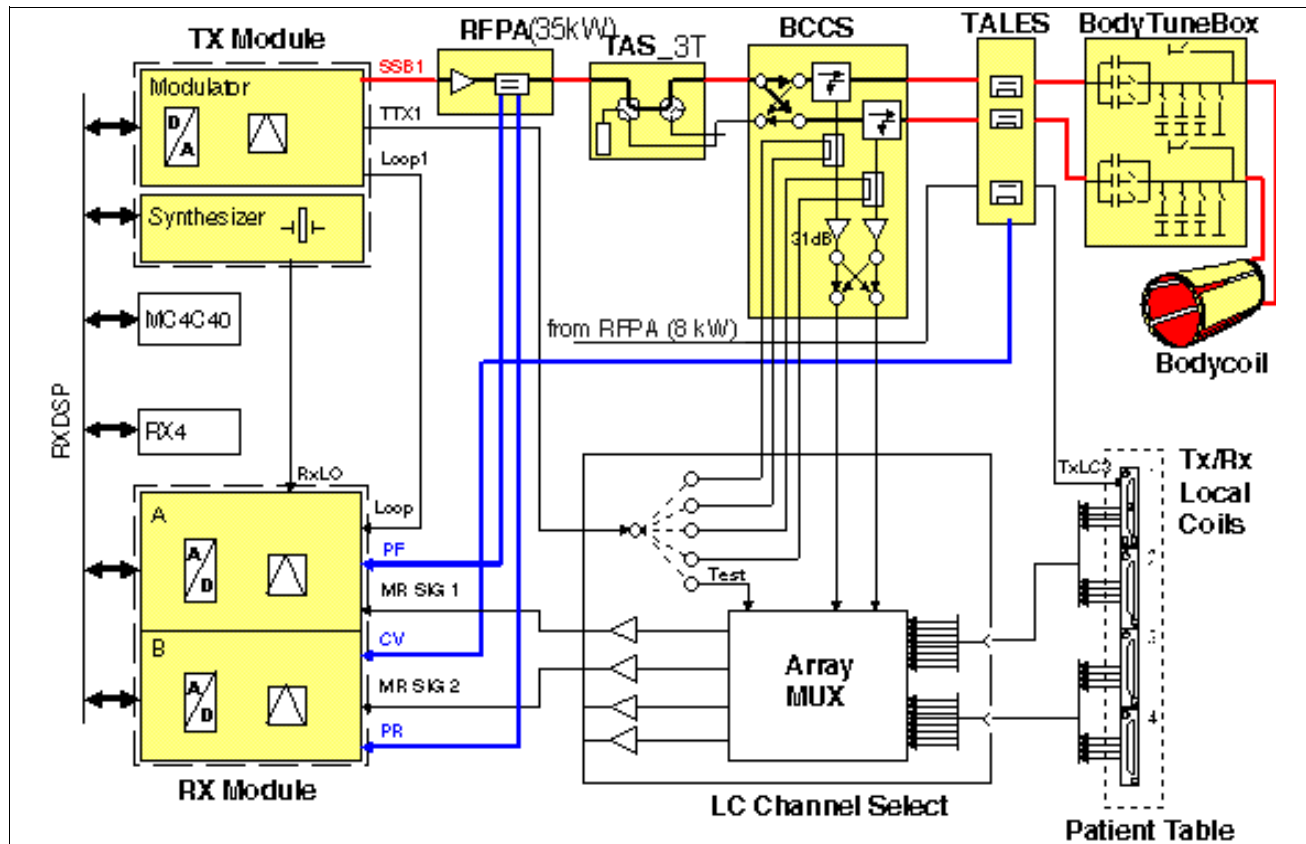
- The forward voltages of both transmitter channels (90° and 0°) measured at the TALES are compared.
- Ratio of the reflected and forward voltage from the TALES.

Conclusions

On the one hand, the deviation of the forward voltage for 90° and 0° serves as a gauge for symmetrical tuning of the 90° and 0° systems of the body coil. On the other hand, the test results should provide information about the BCCS functionality (Hybrid). If the results are not within specifications, first check the tuning of the body coil on both systems (tuning ratio of both systems).

RFPA-TALES (functional test with RFPA 1 - 35 kW Astex; interactive with RFPA 2 - 8 kW Dressler)

Fig. 94 RFPA-TALES



NOTICE Ensure that the loader with spherical phantom is positioned in the body coil.

Signal Path

1. The output signal of the modulator is fed via the RFPA 1 (35 kW)-TAS_3T-BCCS-TALES-BTB to the body coil.
2. The output signal SSB2 of the modulator is fed via the RFPA 2 (8 kW) and SAMI X1/X3 to TALEs X5. The cable to BCCS/X7 must be connected to TALEs/X6 and the service plug or the CP head coil must be connected to coil plug 1. Follow the instructions of the confirmation popup windows.

The test consists of two measurements. One RF pulse is transmitted for each measurement. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALEs.

The body coil must be tuned to evaluate the signals. For this reason, coil tuning is automatically initiated at the beginning of the test.

Evaluations

- Forward Voltage from the TALES within tolerance
- Reflected Voltage from the TALES below limit
- Deviation of the forward voltage from the TALES and the DICO
- Ratio of the reflected and forward voltage from the DICO

Conclusions

This test checks practically the entire transmit path of the system.

The results of the forward voltage at the TALES provide information on whether the actual value acquired by the TALES corresponds to the nominal value determined by the RF characteristic.

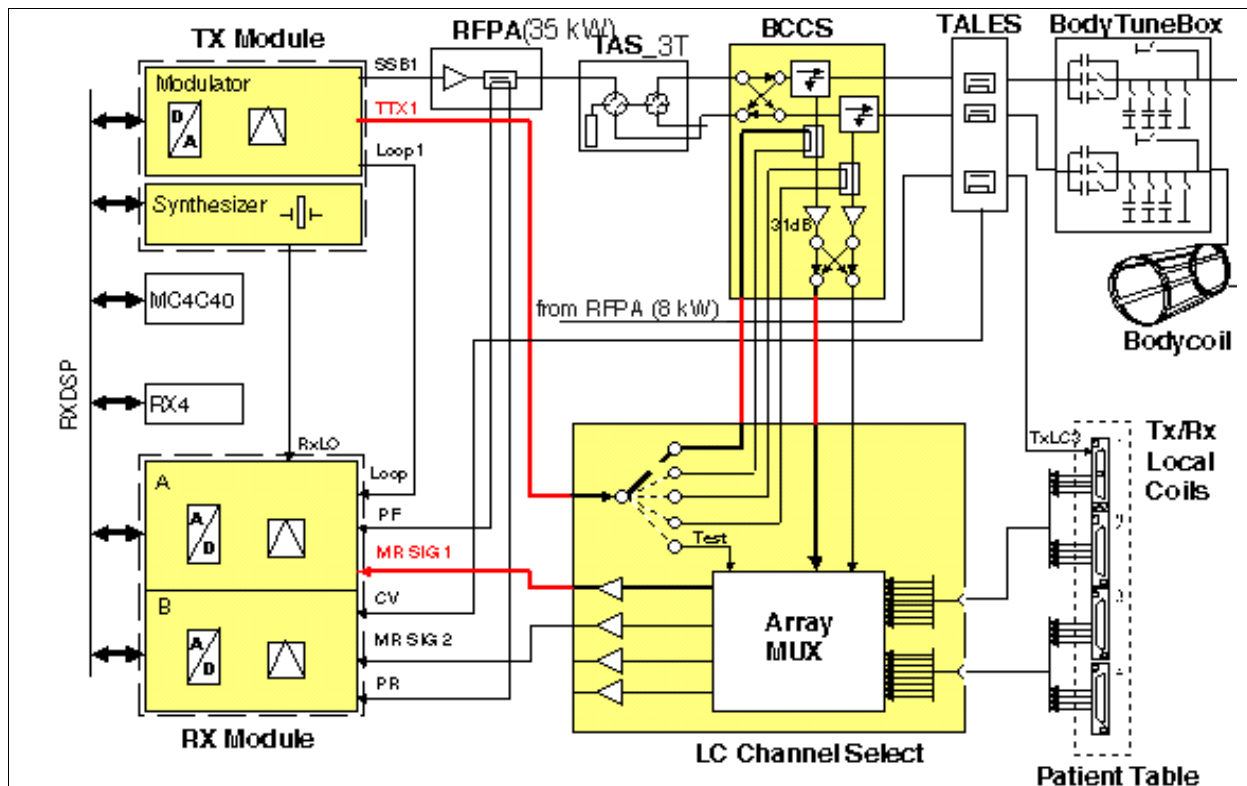
The value of the reflected voltage at the TALES is a gauge for the 50 Ohm adjustment through the body coil.

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for conversion from receiver ADC values into voltage values), a valid characteristic curve must be present.

The ratio of the reflected and forward voltage measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm adjustment of the RFPA. On the one hand, it is determined by the coil tuning, and on the other hand by the characteristics of the components that follow the RFPA in the transmit path such as the TX cable, BCCS, TALES, BTB, etc.

BCCS (preamp1) - LCCS Loop

Fig. 95 BCCS (preamp1) - LCCS

**Signal Path**

The TTX output signal of the modulator is fed via the LCCS forward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver 1A/2A via the LCCS. The tests are performed with high gain and low gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

In a second measurement the TTX output signal of the modulator is fed via the LCCS backward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver 1A/2A via the LCCS. The test is performed with high gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

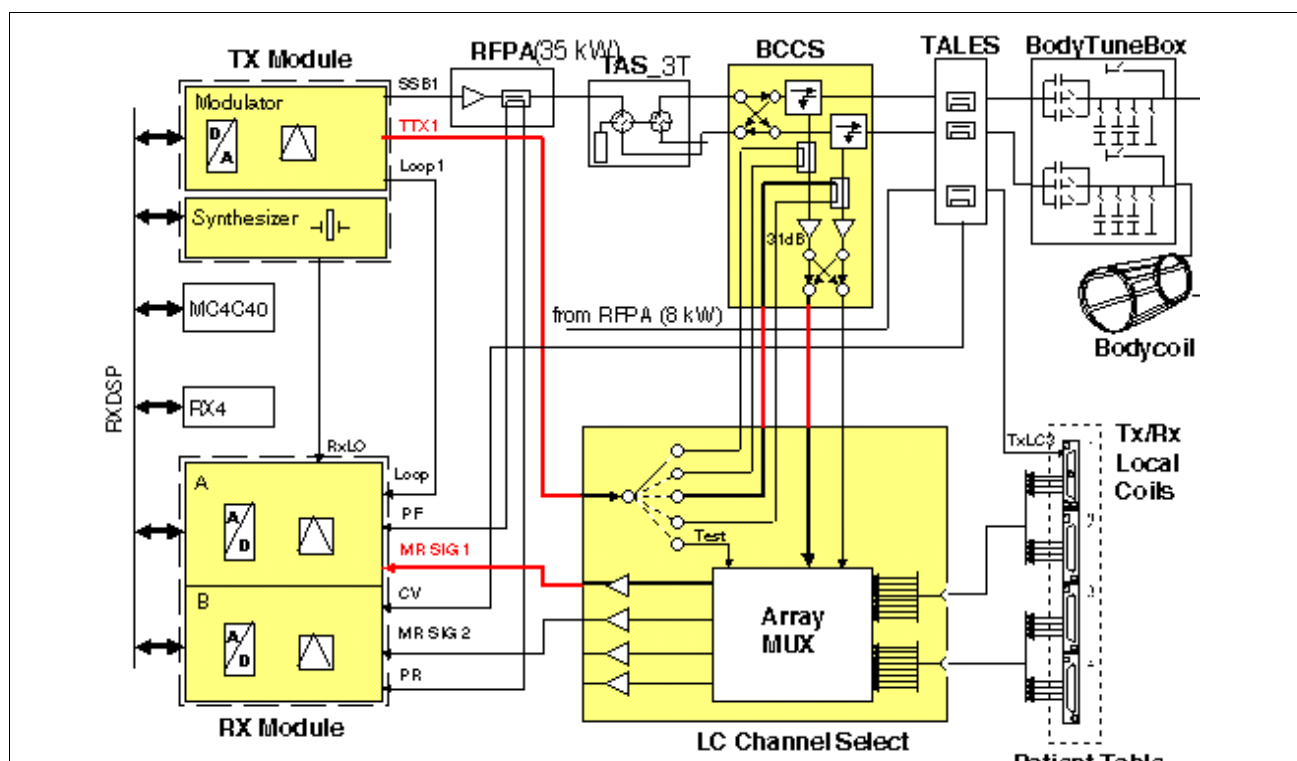
Conclusions

This test is used primarily to check the stability and linearity of the preamplifier in the BCCS and the LCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

Additional the cables and connections between BCCS and LCCS (<W531>, <W532>; <W529>) are checked.

Fig. 96 BCCS (preampl 2) - LCCS

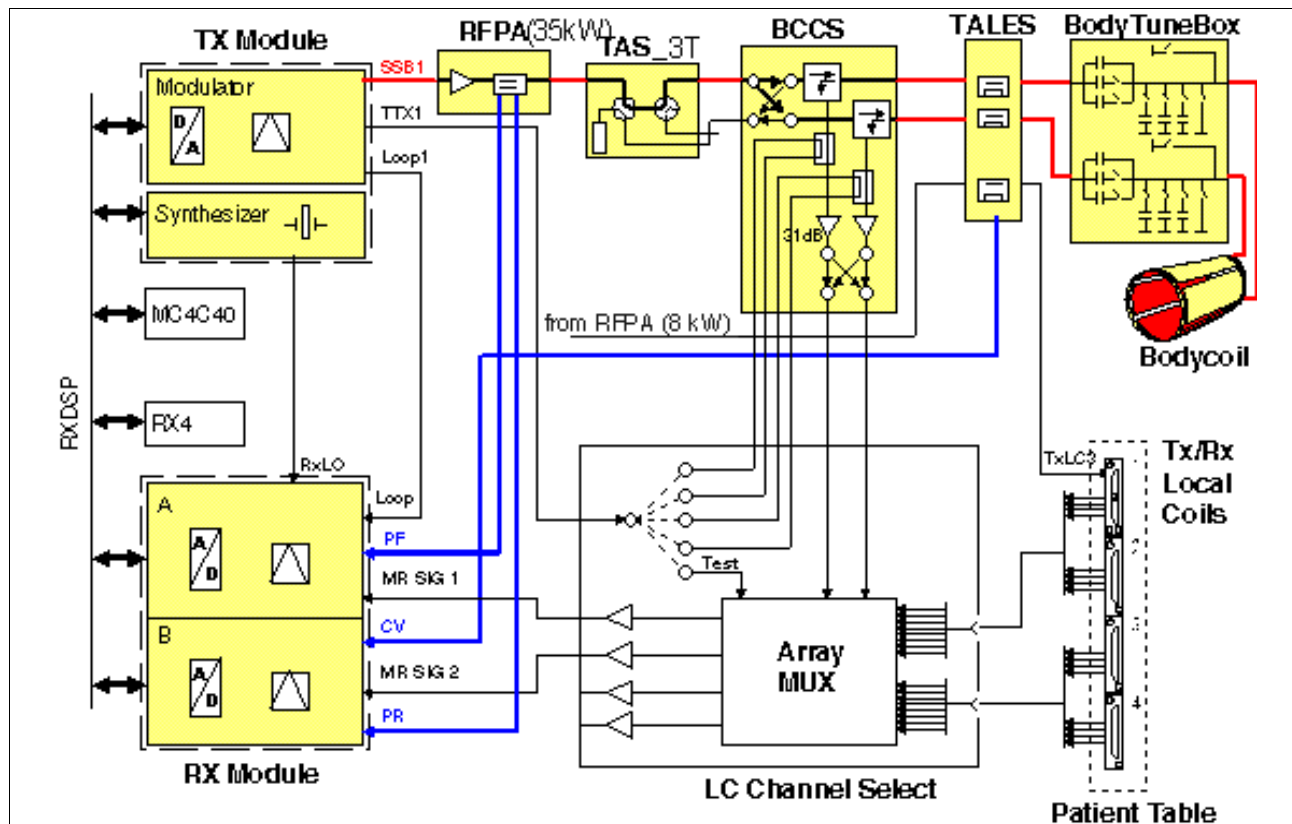
Fig. 96 BCCS (preampl 2) - LCCS



Additional the cables and connections between BCCS and LCCS (<W533>, <W534> and <W528>) are checked.

BC-TX-path (voltage)

Fig. 98 BC-TX-path (voltage)



NOTICE Ensure that the loader with spherical phantom is positioned in the Body Coil.

Signal path

The output signal of the 35 kW RFPA is fed via the BTB to the TX coil. The forward and reflected power is measured via the directional coupler in the RFPA. The signal amplitude used for this measurement is the maximum amplitude read from the RF-characteristic data, varying from the minimum to the maximum.

Evaluation

- The actual forward and reflected voltage (measured via DICO and TALES) at maximum deviation is displayed.
- The maximum deviation of the forward voltage (measured via DICO and TALES) is calculated.
- The first diagram shows the measured power forward (voltage, measured via DICO and TALES) over the transmitter voltage.
- The second diagram shows the deviation (%) of DICO and TALES power forward over the transmitter voltage.

Conclusions

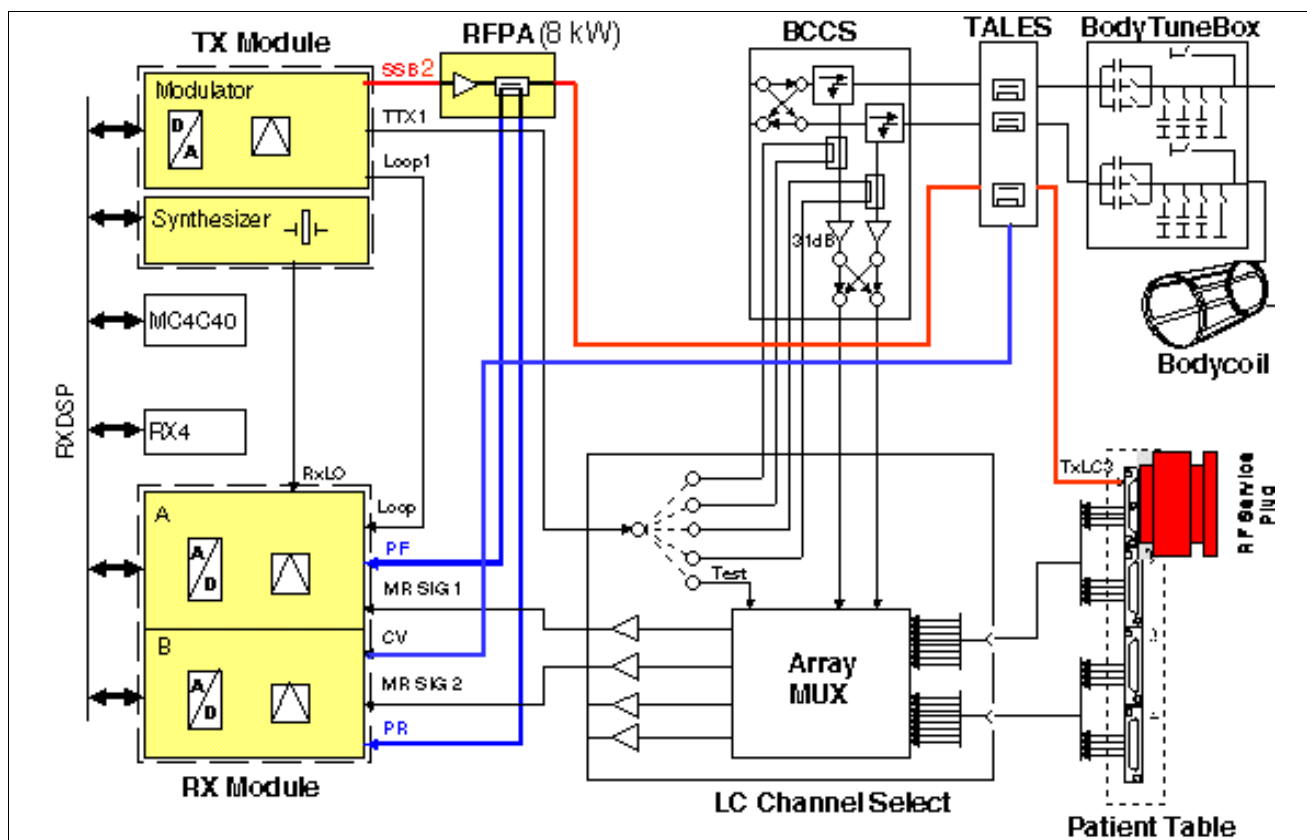
If the deviation of DICO and TALES voltages is out of tolerance repeat first the RF characteristic measurement.

Significant higher voltage values measured by TALES could be caused by an RFPA oscillations on other than the MR-frequency: the DICO measurement is selective at the MR frequency, but the TALES is sensitive for a wider frequency band.

Perform the interactive **Cable** test BC-TX-path (voltage, for experts only).

Local Coil TX Path (interactive)

Fig. 99 Local Coil TX Path



Preparation

The RF Service Plug must be connected to coil connector #1 at the patient table (connector for the extremity coil). Follow the instructions provided during the test run.

Signal Path

The output signal of the modulator is fed to the (8 kW) RFPA–TAS_3T–TALES-Extremity Coil (local coil connector #1).

This test requires that the transmission cable to the extremity coil be terminated at 50 Ohm. The connection must be terminated using the 50 Ohm resistor in the RF Service Plug.

The test consists of two measurements. One RF pulse is transmitted for each measurement. In the first measurement, the signals are decoupled by the directional coupler (DICO) in the RFPA and converted to voltages using a specific factor. The second measurement acquires the signals from the TALES.

Evaluation

- Deviation of the forward voltage from the DICO and the TALES
- Ratio of the reflected and forward voltage from the TALES

Conclusions

The forward voltage measured via DICO must approximately match that of the TALES. Since the correlation between the DICO value and the TALES voltage only occurs when the characteristic curve is generated (determination of the factor for converting the ADC values from the receiver into voltage values), a valid characteristic curve must be present.

With regard to the linear RF-characteristic, i.e. after the software has been installed and before the tune-up has been performed, the deviation may be significantly above the specification.

The ratio of the reflected voltage and the forward voltage measured via DICO in the RFPA is generally used as a gauge for the 50 Ohm termination of the RFPA. In this test, it is determined by the 50 Ohm resistor in the RF Service Plug. If the ratio lies outside of the specifications, the cause can probably be found in the cable connection from the TAS via Tales to coil connector #1 (movable cable in the table) or in the TAS itself (for this test, the TAS is switched to extremity coil path).

If the forward voltage does not exceed a certain level (approx. 100 V), the RF safety watchdog evaluates the measured values within the most sensitive range of the TALES. In this range, the reflected voltage of the TALES can no longer be produced and the value 0 is displayed. In this case, the evaluation of return / forward voltage is no longer valid.

BTB

- BTB Communication/Status
- BTB Test (functional + interactive)

BTB Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

BTB Tests (functional)

NOTICE	Ensure that the loader with the large spherical phantom is precisely positioned in the Body Coil.
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The BTB tests are based on the analysis of reflective voltage factors for the different switching modes of the short circuit switch, the PIN diodes and the tuning capacitors in the BTB.

Prerequisite for this test is the successful tuning calibration of the TALES (tuning calibration is performed during Tune-up, the successful BTB calibration is not necessary for this test). Otherwise, the results of this test are invalid.

The test consists of several individual tests each including a reference measurement and the actual level measurements followed by an evaluation: Static Detune, Dynamic Detune, test of Serial Capacitors and Relays and Parallel Capacitors and Relays.

The first measurement is performed as a "reference" with an abbreviated path, i.e. without using the BTB and the BC. The other measurements are done with a specified setting of the switch in the BTB.

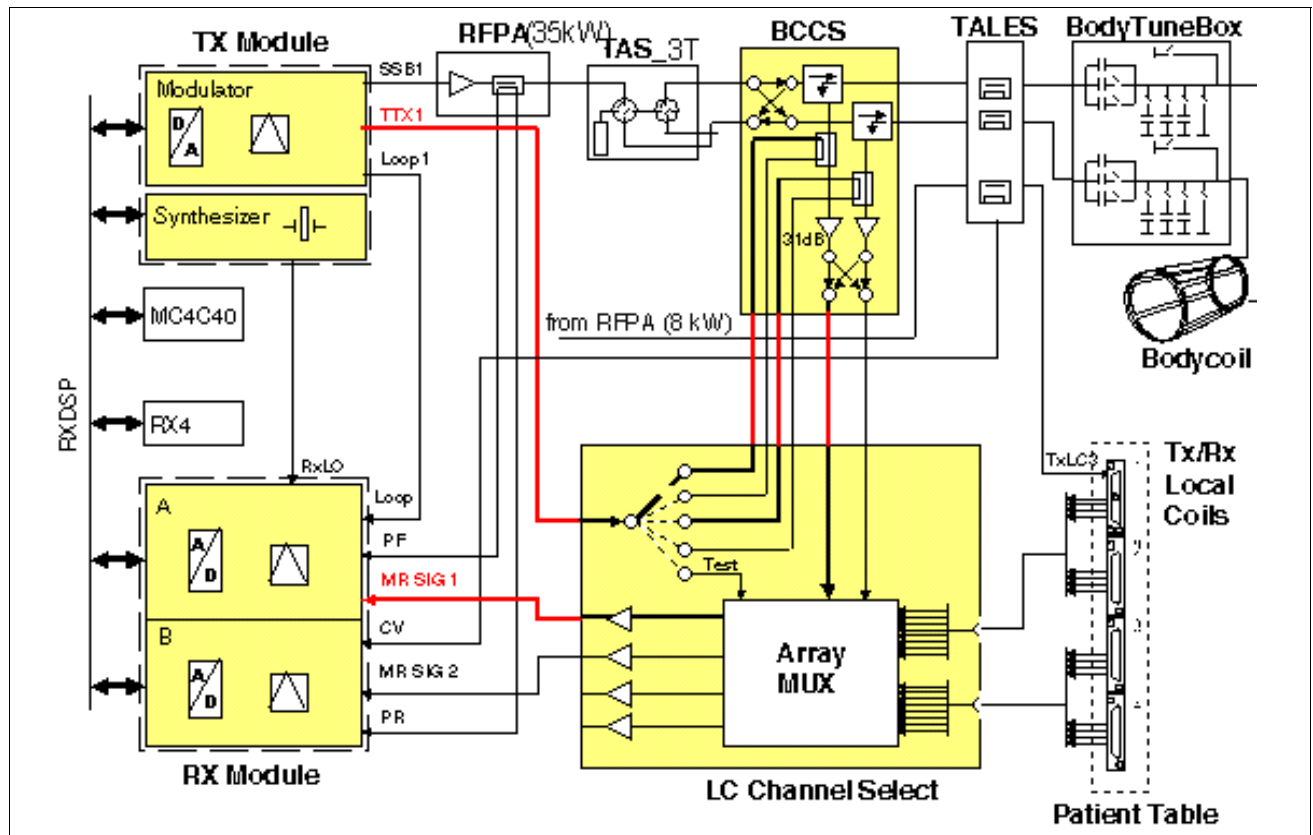
After these tests the measurements are repeated for the BTB 2.

Signal Path for the reference measurement

The TTX output signal of the modulator is fed to the LCCS-BCCS (preamplifier)-LCCS and to the receiver.

The level measured is used as the reference.

Fig. 100 BTB Test, Measurement for Reference



Signal Path for the measurement

The TTX output signal of the modulator is fed to the LCCS-BCCS-TALES-BTB-BC and reflected back to the BTB-TALES-BCCS-LCCS and receiver. The short circuit switch, the PIN diodes and the capacitors in the BTB are switched in sequence.

Test 1: Static detuning active, the serial capacitors are switched in sequence.

Test 2: Dynamic detuning active, the serial capacitors are switched in sequence.

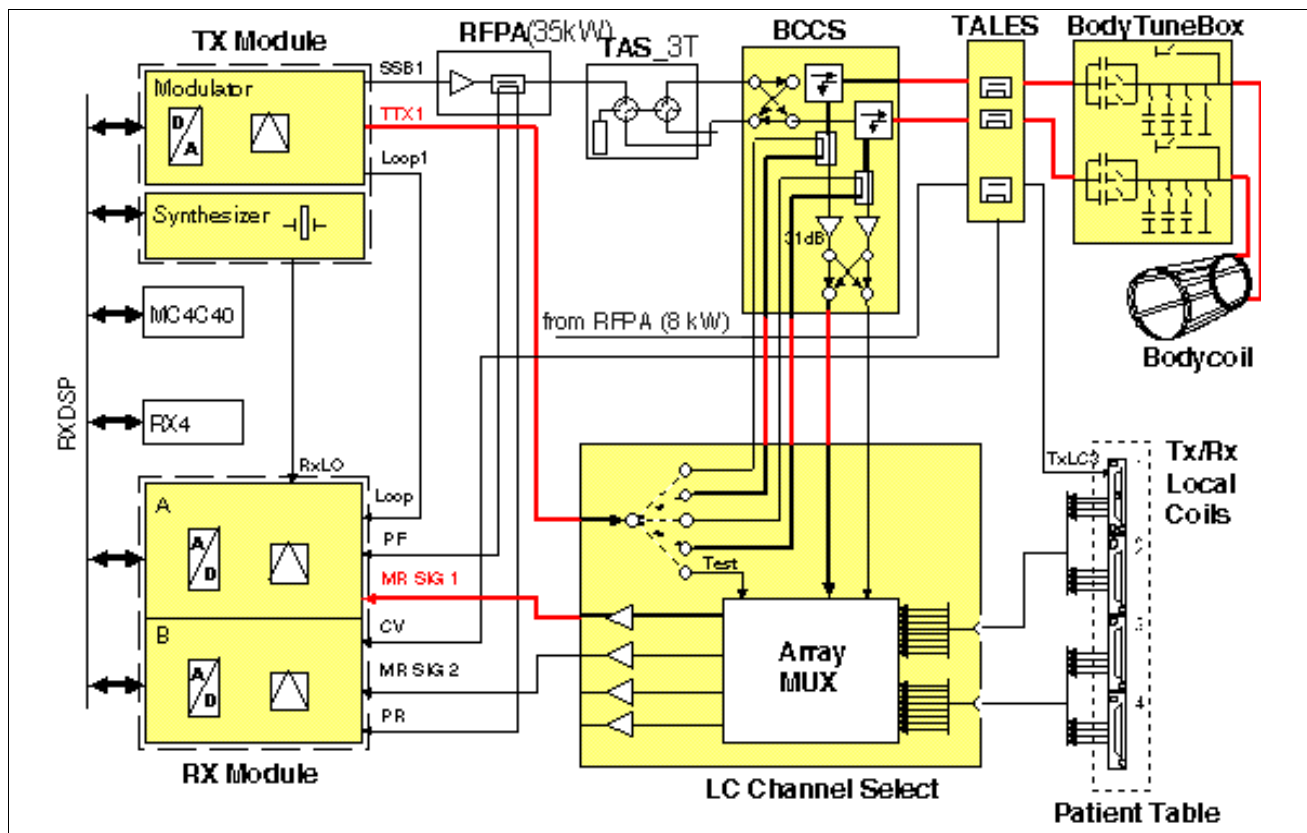
Test 3: The serial capacitors are switched in sequence.

Test 4: The parallel capacitors are switched in sequence.

The tests are repeated for the BTB2.

NOTE Some tests are sensitive to a malfunction of the Body Coil as well.

Fig. 101 BTB Test, Measurement



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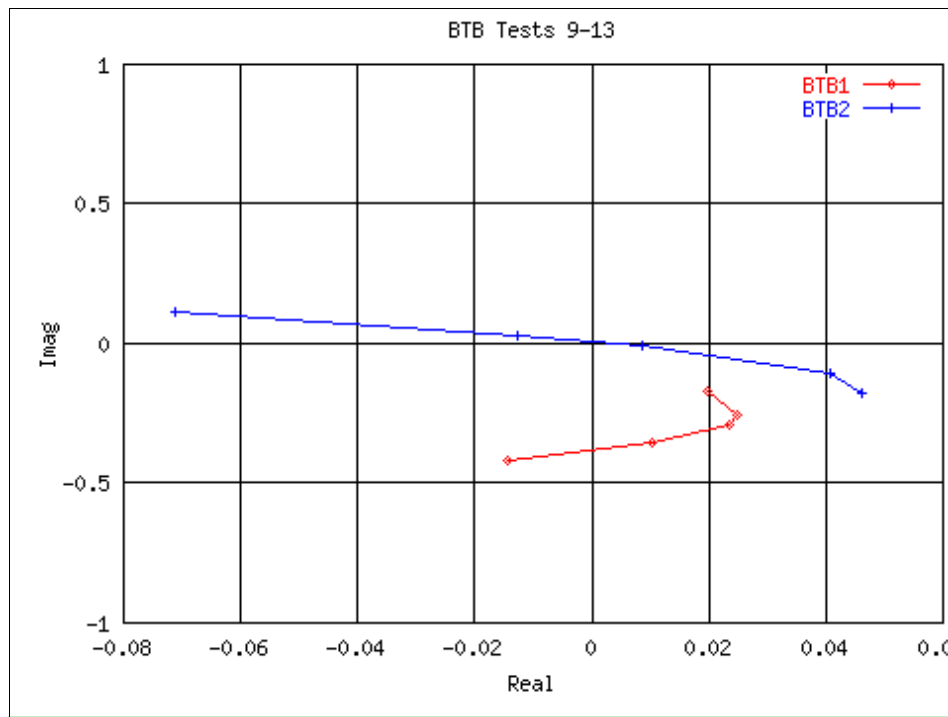
Evaluation

The reflection factors (real and imaginary) are generated for the individual switch settings and checked against the specifications.

Afterwards, several relative values from the individual results are determined. They allow a more detailed evaluation of the BTB functionality.

The results of some measurements are displayed graphically in a x-y diagram (real part on x-axis, imaginary part on y-axis). From deviations of the measurement compared with the normal situation shown in [Fig. 102](#) the error can be analyzed. For further information refer to the Troubleshooting Guide.

Fig. 102 Example of measurement result



Conclusions

In the HTML output, the BTB Diagnosis, possible errors of the BTB/BC are displayed.

In case of doubt whether to replace either the Body Coil or the body tune box (whichever is defective), please contact the MR Hotline (HSC) for further instructions.

In addition, perform the RFCI tests for checking the PIN diodes of the BTB.

BTB Tests (interactive)

For better error localization (TALES/BTB) two interactive tests are used. Pop-up windows inform you about to establish the correct measurement setup (no transmit signal to the body coil).

- Check of Tuning Calibration (TALES outputs X3, (X4) open)
- BTB Reflection factors (BTB outputs X1/1, (X1/2) shorted)

The Tune-up procedure "Tuning calibration" will be checked by measurement of a reflection factor and comparison with a reference value.

Measurement of reflection factors of the BTB systems checks the parallel capacitors C_s and relays of the BTB.

BC

- Body Coil (functional)
- Body Coil (interactive)

Body Coil (functional)

NOTICE Ensure that the loader with the large spherical phantom is **precisely** positioned in the Body Coil.

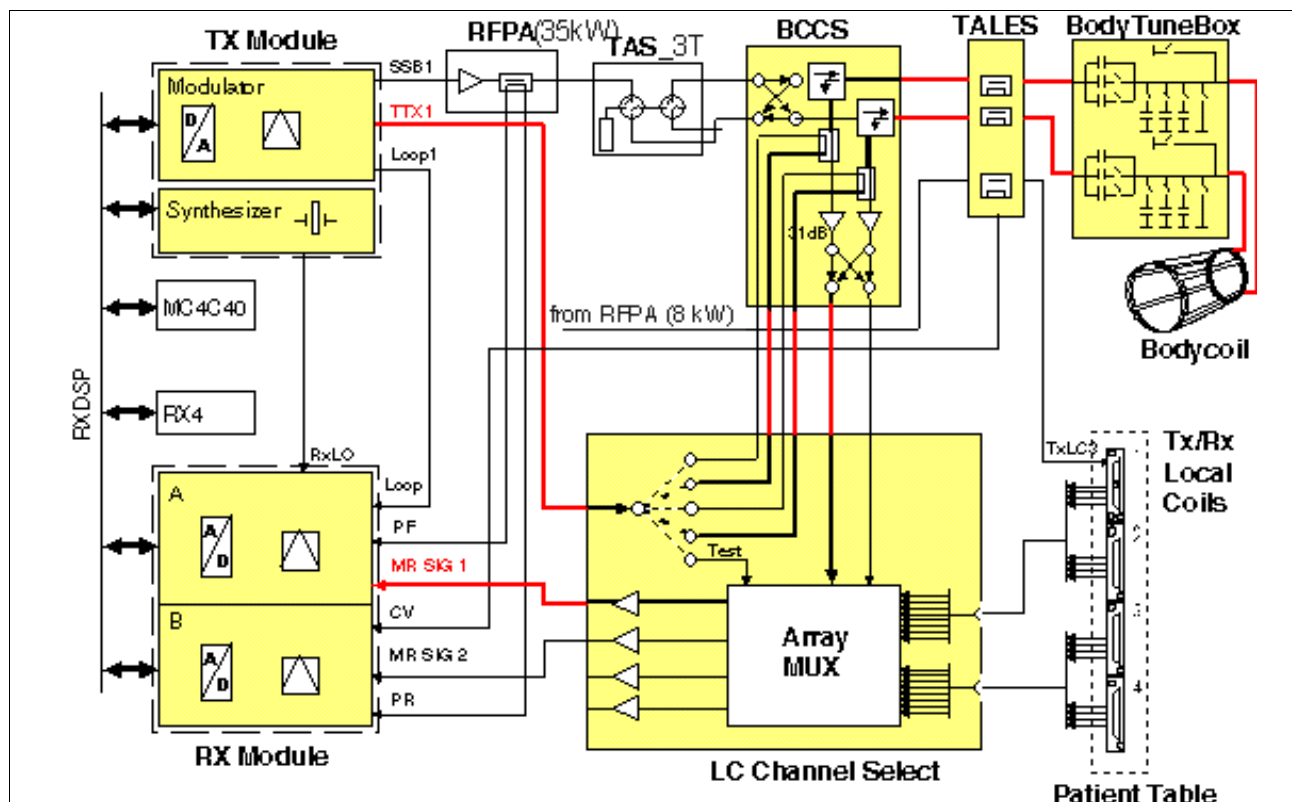
The body coil test is based on the analysis of reflective voltage factors that depend on the frequency. The test simulates a network analyzer that provides the frequency for the smallest reflection factor and the resonance curve of the body coil as results. The available frequency range of approximately 1MHz is stepped through in 10kHz steps. The respective reflection factors are produced and entered in a frequency reflection factor diagram. The frequency with the smallest reflection factor is checked against the specifications. The Q-factor of the coil will be evaluated and checked against specification.

Additionally a power test with increasing transmitter amplitude is performed. The TALES reflection values over the transmitter amplitudes are graphically displayed.

The test is available in two variations:

- As an automatic test with the loader and phantom positioned (functional)
- Load-free, i.e. the table with loader and phantom must be outside the magnet (interactive test)

Fig. 103 Body Coil



Body Coil (interactive/without load)

The first part of the measurement is identically to the functional measurement. Additionally the decoupling of the body coil elements is evaluated.

Fig. 104 Examples: BC - Reflection Evaluation - Left: reflection with load, Right: reflection without load

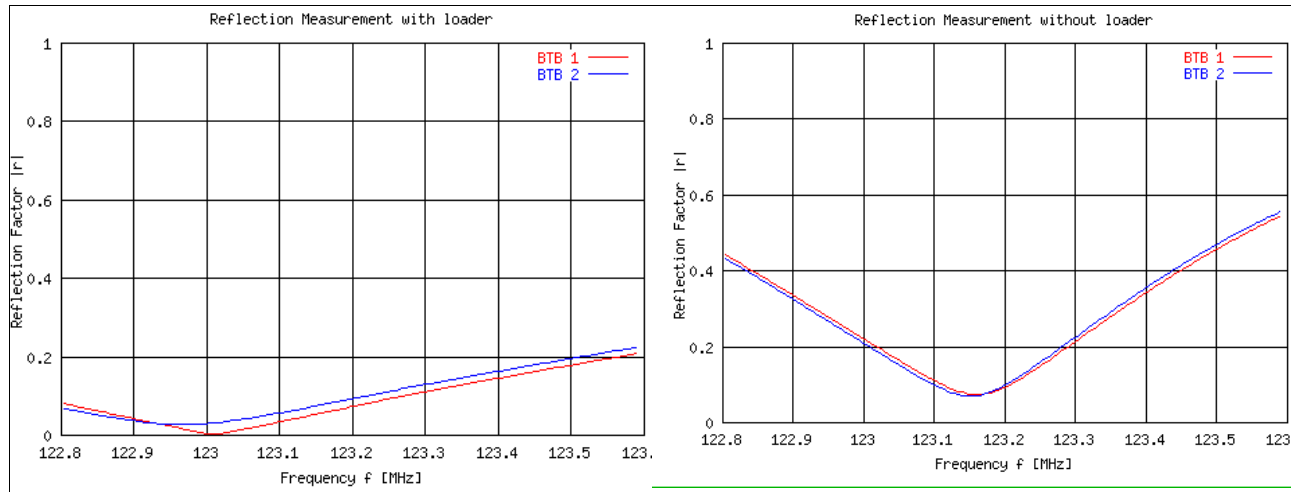
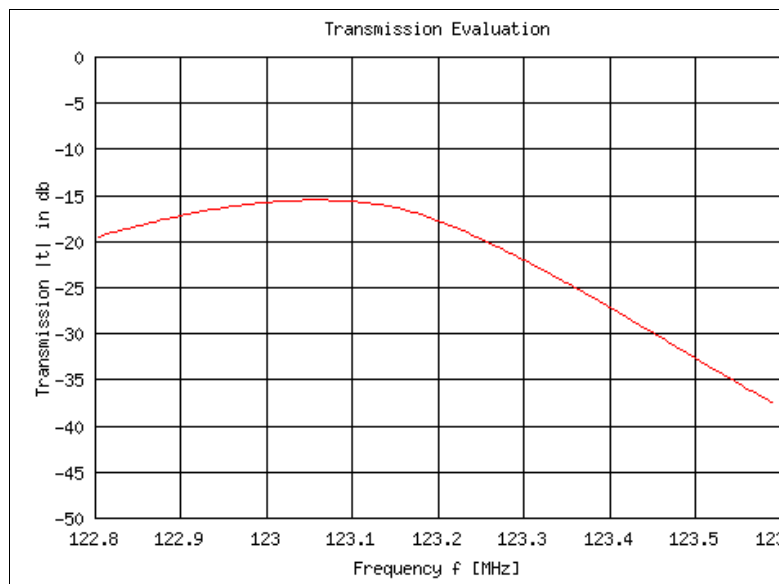


Fig. 105 Example: BC - Transmission Evaluation



Also a power test with increasing transmitter amplitude is performed. The TALES reflection values over the transmitter amplitudes are graphically displayed.

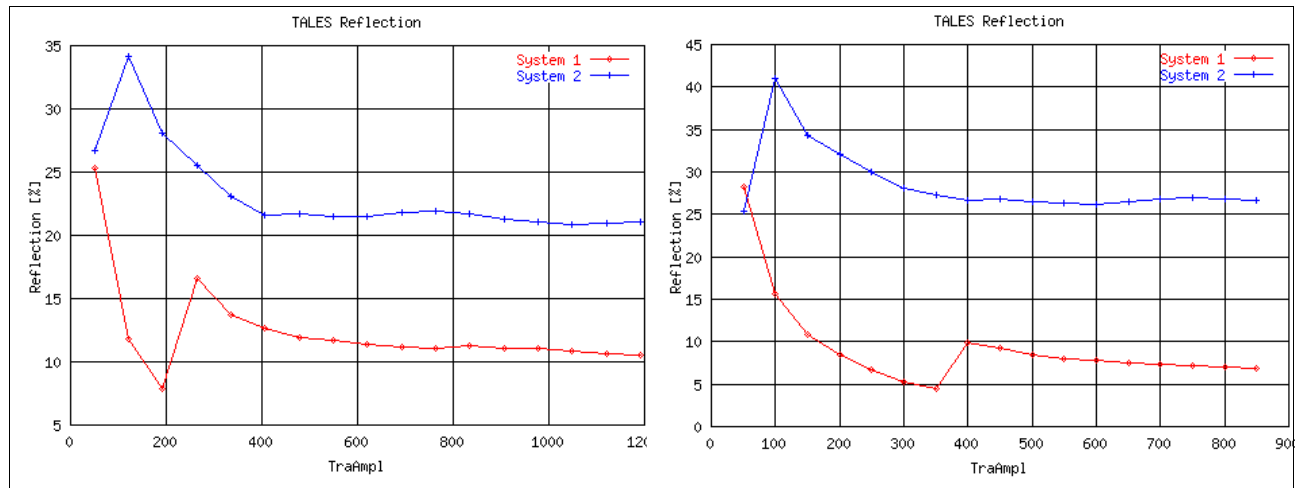
Conclusions

If the tolerances of the reflection measurement evaluation are exceeded and the BTB tests are in specification, the problem is probably located in the BC.

The power test checks the disruptive strength of the BC. The test should run without shut down to the maximum voltage. The reflection of TALEs should be approximately constant at voltages higher than 200 V. Refer to the figure below.

The power test without load is the worst case, i.e. the reflected voltages are significantly higher.

Fig. 106 Examples: BC power test - Left: TALEs reflection with load, Right: TALEs reflection without load



If the results exceed the specification, in case of doubts, please contact the MR Hotline (HSC) for further instructions.

Cable

- BC-TX-path (refl.)
- BC-TX-path (volt.)
- LC-TX-path

NOTE **These tests are for experts only!**
Additionally special equipment is necessary for the tests.

These advanced tests enable you to check the complete TX-path to BC or LC and to make a systematical troubleshooting in the TX-path by connecting a power 50 Ohm terminator at various locations in the TX-path and/or configuring of particular loops.

The test should be used by experienced service engineers only. Special tools (power dummies, RF-cables, connectors, adapters and attenuators) are needed.

Popup windows inform you about the actual measurement configuration.

Safety

WARNING **High RF voltage**

Contact with RF conductors during transmitting of the RFPA will result in death or serious physical injury.

Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input cable at X1 and the unblank line U1 from the RFPA.

Trio with Multi-Nucleus option

Contrary to the MAGNETOM Symphony with Multi-Nucleus option (spectroscopy, X-nucleus) with the MAGNETOM Trio the nucleus selection in the TestTools-platform is not enough when testing the Local Coil Transmit Path for the ^1H -frequency.

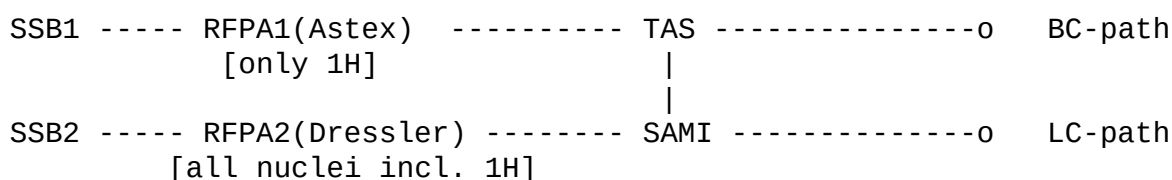
The reason is that the used RFPA's of the Trio do not have completely different frequency ranges: both RFPA's support the ^1H -frequency. The Dressler-RFPA supports all possible frequencies, including ^1H , the Astex-RFPA only ^1H .

With the Trio also the plugged in coil is relevant:

- With a Multi-Nucleus coil capable also of the ^1H -frequency (e.g. Service Plug) always the Astex RFPA is used (RFPA1, RFPA 35kW).
- With a mono-resonant coil capable only of ^1H (e.g. TxRx Headcoil) always the Dressler RFPA is used (RFPA2).

For all other frequencies always RFPA2 and the Service Plug or Multi-Nucleus Coil is used.

A simplified scheme can explain this a little bit:



This knowledge is important to use always the correct RFPA when testing the different TX-paths with ^1H , especially the path RFPA1-TAS-SAMI-LC-path.

The behaviour and the scheme is basically the same like with the MAGNETOM Allegra. It is just that with the Allegra the order of the RFPA's was defined in the opposite way: RFPA1 is RFPA2 and vice versa). Also the RFPA's of the MAGNETOM Symphony use the Trio order.

NOTICE The tests require a valid RF characteristic or the linear RF characteristic.
Ensure that the spherical phantom is positioned in the TX-coil.

NOTE When using the linear characteristic it might be possible that the used transmitter amplitude exceeds the allowed range resulting in an abort of the measurement. This is normal and not a problem of the system.

NOTE For the first measurement of each test also the reflection factors are evaluated and analyzed. If the reflection exceeds 90% the test is aborted with the error message Error: DICO reflection out of spec for first measurement.
In that case check the cabling and the Service Plug/50 Ohm power dummy (if used).

BC-TX-path (refl.) (interactive)

NOTICE When transmitting to the Body Coil, ensure that the loader with spherical phantom is positioned in the Body Coil.
The test requires a valid RF characteristic or the linear RF characteristic.

The objective of the test is to check the components of the TX-path to the body coil with maximum transmitter amplitude for disruptive strength step by step. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (read from the RF characteristic data) are performed.

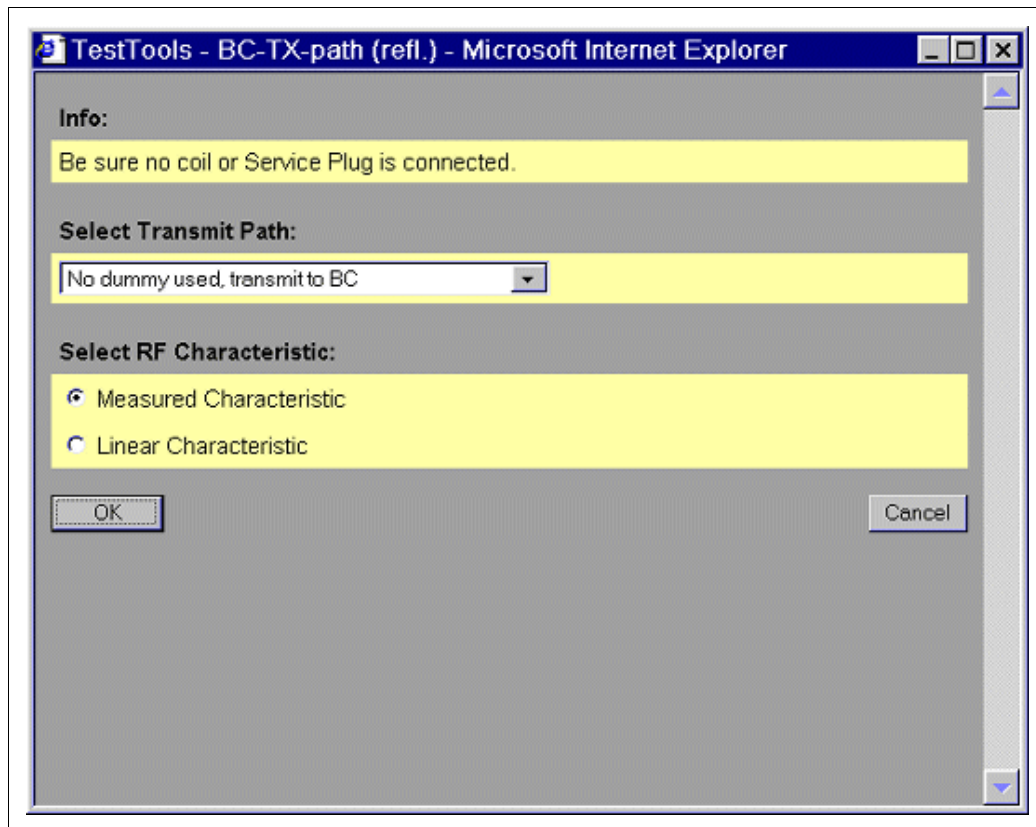
Signal path

The normal TX-path to the body coil is used.

The TX-path can be opened at several locations and terminated by one or two 50 Ohm dummies. In the appearing popup window you can select the desired TX-path (refer to [Fig. 107](#) and [Fig. 108](#)).

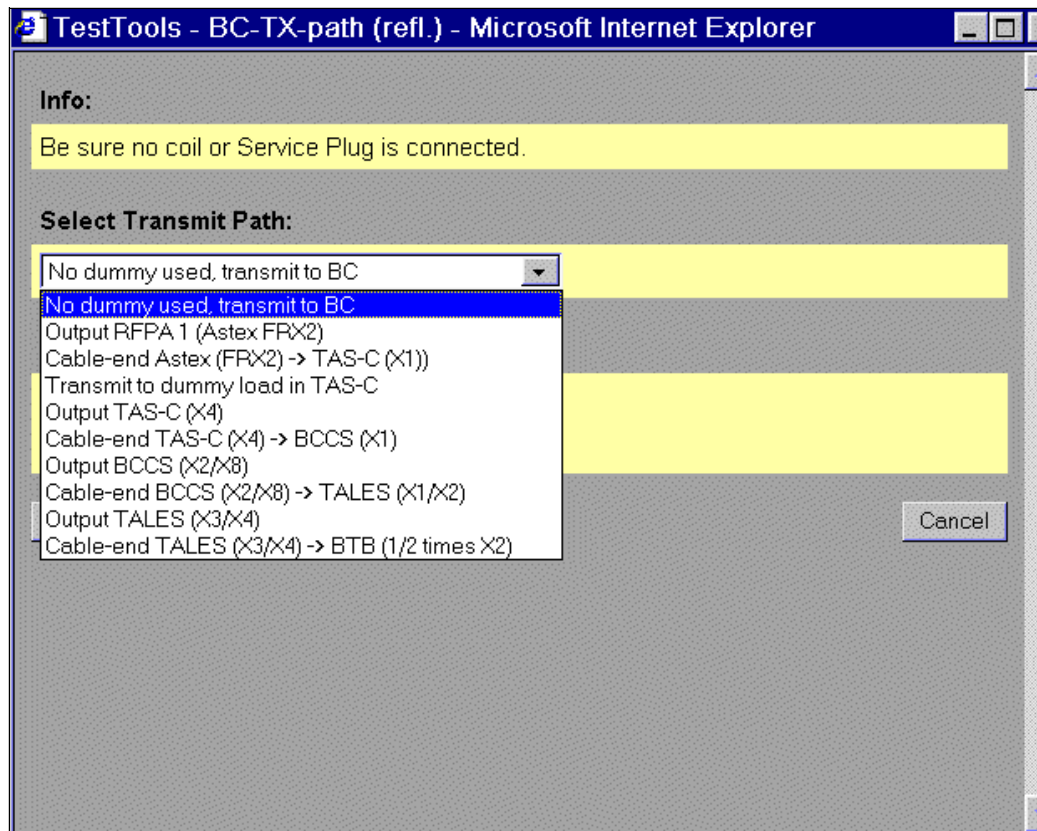
The selection of the location of the termination of the TX path with a dummy load makes no influence on the measurement but selects the heading of the HTML output table. This is e.g. helpful when sending the results to the USC/HSC.

Fig. 107 Pop up window BC-TX-path (refl.) - Selection of used RF Characteristic



The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the popup window.

Fig. 108 Pop up window BC-TX-path (refl.)
Selection of TX-path termination location by 50 Ohm dummies or BC



Evaluation

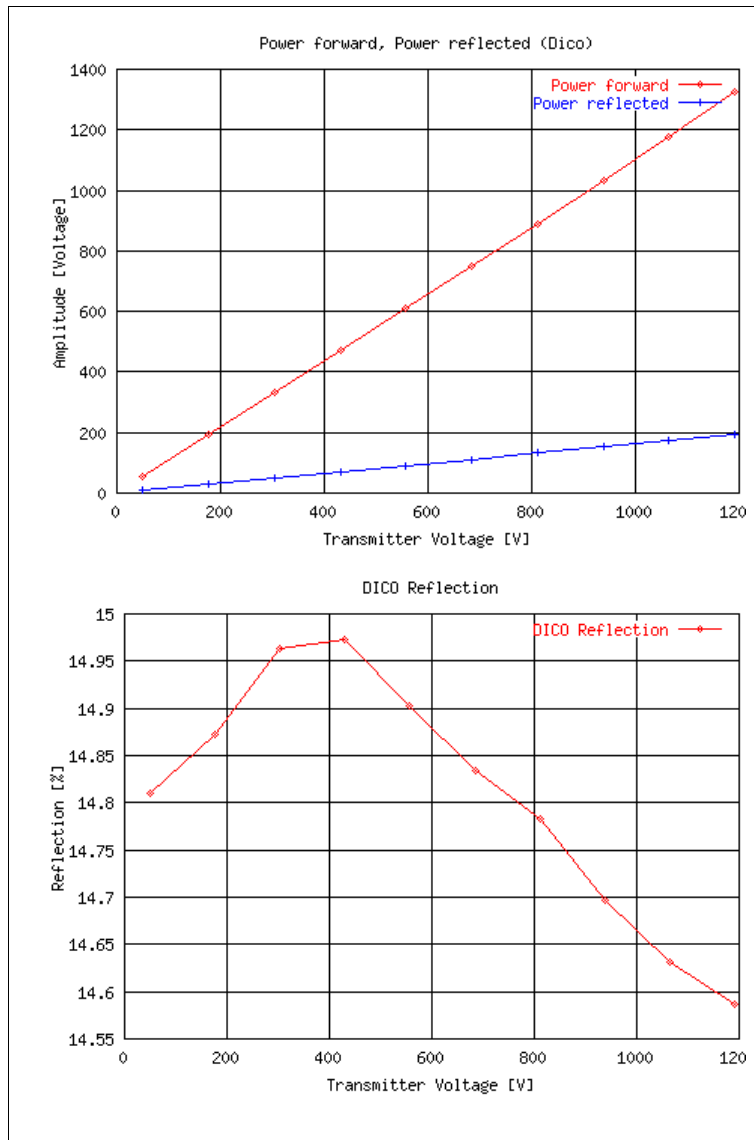
HTML output table:

- PF (DICO) at maximum reflection (Volt)
- PR (DICO) at maximum reflection (Volt)
- PR/PF (DICO) Reflection at maximum ratio (%)

Graphical output:

- PF, PR (DICO/Volt) over TRA voltage
- PR/PF DICO Reflection (%) over TRA voltage

Fig. 109 Example of graphical output



Conclusions

If there are too high reflections in the BC-TX-path you can locate the defective component in the TX-path by repeating the test after connecting the dummy load to a component before it's current position. In particular check for loose connectors.

Examples for errors are low power of RFPA, arcing of capacitors in the BTB or BC, defective cables and/or connectors.

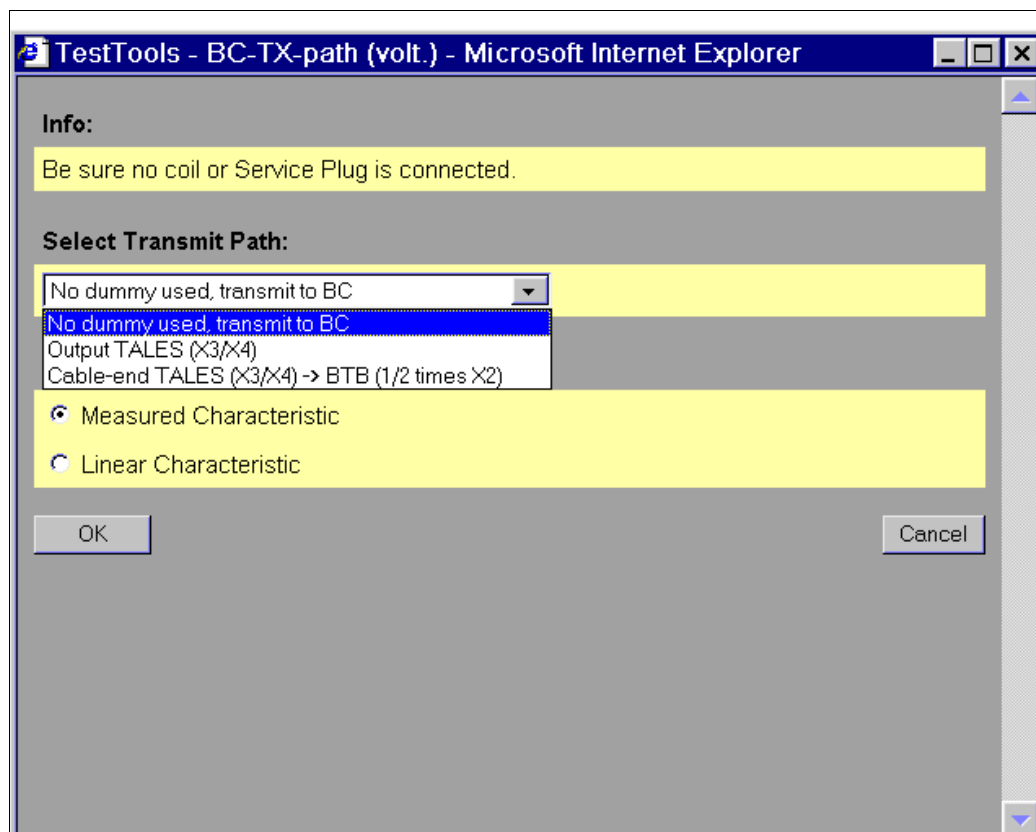
The DICO-Reflection should be lower than 50%. Sporadic reflections up to 70% may be caused by arcing in the BTB (switching of capacitors), reproducible high reflections may be caused by arcing in the Body Coil.

BC-TX-path (volt.) (interactive)

NOTICE When transmitting to the Body Coil, ensure that the loader with spherical phantom is positioned in the Body Coil.
The test requires a valid RF characteristic or the linear RF characteristic.

The objective of the test is to detect oscillations of the RFPA. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (read from the RF characteristic data) are performed.

Fig. 110 Pop up window BC-TX-path (volt.)
Selection of TX-path termination location by a 50 Ohm dummy

**Signal path**

The normal TX-path to the Body Coil is used.

The TX-path can be opened at locations between TALES and BTB and terminated by (one or two) dummy loads. In the appearing popup window you can select the desired TX-path (refer to Fig. 110). The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the popup window.

The selection of the location of the interruption in the TX-path makes no influence of the measurement but selects the heading of the output table. If the interruption is between TALES and RFPA, the TALES values can not be measured and displayed.

Evaluation

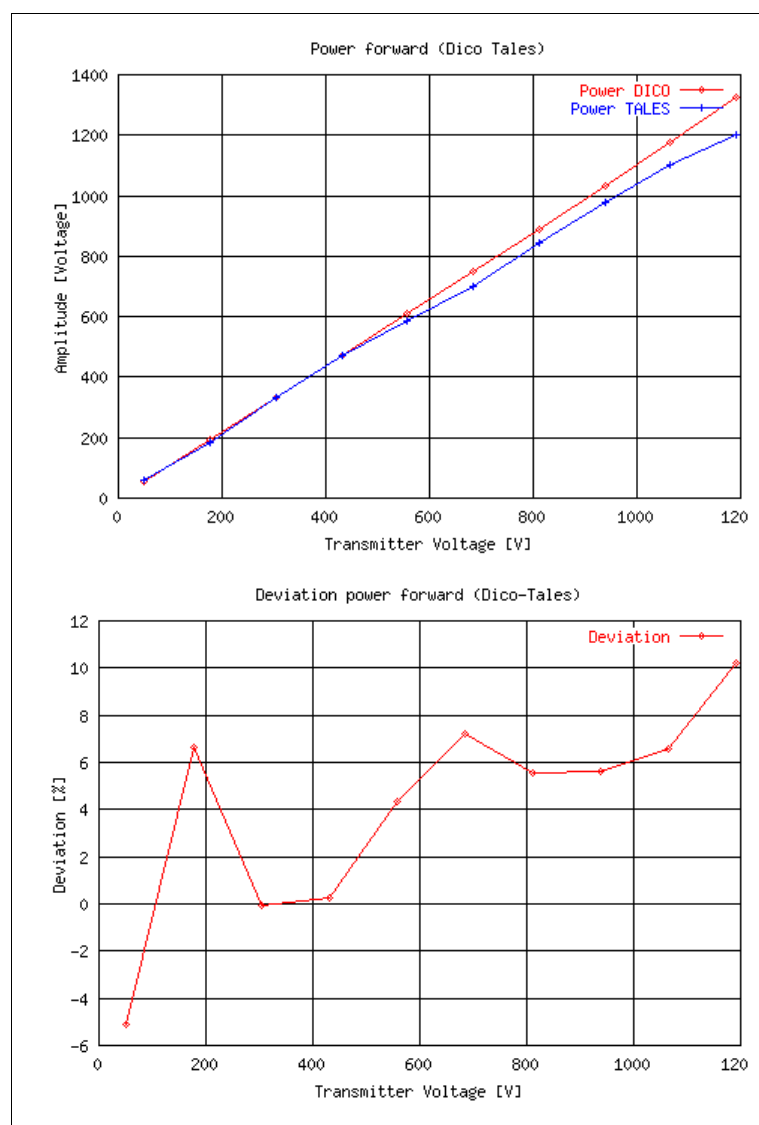
HTML output table:

- PF (DICO) at maximum deviation (Volt)
- PF (TALES) at maximum deviation (Volt)
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ maximum deviation (%)

Graphical output:

- PF(DICO/Volt) and PF(TALES/Volt) over TRA voltage
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ deviation (%) over TRA voltage

Fig. 111 Example of graphical output



Conclusions

If the deviations between DICO and TALES are out of specification, repeat the RF characteristic measurement.

Locate the defective component in the TX-path by repeating the test after connecting the dummy load to a component before it's current position.

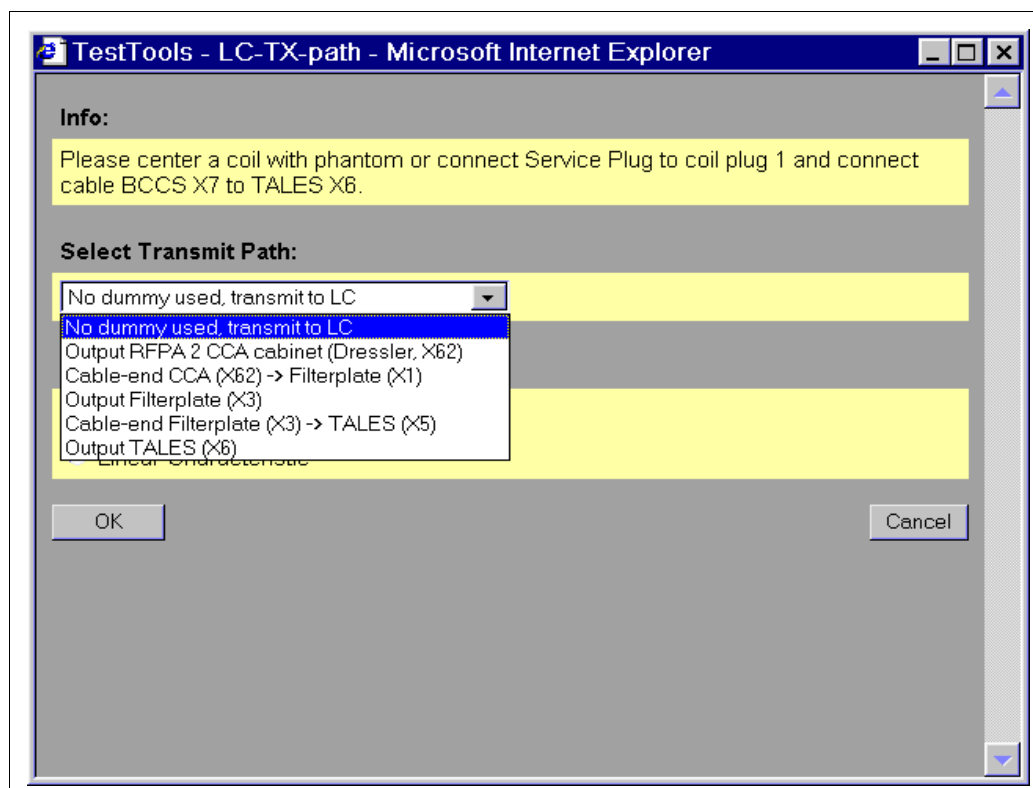
The reason of significantly higher reflection measured by the TALES in comparison to DICO can be oscillations of the RFPA, transmitted on an other frequency besides the MR-frequency.

LC-TX-path (interactive/Multi Nuclei)

NOTICE When transmitting to the Local Coil transmit path, ensure that either the service plug or a local Tx-coil is connected to coil plug 1. The cable to BCCS/X7 must be connected to TALES/X6.
The test requires a valid RF characteristic or the linear RF characteristic.
For spectroscopy/multi nucleus coils select the checkbox Multi Nuclei and the desired nucleus.

The objective of the test is to check the components of the LC-TX-path. Several measurements with an increasing amplitude starting from the minimum (50 V) to the maximum voltage (95 V for service plug and maximum local coil voltage read from the coil file) are performed.

Fig. 112 Pop up window LC-TX-path
Selection of TX-path termination location by a 50 Ohm dummy



Signal path

The TX-path for local coils with the 8 kW RFPA is used.

The TX-path can be opened at locations and terminated by a 50 Ohm dummy. In the appearing popup window you can select the desired TX-path (refer to [Fig. 112](#)). The measurement is started by clicking "OK" or skipped by selecting "Cancel" in the popup window.

The selection of the location of the interruption in the TX path makes no influence of the measurement but selects the heading of the output table. If the TX-path is terminated by a 50 Ohm dummy between TALES and RFPA, the TALES values can not be measured and displayed.

NOTE	Whenever a dummy load is used to interrupt the TX-path BEFORE the TALES, the comparison of DICO- and TALES-values will not be successful as there are no TALES-values available and thus the test will fail. This is normal and not a problem of the system.
-------------	---

Evaluation

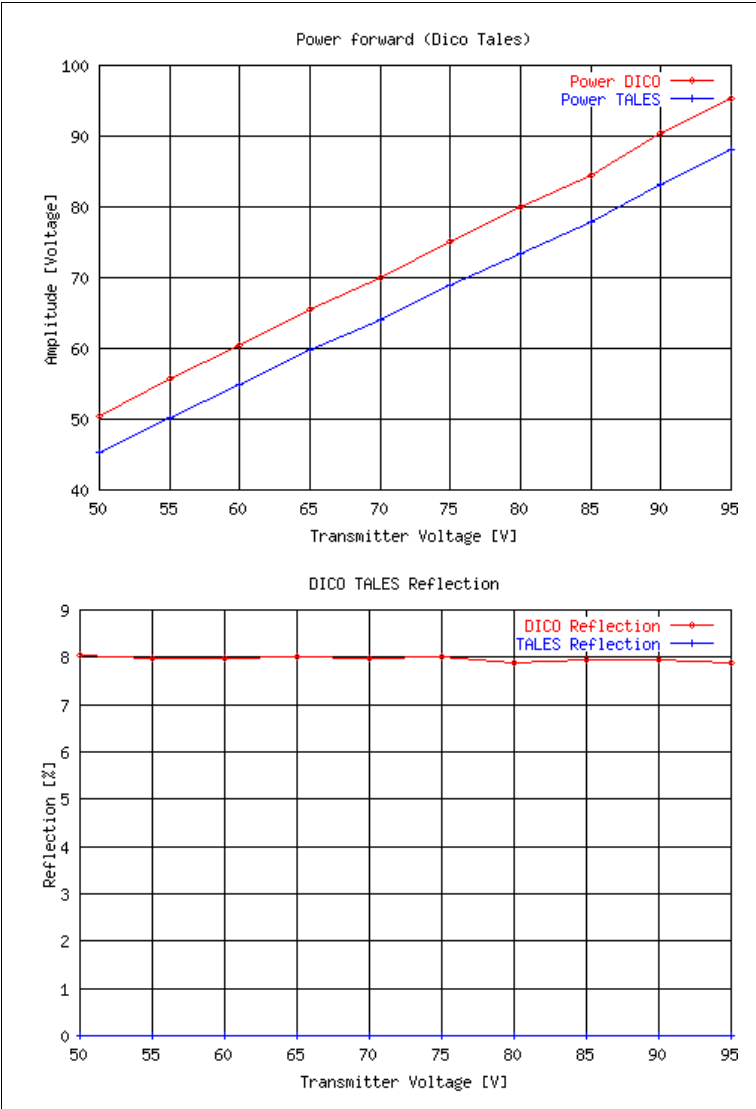
HTML output table:

- PF (DICO) at maximum deviation (Volt)
- PF (TALES) at maximum deviation (Volt)
- $(PF(TALES) - PF(DICO)) / PF(TALES)$ maximum deviation (%)

Graphical output:

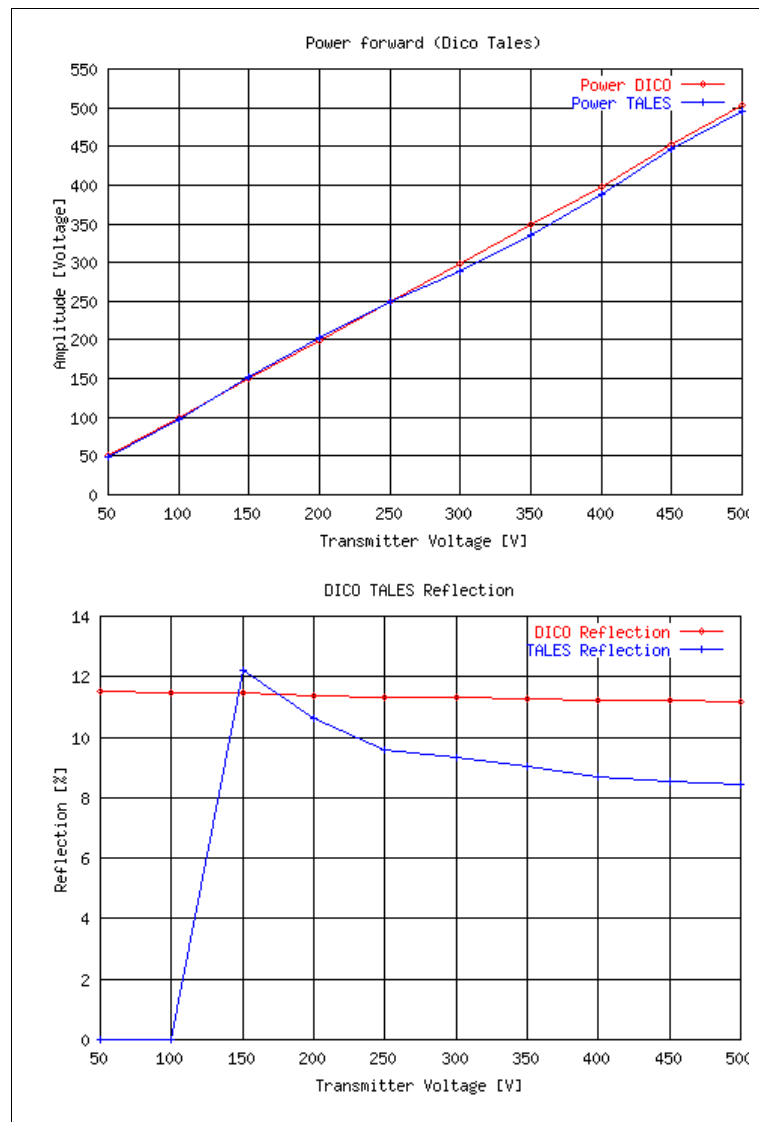
- PF(TALES/Volt) and PF(DICO/Volt) over TRA voltage
- PR/PF DICO reflection (%) and PR/PF TALES reflection (%) over TRA voltage

Fig. 113 Example: Transmit to Service plug



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Fig. 114 Example:
Transmit to CP head coil (Dummy connected to TALES X6)



Conclusions

NOTE For reflected voltages below 100 V the measured/displayed TALES value is 0 V.

If the deviations between DICO and TALES (for voltages higher than 100 V) are out of specification, repeat the RF characteristic measurement for the actual nucleus.

Locate the defective component in the TX-path by repeating the test after connecting the 50 Ohm dummy to a component before it's current position.

LCCS (Functional, Comm/Status)

- LCCS Communication/Status
- BCCS (preamp1) - LCCS Loop
- BCCS (preamp2) - LCCS Loop

LCCS plug 1,2,3,4 (Interactive, Multi Nuclei)

- LCCS (SGA x) Service Plug, pos n
- LCCS SGA x Loop (High Gain/Low Gain, plug 4 only)

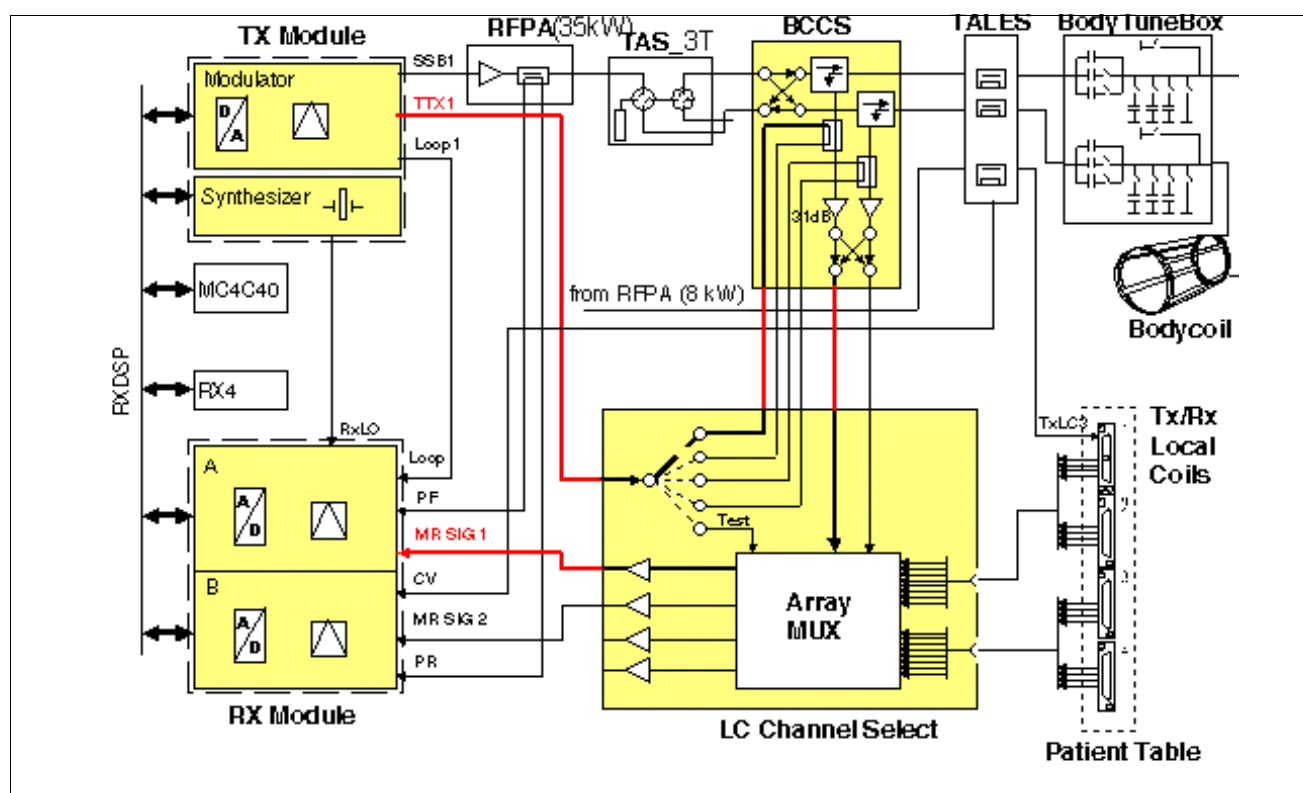
LCCS Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message.

BCCS (preamp1) - LCCS Loop

Fig. 115 BCCS (preamp1) - LCCS



Signal Path

The TTX output signal of the modulator is fed via the LCCS forward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver A/B (MRSIG1/MRSIG2) via the LCCS. The tests are performed with high gain and low gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

In a second measurement the TTX output signal of the modulator is fed via the LCCS backward to the body coil preamplifier input in the BCCS. The signal is amplified in the BCCS and fed back to receiver 1A/2A via the LCCS. The test is performed with high gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

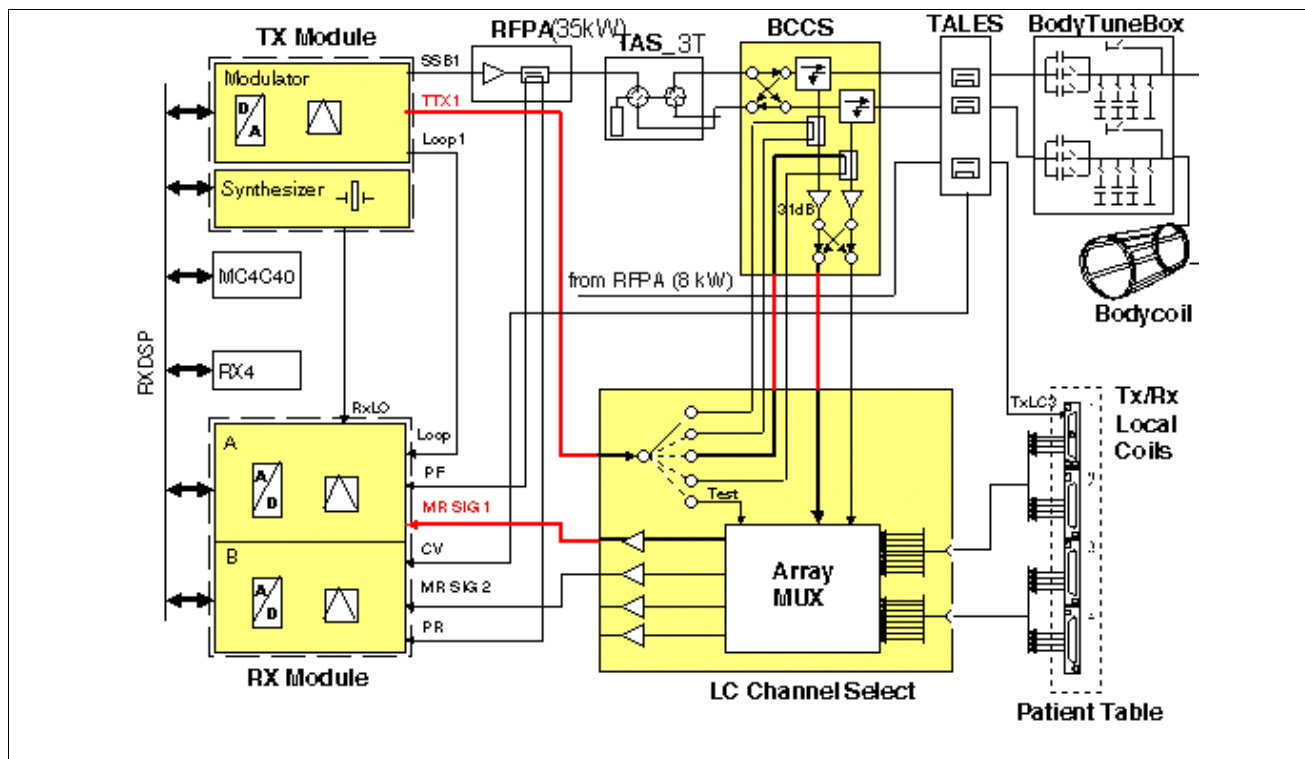
This test is used primarily to check the stability and linearity of the preamplifier in the BCCS and the LCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

The SGA amplifiers (high gain und low gain) are located in the LCCS.

Additional the cables and connections between BCCS and LCCS (<W531>, <W532> and <W529>) are checked.

BCCS (preamp 2) - LCCS Loop

Fig. 116 BCCS (preamp 2) - LCCS



This test is similar to the previous test, but the signal is fed into the second preamplifier of the BCCS. Only the high gain setting of the LCCS preamplifiers (SGA1/SGA2) in the LCCS is used.

Evaluations

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

Conclusions

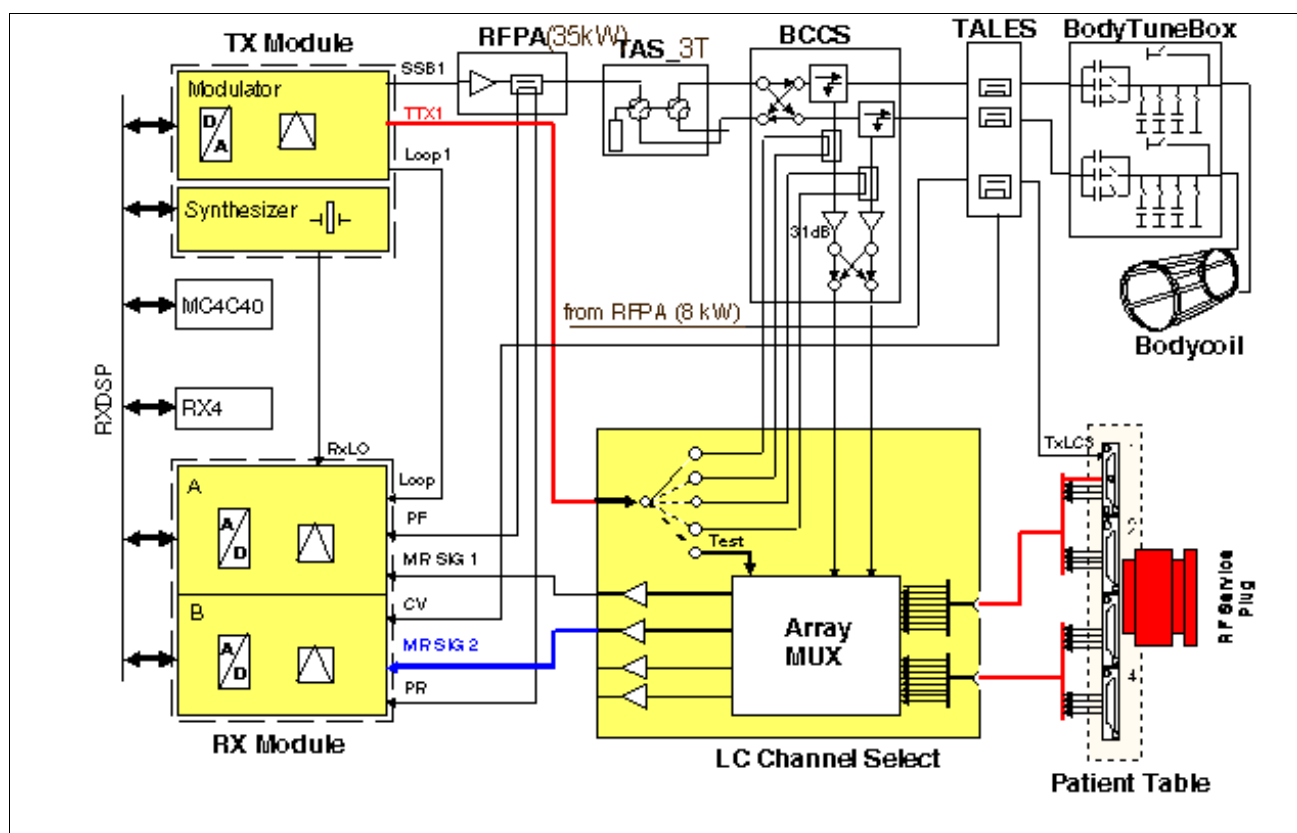
This test is used primarily to check the stability and linearity of the preamplifiers in the BCCS and the LCCS. Modulator control and data acquisition via the receiver must be intact to evaluate the results.

The SGA amplifiers (high gain und low gain) are located in the LCCS.

Additional the cables and connections between BCCS and LCCS (<W533>, <W534> and <W528>) are checked.

LCCS plug 1,2,3,4 - LCCS (SGA x) Service Plug, pos n (interactive)

Fig. 117 LCCS Interactive Test



Preparation

The RF Service Plug must be connected to the coil connector 1,2,3 or 4 at the patient table. Follow the instructions provided during the test run.

Signal Path

The TTX output signal of the modulator is fed via LLCS to the RF Service Plug (attenuator network) and back to the LCCS (via switching matrix) and to the receiver module.

The TTX RF signal from the modulator is applied to one port of the receiver via the internal multiplexer of the LCCS, attenuated via the attenuator network in the Service Plug, and forwarded back to the LCCS and to the receiver module. In the process, the signal path in the LCCS is selected so that all switching points of the matrix are checked.

The service plug position has to be changed accordingly for the local coil receptacles by following the instructions in the interactive dialog.

NOTE	The functional combination of two receiver signals in the LCCS cannot be automatically tested at this time.
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Evaluation

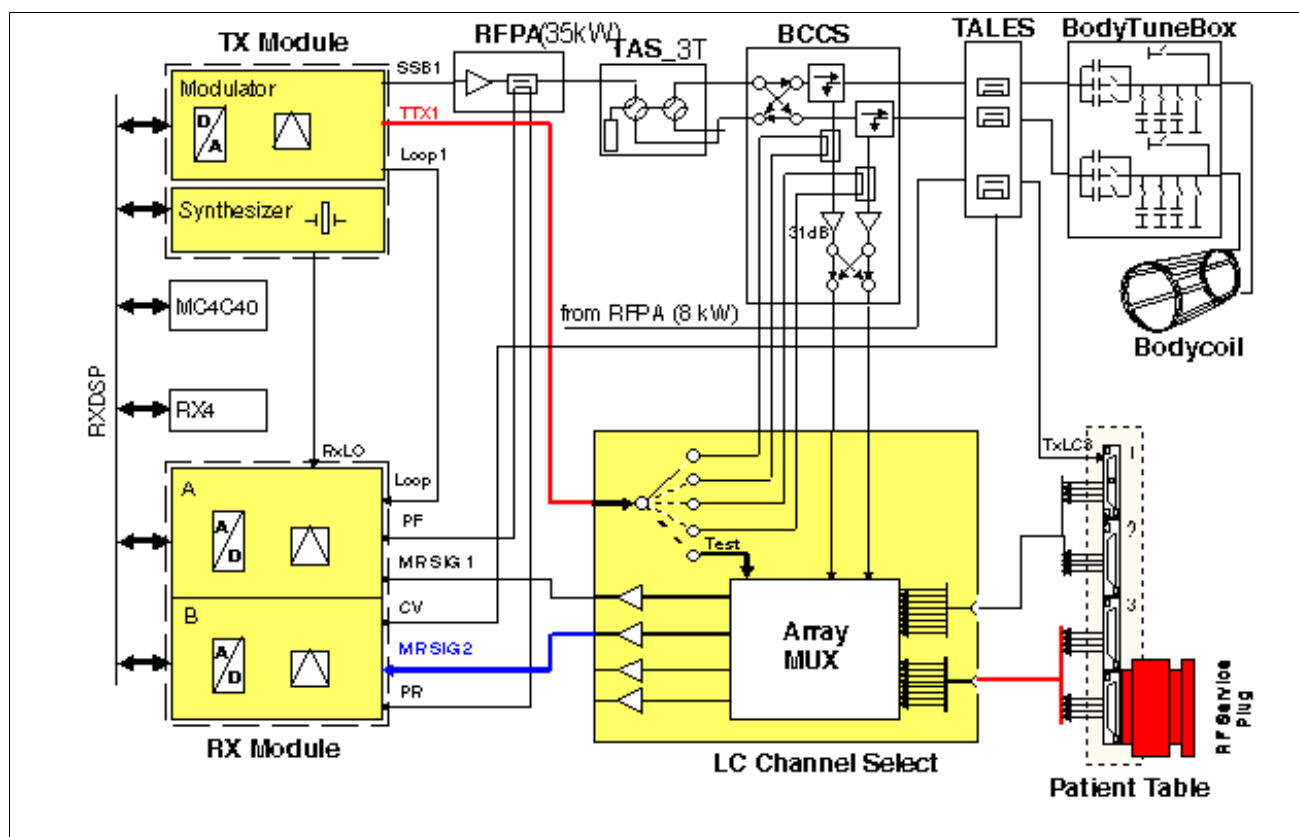
The TTX signal is attenuated in the attenuator network of the RF Service Plug in such a way that the signal level of the first channel is 100 %, the second approx. 50 %, the third approx. 25 % and the fourth approx. 12.5 %. The signal levels are checked against the specifications. The graduated levels allow complete testing of the switch matrix as well as the cabling from the local coil plug to the LCCS and continuing to the receiver.

Conclusions

In case of an error, the cause could be the LCCS. However, it could also result from incorrect or defective cabling from the local coil receptacle to the LCCS (movable cable in the patient table) and the receiver module. Before replacing the LCCS, you should test the RF receiver cable first (local coil receptacle - LCCS).

LCCS plug 1,2,3,4 - LCCS SGA x Loop (Low Gain only at plug4 and High Gain) (interactive)

Fig. 118 LCCS Interactive Test



Preparation

The RF Service Plug must be connected to the coil connector 1,2,3 or 4 at the patient table. Follow the instructions provided during the test run.

Signal Path

The TTX output signal of the modulator is fed via the LCCS to the RF Service Plug and back to the LCCS (SGA) and to the receiver module.

Evaluation

The raw data of the first sinc-pulse are evaluated for stability, the raw data of the second sinc-pulse are evaluated for linearity.

At coil plug 4 the measurement is done with high and low gain setting of the SGA, at coil plug 1,2,3 only with high gain setting.

Conclusions

The test is used primarily to check the SGA stability in the LCCS. Because of the internal circuitry, the SGA 1 cannot be checked using this test. It can only be tested indirectly via the body coil preamplifier in the BCCS (BCCS preampl - LCCS Loop).

RFCI

- RFCI Communication/Status
- RFCI PIN Diodes (Local/Body Coil)
- RFCI PIN Diodes Service Plug, pos n (interactive)
- RFCI PIN Diodes DVM (interactive)

RFCI Communication/Status

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

The status test queries the status of a particular component. It quickly determines whether the component status is OK or shows an error. A faulty status is indicated by three exclamation marks next to the respective status message. As long as no measurement was done with the connected coil the status will be "setting not loaded" and the respective PIN channels and mask "not set". So this is no error condition.

RFCI PIN Diodes (Local coils/PIN-Booster/Body Coil)

NOTICE The local coils and the loaded phantom must not be positioned in the magnet for this test. If additional RF tests are performed afterwards, the local coils must be removed and the loaded phantom positioned in the magnet center to prevent damage to the resonator or to the local coils.

The test of the forward current (Body Coil: 600 mA/5 A, local coils: 100 mA) and reverse voltage (-30/-31/-500 V) supplied and measured by the RFCI/PIN-Booster for the PIN diodes of the Body Coil and all PIN diodes in the connected local coils is done automatically.

NOTE If the PIN diodes in the head coil should be checked, the upper part of the Head Coil must be connected. Without the upper part, the test will show an error on the 3rd channel because of the missing PIN diode in the upper part.

Conclusions

This test verifies the function of the RFCI, the cables between RFCI and PIN-Booster, PIN-Booster and Body Coil and the PIN diodes in the Body Coil and the local coils. In case of an error, the interactive tests should be performed with the service plug to differentiate between the coil and the controller.

RFCI PIN Diodes Service Plug, pos n (interactive)**Preparation**

The RF Service Plug must be connected to the coil connector at the patient table. Follow the instructions provided during the test run.

NOTE	The test does not use the normal communication between the MPCU and the RFCI, but the CAN-bus. In this way it is possible that a defect of the backplane is not detected.
-------------	--

Evaluation

This interactive test checks the RFCI forward current (100 mA) and the reverse voltage (-30 V) driven through the 4 diodes in the RF Service Plug. The plug position has to be changed by following the instructions in the interactive dialog.

Conclusions

If defective PIN diode channels are detected during the test, the measurements with the voltmeter in the following test will provide more detailed information regarding the possible cause of the error.

RFCI PIN Diodes Volt Meter (interactive)

The RFCI switches between current and voltage for all PIN diode channels including the body coil channel in a fixed time pattern of approximately 5 seconds. The RF Service Plug can be connected to the 4 plug positions in sequence without interrupting the test. The voltage drop across the PIN diodes in the service plug and at the test jacks of the RFCI service board is measured using the DVM.

NOTE	The test does not use the normal communication between the MPCU and the RFCI, but the CAN-bus. In this way it is possible that a defect of the backplane is not detected.
-------------	--

Service Plug

- Reverse voltage (UR) at PIN diode: - 27 ... - 33 V DC, typically -30.5 V
- Forward current voltage drop (IF): + 1.9 ... + 2.3 V DC, typically 2.1 V (corresponds to 100 mA)

In addition to the measurement at the PIN diodes, the voltage supply of the internal RX preamplifier (+10 V +/-1 V) of the local coils can be checked at the service plug. The voltage is continuously present independent of the running test, but influenced by the PIN diode switching. For that reason, abort the PIN diode test for an accurate measurement of the preamplifier supply voltage.

RFCI Service Board

Local coil PIN diode control:

- Reverse voltage (UR): - 27 ... - 33 V DC, typically -30.5 V

Forward current voltage drop (IF): + 2 ... + 6 V DC, typically 2.6 V (depending on the cable length between CCA and patient table)

⇒ Error: Open circuit voltage (IF current regulator at limit):
approx. + 14 V DC

Body coil PIN diode control:

- Forward current voltage drop (IF): + 3 ... + 8 V DC
(according to a forward current of 600 mA, dependent on the cable length between CCA and body tune box).

⇒ Error: Open circuit voltage (IF current regulator at limit):
approx. + 15 V DC

Test points at the RFCI

- J10 left side TP 1-4 --> coil connector 1, PIN diode 1-4
- J10 right side TP 5-8 --> coil connector 2, PIN diode 1-4
- J9 left side TP 9-12 --> coil connector 3, PIN diode 1-4
- J9 right side TP 13-16 --> coil connector 4, PIN diode 1-4
- J7 left top TP --> body coil, PIN diode

LEDs at the RFCI

During the test, the LEDs at the RFCI service board light up alternately. There are 4 x 4 groups of LEDs for each of the four PIN diode connections of every coil connector.

- U_ON_I_OF_1 = on: PIN diode control UR = on
- UR_ON_x = on: UR present
- IF_ON_x = on: IF present

Conclusions

Measurement at RF Service Plug

UR	IF	Conclusion
OK	OK	OK.
NOT OK	OK	Error RFCI.
OK	NOT OK	Error RFCI; PIN diode of Service Plug may be defective.
10 V DC (LC preamplifier) not OK		1. Broken or shorted cable between LCCS (X11/12) and coil connector. 2. 15 V to 10 V conversion on LCCS defective. Check error messages!

Measurement at RFCI Service Board

UR	IF	Conclusion
NOT OK	OK	Error RFCI.
OK	NOT OK	Broken cable between RFCI and LC connector.
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and LC connector.

Measurement at RFCI Service Board (body coil)

UR	IF	Conclusion
OK	OK	OK
NOT OK	OK	Error RFCI/defective PIN diode of body coil (short circuit).
OK	NOT OK	Defective PIN diode of Body Coil or broken cable between RFCI and PIN-Booster or PIN-Booster and Body Coil. Check the LED's at the PIN-Booster. Perform TestTools/RF/BTB/BC/Functional.
NOT OK	NOT OK	Error RFCI or short circuit in the cable between RFCI and PIN-Booster, PIN-Booster and Body Coil or defective PIN diode of body coil. Check the LED's at the PIN-Booster. Perform TestTools/RF/BTB/BC/Functional.

In addition, you can measure several supply voltages at the RFCI service board and the -500 V static detune voltage at PIN-Booster (KA).

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Gradient - All systems

NOTE This section describes all MR systems except the MAGNETOM Concerto.

The stability of the gradient system is essential to the image quality obtained in MR. The gradient tests are a tool for verifying the performance of the gradient system as well as assisting you in locating the cause for errors.

Before running the gradient tests, perform the MMC test to ensure that the components GCTX and RX4 are functioning as specified.

General

The gradient tests have been developed for locating the source of instability/nonlinearity in the gradient system. For example, when the instability/nonlinearity measured using the Stability Check or CalcAr is a function of the orientation, there is every indication that the instability/nonlinearity is caused by gradient instability/nonlinearity rather than by RF or magnetic field instability/nonlinearity.

In preparation for the gradient linearity tests, communication of the GPA with the MMC will be tested using the GPA status test and the first of the DAC Functional tests (GCTX-DAC).

The results of the tests will be saved automatically in the **Report** File.

Test Evaluation

Both evaluations, Stability and Linearity, analyze two sets of data that are extracted from the raw data. One set of data is the **center column**, which is located in the middle of the gradient pulse plateau. The second set is the **integral of each line of data** of the pulse. The integral is the summation of the data in each line divided by the number of data points.

Stability

The gradient amplitude is constant. The **center column result** is an evaluation of the stability of the center column which is located in the middle of the gradient pulse plateau. The numerical result is the maximum minus the minimum value divided by the mean value. The **integral result** is calculated by using the mean value of each line instead of the center column and using this data to evaluate the stability. Again, the numerical result is the maximum minus the minimum value divided by the mean value. Both center column and integral results are displayed in percent (%).

Linearity

To evaluate the linearity, the gradient amplitude is stepped from a negative to a positive pulse amplitude. The **center column data** and **integral data** of each line are fitted to a linear function. The numerical result is the maximum deviation from the linear fit divided by the maximum value making up the data set. Both center column and integral results are displayed in percent (%)

Test Results/Graphical Output

The numerical results of the Gradient Tests will be displayed in the right frame of the window which appears after completion of the tests. A label identifying the test, the center column result, the specification, the integral result and its specification is printed out. If one or both of these results are out of specifications, the respective field in the table appears in color.

In addition to the numerical results that are displayed for each test, two graphs are plotted. The one graph is a plot using the center column data, the other graph is using the integral data.

With respect to the stability evaluation, the two graphs are simply the plot of the two sets of data (with subtracted mean value), one for the central column data, the other for the integral data. Usually, no pattern is seen in the data, and the data shows characteristics typical for noise or ADC sampling errors (changing between a few amplitude values due to limits of the ADC resolution). However, if a non-constant pattern is seen, this may be indicative of interference at the gradient output.

For the linearity test, the two graphs are the difference between the data and the linear fit that is determined from the data. The first graph uses the central column data, and the other uses the integral data. Ideally, the difference should be zero since the gradient is linearly stepped for each line in the raw data. Usually, no pattern is seen in the graphs, and the data shows characteristics typical of noise or ADC sampling errors (changing between a few amplitude values due to limits of the ADC resolution). However, if a non-constant pattern is seen, this may be an indication of a problem in the linearity of the gradient output.

GPA Communication/Status

- GPA Communication Test
- GPA SetPowerSwitch
- GPA Status

GPA Communication

The communication between MMC/PCI-CAN and GSSU/D6 K2209/D16 K2217 Service board via CAN-Bus will be tested.

In case of error, the system cancels all other tests selected.

If the test fails, run the MMC/PCI/CAN test and check the cable connection between MMC/PCI/CAN and GSSU/D6 K2209/D16 K2217 followed by the GSSU/D6/D16 service board (with CAN module D1).

GPA Status

The MPCU requests that the GPA provides its present (error) status.

The test provides you with a listing of the present status of the GPA, i.e. all possible error/status conditions, which will be recognized and stored by the local application software of the GPA CAN-unit. If an error condition exists, the status of the GPA will be "not ok". Refer to the System error report (Service software event log or System/Report) and to the Troubleshooting Guide charts for the respective error conditions.

In case of an error, the system cancels all other tests selected.

If the status is "not ok", search for the corresponding error messages. For remedy actions, refer to the system error report details and the Troubleshooting Guide (GPA charts).

The following GPA-Status messages cause "Status of GPA not ok", i.e. not ready for measurement:

- All messages marked with "!!!" represent errors
- Resonant converter standby = yes
- X and/or Y and/or Z Modulator inhibit = yes

If a non-persistent error marked with "!!!" exists in the first run of GPA status, the SetPowerSwitch function resets the GPA to "ok" status.

The test is initiated a second time after the SetPowerSwitch to clear possible previous error messages which do not come from the GPA. Check the second status test, if it is okay the first test was most probably affected by previous tests only and the GPA has no defect.

GPA SetPowerSwitch

This function, performed from the MPCU via the CAN-Bus, initiates a reset of the GPA status "GPAK2217 not ok". All non-persistent error messages will be reset.

For error analysis, the first run of GPA status should be evaluated. Running the MMC tests before the GPA status test will always indicate a stimulation danger error (no real error). This error will be reset by the SetPowerSwitch function.

DAC (functional)

- Gradient data loop test (Loop MMC/GCTX - D1 K2209/D11 K2217/DAC GPA)
- DAC (nominal value) test for X, Y, Z-Axis (stability/linearity)

Select the test using the DAC checkbox.
Deselect the X, Y, Z-Axis checkboxes.

MMC Gradient Loop GCTX-DAC

The register GRAD1_X (GCTX) is selected and a test pattern written in the register. As a consequence, the receiver register of the GPA is filled via the data transmission line from GCTX to D1 K2209/D11 K2217 (DAC). The 18 bit test data pattern at the input of the DAC (X-axis) is read back and compared with the originally transmitted data set for consistency.

This test checks the data transmission line MPCU - RS485-Interface (GCTX) - plug (GCTX) - cable - plug (GSSU/D1/D11) - RS485-Interface (Grad).

NOTE In case of an error condition, the remaining tests will not be performed.

Conclusion

In case of error, perform the MMC/ MC4C40 and GCTX tests and check the cable connection between GCTX and GSSU/D1 K2209/D11 K2217. In addition, check the D1 K2209/D11 K2217 DAC board.

Nominal Value X/Y/Z (High/Low Ampl)

The measured signal is the nominal value for the gradient current used to regulate the actual value. The signal is measured at the output of the DAC's of D1 K2209/D11 K2217, routed to D6 K2209/D16 K2217, there selected by the loopmux, amplified and sent to the DC-inputs of the Receiver Channel 1 (RFSU) via cable connection. The raw data received (ADC-data) will be evaluated.

All three axes will be measured one after another using two different gradient amplitudes, i.e. with the maximum and a medium gradient strength.

Selection of a single axis is not possible. In case of an error condition, the remaining tests (DAC,X,Y,Z-Axis) will be performed.

Conclusions

In case the signal is too low (stability/center column), check the QLA cable connections between GSSU/D6 K2209/D16 K2217 and RFSU/Receiver channel 1 DC inputs.

If the test fails, check D1 K2209/D11 K2217 (DAC); replace D1/D11 if a particular axis fails in every case. Check D6 K2209/D16 K2217, the QLA cable connection to the RFSU Receiver Channel 1, the DC-input and the receiver board.

Perform the MMC tests.

Stimulation Monitor Test (functional)

TestMode measurement

With the stimulation monitor switched to test mode, one measurement is performed for each gradient axis. Because of the high gradient amplitudes used, the GPA must be switched off by the stimulation monitor.

Conclusions

If the GPA is not switched off, the stimulation monitor or the connection between stimulation monitor and GPA D17 (I/O board) or D17 may be defective.

Perform the MMC/MC1C40 test. Check the cabling and connections between MMC and D17.

NOTE	Internal errors can cause the GPA to switch off as well, e.g. wrong regulator adjustment (-> resulting error: GPA loop error). Therefore, check for all error messages from the GPA.
-------------	--

ServiceMode measurement

One measurement is performed in service mode (GPA switch off by stimulation monitor not possible) with each gradient axis. The maximum values of the stimulations will be recorded and compared to the specification values.

Conclusions

If the stimulation values are out of specifications, the stimulation monitor may be defective. Run the MMC/MC1C40 test. If zero stimulation values are measured, the connection to the stimulation monitor may be defective. If the stimulation values are too small, check the jumper setting on D14 (modulator board). Check for other GPA error messages and refer to the corresponding Troubleshooting Guide charts.

Actual Value X/Y/Z (High/Low Ampl) (functional)

- Actual value test for X, Y, Z-Axis (stability/linearity)

Select the test using the X and/or Y and/or Z Axis test type Functional checkboxes. Deselect the DAC checkbox.

X, Y, Z Actual Value test

The measured signal is the actual value of the gradient current sent to the gradient coil. The signal is measured by the current sensors (SENS) and routed via D7 K 2209/D12 K2217 (FOC/Regulator board) to D2 K 2209/D12 K2217 (Regulator board). After amplification on D2/D12, it is routed to D6 K2209/ D16 K2217, there selected by the loopmux, amplified and sent to the DC-input of the Receiver Channel 1 of the RFSU via cable connection. The raw data received (ADC-data) will be evaluated.

If selected, all three axes will be measured one after another using two different gradient amplitudes, i.e. with the maximum and a medium gradient strength.

Selection of a single axis is possible. In case of an error condition, all remaining tests (X,Y,Z Axis Functional) will be performed.

Conclusions

In case the signal is too low (stability/center column), check the QLA cable connections between GSSU/D6 K2209/D16 K2217 and RFSU/Receiver channel 1 DC inputs.

If some of the test fail, check D2 K2209/D12 K2217 (regulator board) of the corresponding axis using the "Regulator Check". Check the D6 K2209/D16 K2217 (service board), the cables to the RFSU and the receiver board.

Perform the MMC tests.

Also check the following GPA components:

D4/5 K2209/D14 K2217 (modulator board), D3 K2209 (synchronization board), D7 K2209 (FOC board) and the FOC's (GPA K2209)/cabling (GPA K2217) to the power stages.

In addition, check for error messages concerning the D10 K2209 (GPA K2209, driver supply board) and the RC (GPA K2209). Check the output voltage of the RC in case linearity tests/high amplitude fail. Check at least the current sensors (SENS) and the power stages. Check the line voltage and the output voltage of the rectifier board D32 (GPA K2217).

In case you find error messages from the GPA in the **System/Report** or **GPA_Status** list, refer to the corresponding Troubleshooting Guide GPA charts.

Linearity Check (functional)

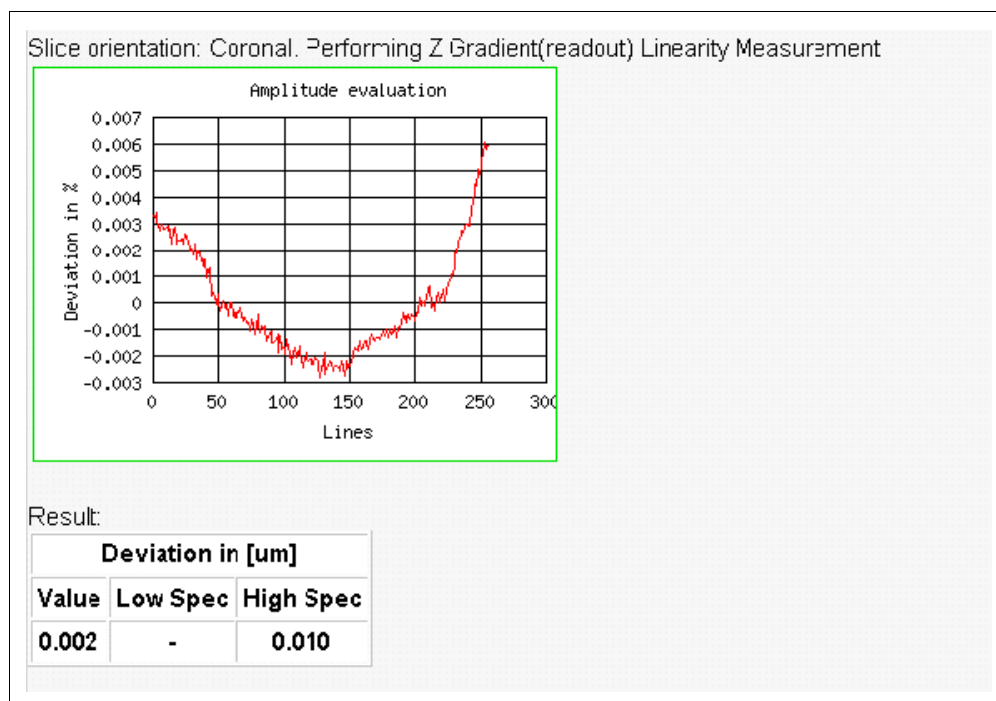
- Gradient amplitude linearity check using MR signal measurements to determine the linearity of the gradient system. All three gradients are used to produce the spin echo signals which are evaluated over the complete gradient range.
- Position the large spherical phantom with loader in the magnet center.
Select the test using the Grad.linearity checkbox.

Test results for gradient linearity/Graphical HTML Output

The numerical results of the gradient linearity tests and a corresponding graph will be displayed in the HTML window after the tests have been completed. A label identifying the test including slice orientation and gradient axis is printed out. If the result is out of specifications, the respective field in the table appears in color.

Each point of the graph represents one amplitude value of the gradient axis tested (line 0-255). The deviation value means the deviation of the integral value of the gradient pulse from the nominal value, relating to the integral value of the maximum gradient pulse.

If a non-constant pattern is seen, especially at the beginning or end of the plots (near line 0 or 255), this may be an indication of a problem with the linearity of the gradient output at high amplitudes.



Log-File (section of the HTML output page)

Clicking **Log-File** in the HTML output page lets you open another window displaying all measured maximum gradient amplitudes and some additional information. The example below shows only a section of this window.

```

tt_2000-01-01-000000_1.log - Editor
Datei Bearbeiten Suchen ?

Line: 238      Maxround: 257.000      Maxinter: 256.985
Line: 239      Maxround: 257.000      Maxinter: 256.986
Line: 240      Maxround: 257.000      Maxinter: 256.986
Line: 241      Maxround: 257.000      Maxinter: 256.985
Line: 242      Maxround: 257.000      Maxinter: 256.986
Line: 243      Maxround: 257.000      Maxinter: 256.986
Line: 244      Maxround: 257.000      Maxinter: 256.987
Line: 245      Maxround: 257.000      Maxinter: 256.988
Line: 246      Maxround: 257.000      Maxinter: 256.988
Line: 247      Maxround: 257.000      Maxinter: 256.988
Line: 248      Maxround: 257.000      Maxinter: 256.988
Line: 249      Maxround: 257.000      Maxinter: 256.989
Line: 250      Maxround: 257.000      Maxinter: 256.988
Line: 251      Maxround: 257.000      Maxinter: 256.989
Line: 252      Maxround: 257.000      Maxinter: 256.990
Line: 253      Maxround: 257.000      Maxinter: 256.989
Line: 254      Maxround: 257.000      Maxinter: 256.990
Line: 255      Maxround: 257.000      Maxinter: 256.989
Results of maximum fit: Offset = 256.984467, Slope = -0.000006, stddev = 0.002317, Maxdev =
End result: Deviation      0.0009 %
Specification:              0.0100 %
Slice orientation: Transversal. Performing X Gradient(readout) Linearity Measurement
Set slice for transversal orientation (Readout gradient: X).
Specification of the deviation: 0.010
line #1 in X: Max Amplitude of 3740.42 and SN of      20.82
line #2 in X: Max Amplitude of 3522.22 and SN of      20.93
line #3 in X: Max Amplitude of 3461.04 and SN of      20.94
line #4 in X: Max Amplitude of 3471.40 and SN of      20.88
line #5 in X: Max Amplitude of 3469.93 and SN of      20.90
line #6 in X: Max Amplitude of 3467.32 and SN of      20.89
line #7 in X: Max Amplitude of 3459.07 and SN of      20.91

```

Conclusion

In case of an error, check the input voltage of the power stages and the stability of the main voltage. Swap power stages. Replace defective power stages. Check/replace defective SSUs.

Gradient - MAGNETOM Concerto

NOTE This section describes the MAGNETOM Concerto only.

The stability/linearity of the gradient system is essential to the image quality obtained in MR. The gradient tests are a tool for verifying the performance of the gradient system as well as assisting you in locating the cause of errors in the gradient amplifier (GPA).

Before running the gradient tests, perform the MMC tests to ensure that the components GCTX and RX4 are working as specified.

General

The gradient tests have been developed for locating the source of instability/nonlinearity in the gradient system. For example, when the instability/nonlinearity measured using the Stability Check or CalcAr is a function of the orientation, there is every indication that the instability/nonlinearity is caused by gradient instability/nonlinearity rather than by RF or magnetic field instability/nonlinearity.

In preparation of the gradient linearity tests, communication of the GPA with the MMC will be tested using the GPA status test and data transfer from MMC to the GPP using the DAC functional tests (GCTX-DAC).

The results of the tests will be saved automatically in the **Report** File.

Gradient linearity

NOTE Refer also to the on-line help concerning QA/Linearity check.

The linearity of the gradient amplitudes is measured using an MR measurement. The "Gradlin" sequence excites the phantom with a slice-selective 90 degree pulse. The gradient pulse under test is switched in readout direction with varying amplitudes. After a slice-selective 180 degree pulse, the echo is measured under a weak readout gradient. The design of the sequence ensures that the spin echo always appears in the center of the acquisition interval (symmetric 90-180 degree echo).

After the measurement, each raw data line contains a narrow echo. The maximum position of the echo will be determined.

If nonlinearities are present in the gradient system, the position of the echo maximum changes depending on the gradient amplitude.

To be sensitive to the nonlinearities, a linear fit is calculated through the measured maximum echo positions. After that, the deviations of the measured maximum echo positions (standard deviation) are calculated. This value, converted into a percentage, is compared to a specification.

GPA Communication/Status

- GPA Communication
- GPA Status

GPA Communication

The communication between MMC/PCI/CAN and SSU/D1 K2229 via CAN-Bus will be tested. In case of error, the system cancels all other tests selected.

If the test fails, run the MMC/PCI/CAN test and check the cable connection between MMC/PCI/CAN and SSU/D1 K2229. Replace defective SSU's.

GPA Status

The MPCU requests that the GPA provides its present (error) status.

The test provides you with a listing of the present status of the GPA, i.e. all possible error/status conditions, which will be recognized and stored by the local application software of the GPA CAN-unit. If an error condition exists, the status of the GPA will be "not ok". Refer to the System error report (Service software event log or System/Report) and to the Troubleshooting Guide charts for the respective error conditions.

In case of an error, the system cancels all other tests selected.

If the status is "not ok", search for the corresponding error messages. For remedy actions, refer to the system error report details and the Troubleshooting Guide (GPA charts).

The following GPA-status messages cause "Status of GPA not ok", i.e. not ready for measurement:

- All messages marked with "!!!" represent errors
- Status Modulator locked X,Y,Z = fault.

If there are non-persistent errors marked with "!!!" (except for the status "GPA ready for measurement = no !!!"), the next load/start of a sequence should reset the GPA to "ok" status. In some cases, a reboot of the scanner will help.

DAC (functional)

- Gradient data loop test (Loop MMC/GCTX - SSU/D2 K2229/DAC).

Select the test using the DAC checkbox.

MMC Gradient Loop GCTX-DAC

The register GRAD1_X (GCTX) is selected and a test pattern written in the register. As a consequence, the receiver register of the GPA is filled via the data transmission line from the GCTX-board to SSU/D2 K2229 (DAC). The 18 bit test data pattern at the input of the DAC (X-axis) is read back and compared for consistency with the originally transmitted data set.

This test checks the data transmission line MPCU - RS485-Interface (GCTX) - plug (GCTX) - cable - plug (SSU/D2) - RS485-Interface (Grad).

NOTE	In case of an error condition, the remaining tests will not be performed.
-------------	---

Conclusion

In case of error, perform the MMC/MC4C40 and GCTX tests and check the cable connection between GCTX and SSU/D2 K2229. Replace defective SSU's.

Linearity Check (interactive)

- Gradient amplitude linearity check using MR signal measurements.

Position the CP Body Spine coil together with the spherical phantom in the magnet center. Select the test using the Grad.linearity checkbox.

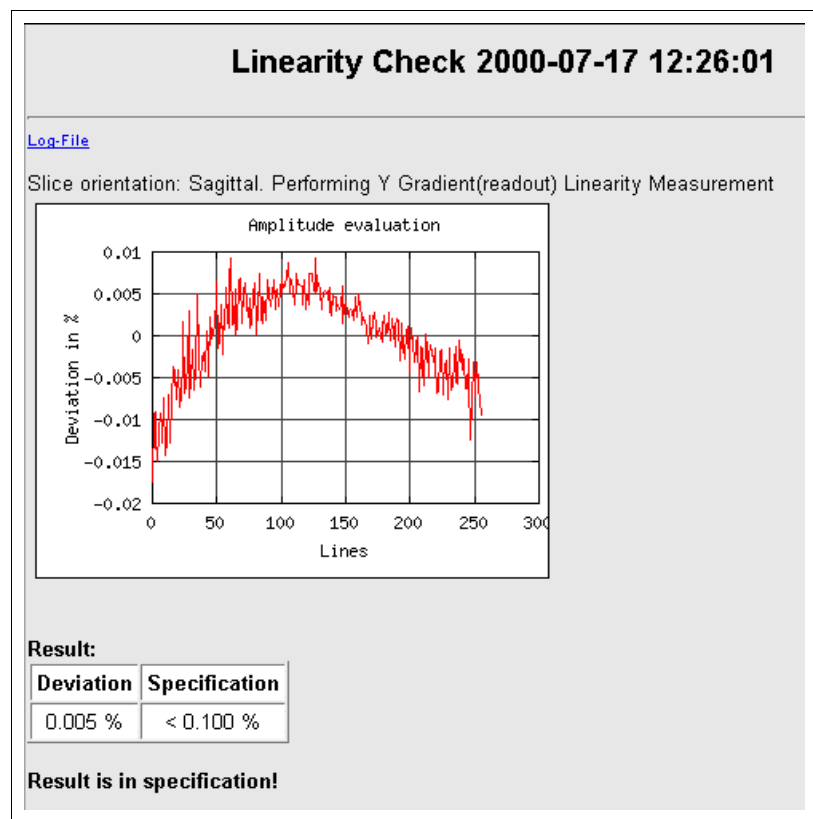
NOTE Refer also to the on-line help concerning QA/Linearity check.

Test results for gradient linearity/Graphical HTML Output

The numerical results of the gradient linearity tests and a corresponding graph will be displayed in the HTML window after the tests have been completed. A label identifying the test including slice orientation and gradient axis is printed out. If the result is out of specifications, the respective field in the table appears in color.

Each point of the graph represents one amplitude value of the gradient axis tested (line 0-255). The deviation value means the deviation of the integral value of the gradient pulse from the nominal value, relating to the integral value of the maximum gradient pulse.

If a non-constant pattern is seen, especially at the beginning or end of the plots (near line 0 or 255), this may be an indication of a problem with the linearity of the gradient output at high amplitudes.



Log-File (section of the HTML output page)

Clicking **Log-File** in the HTML output page lets you open another window displaying all measured maximum gradient amplitudes and some additional information. The example below shows only a section of this window.

```

Line: 251      Maxround: 257.000      Maxinter: 257.044
Line: 252      Maxround: 257.000      Maxinter: 257.054
Line: 253      Maxround: 257.000      Maxinter: 257.047
Line: 254      Maxround: 257.000      Maxinter: 257.042
Line: 255      Maxround: 257.000      Maxinter: 257.038

Results of maximum fit: Offset = 257.079956, Slope = -0.000070, stddev

End result: Deviation      0.0051 %
Specification:              0.1000 %

Result is in specification!
Slice orientation: Coronal.
Performing Z Gradient(readout) Linearity Measurement

Selected Coil Elements: 1: CP_BodySpine BS_290 | 2: CP_BodySpine BS |
Set slice for coronal orientation (Readout gradient: Z).

Series Loid: Local 4.0.247233

line #1 in Z has Max Amplitude of 1102.42 and SN of 32.84
line #2 in Z has Max Amplitude of 1106.64 and SN of 32.49
line #3 in Z has Max Amplitude of 1097.59 and SN of 31.77
line #4 in Z has Max Amplitude of 1054.84 and SN of 31.71
line #5 in Z has Max Amplitude of 1042.13 and SN of 31.57
line #6 in Z has Max Amplitude of 1065.39 and SN of 31.79
line #7 in Z has Max Amplitude of 1084.70 and SN of 31.66
line #8 in Z has Max Amplitude of 1008.38 and SN of 30.88
line #9 in Z has Max Amplitude of 1002.20 and SN of 30.79

```

Conclusion

In case of an error, check the input voltage of the power stages and the stability of the main voltage. Swap power stages. Replace defective power stages. Check/replace defective SSUs.

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Imager

NOTE This section describes all MR systems.

The purpose of the Imager tests is to find the error causes in the Imager and the raw data link from the RX4 to the Imager.

NOTE Do not abort the Imager tests, the Imager may remain in an undefined state (simulation mode) where only a power off can help.

General

The imager tests include communication, raw data handling, image calculation and feedback loop Imager-MMC (timing-critical).

All Imager tests are **functional tests**.

Test sequence

Initialization	MMC_IMAGER_Init
Test 1: Data Transfer	RX4 - Imager (Raw data link)
Re - Initialization	RX4_ReInit
Test 2: Data acquisition	Data acquisition (PCI-Receiver)
Test 3: Image Calculation	Image Calculation (Source: File)
Test 4: Image Calculation	Image Calculation (Source: Receiver)
Test 5: Image Calculation	Image Calculation and Transfer to Host
Feedback Loop Imager-MMC	Feedback Loop Imager-MMC (Performance)

Initialization

This step initializes the MMC and the Imager for the tests.

Fig. 119 MMC_IMAGER_Init

MMC_IMAGER_Init

```
[Wed Apr 03 14:55:15]
Start MMC_IMAGER_Init ... nucleus 1H ...
preparation ...measurement ... ok (series-loid: Local 4.0.34119874)
```

Test 1: Data Transfer

This step checks if the fiber optics link between RX4 in the MMC to the PCI-Receiver in the Imager is OK.

Fig. 120 Imager Test 1

Imager Test 1

[Wed Apr 03 14:55:28]

Simulation Measurement

Starting step *Data transfer*
 Error: ttf_SFMeasIF::start() failed: Scanning stopped by Image Reconstruction.



In case of an error, one or more of the following units might be defective:

- RX4
- Fiber optics link between MPCU and Imager
- PCI-Receiver
- Memory in Imager

Re - Initialization

This step re-initializes the MMC and the Imager for the tests.

Fig. 121 Imager Test 2

RX4_ReInit

[Wed Apr 03 14:39:58]

Start RX4_ReInit ...call (long)fSendDataRX412RawLinkToBR 0 4 0 ... ok



Test 2: Data acquisition

PCI-Receiver and Memory in Imager are tested.

Fig. 122 Imager Test 2

Imager Test 2

[Wed Apr 03 14:39:59]

Simulation Measurement

Starting step *Data acquisition*

Starting comparison (1 set)

Comparing set no. 1 ... done

Test was **successful**



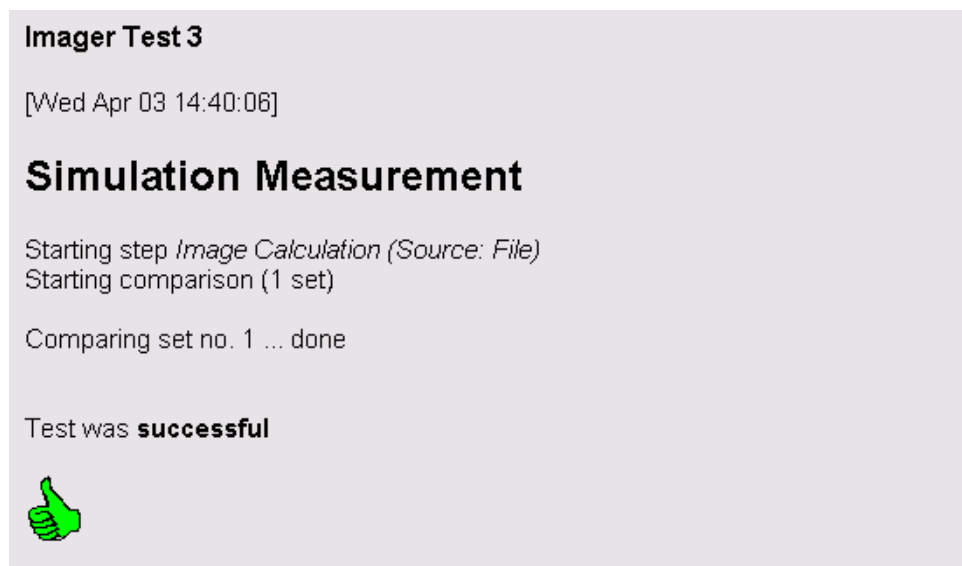
In case of an error, one or more of the following units might be defective:

- PCI-Receiver
- Memory in Imager

Test 3: Image Calculation

(Raw data source: File.)

Fig. 123 Imager Test 3



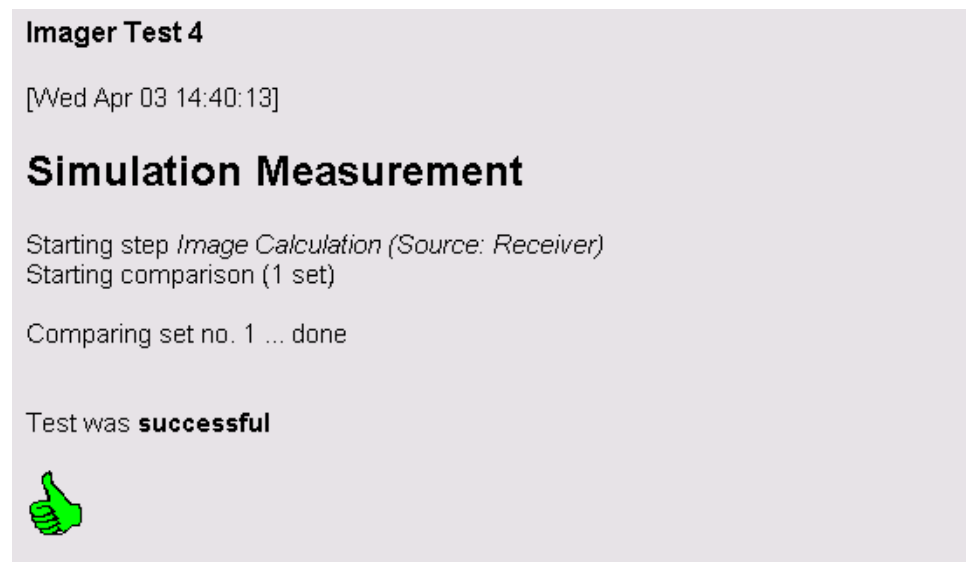
In case of an error, one or more of the following units might be defective:

- Memory in Imager
- Hard disk for preprocessed measurement data

Test 4: Image Calculation

(Raw data source: Receiver.)

Fig. 124 Imager Test 4



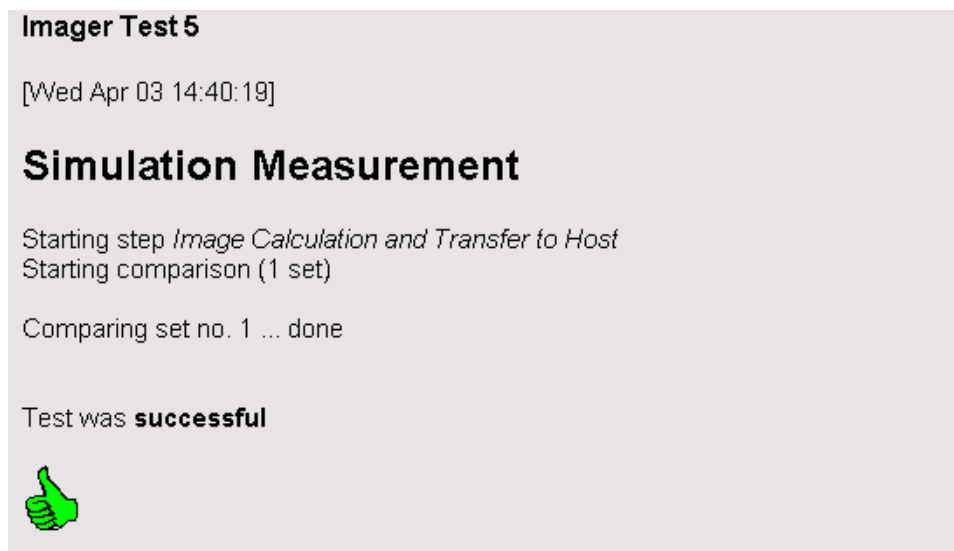
In case of an error, one or more of the following units might be defective:

- PCI-Receiver
- Memory in Imager
- Hard disk for preprocessed measurement data

Test 5: Image Calculation

(Source: Receiver.)

Fig. 125 Imager Test 5



In case of an error, one or more of the following units might be defective:

- PCI-Receiver
- Memory in Imager
- Hard disk for preprocessed measurement data
- Network performance problem between Imager and Host

Feedback Loop Imager-MMC

Feedback Loop Imager-MMC is a performance test, a normal feedback time result is **2.000 to 10.000 ms**.

Fig. 126 Imager Feedback Loop Imager-MMC

Feedback Loop Imager-MMC
 [Wed Apr 03 14:40:29]
 Start Feedback Loop Imager-MMC ... nucleus 1H ...
 preparation ...measurement ... ok

 Determined time for feedback: **7.000** ms

Feedback Loop Imager-MMC		
	Value [ms]	Specification [ms]
Feedback time	7.00	0 - 10.00

Result is in specification!



TestSuite completed

Prerequisite: Imager Tests 1-5 have been completed successfully.

In case of an error, one of following connections might be defective:

- Ethernet between Imager and MMC (performance problem)
- Fiber optics link between MPCU and PCI-Receiver (performance problem)

NOTE The test 'Feedback loop Imager - MMC' has a critically set time limit, this test might fail in 1 of 20 performed tests at Imagers without any defects!
 Before replacing hardware: reboot the Imager and repeat the **Feedback Loop Imager-MMC** test.

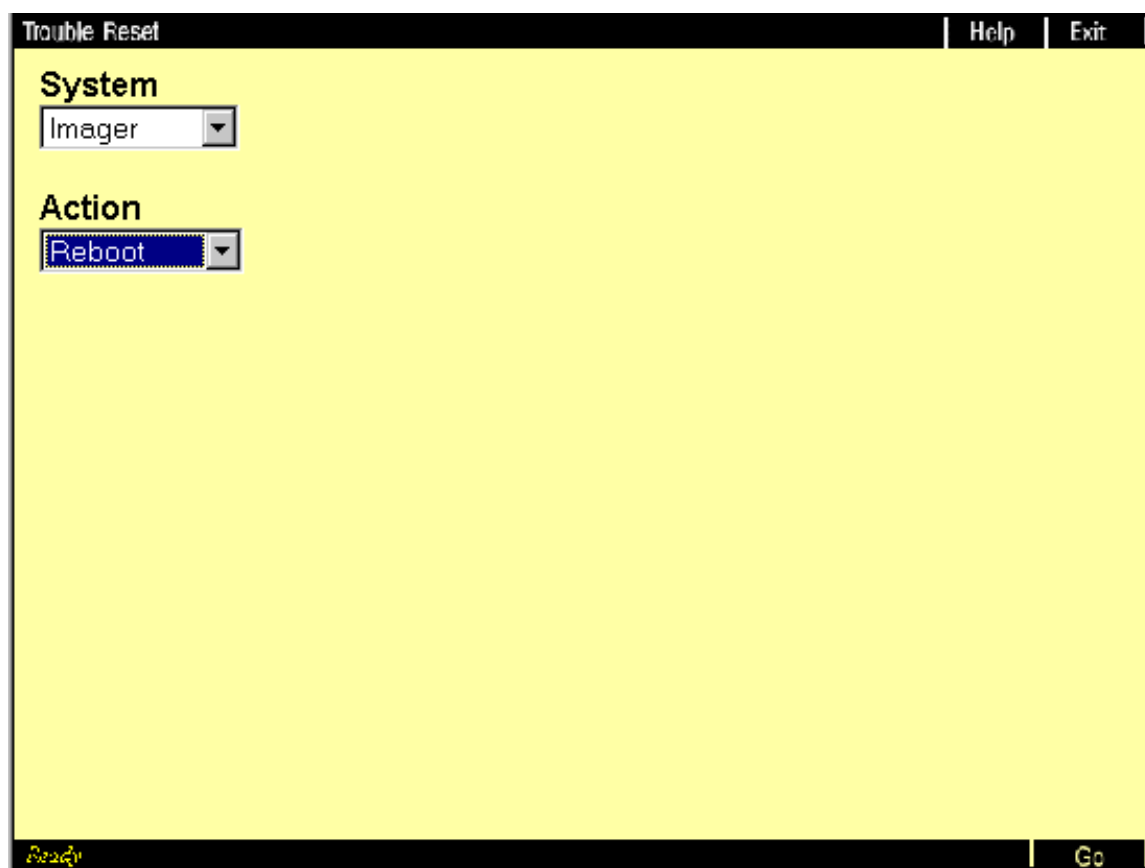
In case of errors

Before performing any other service action, reboot the Imager using the **TroubleReset**, Imager, Reboot function. This procedure is also possible when using RDIAG.

NOTE The reboot takes approximately 4 minutes.

Repeat the Imager Tests.

Fig. 127 Reboot Imager



NOTE For detailed troubleshooting information regarding the Imager, refer to TSG Imager.

Host

The purpose of the Host tests is to find errors in the network interfaces hardware or at the dongle.

General

The host test includes network tests of the both NIC's (measurement network NIC and customer network NIC).

The dongle is tested for intermittent read out errors trough multiple reads of the dongle ID.

Test sequence

The host tests are **functional and Interactive tests**.

- Note your dongle ID before you start the test, the dongle ID is needed for the interactive dongle test. (the dongle ID is labeled physically at the dongle)
- Select the check-box Host, Network and Dongle is then automatically selected.

The screenshot shows a software interface for configuring tests. At the top, there are three checkboxes: 'Host', 'Network', and 'Dongle', all of which are checked. Below these, there is a 'Testtype' section with four checkboxes: 'Functional' (checked), 'Comm/Status' (unchecked), 'Interactive' (checked), and 'Multi' (unchecked). At the bottom, there is a status bar with the following elements: 'Status' followed by a text input field, 'Timer' followed by a numeric input field containing '0', 'Estimated Time' followed by a time input field containing '00:10:02', and three buttons: 'Go', 'Abort', and 'Re'.

Network Dongle

The internal hardware communication is tested to the NIC's
Test of multiple reads of the dongle ID

Network

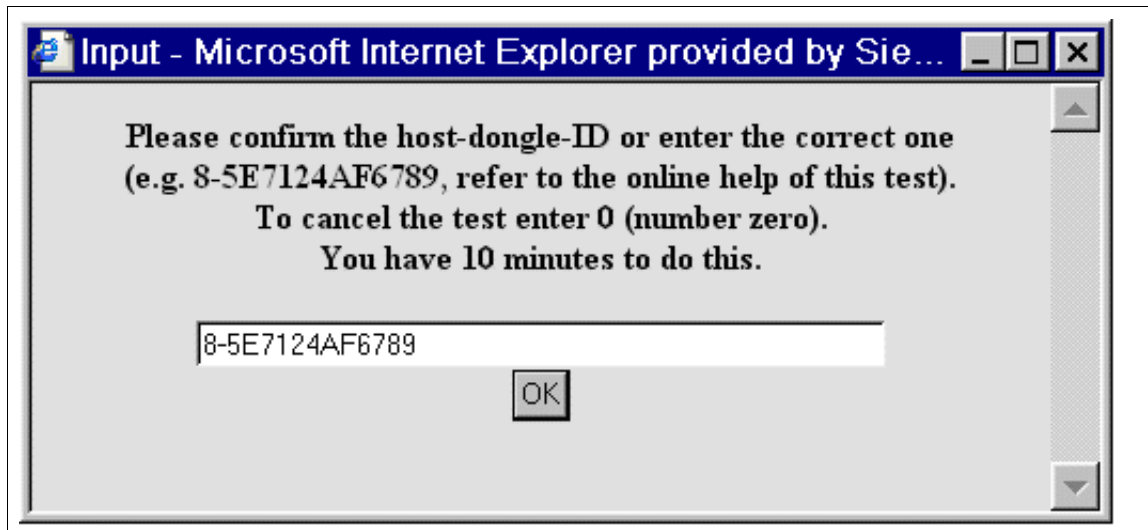
This test displays global TCP/IP and Net BIOS parameters that are read from the NIC's (**N**etwork **I**nterface **C**ard).

TestTools		Exit	
TestTools			
Network Functional Host Network OK 2003-09-16 17:11:38-01:00 2003-09-16T17:11:39-01:00			
Dongle Interactive Host Dongle OK 2003-09-16 17:11:39-01:00 2003-09-16T17:21:54-01:00			
Testsuite completed			
TestTools			
Network Functional			
Host Network			
Global TCP/IP parameters			
	Value		
Host name	MRC6		
Database path	%SystemRoot%\System32\drivers\etc		
Routing	0 (disabled)		
Global NETBT parameters			
	Value	Unit	
LMHOST name resolution	1 (enabled)		
DNS name resolution	0 (disabled)		
Name proxy server	0 (disabled)		
Name server port number	137 (89 hex)		
Name table size	1 (small)		
Name cache timeout	600000.000	ms	
Broadcast name query count	3.000		
Broadcast name query timeout	750.000	ms	
Name server query count	3.000		
Name server query timeout	1500.000	ms	
Session keep-alive time	3600000.000	ms	

Dongle

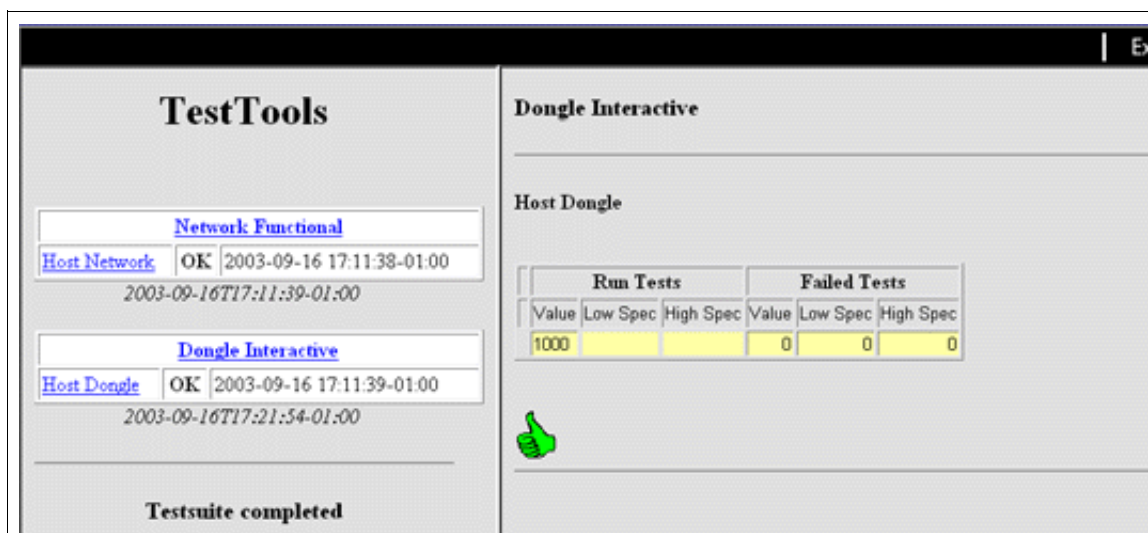
This test performs multiple reads of the dongle ID.

- Enter your dongle ID, click OK (To cancel this test enter 0, click OK).
Please note that the test itself is not case-sensitive and e.g. 8-5E7124AF6789 is treated the same like 8-5e7124af6789.



- The multiple dongle read out test will be repeated for 1000 times. In case an error is detected the test is aborted immediately as no read errors are allowed.
At the end of the test both the entered dongle ID and the read ID from the dongle is displayed. In this way mistakes in entering the dongle ID can be easier detected.

Result of dongle test



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Magnet Supervision

All systems

NOTE This section describes all MR systems except the MAGNETOM Concerto.

MSUP Communication

During the communication test, a command is sent to the MSUP and to the SHIM power supply (if installed). Successful completion of this test assures satisfactory communication with these components.

The communication test returns the following messages:

- MSU Communication "OK" or "FAILED"
- SHIM Communication "OK" or "FAILED"

If the test "failed", check the cable connections and the communication board with the CAN module.

MSUP Status

During the status test, the status of the MSUP and the SHIM power supply is queried. The status test is a fast and reliable way to determine whether the self-test of the respective component failed or not.

This test provides you with a listing of the present status of the MSUP and the SHIM including all possible "ok/not ok" conditions which will be recognized and stored by the local application software of the MSUP and the SHIM CAN-unit.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status will therefore be "NOT OK" .

MAGNETOM Concerto

NOTE	This section describes the communication test with the Magnet Temperature Unit (MTU) for the MAGNETOM Concerto.
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MTU Communication/Status

During the communication test, a command is sent to the MTU (Magnet Temperature Unit). Successful completion of this test ensures satisfactory communication with this component.

During the status test, the status of the MTU is queried. The status test is a fast and reliable way to determine whether the self test of the respective component has failed or not.

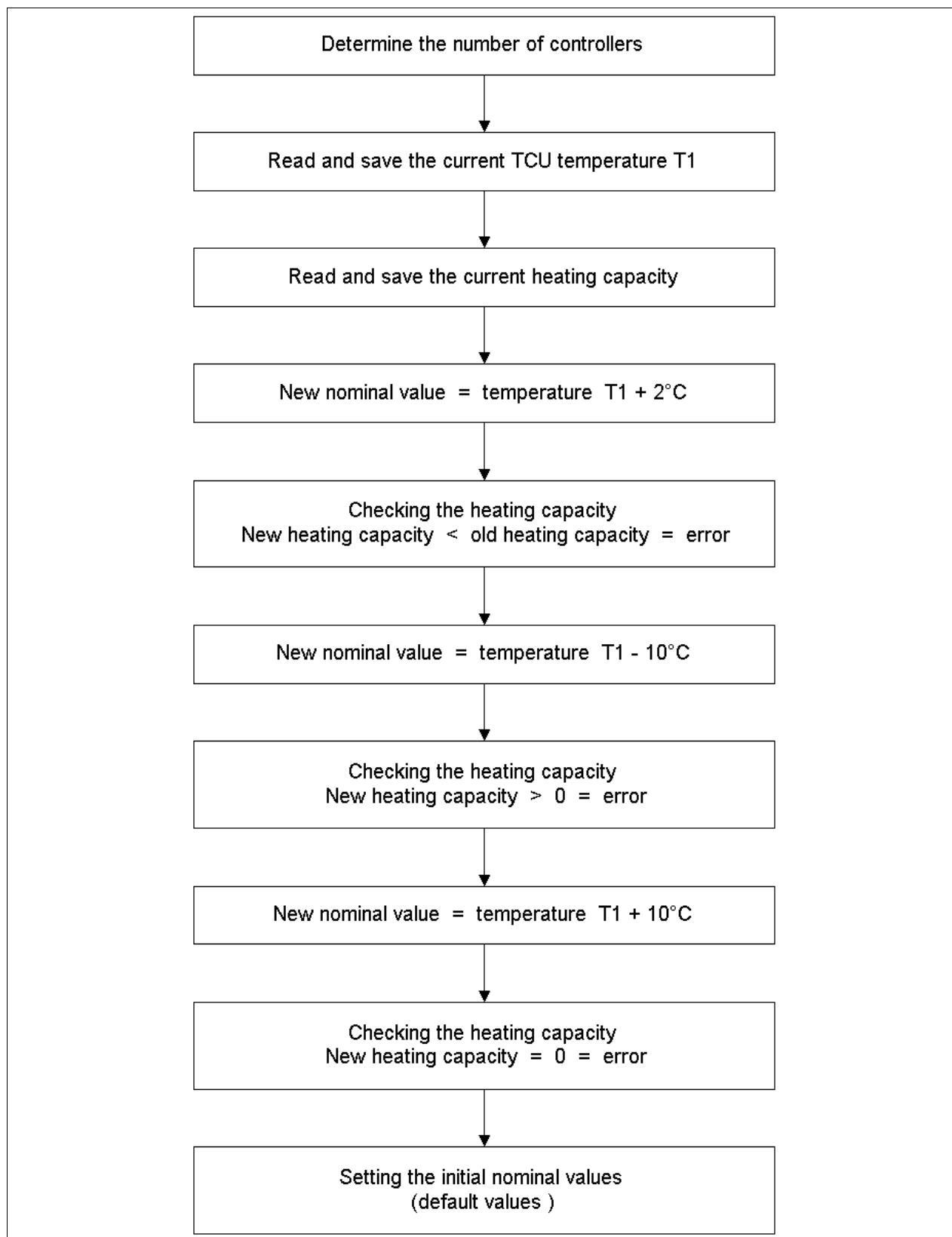
MTU Test

This test is used to check whether the regulators are working correctly or not.

This is an interactive test that requires you to monitor the TCU controller display inside the electronic cabinet (ECA).

The nominal values of all four controllers will be changed simultaneously during the test to higher and lower values and finally back to the original (actual) settings. The sequence is +2, -10 and +10 degrees.

Check the "set nominal values" at the TCU and compare to the default values under SESO Magnet & Cooling Initialization after the test.

Test sequence:*Fig. 128 Test sequence of MTU Test*

RCA

All systems

NOTE This section describes all MR systems except the MAGNETOM Concerto.

RCA Communication

During the communication test, a command is sent to the RCA. Successful completion of this test assures satisfactory communication with this component.

The communication test returns the following message:

- RCA Communication "OK" or "FAILED"

If the test "failed", check the cable connections and the communication board with the CAN module.

RCA Status

During the status test, the status of the RCA is queried. The status test is a fast and reliable way to determine whether the self-test of the respective component failed or not.

This test provides you with a listing of the present status of RCA including all possible "ok/not ok" conditions which will be recognized and stored by the local application software of the RCA CAN-unit.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status will therefore be "NOT OK" .

PMU

The PMU tests consist of a communication/status request and a transmit (interactive) test.

All systems

NOTE This section describes all MR systems except the MAGNETOM Concerto.

PMU Communication

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

PMU Status

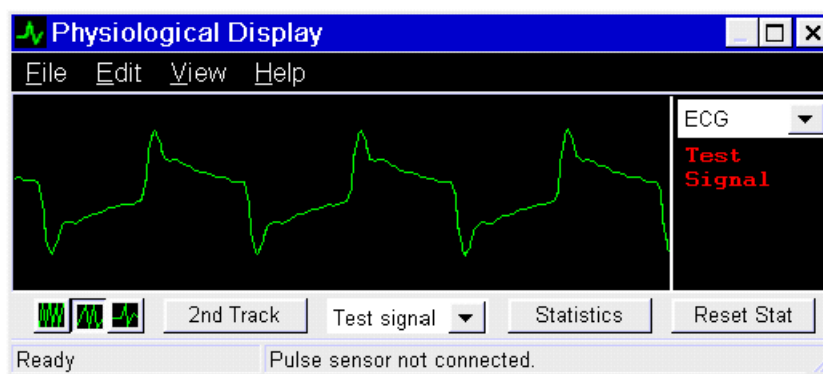
This test provides you with a listing of the present status of the PMU, i.e. the supply voltage, data receiving, memory tests, any leads and sensors are connected, including all possible "ok/not ok" conditions which will be recognized and stored by the local application software.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status will therefore be "NOT OK" .

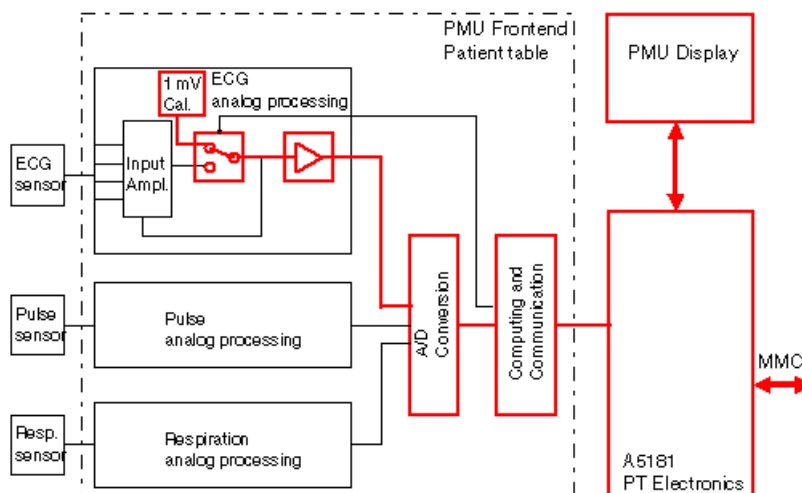
PMU Transmit (interactive)

A physiological submenu is displayed on the PMU display at the magnet front and on the physiological display at the console. After a learn phase, the test signal will show on the PMU display and on the physiological display. In addition, you will hear a beep sound from the intercom (if activated with the trigger switch on the left side of the intercom).



Signal Path

At the PMU Frontend, a 1 mV test signal is generated and sent to the MPCU Box via the electrical/optical converter on the PMU Converter.



Evaluation

If both pulse and beep sound appear, the message window should be acknowledged by clicking **OK**.

MAGNETOM Concerto

NOTE This section describes the MAGNETOM Concerto only.

PMU Communication

The communication test involves sending a command (status and version query) to an individual component which, if successfully executed, verifies communication with this component.

PMU Status

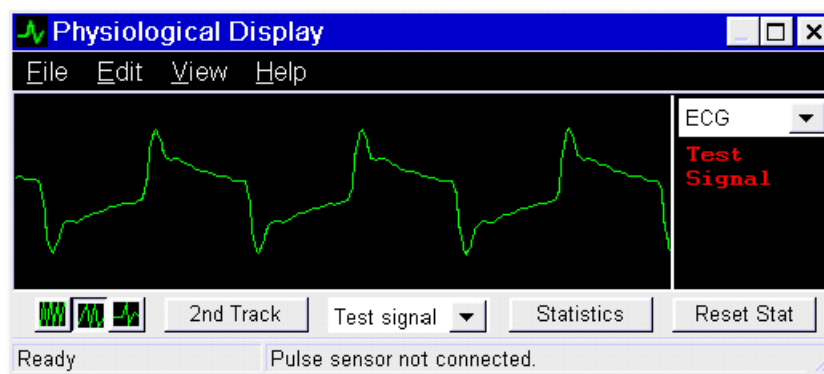
This test provides you with a listing of the present status of the PMU, i.e. the supply voltage, data receiving, memory tests, any leads and sensors are connected, including all possible "ok/not ok" conditions which will be recognized and stored by the local application software.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status will therefore be "NOT OK" .

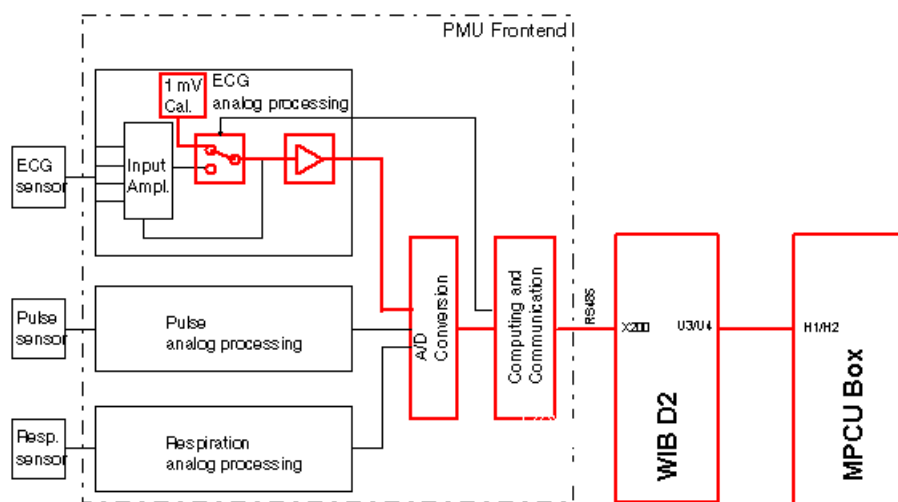
PMU Transmit (interactive)

A physiological display on the console will show the test signal after a short learning phase. In addition, you will hear a beep sound from the intercom (if activated with the trigger switch on the left side of the intercom).



Signal Path

At the PMU Frontend, a 1 mV calibration pulse is generated and sent to the MPCU Box via the electrical/optical converter on the wiring board WIB.



Evaluation

If both pulse and beep sound appear, the message window should be acknowledged by clicking **OK**.

Conclusions

This test checks the signal path from the PMU Frontend to the MPCU Box and the signal path from the MPCU Box to the intercom.

Patient Table

All systems

NOTE	This section describes all MR systems except the MAGNETOM Concerto.
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The patient table test consist of a communication/status request only.

PTAB Communication

The test provides you with a listing of the present status of the selected component, i.e. all possible error/status conditions, which will be recognized and stored by the local application software of the component CAN-unit.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status of the communication test will therefore be "NOT OK".

PTAB Status

This test provides you with a listing of the present status of the patient table, i.e. the power supplies, switch status, movement control, display and control panels, including all possible "ok/not ok" conditions which will be recognized and stored by the local application software of the CAN-unit.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status will therefore be "NOT OK" .

NOTE	If no PTAB status is reported, check the CAN-bus connection to the patient table controller.
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MAGNETOM Concerto

NOTE This section describes the MAGNETOM Concerto only.

The PTC tests consists of a communication/status request and an interactive test.

PTAB Communication

The test provides you with a listing of the present status of the selected component, i.e. all possible error/status conditions, which will be recognized and stored by the local application software.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status of the communication test will therefore be "NOT OK".

PTAB Status

This test provides you with a listing of the present status of the PTC.

In case the status is "not ok", perform a "fault" search for the corresponding error message in the system report. For remedy actions, refer to the Troubleshooting Guide.

An "error" status is identified by 3 exclamation marks to the right of the status message. The overall status will therefore be "NOT OK" .

PTAB interactive tests

As the table of the MAGNETOM Concerto does not include any motor driven components, also no test functions are available. Instead, just the display at the magnet (DTP, Display Tune Panel) is tested. The test requires you to push some buttons at the DTP and monitor the DTP itself. The sequence of the single test steps will be displayed at the DTP.

The single tests steps are started 20 seconds after invoking the test so that there is enough time to walk to the magnet.

Test sequence	Text at the DTP
Set Startup Text	"Starting Display Test"
Set Text	"1234567890ABCDEF"
Set Flashing Text	"Flashing Text"
Test Display	All 16 digits are on
Key Test	"Press POS"
Key Test	"Press AT"
Key Test	"Press SEL"
Key Test	"Press LAMP"

Possible error sources are the DTP, WIB (wiring board), PTC (patient table controller) and the cabling including the intercom.

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